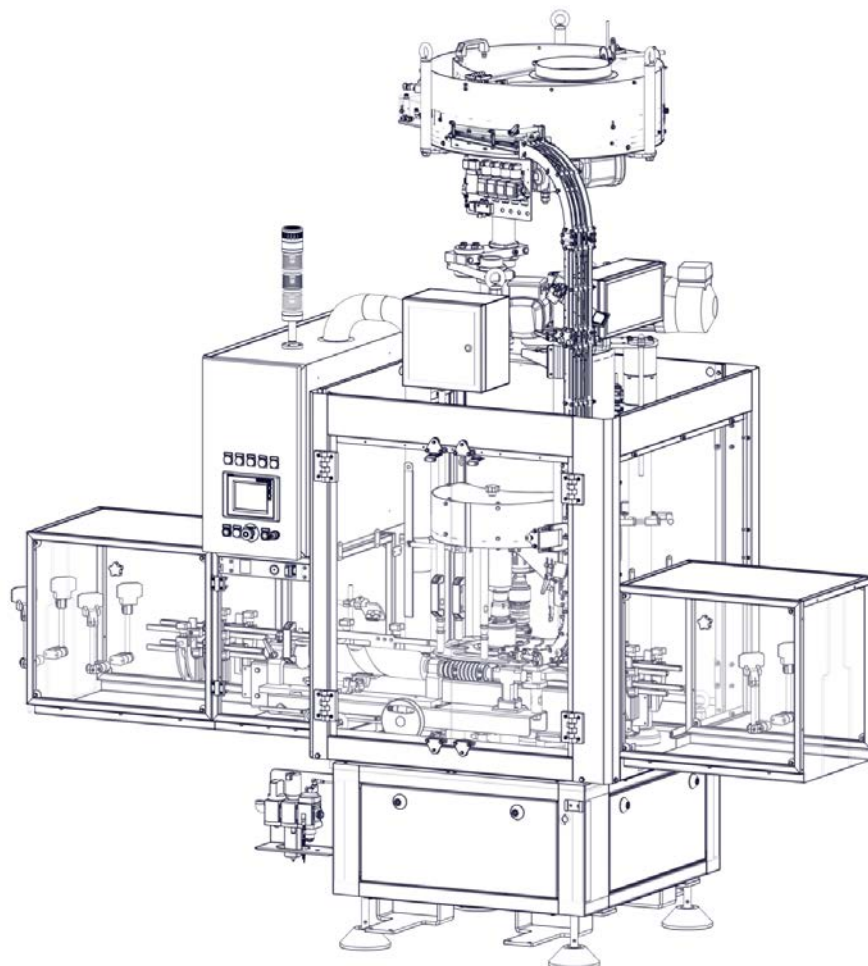




Model
M - EURO VP - IES

Serial number
17478

Pitch diameter
288 mm.

Nr of heads
3

Manufacturing year
2019

USE AND MAINTENANCE

TRANSLATION OF THE
ORIGINAL INSTRUCTIONS

Code
Z17478ABSEN001

Revision
0



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BEFORE START UP THE MACHINE, READ THE TECHNICAL INSTRUCTIONS IN THIS MANUAL AND FOLLOW THE INDICATIONS SHOWN.

KEEP THE USE AND MAINTENANCE MANUAL IN PLACE KNOWN AND ACCESSIBLE TO OPERATORS AND THE SERVICE PERSONNEL.



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1 PURPOSE OF THE MANUAL - EDITION - UPDATING METHOD

The manual is integral to the machine M - EURO VP - IES, serial number 17478 model; it should be kept all the life of machine; It must accompany the same machine when it is moved to other premises or if it sold.

It is in Italian and translated in target language whenever required.

In case of doubts about the interpretation of the text, always refer to the original language and contact Customer Service in case of need:

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The text of the manual will refer to the "closing machine" as an abbreviation of the standard name **"Closing machine which uses a closure"** as stated in the UNI EN 415-1 standard (Terminology and classification of packaging machines).

The manual contains information and instruction about all the operating cycle of the closing machine:

- Transport and moving.
- Installation.
- Start up.
- Use.
- Size change.
- Maintenance.
- Dismantling and disposal.

The manual should be kept so that:

- avoid any damage.
- Arrange for its availability to the professional operators working on the closing machine.
- Prevent it from being modified or damaged during the use.
- Make it available in case of need (training, changed ownership of the closing machine, etc.)



Ask for a replacing copy if damaged.

This manual was written with care; if some of the sections needs to be reviewed to add information or instructions, or simply for clarity reasons, *AROL S.p.A.* will proceed as follows:

- *should the modification involve only a few pages*: the replacement pages will be sent.
- *should the modifications involve more pages of different chapters*: all the chapters involving modification will be sent again.
- *should the modification involve many chapters* : the whole replacement manual will be sent.

Should the manual be supplied in electronic form, the supplied manual will always be the complete and updated one.

The consultation of this manual is made easier by the addition of the initial pages of the general table of contents, for an immediate tracking of the topic concerned.

The chapters are organised a hierarchy structure for a fast tracking of the info required.



The manual shows graphic symbols that are used to distinguish:

- Informations and prescriptions.
- Obligations.
- Warning about residual risks.

The following documents are attached to this manual:

- Electrical diagram.
- Pneumatic diagram.
- Closing machine installation diagram.
- Identification tables for size change.
- Instructions manuals for accessories and/or equipment installed on the closing machine and not integrated in this manual.



Before any operation with or on the closing machine read the procedures and warnings herein carefully.



IMAGES OF ANY GENERAL KIND ARE NOT USED IF THE DETECTION OF SPECIFIC STRUCTURE OF CLOSING MACHINE IS ESSENTIAL IN THE CONTEXT TO THE SUBJECT MATTER. The reported instructions and use modes for all described components anyway remain valid.



2 IDENTIFICATION OF THE CLOSING MACHINE

The closing machine has an identification plate, placed in the rear part of the base (See Figure).

On the plate are reported:

- Closing machine model (**MODEL**)
- Type of closing machine (**TYPE**)
- Pitch diameter (**P.D. in mm**)
- Number of closure elements (**HEADS**)
- Serial number (**SERIAL N.**)
- Manufacturing year (**YEAR**)
- Marking (**CE**)

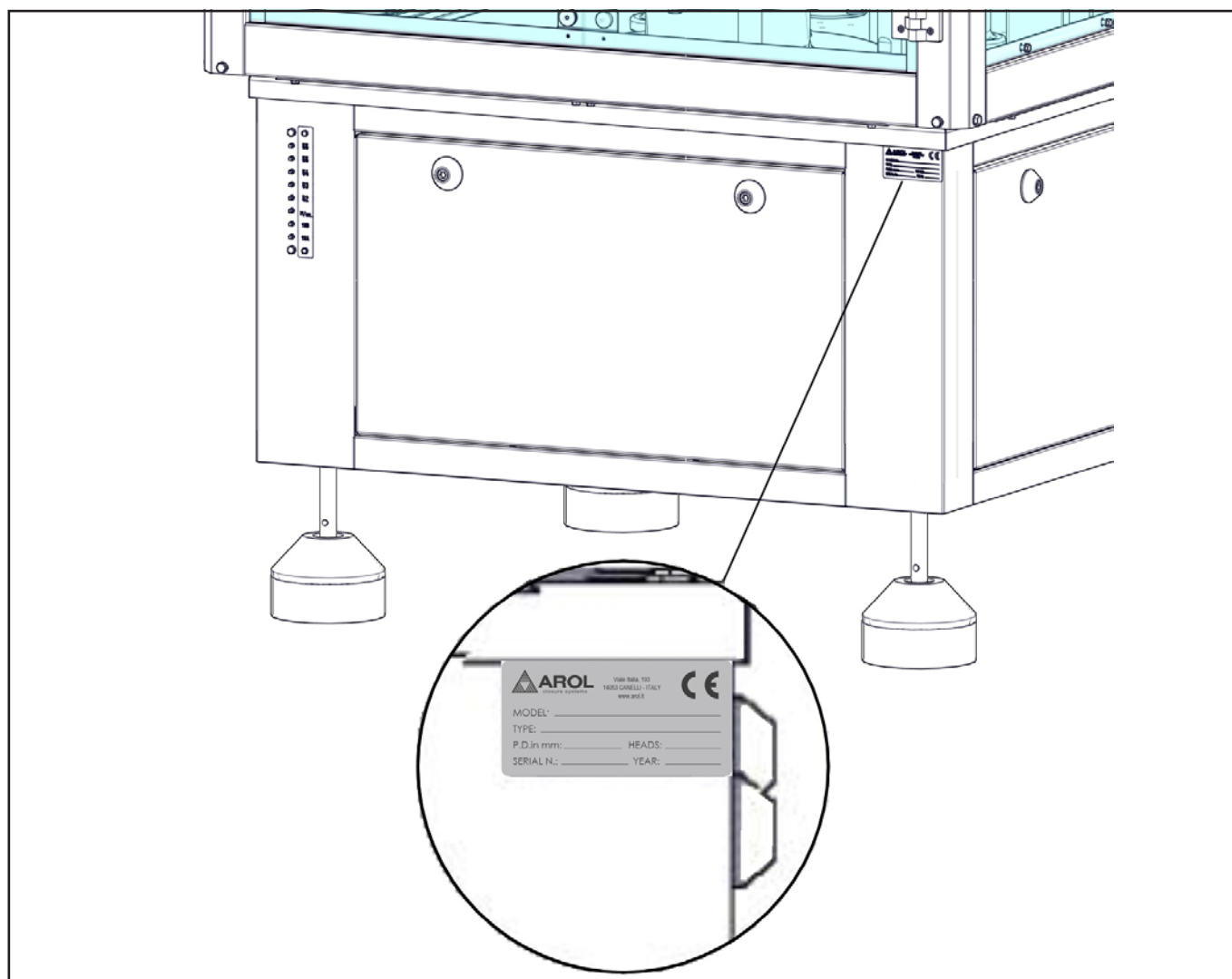


Fig. 1

The data plate is applied with 4 rivets and should not be removed for whatever reason.

In case of damage, ask for a spare copy to *AROL S.p.A.*.

Clean with neutral detergents paying attention in order not to damage the warning instructions.



3 DECLARATION OF CONFORMITY

The closing machine described in this manual is defined as "Machinery" since it is realized in accordance with the relevant EU Directives, applicable at the time of its marketing. *AROL S.p.A.* certifies the compliance with the European Directive 2006/42/EC ANNEX II A.

The original *DECLARATION OF CONFORMITY CE* is kept at the *AROL S.p.A.*. A true copy of the *DECLARATION OF CONFORMITY* is delivered attached to the documentation of the closing machine and must be kept up to its disposal.



The *AROL S.p.A.* highlights that such declaration is relative **exclusively** to the machine described above, in the state and with the use declarations for which it was marketed, excluding any parts added and/or operations carried out subsequently without the permission of the manufacturer.

In particular, any modification or replacement of parts of the closing machine as well as, by way of example, changes to the software of programmable components, not authorized by *AROL S.p.A.* as well as excluding any liability of the manufacturer, may cause the loss of validity of the CE declaration of conformity of the machine.



Below you can find the copy of the *DECLARATION OF CONFORMITY*.



EC DECLARATION OF CONFORMITY
(Attachment II A of Directive 2006/42/EC)

The manufacturer

AROL S.p.A.
VIALE ITALIA 193
14053 CANELLI (AT) - ITALY

DECLARES

under his responsibility that the new machine described below:

Model: CAPPING MACHINE

Type: EURO VP IES

Serial no.: 17478

Construction year: 2019

suitable to apply caps to rigid preformed containers is compliant with the following directives:

2006/42/EC
2014/30/EU

The person authorized to prepare the relevant technical construction file is:

AROL S.p.A. - Viale Italia 193, 14053 CANELLI (AT) - ITALIA

Canelli, 05/04/2019



CIRIO Sergio
Managing Director

TRANSLATION OF THE ORIGINAL STATEMENT

AROL S.p.A. - Viale Italia, 193 - 14053 CANELLI (AT) - ITALY
Tel.+39 0141 820.500 - Fax +39 0141 820.555 - www.arol.it



4 CLOSING MACHINE DESCRIPTION

The closing machine with pre-threaded cap EURO VP - IES has been designed and built for the application of pre-threaded caps (namely screw caps) on bottles/containers.

4.1 MAIN UNITS OF THE CLOSING MACHINE

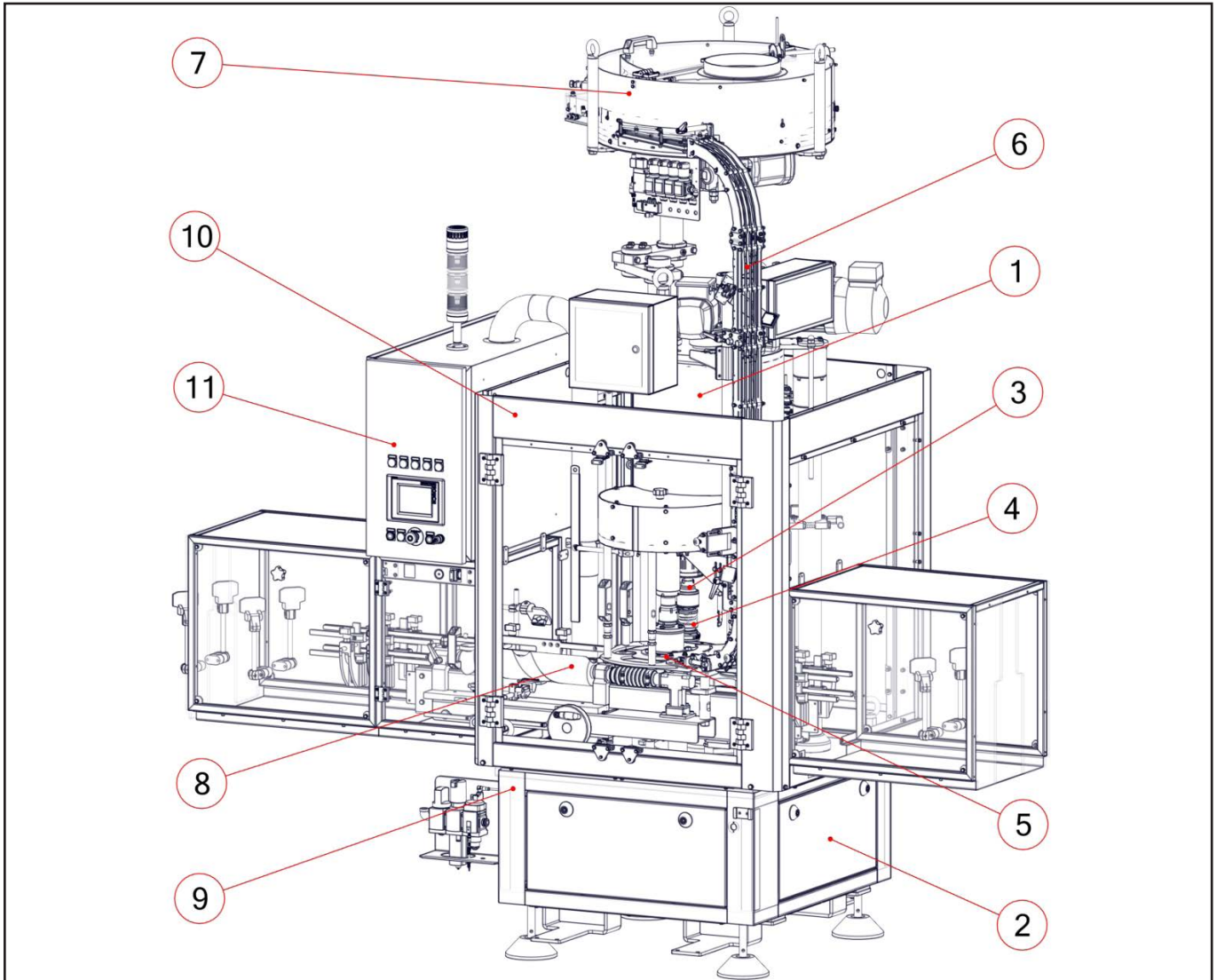


Fig. 2

Legend:

- 1 Upper closure carousel.
- 2 Lower closure carousel.
- 3 Closure heads.
- 4 Closure gripper.
- 5 Transfer disk for closure systems.
- 6 Chute for closure systems.
- 7 Closure system sorter.
- 8 Container conveying equipment.
- 9 Closing machine basement.
- 10 Accident prevention guards.
- 11 Electric board and control panel.



The components forming the closing machine are located inside the main assemblies:

- Closure heads.



Fig. 3

- Closure gripper.

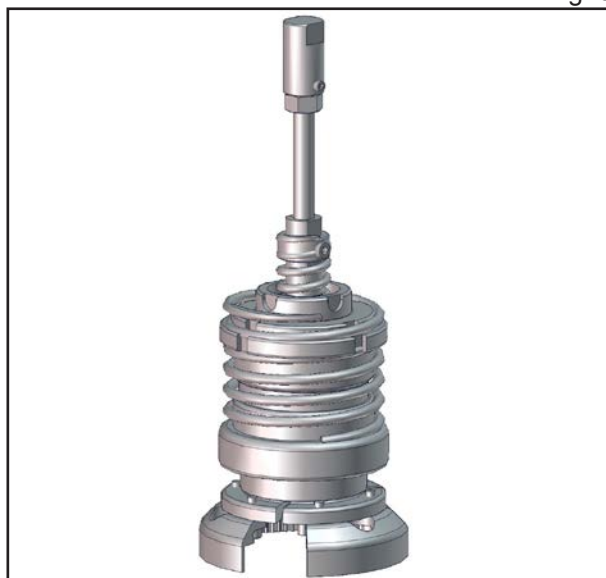


Fig. 4

- Height electrical adjustment of the closure carousel.

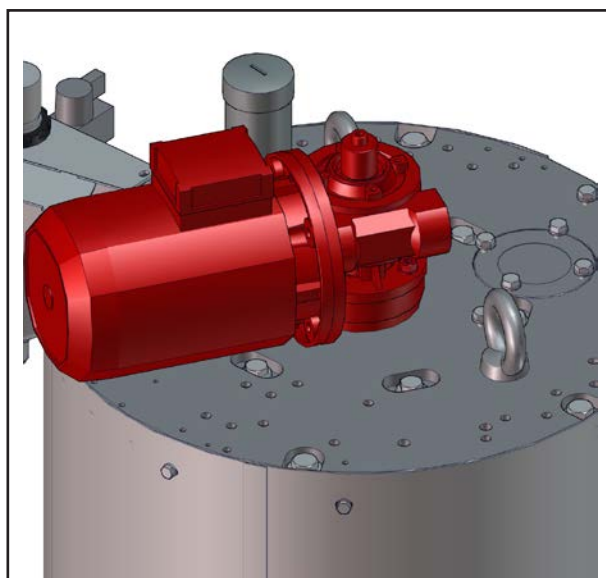


Fig. 5

4.2 DESCRIPTION OF THE PRODUCTION CYCLE

The application and sealing of closure systems on the containers is the following:

- An orientation device selects the caps inside the hopper.
- A transport system, conducts the caps from caps sorter to transfer disk.

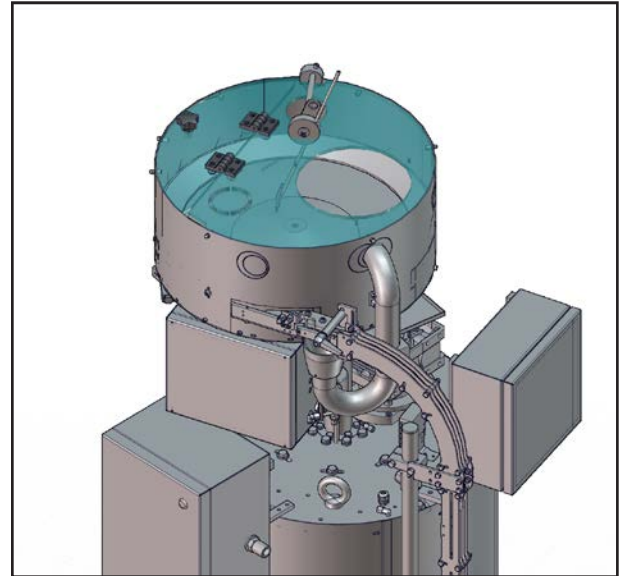


Fig. 6

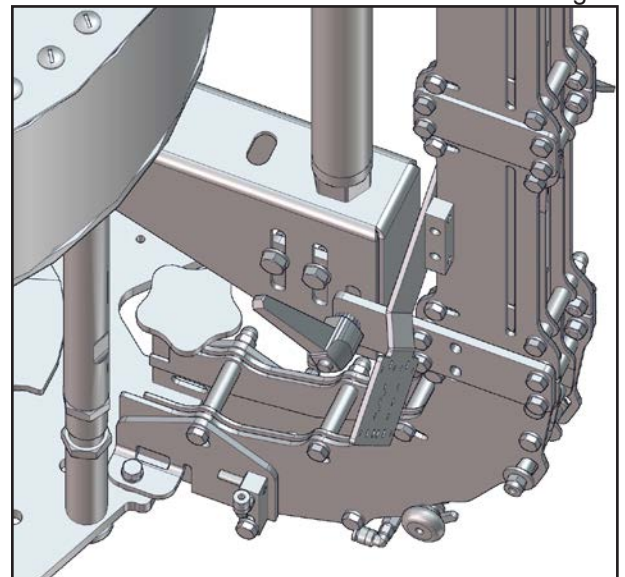


Fig. 7



- A transfer disk, called "pick & place", carries the caps under the closure head.

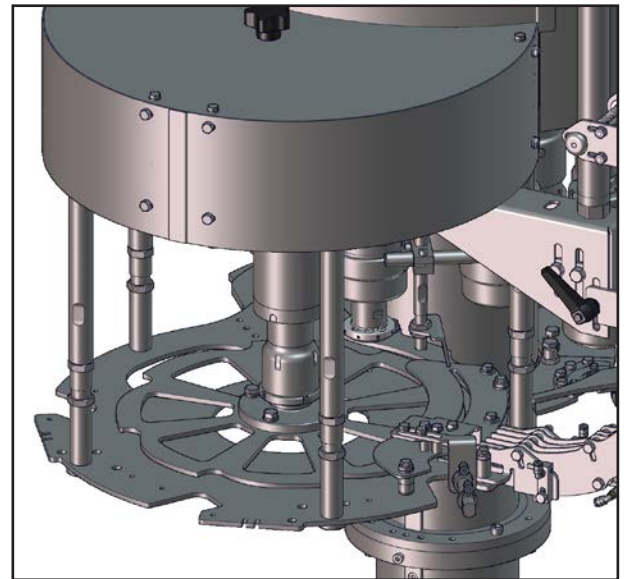


Fig. 8

- The closure head, follows the profile of a cam and picks the cap from the transfer disk by a mechanical gripper.

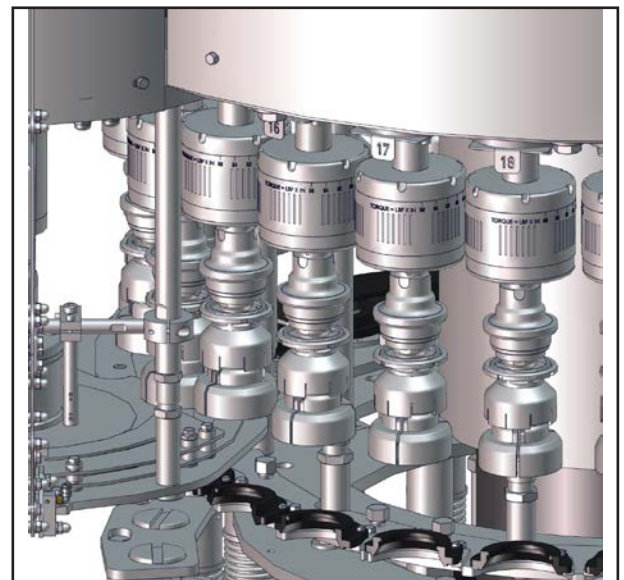


Fig. 9

- The containers are driven by a conveyor belt inside the closing machine.

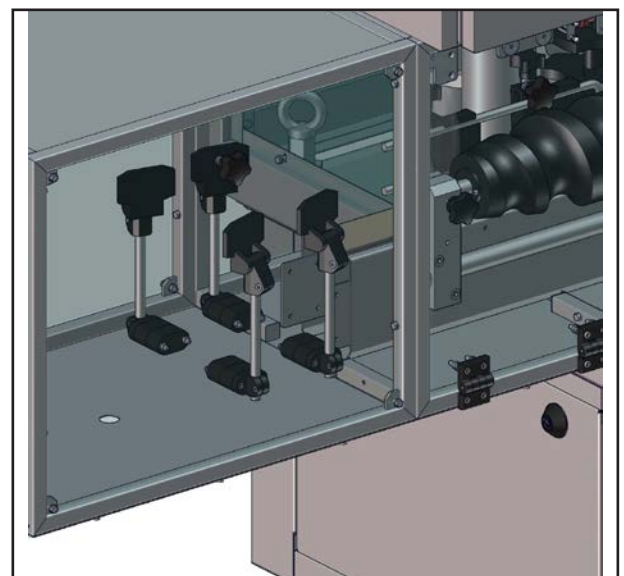


Fig. 10



- The rotation of the entry star-wheel conveys the bottles on the closure carousel where they rotate till being under the closure head, that provides to the application of the cap on the container. Then, the application on the containers follows, by screwing the cap.

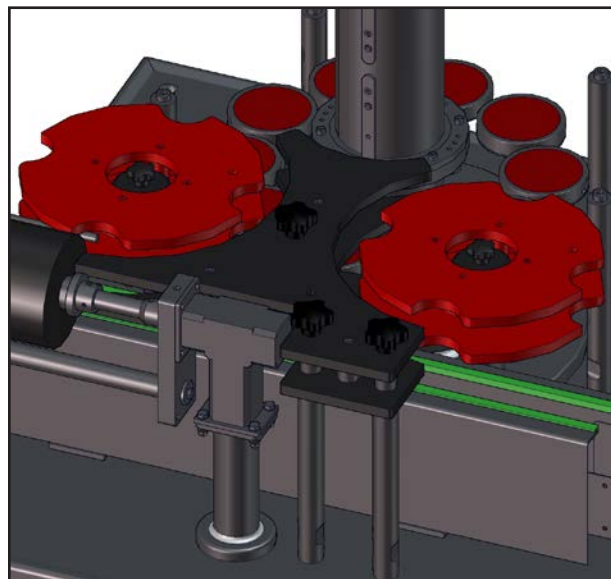


Fig. 11

- After the closure, the exiting container is released, then there is a reset for the following closure.

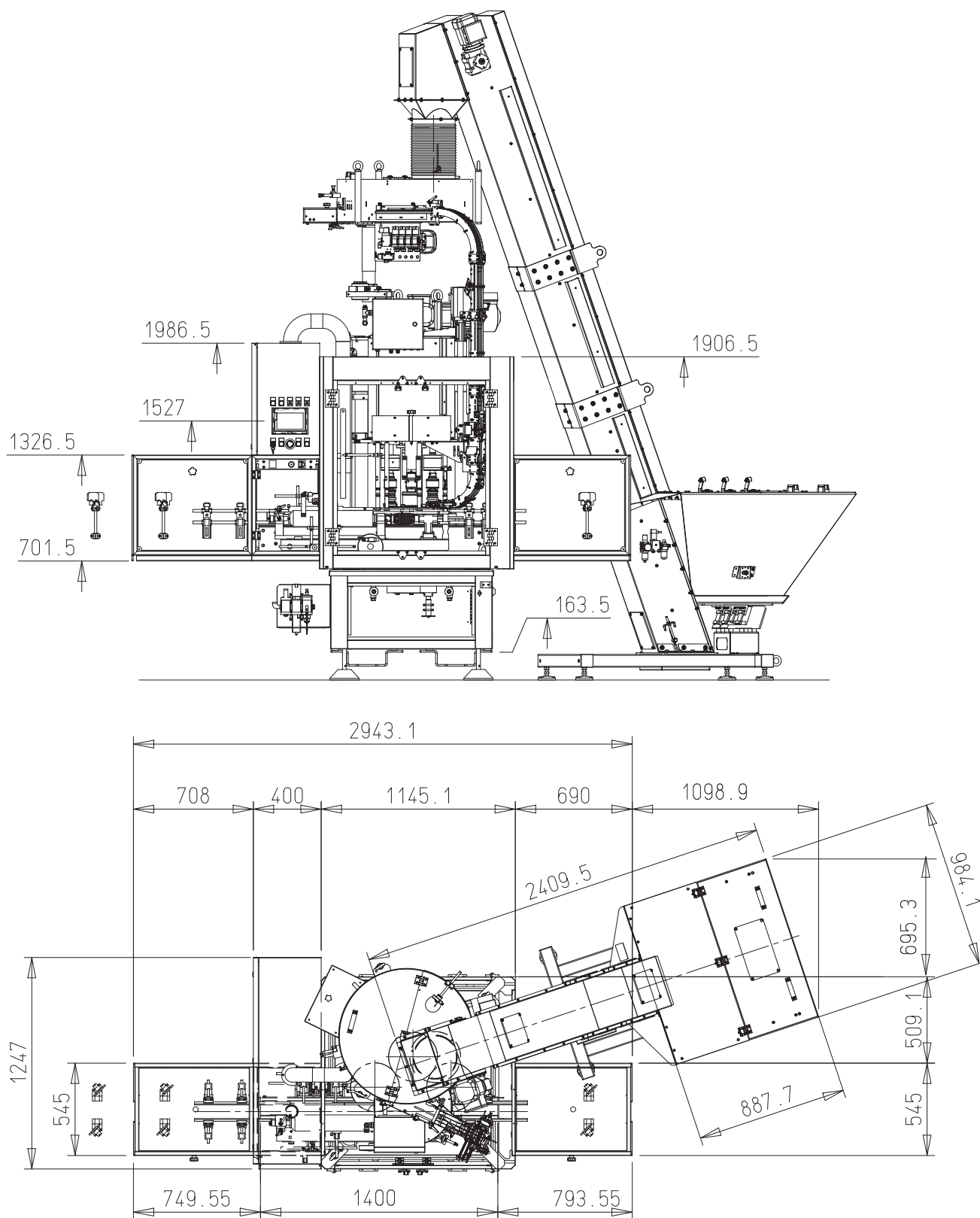


4.3 CLOSING MACHINE TECHNICAL DATA

TECHNICAL SPECIFICATIONS		
Model	EURO VP - IES	
Serial number	17478	
Pitch diameter	288 mm.	
Nr of heads	3	
Direction of rotation	Clockwise	
Manufacturing year	2019	
NOMINAL PRODUCTION		
Pieces/hour	4500	bph
CLOSING MACHINE WEIGHT		
Weight	1450	kg.
ELECTRICAL DATA		
POWER SUPPLY VOLTAGE:		
main	220	V ~
	60	Hz
auxiliary	24	V =
Installed power:		
Electric board	0,40	kW
closing machine rotation motor	2,20	kW
closing machine lifting motor	0,37	kW
heads rotation motor	1,50	kW
caps sorter motor	0,37	kW
Total installed power:	5,21	kW
PNEUMATIC DATA		
Sterile air capacity	200ca	l/min (1 bar 20°C)
Sterile air min. pressure	6	bar
Sterile air max. pressure	8	bar



LAYOUT DIMENSIONS





4.4 USE RESTRICTIONS

4.4.1 PRESCRIBED USE AND FORBIDDEN USES

The closing machine has been designed and made to apply closure systems (caps/corks) on preformed rigid containers with production limits indicated in chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 12.

The closing machine must be used to apply closure systems (caps/corks) on preshaped rigid container already produced according to the order, planning and building stages, as described in the tables attached to this manual.

The closing machine receives the closure systems correctly orientated by its orientation system.

Container charge and discharge operations are automatically performed by a conveyor belt.

Besides the conduction of the closing machine out of what underlined in the chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 12, just as an example and not limited to, the following uses are forbidden and not expected:

- use of closure systems (caps/corks) and/or preshaped rigid containers whose specifications are different to the ones described in the expected use.
- supply to the closure systems (caps/corks) and/or preshaped rigid hand containers or other means.
- revert the rotation direction of motors.
- tamper guards and/or protection devices.
- overlook the protection means using means to rise on the top or climb on parts of the closing machine.
- perform maintenance operations without insulating and locking all the energy sources and dissipating the residual energies.
- do not comply with the specifications of power sources listed in the table page 12.
- Treat products with characteristics other than those communicated at the order, changing the environmental conditions here the closing machine must be fitted (example: temperature of the treated product; with environmental characteristics other than those communicated at the order).

4.4.2 OPERATING MODES

The closing machine can operate, according to the situations, in automatic or manual mode.

Switching between the two operating modes is operated by the key switch with two positions *AUT/MAN*:



During the operation in *AUT* mode, the key mode must be removed from the control panel and kept far from the working place.



4.4.3 NUMBER OF OPERATORS

The operations described in this manual, referring the cycle of life of the closing machine, have been analysed by AROL S.p.A.. Therefore, the number of phases and operators indicated is the one most suitable to carry out the functions in a optimal way.

4.4.4 PROFESSIONAL ROLES OF USERS

The closing machine User shall check that the operators have the following requirements:

- had read and understood the indications that are in the manual.
- had received job instruction and training appropriate to their tasks;
- have been properly trained about the correct use of the closing machine, about the residual risks, on the closing machine controls and on the PPM.

Only the operators and the person charged with the maintenance with a suitable training can use the closing machine.

The professional terms used are specified here below to understand the manual fully:

TERM	DESCRIPTION
EMPLOYER	Person legally responsible for the use of the closing machine described herein.
OPERATOR	Person charged with the standard closing machine operation.
MECHANICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary mechanical maintenance of the closing machine.
ELECTRICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary electrical maintenance of the closing machine.

4.4.4.1 OPERATOR

The Operator must have:

- familiarity with the use of the check system (operator interface), with capacity of checking the correlation between the set data and the effect of these on the production;
- knowledge of the technology and specific experience in operating the closing machine acquired through the basic training of this manual;
- base general knowledge and sufficient technical knowledge to read and understand the manual content and the annexes, included the correct interpretation of the drawings.
- Sufficient technical knowledge to carry out intervention specified in the manual, in safe operation condition.
- Knowledge of the accident-preventing norms.
 - General Health and safety, prevention of the accidents on the work place.
 - specification (for the type of product processed by the closing machine, according the laws in force of the country in which the machine is installed).



Besides the conduction of the closing machine from the productive point of view the operator must:

- Know the operation of all the safety devices (for the operator and for the closing machine) so as to be able to check their efficiency and signal any possible fault to the personnel in charge according to the procedures described herein.

When the operator locates a failure that might compromise safety, he must stop the closing machine using the emergency stop controls.



it is forbidden to execute any intervention on the electric and/or pneumatic equipment.

4.4.4.2 MAINTENANCE MAN

The Maintenance Man (mechanical/electrical), besides the above characteristics, must also have a good technical preparation and a suitable working experience in the relevant field. Further, he must have specific and special (mechanical and electrical) technical knowledge needed for the operations described in this manual.

4.4.5 EMPLOYER'S OBLIGATIONS

the employer shall guarantee appropriate training and information of all the professional roles of the closing machine on the following topics:

- Instruction and specific procedures for use and maintenance of the closing machine.
- The user of the closing machine should comply with the safety and health prescriptions on the workplace, in force in the premises where the closing machine is started; all the specific risk conditions should be taken into account.

He must have acquired a good knowledge of the contents of this manual so that he can correctly process the training procedures of the professional roles of the closing machine.

At the end of the training, the employer must make sure that operators and maintenance technicians have understood the above points.

Assess the risks resulting from installing the closing machine in the workplace.

The employer must supply the PP (Personal Protection Means) required for all operations linked to the useful life of the closing machine.



4.4.6 INFORMATION - JOB INSTRUCTIONS - TRAINING

4.4.6.1 INFORMATION - JOB INSTRUCTIONS

The employer shall inform and train the personnel charged to use the closing machine and make sure that:

- On the work place there are suitable information concerning dangers, risks and prevention and protection solutions including the use of P.P.M., if provided.
- the workers the task of using of work equipment receive a suitable training accordingly.
- The workers the task of using of equipment requiring special knowledge and responsibility receive a suitable and specific training, so that they can use such equipment safely and suitably, also as far as risks caused to other people are concerned.

4.4.6.2 TRAINING

Training serves to explain the correct use of the closing machine, in compliance with the obligations prescribed by national laws transposing 2009/104/EC directive concerning minimum safety and health requirements for the use of working tools by the operators while working.

4.4.7 ARRANGEMENTS BY THE CUSTOMER

Excluding special contract conditions, the user must arrange the following:

- Connect with the power supply mains, including the protection lead commonly called "EQUIPOTENTIAL LEAD".
- The protection of the power line from overcurrents and indirect contacts (it is recommended to install a differential switch).
- Connection of the bottle transport belt section to the bottling line bottle transport belts.
- Compressed air mains connection.
- Generic tools for the closing machine maintenance and consumables.
- Lifting means suitable for the closing machine handling.



4.4.8 SIGNS FOR WARNING, OBLIGATION, PROHIBITION AND DANGER

SYMBOLS CONTAINED IN THE MANUAL

The graphic symbols inside the manual are described and shown here below.

GRAPHIC SYMBOL	DESCRIPTION
	It indicates a generic warning. Pay utmost care to the text blocks marked by this symbol.
	Indicates the danger of hot surfaces. Pay utmost care to the text blocks marked by this symbol.
	It refers to a warning about key function or useful information. Pay utmost care to the text blocks marked by this symbol.
	Notes, usage tips, or additional information
	Part of the manual dedicated to the OPERATOR.
	Part of the manual dedicated to the MECHANICAL MAINTENANCE OPERATOR.
	Part of the manual dedicated to the ELECTRICAL MAINTENANCE OPERATOR.
	Obligation to insulate all energy sources. The symbol placed close to this description means that all the energy sources must be insulated, dissipating the residual energies by the user, because this involves the risk of an accident.
	Mandatory use of <i>PROTECTIVE GLOVES</i> . The presence of the symbol close to the descriptions informs that the operator must wear safety gloves due to the risk of injury.
	INSPECTION AND OPERATION Visual inspection.
	INSPECTION AND OPERATION Inspection hearing.
	INSPECTION AND OPERATION Intervention with tools
	INSPECTION AND OPERATION Manual intervention.
	INSPECTION AND OPERATION Intervention of lubrication.



SYMBOLS ON THE CLOSING MACHINE

A set of pictograms describing procedures, obligations, cautions and prohibitions are arranged on the closing machine to attract the operator's attention.



The warning and/or prohibition pictograms on the closing machine have an extremely important safety function; the operators must follow the warning indications.



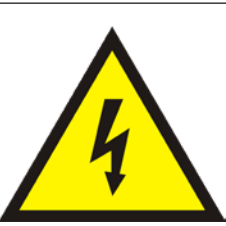
The pictograms of the closing machine cannot be removed.

Should the pictograms wear out occasionally and become illegible, or if they are removed, they must be restored with new ones that must be asked to AROL S.p.A..

The pictograms on the closing machine are described and shown here below.

PICTOGRAM	DESCRIPTION
	Indicates the duty to see instruction manual before to operate on the machine.
	Indicates the lifting points, indicating the necessity to use suitable means.
	Indicates the duty to isolate in case of access at any height, to make operations of cleaning, size change, maintenance.



PICTOGRAM	DESCRIPTION
	It indicates the prohibition to reach inside the accident prevention guards during the motion.
	It indicates the prohibition to extinguish fires with water or other liquids.
	It indicates the prohibition to touch as there are equipment under voltage.
	Indicates the prohibition to adjust, clean or lubricate moving parts.
	It indicates a generic warning. Pay attention to the specific pictogram that accompanies it.
	Indicates the danger to knock against the door of the electrical board during the opening.
	Indicates the presence of voltage with dangerous values.



PICTOGRAM	DESCRIPTION
	Instructs to unlock the handles holding the anti-torsion columns and the central column fastening screw.
	Means that the lubricant cartridge must be activated before starting the machine.
	



The following figures indicate the pictograms on the closing machine and their position.

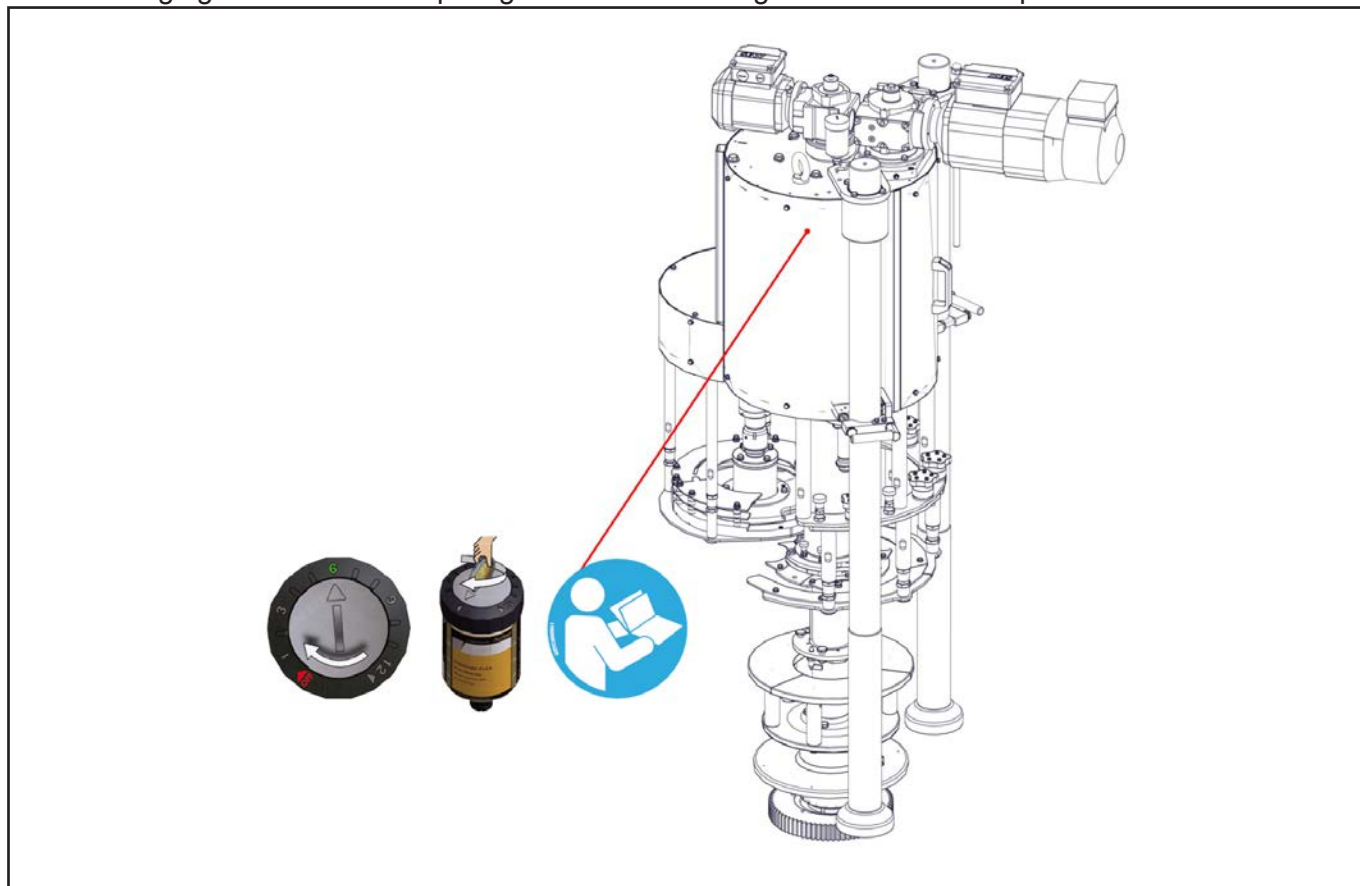


Fig. 12

Re-ordering code: GETIC0000061

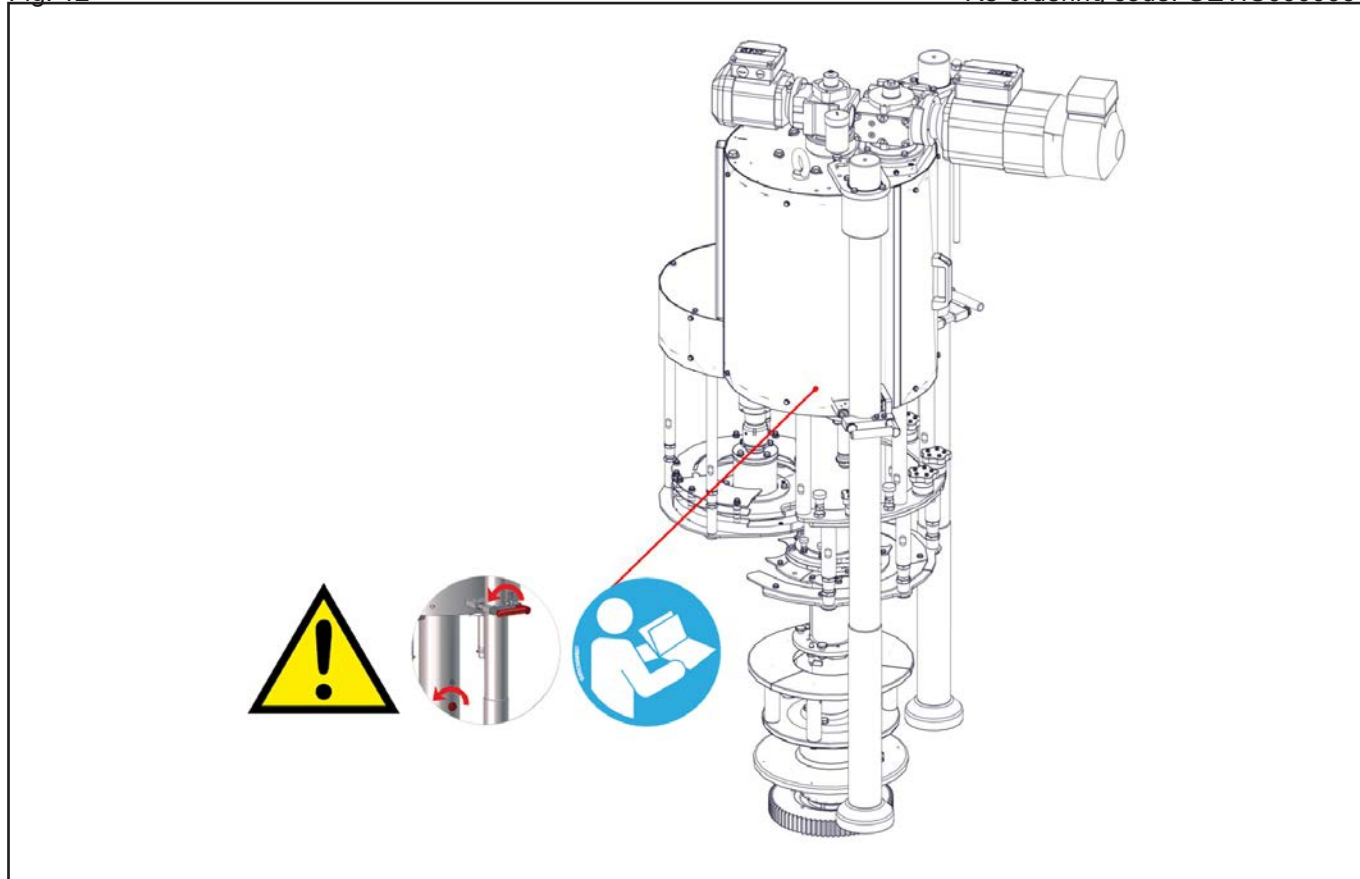


Fig. 13

Re-ordering code: GETIC0000021

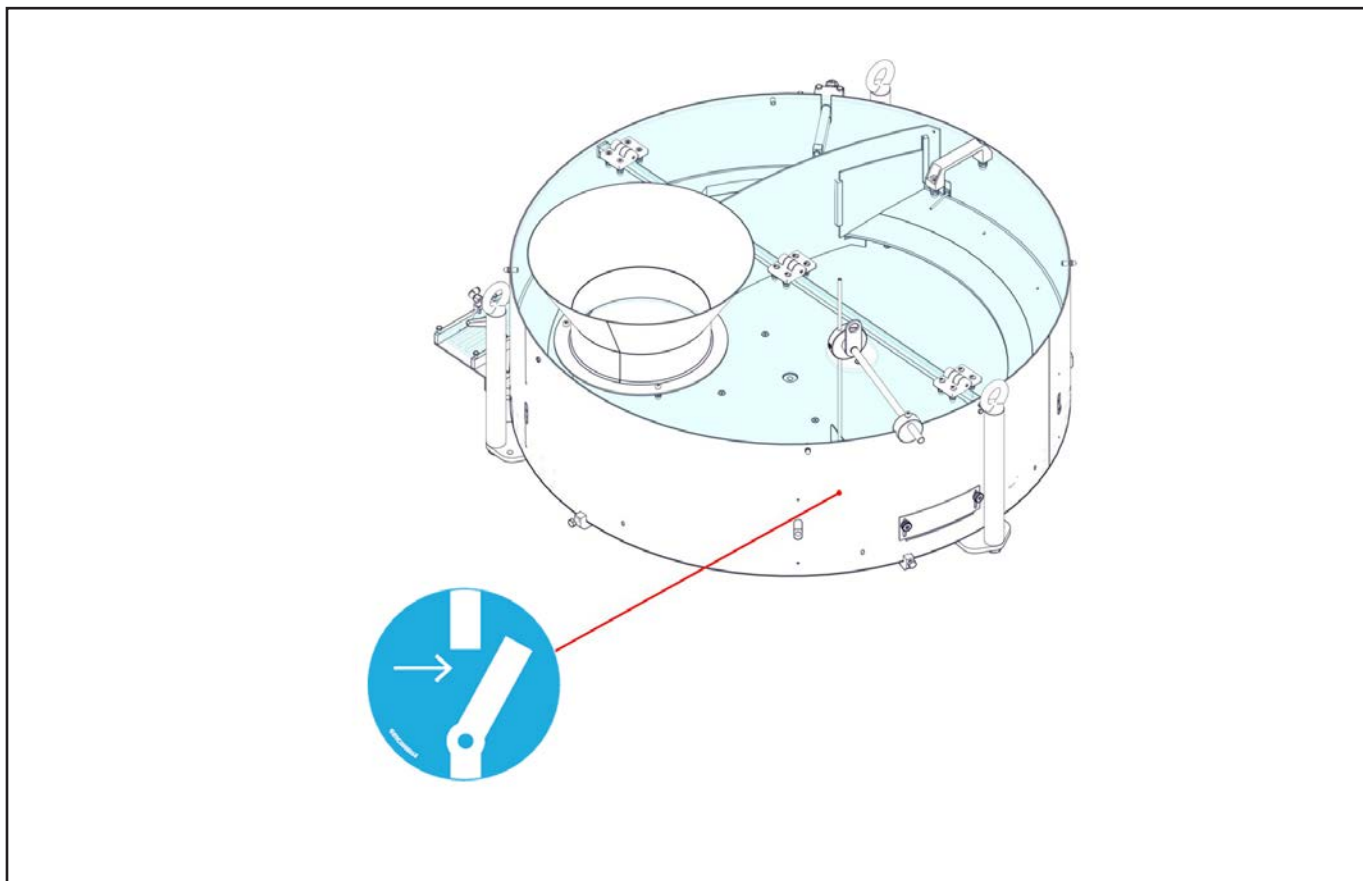


Fig. 14

Re-ordering code: GETIC0000065

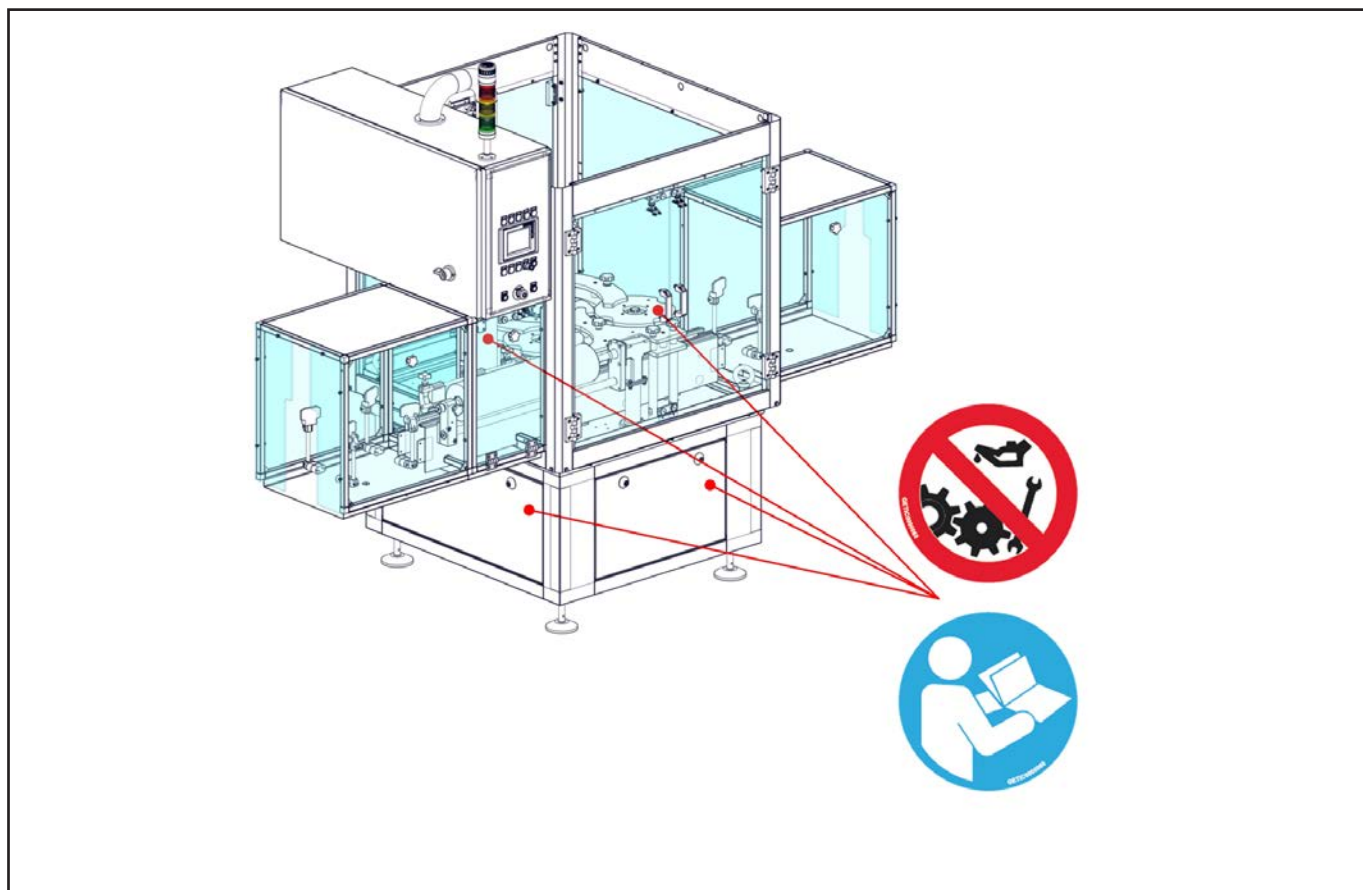


Fig. 15

Re-ordering code: GETIC0000060

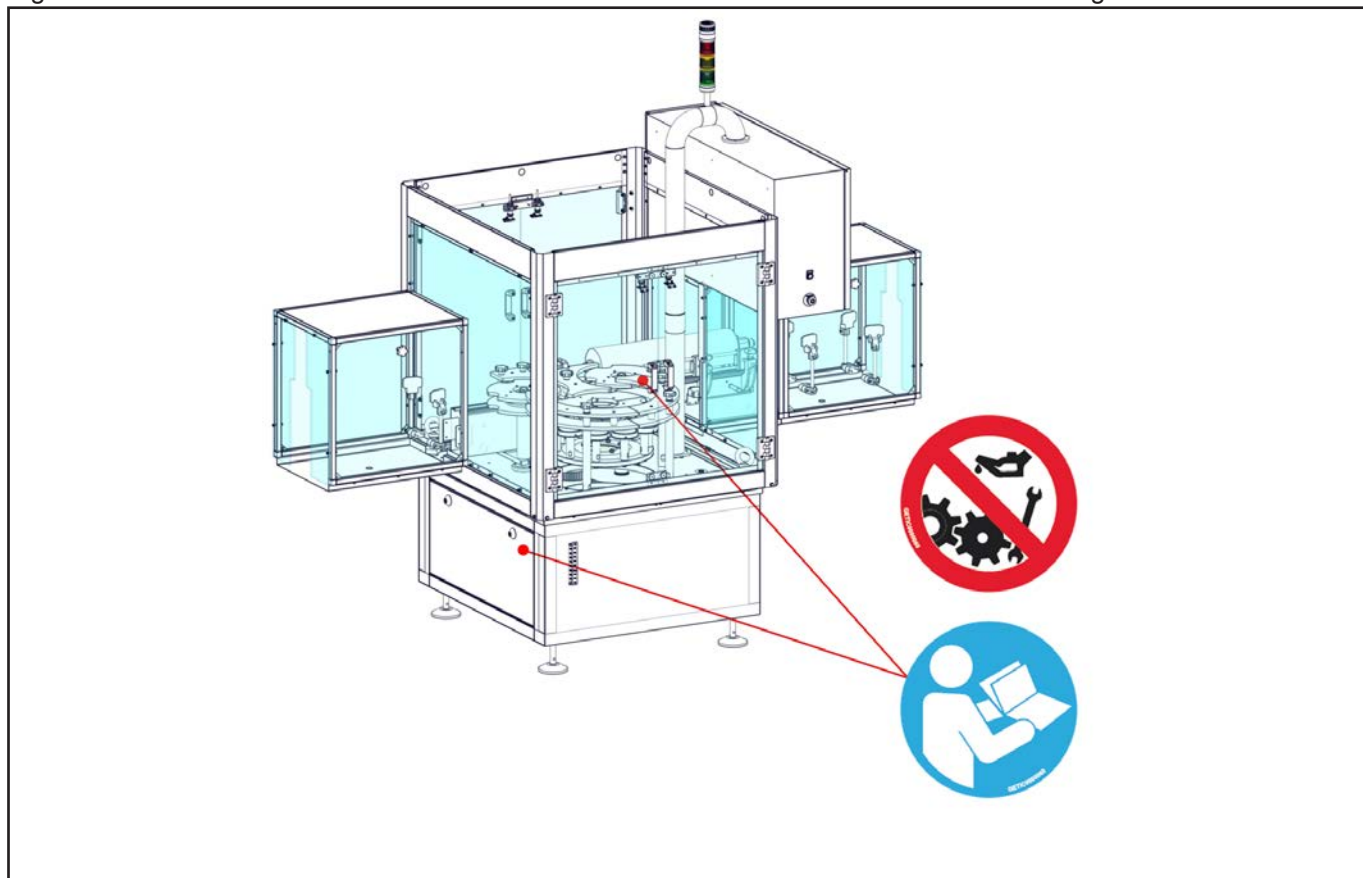


Fig. 16

Re-ordering code: GETIC0000060

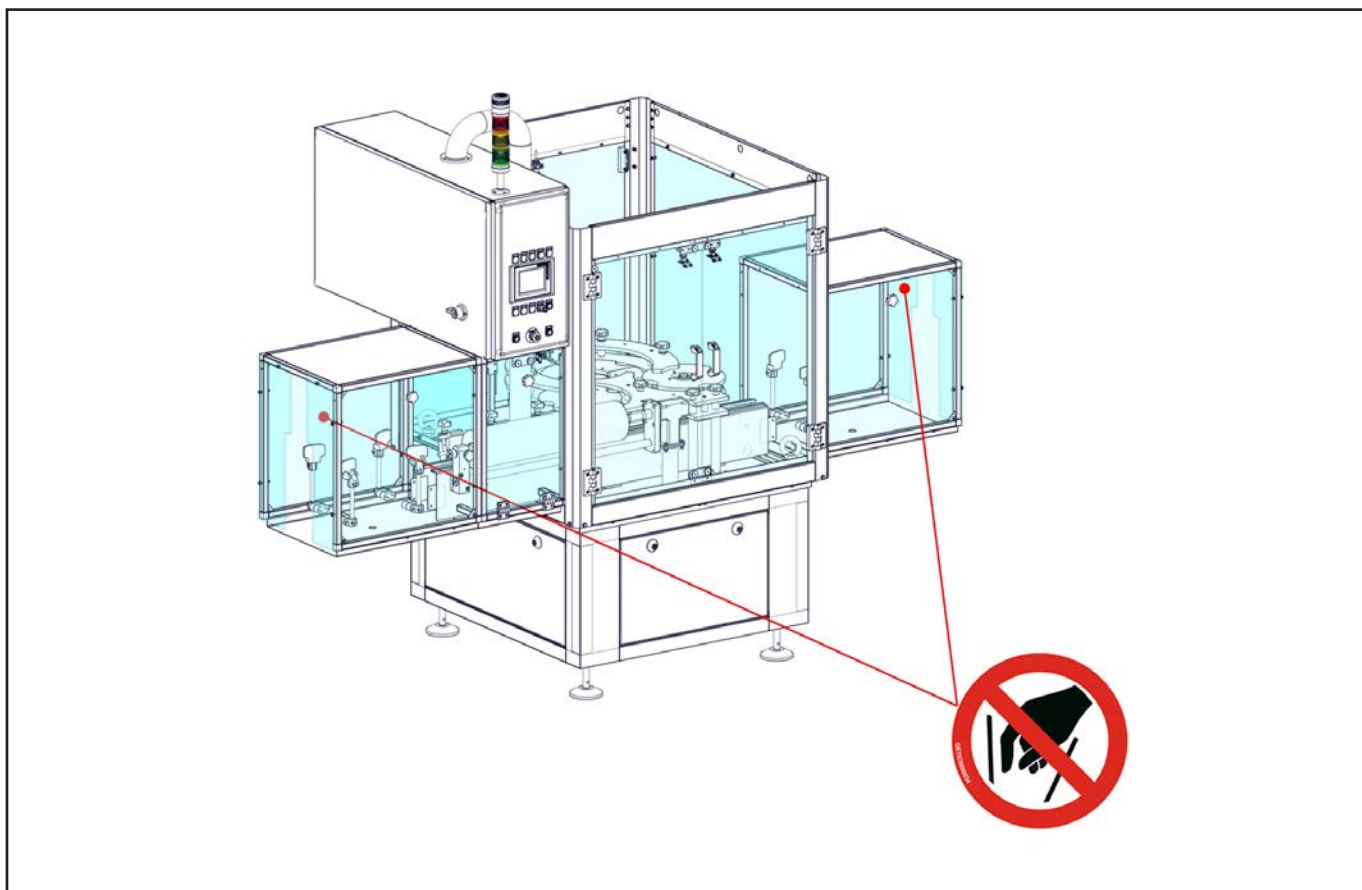


Fig. 17

Re-ordering code: GETIC0000034

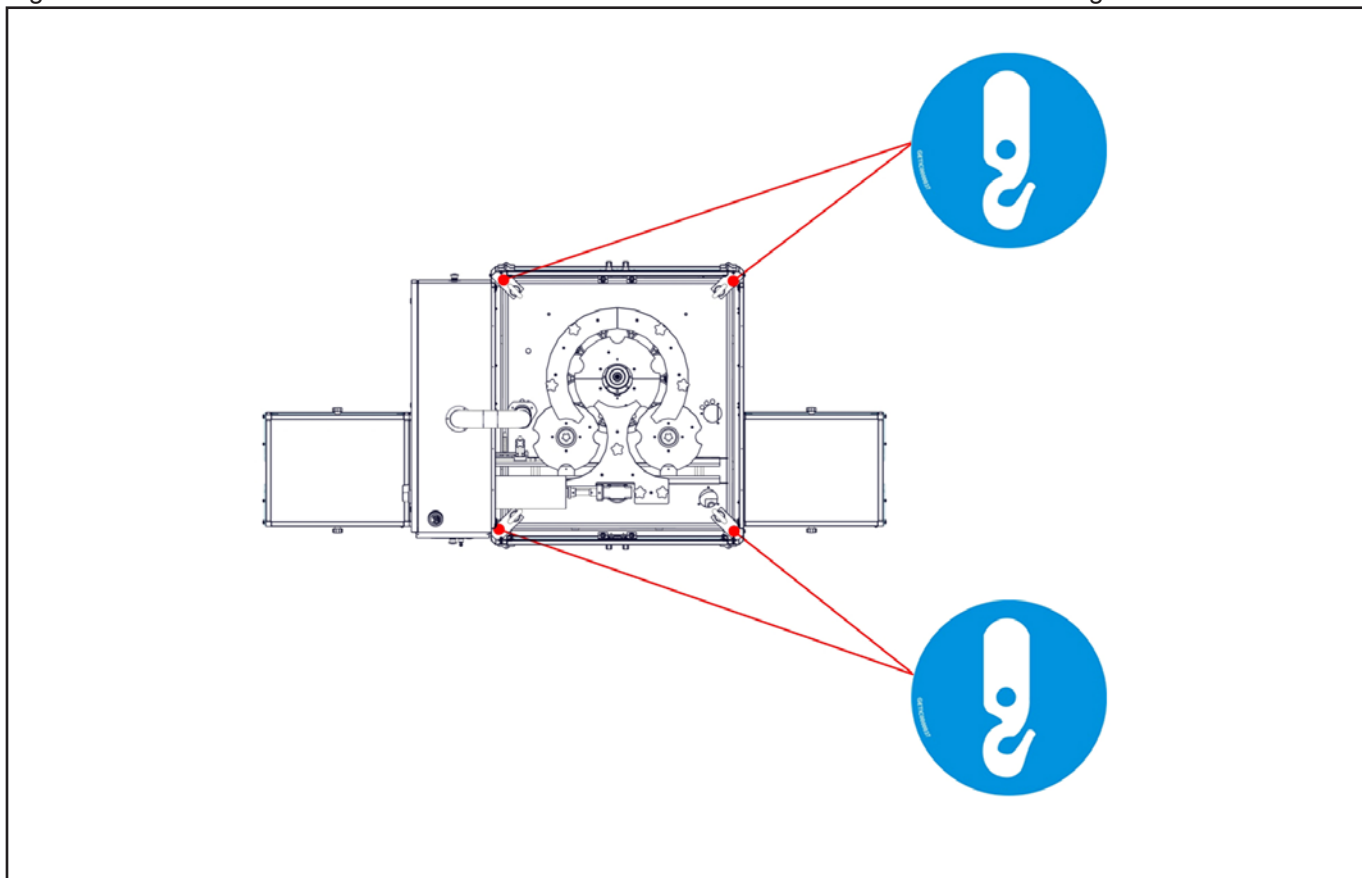


Fig. 18

Re-ordering code: GETIC0000037

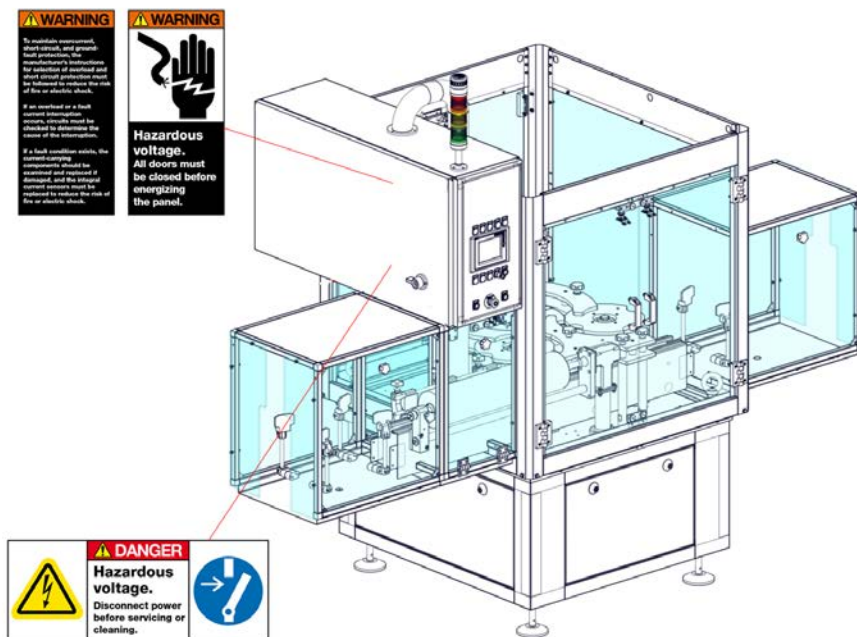


Fig. 19



4.5 SPACE LIMITATIONS

4.5.1 WORKING AND CONTROL AREA

The terms “**Working area**” “**Control area**” mean the areas in which the operator can stop making his tasks.

WORKING AREA: is the place where the operator can stay to check the production and has the possibility to intervene in case of necessity according to his duty.

CONTROL AREA: is the place where the operator can carry out the controlling and check operations on the closing machine, carrying out on the suitable controlling panels.



The work area shall NEVER be occupied by tools or other objects which may create obstructions. Nothing shall interfere with the freedom of movement of the operators. Moreover, in case of emergency, the personnel in charge must be able to freely and quickly access the line. The operators shall enforce the prescriptions and notice the persons in charge about any non compliance.



Closing machine control and operation in standard working conditions (automatic mode) must occur in the specific operating area only.

The figure below indicates the work and control areas on the closing machine.



The control panel, in design stage, can be positioned either at the entry or at the exit of the closing machine.

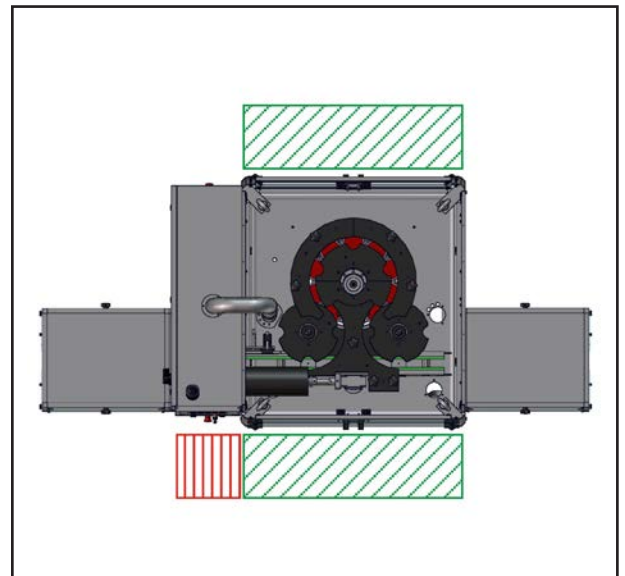
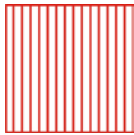


Fig. 20

Legend:			
	Control Area		Working Area



The maintenance operator shall operate around and inside the closing machine only after the activation of all the safety devices foreseen by the Manufacturer to put the closing machine in safe conditions. Moreover, when the closing machine undergoes maintenance, all the surrounding areas must be forbidden.



4.6 ENVIRONMENTAL LIMITS

4.6.1 ENVIRONMENTAL OPERATING RISKS

USE ENVIRONMENTAL CONDITIONS

The closing machine must work in premises protected by weather agents, with a temperature ranging between 5°C and 40°C, a maximum relative humidity of 80% and height above sea level not over 2000 m.

LIGHTING

The closing machine can work with the standard lighting of the workplace (about 250 lux); local lighting of areas concerned with maintenance is not required very often and must be performed with low voltage lights.

EXPLOSIVE ATMOSPHERE

It is strictly forbidden to operate the closing machine in an area with potentially explosive atmosphere, even if only momentarily.

4.6.2 RESIDUES AND ENVIRONMENTAL CONTAMINATION

No toxic substance is released if the closing machine is used and maintained according to the instructions herein.

The disposal of spent lubricants must respect the norms in force for environment protection in the country of use.



4.6.3 NORMS COVERING THE RISK OF EXPOSURE TO NOISE

The closing machine is designed to operate in industrial environments. In such plants the noise level can be very high, and must be managed bearing in mind the significant risks to hearing and other parts of the body, in compliance with all of the standards issued by the various national and local bodies.

4.6.3.1 INFORMATION ON NOISE EFFECTS

As defined in section 4.7 of EN 415-10: 2014 (Safety of packaging machines - Part 10: General requirements), the noise generated by the packaging machines can cause:

- permanent hearing loss;
- hum;
- fatigue, stress, etc;
- other effects, such as loss of balance, loss of consciousness;
- disruption of verbal communication;
- inability to hear warning signals.

4.6.4 THE EFFECTS ON HEARING

More is known about the effects on hearing, and it is the problem seen most often in the work environment. Damage from noise pollution has the following characteristics:

- increases with exposure time;
- has a greater effect on subjects with reduced resistance and those who are not of working age;
- can disrupt sleep and rest.

For the purposes of the forecast of hearing loss, it is necessary to refer to the maximum values of daily exposure.

When determining this, one must bear in mind other factors such as pulsing, the tone components, etc.

In short, the main factors that determine the presence of damage are:

- the level of sound pressure.
- The exposure time.
- The spectral composition of the noise.



4.6.5 CLOSING MACHINE SOUND EMISSION

The measure of the intrinsic noise of the closing machine has been made according to EN 415-9 at the AROL S.p.A., according to the following criteria:

- Rated production speed 4500
- Containers in use. (See the table accompanying this manual).
- Corks/caps used. (See the table accompanying this manual).
- Complete of conveyor belts.
- Automatic operation.



The values below represent the level of noise emission measured at the manufacturing site. It is the user responsibility to take the appropriate measurements at the installation site taking into account that the machine does not require the constant presence of an operator. The factors that influence the level of exposure of the operators include the characteristics of the working environment, the other noise sources, etc., i.e. the number of machines and/or the performance of other processes in adjacent areas. The allowed exposure level may also vary based on the Country, therefore the employer is requested to comply with the current regulation of the site where the machine will be used.

Nevertheless, the information below allows the user to carry out a previous risk assessment.

INFORMATION ON NOISE EMISSION LEVEL			
Weighted sound pressure level (C) - if required		Not determined	dB(C) _{peak}
Correction of background noise (average value)	K_{1Aavg}	1	dB(A)
Environment correction	K_{2Aavg}	Not determined	dB(A)
Local environment correction (average value)	K_{3Aavg}	1	dB(A)
A-weighted sound pressure level at the workstation	L_{pAavg}	78	dB(A)
Uncertainty	K_{pA}	from standard 415-9: max 8	dB(A)
Other information, if required			
A-weighted superficial sound pressure level	L_{pfA}	Not determined	dB(A)
Superficial measurement	L_s	Not determined	dB
A-weighted sound power level (ref. 1pW)	L_{wA}	Not determined	dB(A)
Uncertainty	K_{wA}	Not determined	dB
Reference standards:			
for sound pressure level at the workstation	UNI EN 415-9, UNI EN ISO 11202		



5 SECURITY AND ACCIDENT PREVENTION

5.1 GENERAL INFORMATION

The employer will must provide for the training of the staff about dangers caused by accidents, about equipments arranged for the operator safety, about dangers of noise emission and about the general accident rules provided by international directives and by legislation of the country of designation of the closing machine.

Anyway the behaviour of staff, maintenance operator, cleaning, control, etc. will must scrupulously respect the accident norms of the country of designation of the closing machine.



Before starting the work, the operator must be thoroughly familiar with the location and operation of all controls and characteristics of the closing machine must also have fully read this manual.



The closing machine must be used only from operators that had received a training and understand totally the instructions of this manual.



It is necessary to respect totally instructions, cautions, general accident rules that are in this manual.



Before any maintenance and/or size change operation, the closing machine must be safety conditions:

- **insulate the closing machine from all energy sources dissipating residual energies.**
- **divide and lock all the cutting off devices;**
- **Prevent access to the closing machine to unskilled staff, not charged with maintenance operations, using suitable signs.**



5.2 RISKS - PROTECTIONS - CAUTIONS AND PRECAUTIONS

5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE

To guarantee health and safety of the people exposed, the closing machine is provided with this safeties:

FIXED GUARDS

All the fixed guards installed on the closing machine are connected to the structure by means of fastening devices which do not allow manual removal, since they necessarily require the use of specific tools.

It is forbidden to use the closing machine after having disassembled the fixed guards; operation can resume after the reassembly of guards.

The housing of the fixed guards in vertical position in the structure of the closing machine has been designed so that the guards, once the fastening devices have been removed, do not remain in closed position, falling by gravity.



To access the area protected by the guard, the closing machine must be insulated from all power sources. (See sub-chapter 5.4 SAFETY PROCEDURES FOR THE MAINTENANCE page 38).

INTERLOCKED MOBILE GUARDS

The closing machine is provided with mobile guards to protect the inner elements, connected to the structure by means of fastening systems which enable its opening, but not its detachment.

Each mobile guard is provided with interlock device to perform the check of the opening, stopping all mobile organs.

EMERGENCY BUTTONS

Pressing one of the emergency red mushroom buttons placed on the control panel, the following occurs:

- Immediate stop of the closing machine
- Consent inhibition of the auxiliary organs.
- Cutting power off at the actuators.

5.2.1.1 POSITION OF THE SAFETIES

FIXED GUARDS

- Head protection guard.

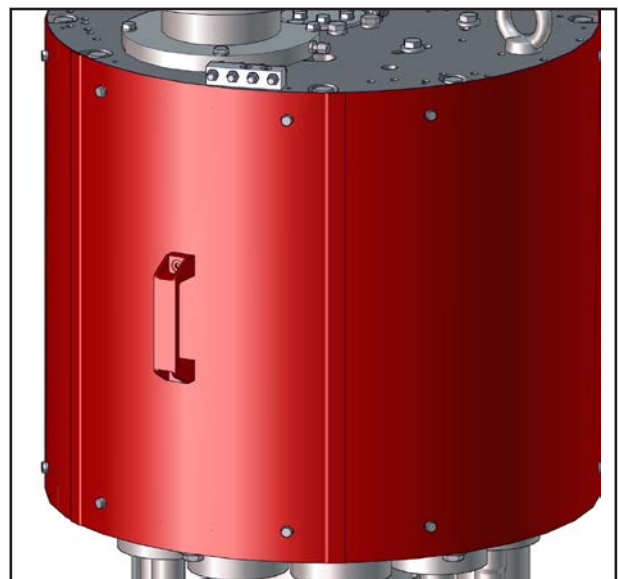


Fig. 21



- Basement guard.

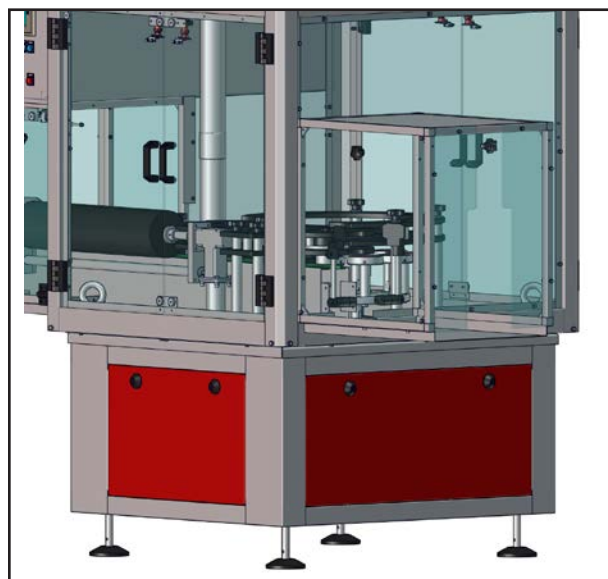


Fig. 22

- Entry tunnel.

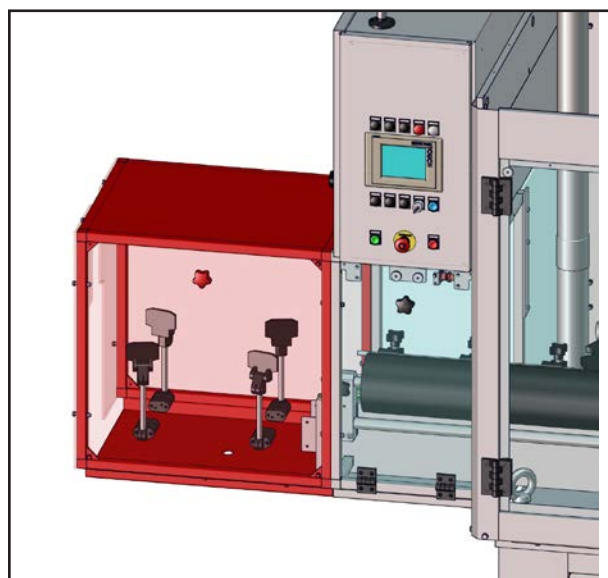


Fig. 23

- Exit tunnel.

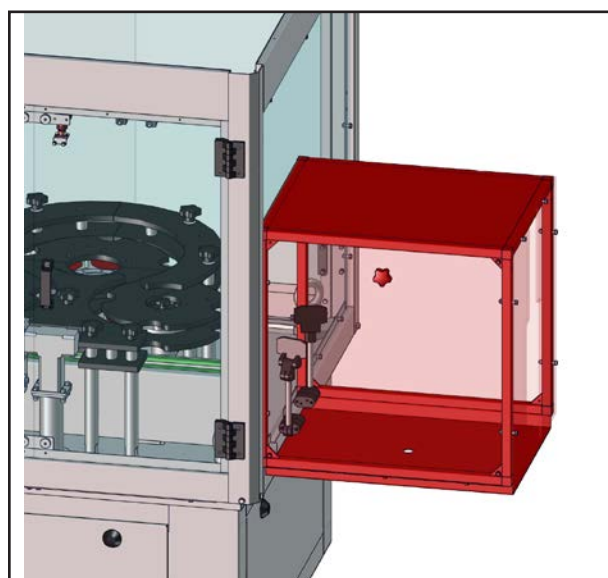


Fig. 24



- Caps stop device door.

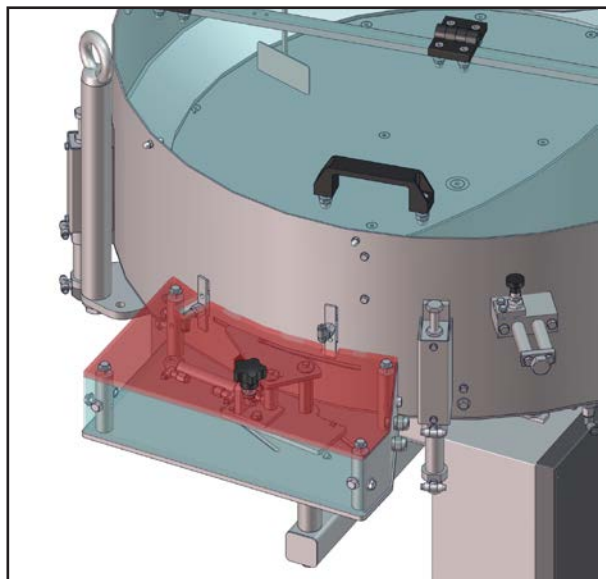


Fig. 25



INTERLOCKED MOBILE GUARDS

- Worm protection tunnel.

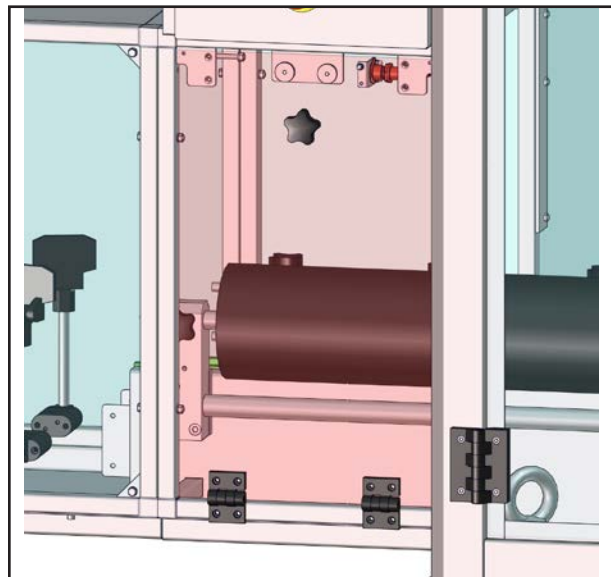


Fig. 26

- Front and rear safety doors.

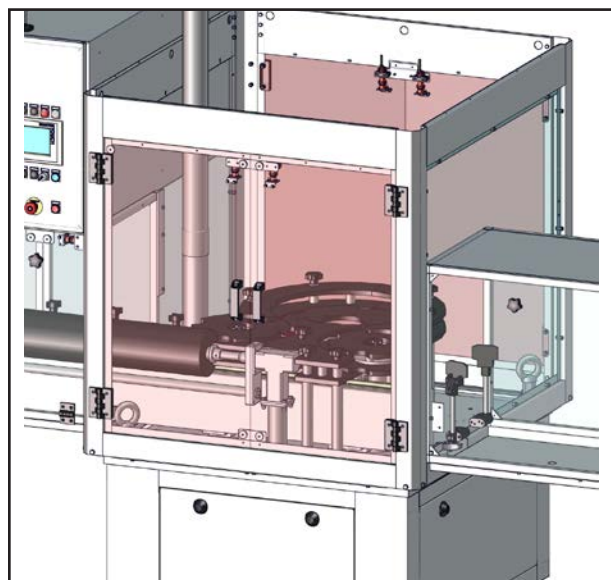


Fig. 27



EMERGENCY BUTTONS

- Mushroom button on the control panel front.

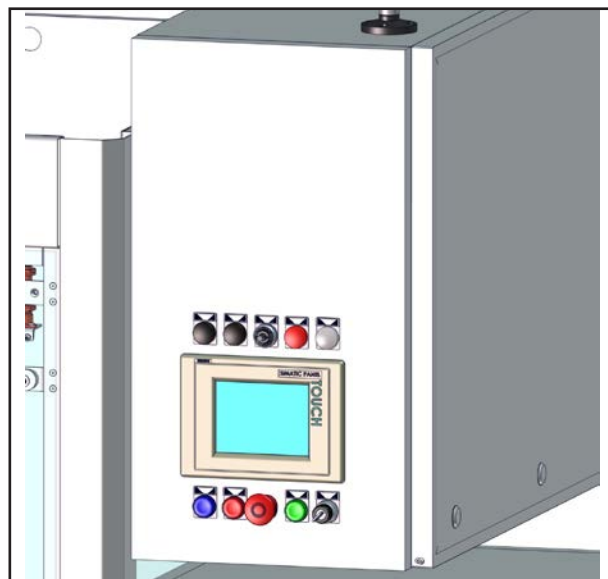


Fig. 28

- Mushroom button on the control panel back.

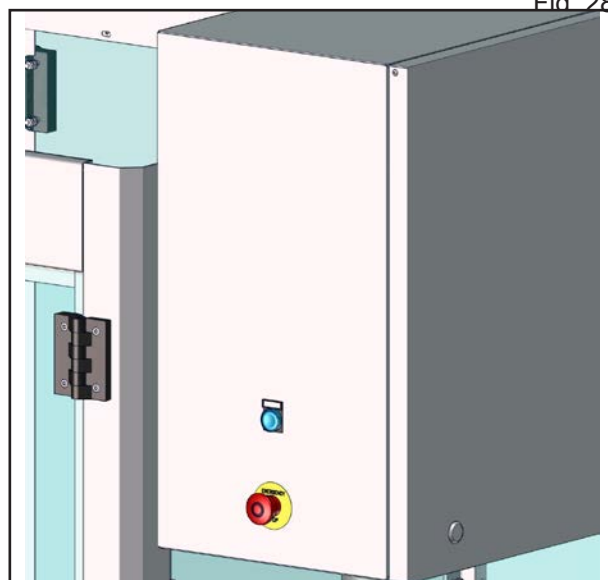


Fig. 29



5.3 SAFETY PROCEDURES FOR THE OPERATION



Wrenches and tools used for the maintenance or the access to the closing machine hazardous sections, as well as the key of the mode switch *AUT/MAN* on the electric panel, must not be given to operators not suitably trained for the size change or the maintenance.

All check, adjustment, maintenance and lubrication steps must be carried out by staff previously trained and instructed.

All check, adjustment and maintenance steps of the electric system must be carried out by staff previously trained and instructed.

People on drugs, alcohol or drowsiness-causing medicines are not allowed to use or maintain the machine.

All check, adjustment, maintenance and lubrication steps must be carried out with the closing machine stopped and separated and split electric power supply.



The cap/cork feeding system has been designed and manufactured to be loaded by an automatic elevator with ground loading.

If the cork/cap loading is manual, it should be performed leaving the closing machine at a standstill, with separated and split electric power supply.



Before accessing the top of the machine, separate and split the energy sources and dissipate the residual energies.

Access at any height must occur using certified means according to the norm UNI EN ISO 14122.



5.4 SAFETY PROCEDURES FOR THE MAINTENANCE



BEFORE CARRYING OUT ANY MAINTENANCE OPERATIONS, the closing machine must be in safety conditions to avoid untimely screwing and/or electrocution.



It is therefore compulsory to insulate all energy sources dissipating residual energies.



The components accumulating electric charges (e.g. frequency converters) can generate electrical discharges for approximately 5 minutes. Refer to the manuals of the electric equipment attached to this manual.

The insulation for power sources must be charged to qualified technicians and be performed as follows:

- ELECTRICAL ENERGY - the door disconnecting switch must be set "O" (OPEN), then locked in this position by a lock - not supplied - to be fit in the suitable slots of the disconnecting switch.



The lock key must be removed and kept by the responsible maintenance technician who is performing maintenance, for the whole intervention.



Wrenches and tools used for the maintenance or the access to the closing machine hazardous sections, as well as the key of the mode switch *AUT/MAN* on the electric panel, must not be given to operators not suitably trained for the size change or the maintenance.

A sign "Maintenance in progress" must be placed on the control board.

The area where maintenance is being performed must be suitably fenced and surrounded, to inform bystanders about the forbidden entry. Special warning plates should be used as follows: "Danger! Maintenance in progress".

After the maintenance works and before starting the closing machine again it is always necessary to:

- check that the possibly replaced parts and/or used tools have been removed from the closing machine.
- check that all guards, protections and safety devices possibly removed during the intervention have been re-installed correctly and are operating.



Replace the electrical safety components within their life cycle, approximately 20 years.



The standard operating conditions will be restored only after the intervention and the checks.



5.4.1 GENERAL SAFETY WARNINGS

The safety for the closing machine and the operators also depends on a regular maintenance according to the manufacturer's indications.

The routine and extraordinary maintenance interventions must be exclusively charged to skilled staff, that is to technicians qualified according to the specifications in this manual.

If outsourced operators intervene, they must ensure the same performance of the inhouse qualified technicians and must work in safe operating conditions.

The access to high levels is allowed only after separating and disconnecting any residual energy and using means certified according to UNI EN ISO 14122 standard.

The qualified technicians must:

- respects the limits of their tasks (mechanical, electrical, etc.).
- Follow the procedures and warnings in the manual, within their tasks.
- Follow the times and intervals indicated in the manual for scheduled maintenance interventions.



The following general warnings must be noted:

- Electrocution hazard due to direct contact with the closing machine power supply terminal boards and in the electrical system connector blocks (the hazard is signalled by plates with the suitable triangle with yellow bottom).
- Be extremely careful during the interventions on the electrical system, if the system must be left live. Such interventions must be carried out by staff skilled in this type of maintenance only, trained and correctly equipped for the steps to be made. The intervention with live system must always be authorized in advance by the employer who must oversee its operations by assessing the specific risk situations.
- Carry out maintenance operations on pneumatic systems and components only after cutting off and separating the electric power and pneumatic sources and checking that the residual pressures have been relieved.
- It is forbidden to carry out maintenance and lubrication interventions on moving components.
- While working on heated parts, the operator must always wait until they cool to a temperature below 55°C to prevent serious burns.

5.5 RESIDUAL RISKS

The residual risks, where present, are identified through triangular shape pictograms on yellow background installed on the closing machine as described in subchapter 4.4.8 SIGNS FOR WARNING, OBLIGATION, PROHIBITION AND DANGER page 18.



6 HOW TO ASK FOR INTERVENTIONS

AROL S.p.A. puts its own Technical Service at the disposal of its customers to solve any problem regarding the machine use and maintenance.

The interventions must be required after carefully evaluating the faults and their reasons, and stating:

- The details on the occurred faults.
- The performed checks.
- The performed adjustments and resulting effects.
- Any further information deemed useful.

7 HOW TO ORDER SPARE PARTS

All orders for spare parts must be sent to AROL S.p.A..

The installation of not original spare parts will invalidate the warranty.

To order spare parts, fill in the suitable form in the “*SPARE PART CATALOGUE*” for the machine.



The replacement of safety devices must be performed with the same components. The 2006/42/EC directive provides that the components supplied by the machine genuine manufacturer do not need the Declaration of Conformity. Vice versa for all components purchased on the market, the Declaration of Conformity is required, along with the control that the replacement component has the same features and performance of the replaced one. Unsuitable components might cause the decay of the safety performance expected by AROL S.p.A.. It is therefore recommended to contact AROL S.p.A..



8 TRANSPORT, INSTALLATION AND CONNECTIONS

8.1 GENERAL WARNINGS

The transport of the closing machine must always be made by highly qualified staff; they should take care to prevent any damage to parts of the machine, due to bumps or strains caused by non-allowed lifting methods.

For a correct performance of the operations, comply with the following general instructions:

- always use the most suitable equipment as for specifications and capacity, well maintained.



The information about weight, centre of gravity, addressee, packing list and handling instructions can be found in the closing machine package.

- Before starting the operations, to reach to everyone involved in the safety position and prevent access to strangers in handling areas.
- Ensure that a correct overall overlook of the path to be run is always available.
- Never deposit loads on irregular surfaces, unstable supports or supports having a different resistance.

8.2 HANDLING MODE

Handle the case as follows:

- Use belts, chains or ropes having a rated loading capacity corresponding at least 4 times the weight of the closing machine (weight shown on the packing list) to sling the case in the points shown on it.
- Lift the case and convey it close to the installation place.



Consider the position of the center of gravity to handle the case correctly. Lift from the ground max. 10 cm; if the load is parallel to the floor, continue lifting; otherwise adjust sling belts/chains.

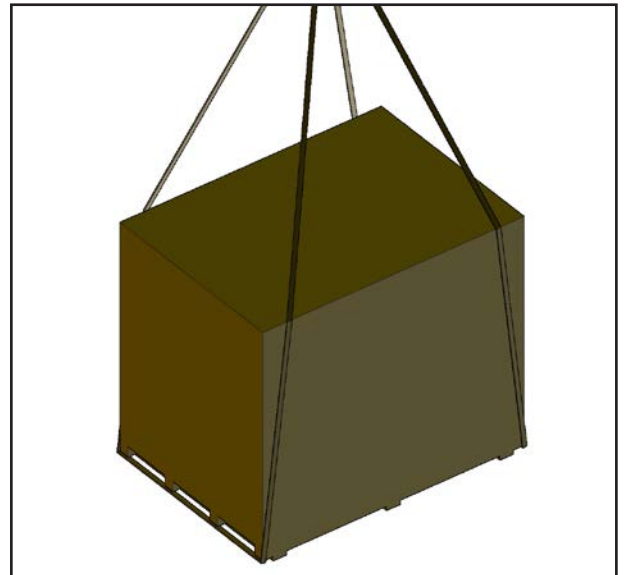


Fig. 30



8.3 PREDISPOSITION FOR THE INSTALLATION

The user must proceed as follows:

- presetting of the installation premises of the closing machine.
- Power supply to the closing machine according to the regulations in force in the utilisation country.
- Pneumatic power supply of the closing machine.
- Connection and washing fluid collection system.

8.4 ENVIRONMENTAL INSTALLATION REQUIREMENTS

The closing machine does not have special environmental requirements for installation; nevertheless it should be placed in well lit, ventilated industrial building, with a strong and level service plane.



8.5 CLOSING MACHINE INSTALLATION

After having conveyed the closing machine in the prescribed installation, remove the external package and proceed as follows:

- Remove the fixings that fasten the capping machine to the transfert platform.

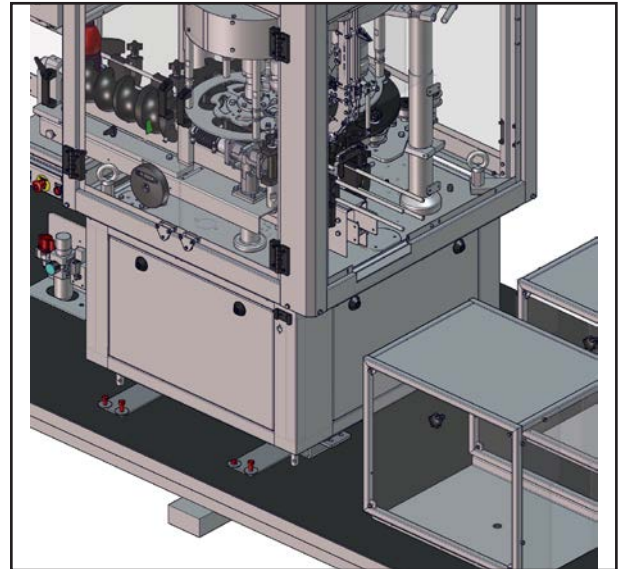


Fig. 31

- Insert the forks of the forklift in the proper overturning preventions seats.

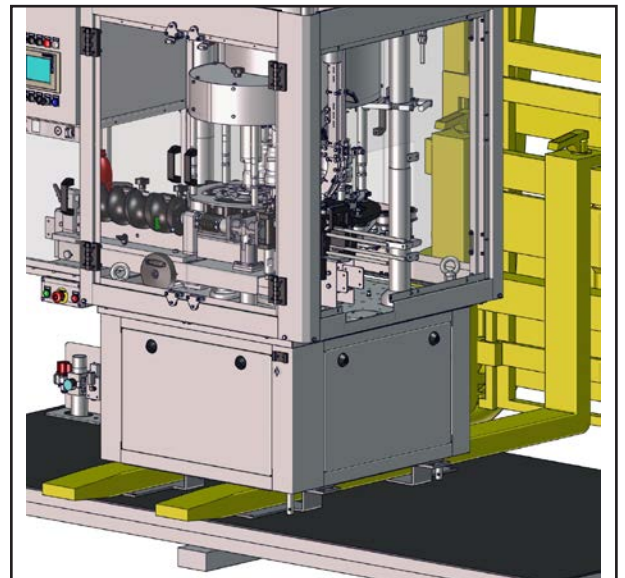


Fig. 32



- Screw the threaded supporting feet to the suitable threaded bushes.

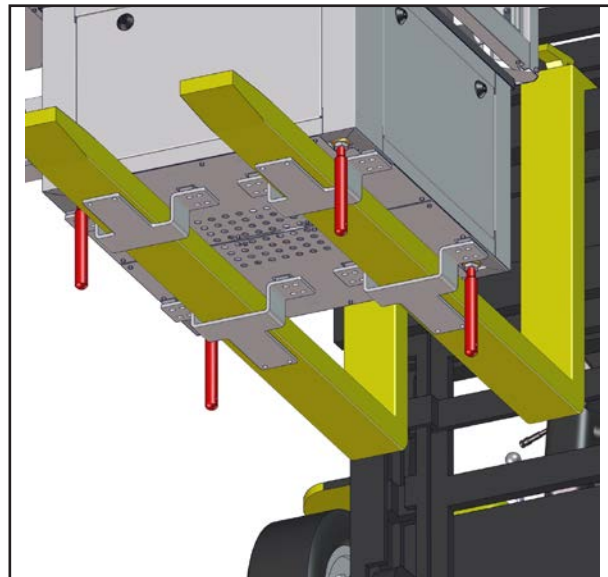


Fig. 33

- Lay the closing machine on the floor on the supporting plates previously positioned.

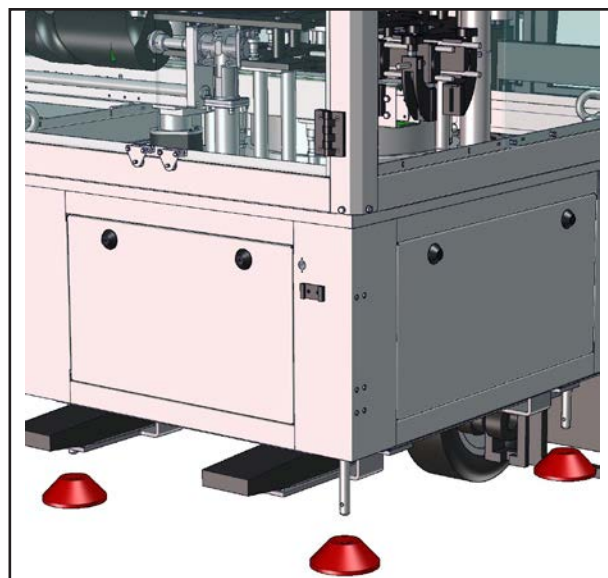


Fig. 34

- Level the closing machine with respect to the floor screwing or unscrewing the threaded supporting feet until the according leveling.



The disposal of materials must be carried out according to the norms in force of the country where the machine is installed, taking note of the nature of the materials:

- Nails and metal parts.
- Wooden boards.
- Plastic materials (protection sheet, barrier bag).

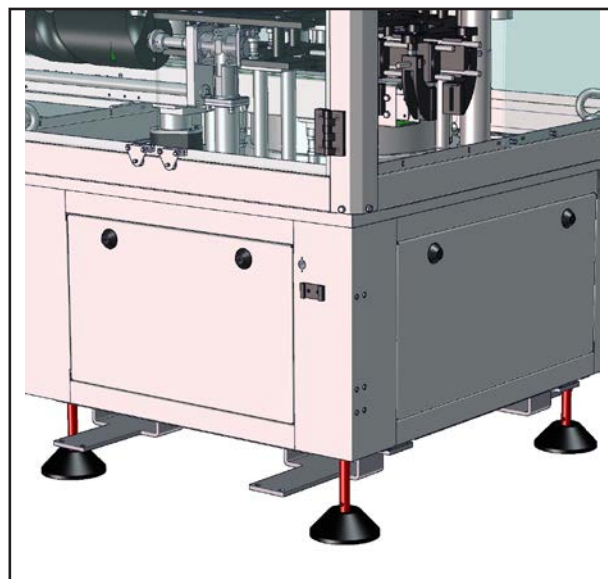


Fig. 35



8.5.1 CONNECTION OF THE CLOSING MACHINE WITH THE LINE CONVEYOR BELTS

Connect the conveyor belt with the closing machine as follows:

- Fit the line conveyor belts on the plates on the body of the closing machine belt.
- Drill two holes on each side of the components to be connected.

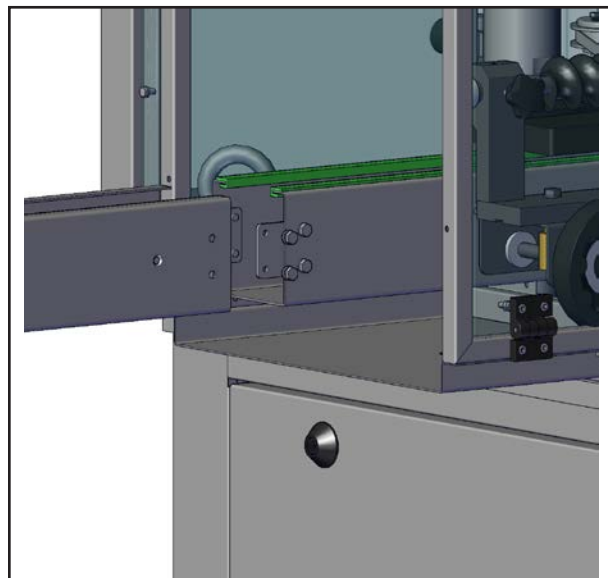


Fig. 36

- Fasten the various elements with bolts.

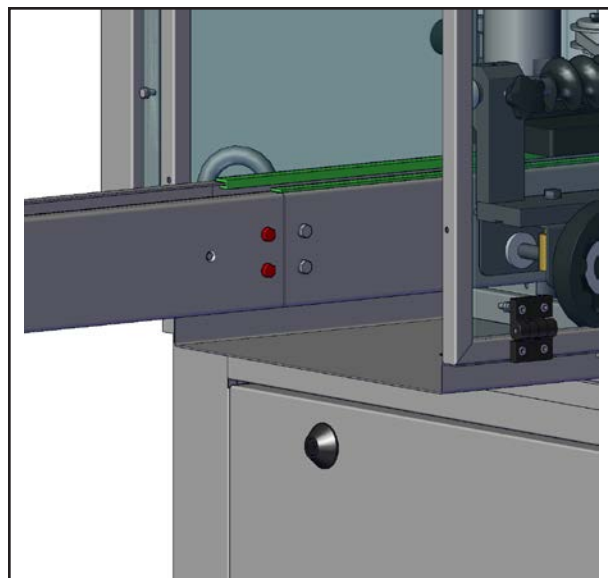


Fig. 37



8.5.2 PHOTOCELL POSITIONING ON THE CONVEYOR BELT

The automatic start photocell and the photocell which stops the closing machine due to overflow at the outlet should be positioned in the following way:

- The idle speed photocell must be positioned in such a way that the closing machine rotates when at the entry of the auger there is a number of containers sufficient to guarantee the thrust of the containers on the worm.

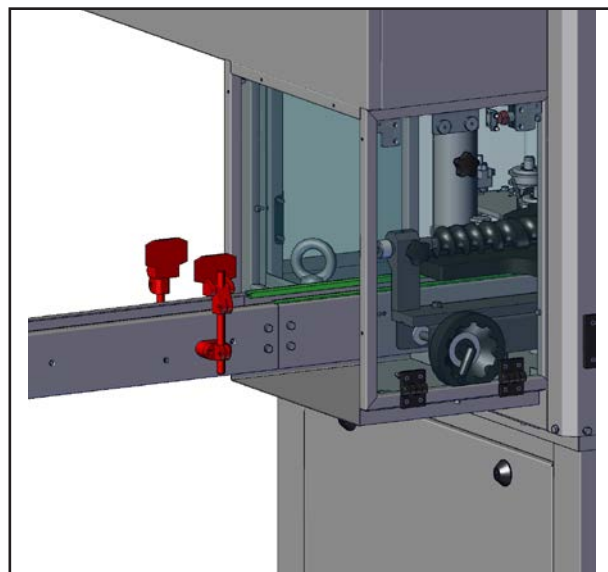


Fig. 38

- The maximum speed photocell must be positioned at about 1 m from the minimum speed photocell.

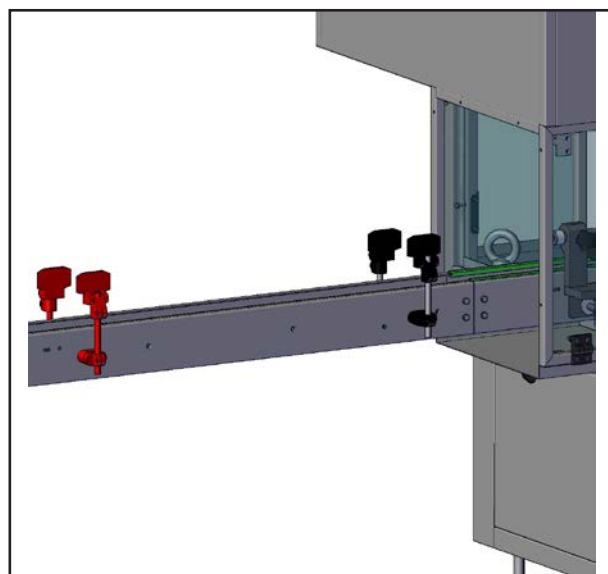


Fig. 39

- The stop of closing machine for overflow output photocell must be positioned so that from the center of the exit star-wheel to the photocell there is about 1 meter.



The supports of the photocells and their reflectors must be appropriately fixed to the bottles transport belt.

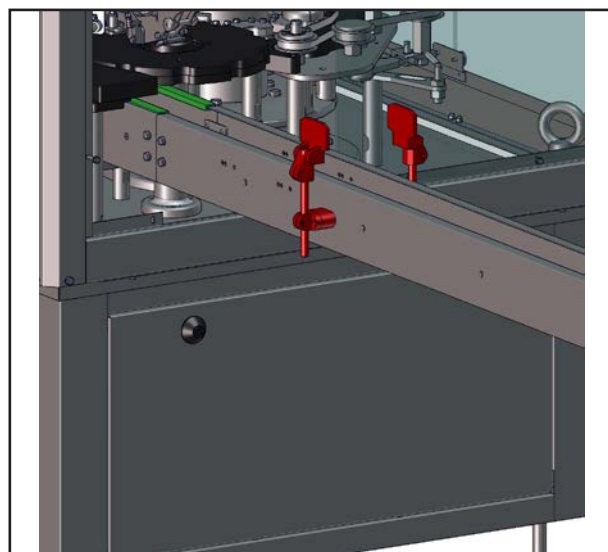


Fig. 40



8.5.3 CONNECTION OF THE PROTECTION TUNNELS AT THE ENTRY AND EXIT CLOSING MACHINE



It is mandatory to install the conveyor belts protection tunnels supplied with the closing machine.

For the installation, operate as follows:

- Unscrew the fixing screws and remove the lower guard of the tunnel.



Fig. 41

- Position the tunnel on the entry belt and fasten it with the bolts supplied with the closing machine.

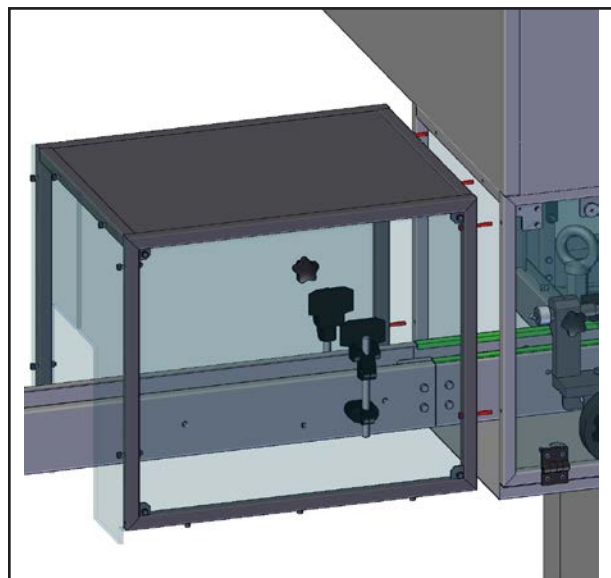


Fig. 42



- Refit the bottom of the tunnel and secure it with the suitable fixing screws.

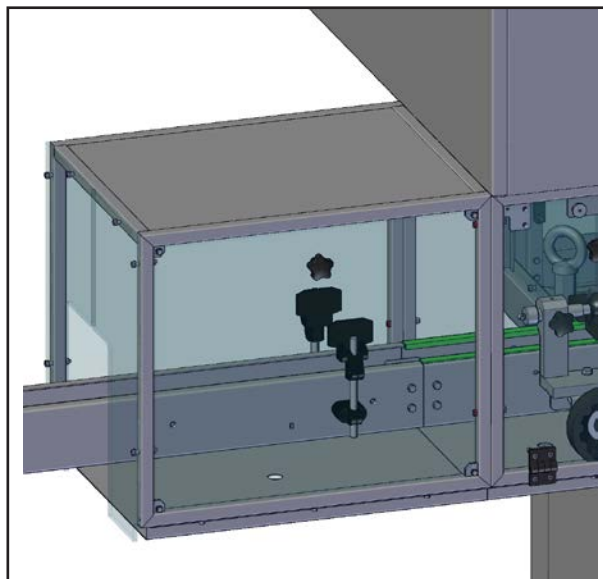


Fig. 43

- Position the tunnel on the exit belt and fasten it with the bolts supplied with the closing machine.

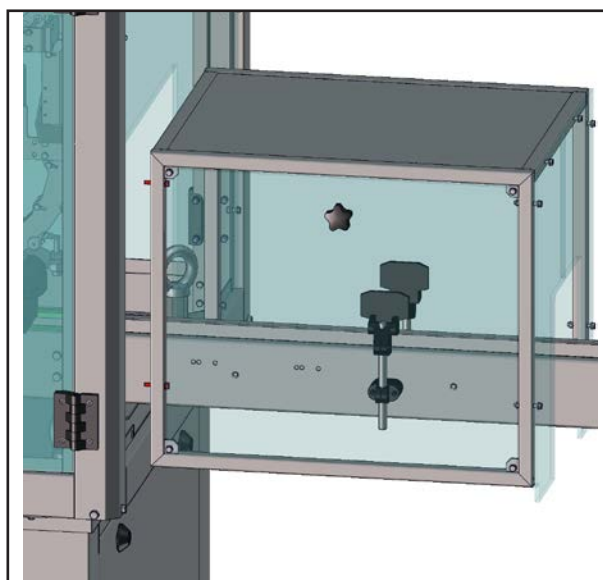


Fig. 44

- Refit the bottom of the tunnel and secure it with the suitable fixing screws.

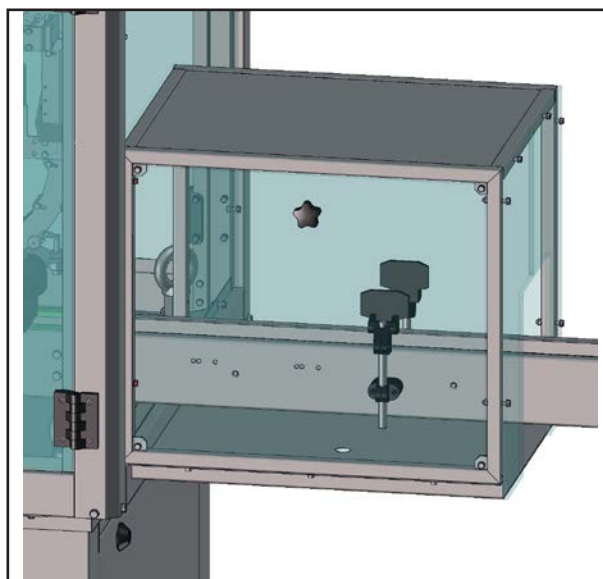


Fig. 45



8.6 CAPS CHUTE INSTALLATION



To install the caps chute on the machine it is necessary:

- Loosen the threaded fastening placed on the caps sorter

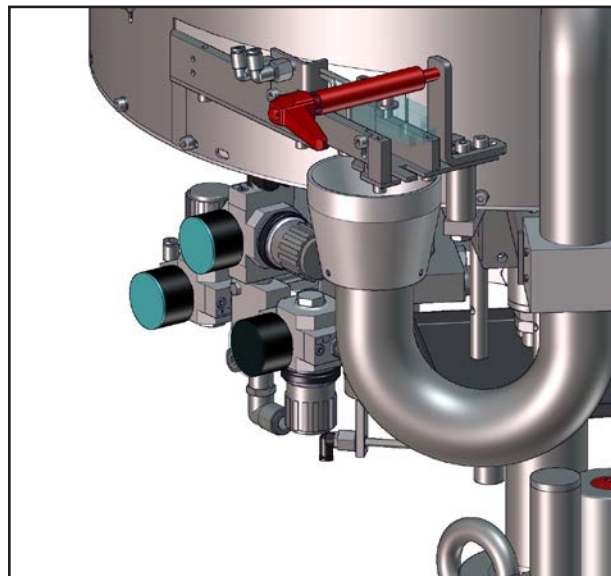


Fig. 46

- Loosen the threaded fasteners located on the support chute.

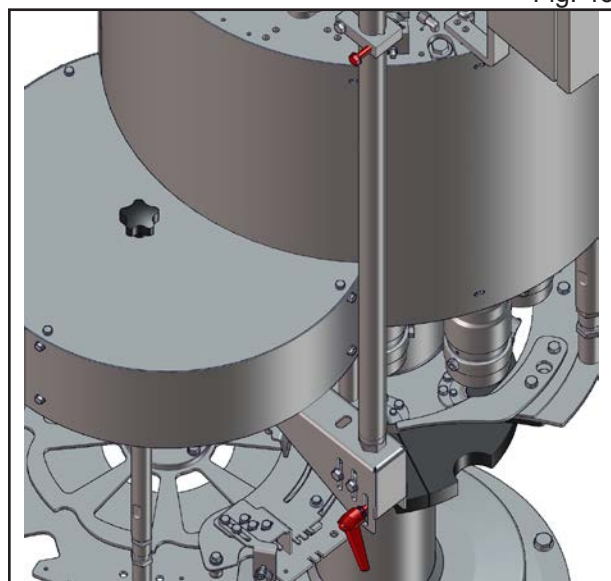


Fig. 47



- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.

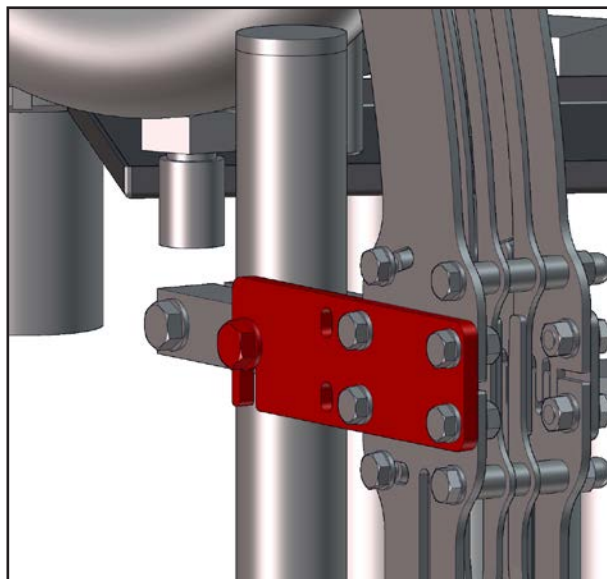


Fig. 48

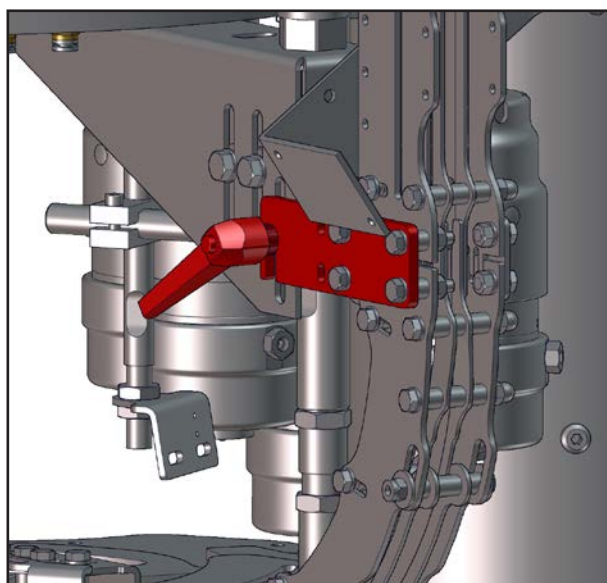


Fig. 49

- Check that the centering pin of the caps chute fits into the seat of a basic disk.

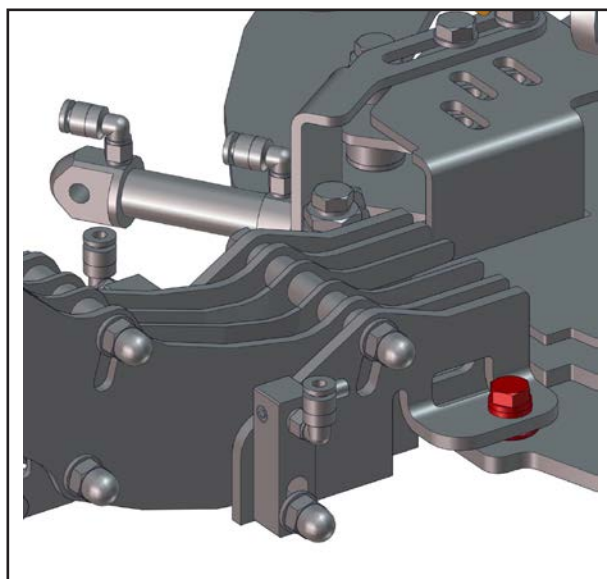


Fig. 50



- Fasten the caps chute to the base disk with the fastening bolt.

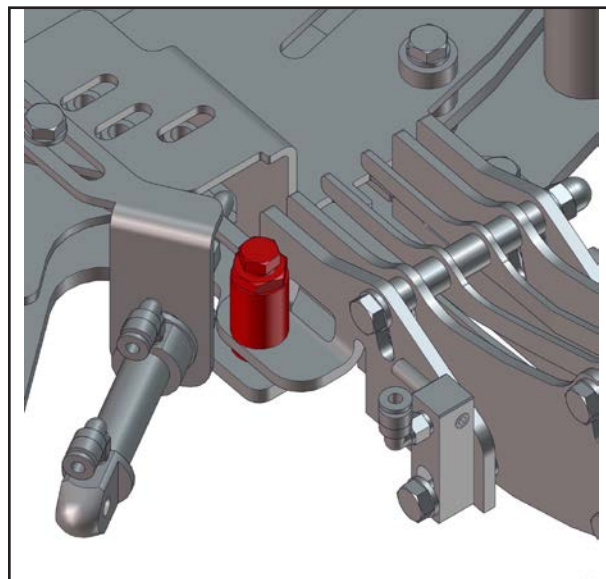


Fig. 51

- Lock threaded fasteners located on the support chute.

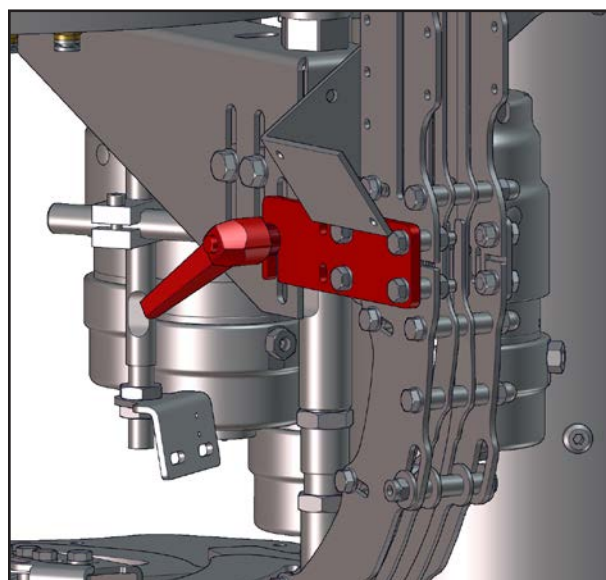


Fig. 52

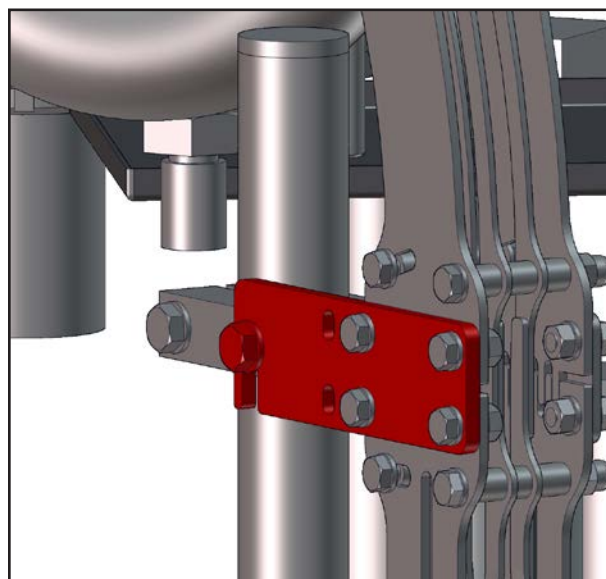


Fig. 53



- Connect the electrical connection of the caps presence photocell (one or more).

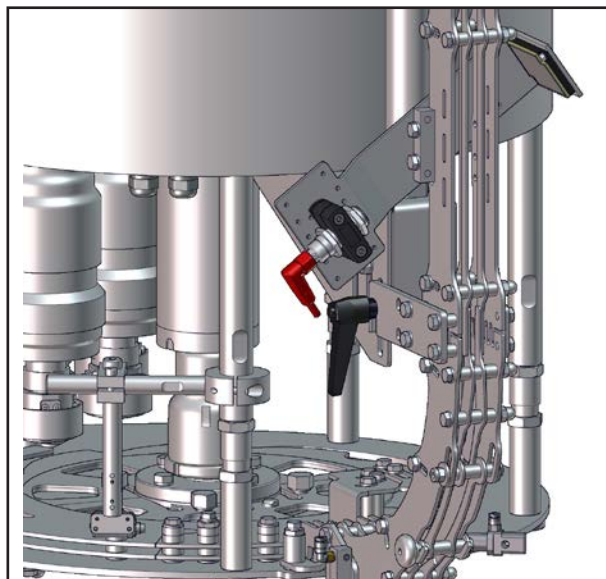


Fig. 54

- Connect the air compressing fitting to them connections.

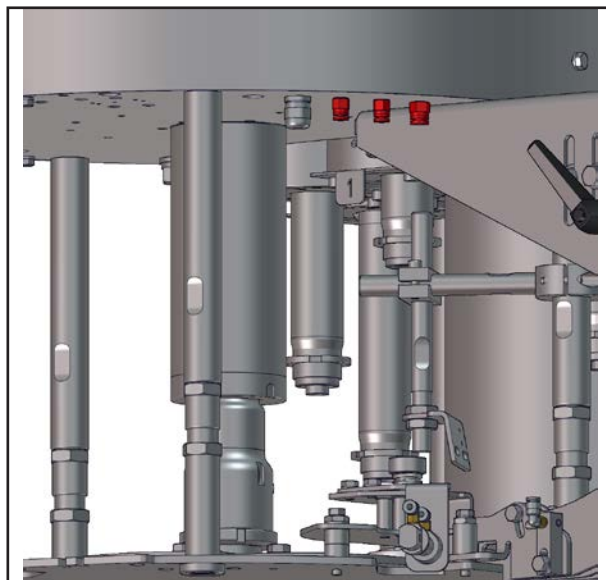


Fig. 55



8.7 INSTALLATION OF THE "FLAT" CENTRIFUGAL CAPS SORTER



For the installation of the centrifugal caps sorter see the listed below operations:

- Fasten belts or chains with hooks to the eyebolt outside the hopper.



Fig. 56

- Unscrew the bolts that fasten the caps sorter to the transport platform.
- Lift the caps sorter with adequate lifting means.



The caps sorter mass is not symmetrically distributed so it overturns when lifted up. Do not stand or walk under the caps sorter when it is hoisted!

- Position the caps sorter complete with the support in such way that the references on the support and the references of the flange are aligned and that the exit mouth is aligned with the caps chute.
- Fix the caps sorter by bolts.

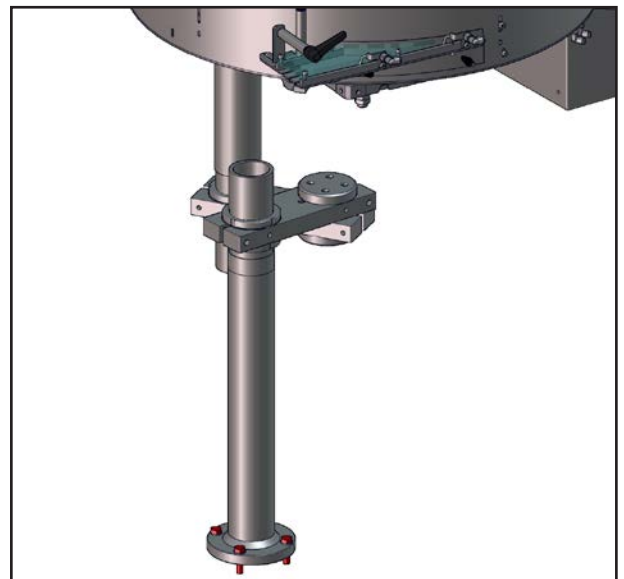


Fig. 57



- Unhook the belts or the chains with hooks from the eyebolts.

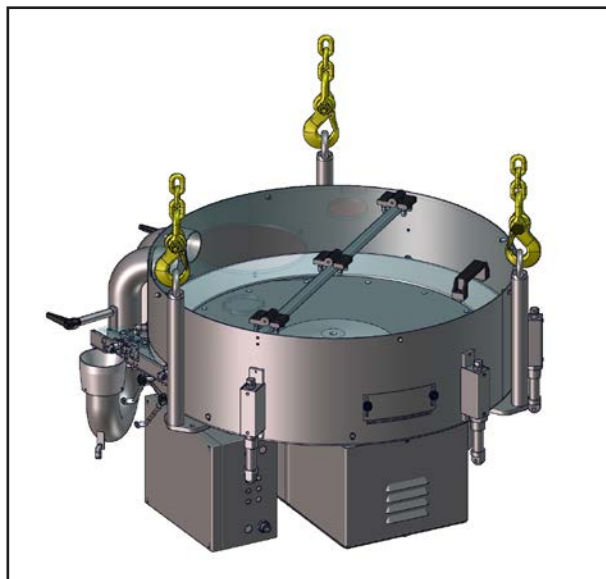


Fig. 58

- Lock the return handles located on the caps exit mouth.

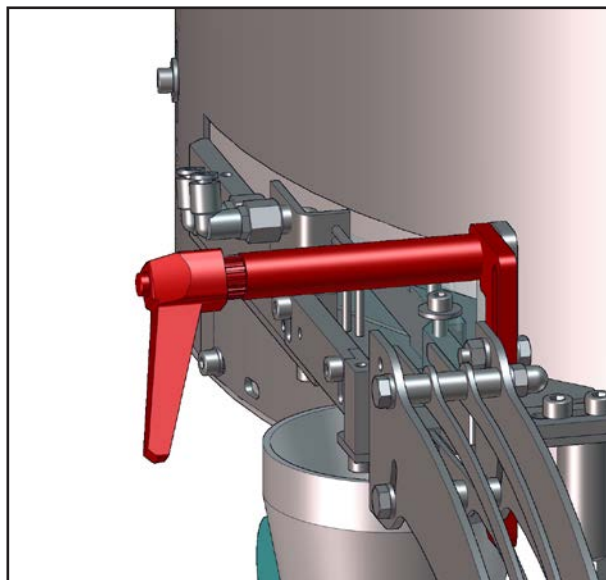


Fig. 59

- Place the caps level device on the caps sorter proceeding as described below:
 - Place the level blade in the hole of the cover and the rod fixing bush on the hopper.

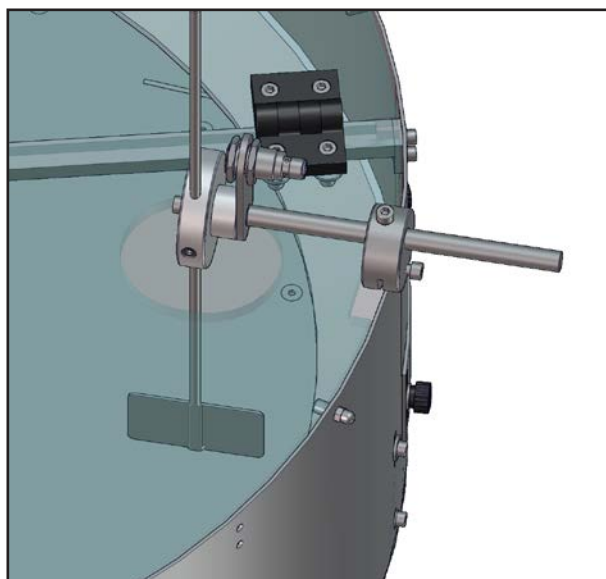


Fig. 60



- Fasten all things locking the screw against the hopper.

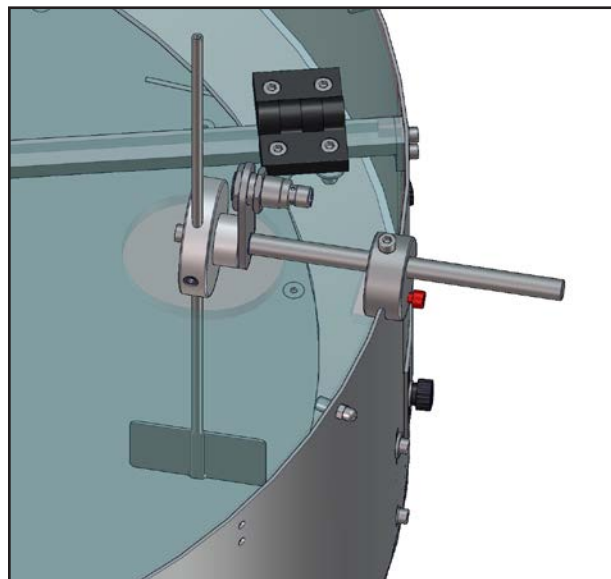


Fig. 61



8.8 PNEUMATIC CONNECTION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The air used in the pneumatic installation must be filtered, so without impurities, and dried.

For pneumatic data see the table page 12.

Connect with the mains as follows:

- Connect the flow rate pipe coming from the main pneumatic unit to the caps orientation unit coupling

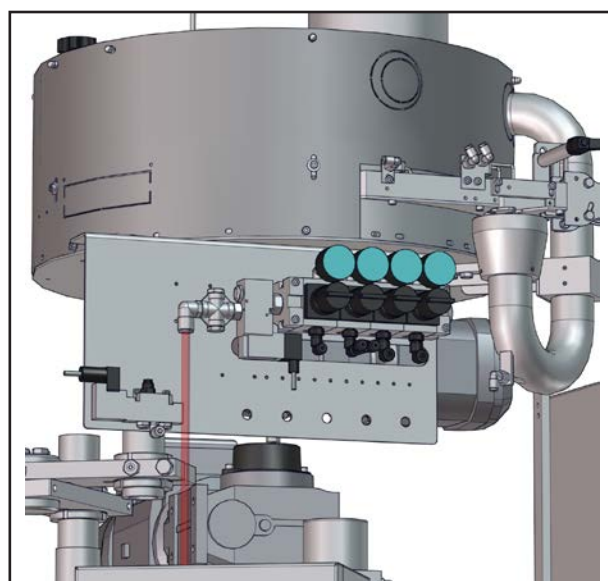


Fig. 62

- Connect the coupling of the air treatment unit to the main distribution network.

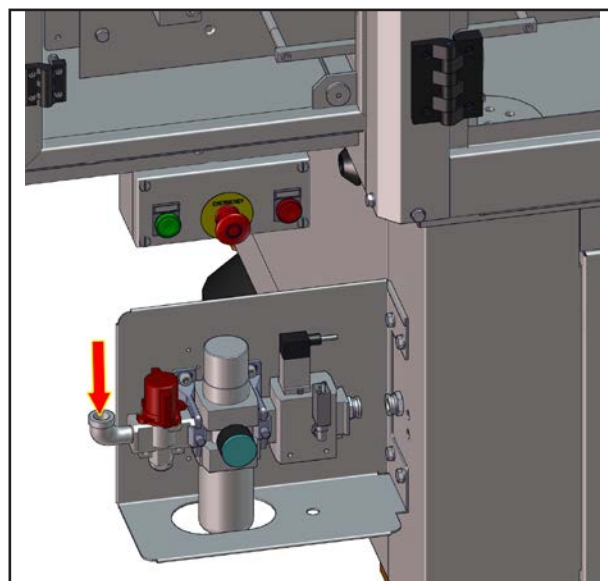


Fig. 63



- Enable the sectioning device to allow air to flow in.

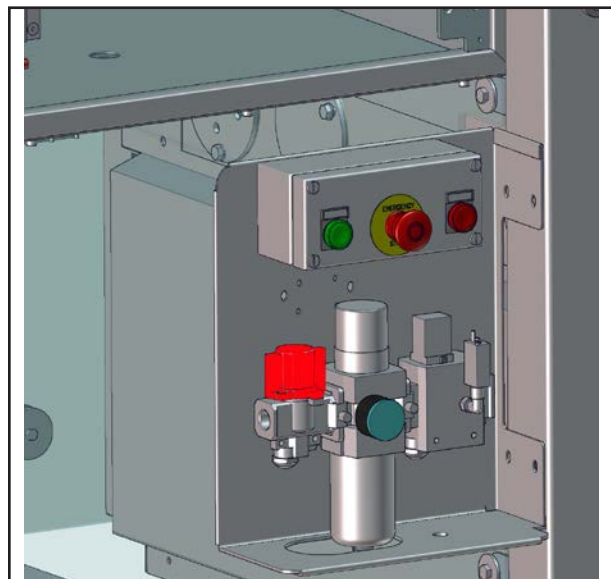


Fig. 64



8.9 ELECTRICAL CONNECTION



Before making the connection to the power supply, check that the characteristics of the power line match the data plate of the closing machine (voltage, frequency, etc.) on the door of the electric panel.



Before the connection, disconnect the electrical feeding in the installation at the bottom of the general switch.



The electrical connection of the closing machine is made by and under the responsibility of the user, who must always keep in mind:

- *laws and technical norms of the country where the machine is installed.*
- *Data reported on the plate on the door of the main electrical board (see the fac-simile example).*
- *Data reported on the electric diagram that is enclosed to this manual.*

Restore the electrical connections of all components that are supplied removed from the closing machine. (See the electrical diagram supplied with the closing machine.)

If the closing machine is not provided with plug connection, the entry of the cable into the electric cabinet from above is provided through a special cable gland and its connection is provided via the terminals marked on the main switch. Connect the three phases (**L1**, **L2**, **L3**), of the ground wire (**PE**) and, where its terminal is present, also the neutral wire (**N**).



The Figure on the side is just an example and can be lightly different, according to the used components.

The data plate is a rectangular form with the AROL logo and contact information at the top. It contains several fields for technical specifications, each with a label and a corresponding input box. The fields are: Power voltage (Vac), Phase, Frequency (Hz), Auxiliary voltage, Total load current (A), Short circuit rating (kA), Enclosure environment type, Field wiring diagram / sn, and Year. At the bottom, it mentions European Standards EN 60204-1 and EN 60439-1.

AROL closure systems	
viale Italia, 193 14053 Canelli - ITALY tel. +39 0141 820 500 fax +39 0141 820 555	
Power voltage	Vac
Phase	
Frequency	Hz
Auxiliary voltage	
Total load current	A
Short circuit rating	kA
Enclosure environment type	
Field wiring diagram / sn	
Year	
European Standard: EN 60204-1 EN 60439-1	

Fig. 65

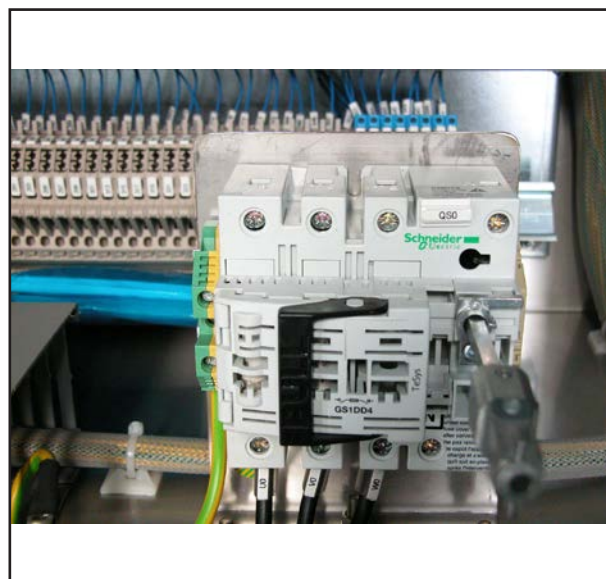


Fig. 66



To avoid damages and let the closing machine operate in a correct way, it is necessary to connect the machine to an efficient grounding.

A good grounding of the closing machine it is necessary for the protection against indirect contacts. Moreover it decreases the possibility of inconveniences that may affect on the control circuit.

Inside the main electrical panel there is an equipotential bonding (ground bar) to which all the grounds on the machine and all the grounding points of the equipment located within the same panel are connected. To ease the connection of the ground wire to the line, was derived from a yellow/green terminal next to the main switch, which acts as "grounding point"; the ground wire shall be connected to this terminal.



*When connected to a system of type **TN**, in order to guarantee effective protection of persons against indirect contact, you should always make sure that the impedance of the fault loop in the supply point is such as to ensure proper coordination with the protective device installed upstream of the closing machine. This is essential to prevent, in case of fault, contact voltages persistence for dangerous times.*

A differential current device, placed at the bottom of the closing machine, is recommended.



*When connected to a system of distribution of type **TT**, please note that a residual current device (appropriately coordinated with the ground system) **MUST** be installed upstream of the electric panel of the closing machine.*



9 CLOSING MACHINE USE

9.1 SETTING UP, CHECKS AND TEST FOR THE FIRST START UP



The setting up operations of the closing machine for the first start up must be carried out by qualified technicians.

The closing machine has been tested by AROL S.p.A. premises before the shipping; all the calibration and required adjustments have been carried out during the test.



Before carry out operations on the closing machine, it is necessary to install and activate the lubricant cartridge, proceeding as described in sub-chapter 11.6.6.2 CARTRIDGE REPLACEMENT page 221.

9.1.1 CHECKS BEFORE THE START UP



In order to prevent errors and/or accidents, before the start up of the closing machine it is necessary to carry out some checks:

- Check that all the electric power supply phases are correctly connected.
- Check that the electric power supply corresponds to the one shown on the identification data plate located on the electric board and that the power draw is consistent with the value shown on the data plate.
- Check the sequence of the motor supply phases. (See sub-chapter 9.1.2 VERIFICATION OF CORRECT PHASE SEQUENCE page 61.
- Check that there are no leaks from the pneumatic circuit.
- Make sure that the closing machine stops immediately if one of the interlocked guards opens.
- Check that all the buttons are operating.
- Check the correct connection of the closing machine with the protection equipotential lead.



9.1.2 VERIFICATION OF CORRECT PHASE SEQUENCE

After carrying out the electric connections, turn the main switch on and, before starting the closing machine, operate as follows:

- rotate the switch to the position *MAN*.
- Insert the *DOOR LOCK* disconnecting knife switch. The luminous white lamp *POWER ON* turns on.
- Check that the *EMERGENCY* button is not pressed.
- Press the *RESET* button.
- Press the control  to lift the closure carousel.
- Verify that the closure carousel raises.
- If instead the closing carousel lowers, it is necessary:
 - open the door disconnecting knife switch.
 - Turn the switch off from the main board of the supply line.
 - Check there is no voltage by a voltmeter.
 - Reverse two phase cables.



9.2 MACHINE CONTROLS



In the following pages, in addition to the layout of the control panel, you can find a table that identifies each button or signal lamp and the description of its operation.

The complete and correct layout can be seen in the wiring diagram attached to this manual.

9.2.1 CONTROL PANEL LAY-OUT

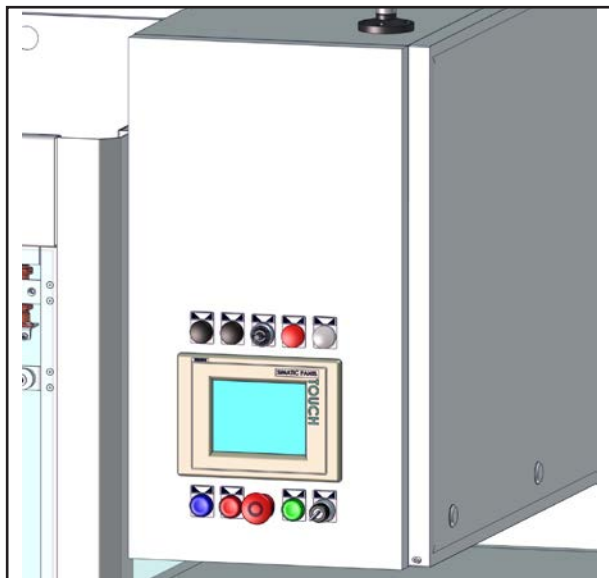


Fig. 67

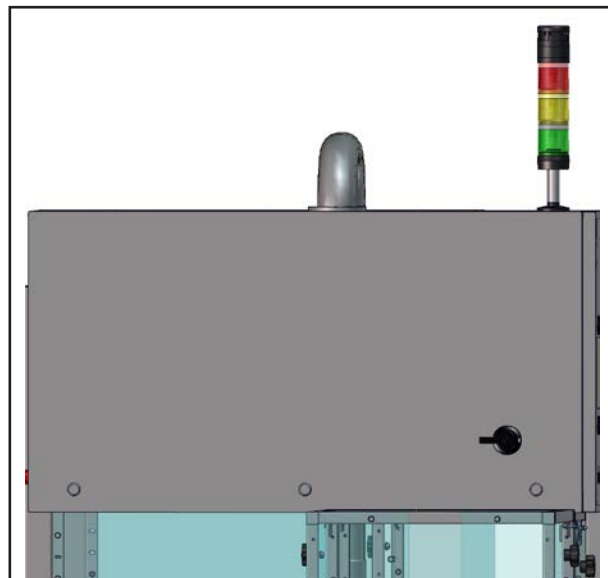


Fig. 68 SIDE VIEW

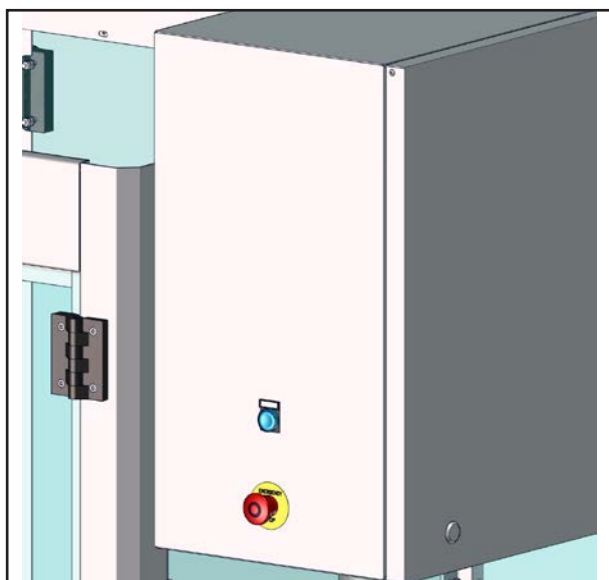


Fig. 69 REAR VIEW

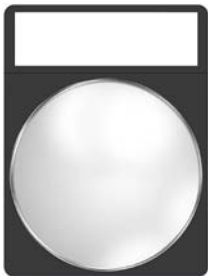


9.2.2 SIGNALLING AND CONTROL DEVICES OF THE CLOSING MACHINE

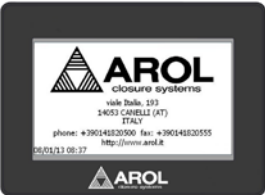
Please find below the descriptions of the control and signalling devices of the closing machine.

SYMBOL	FUNCTION	DESCRIPTION
	DOOR LOCK SWITCH	
	EMERGENCY Red push button on yellow background.	It stops the closing machine immediately.
	JOG Floating black push button.	It controls the jog rotation of the closing machine.
	AUT/MAN SWITCH Rocker switch with key.	AUT - activates the closing machine automatic rotation. MAN - activates the closing machine jog rotation.
	START Floating green push button.	The closing machine is started pressing the push button.



SYMBOL	FUNCTION	DESCRIPTION
	STOP Floating red push button.	The closing machine is temporary stopped pressing the push button.
	SAFETY DEVICE RESET Floating blue lighted push button.	The safety devices are reset pressing the push button. The push button is active only when the light indicator is on.
	SAFETY DEVICE TRIPPED Luminous red lamp.	When on, it indicates that one or more safety devices have activated stopping the closing machine.
	POWER ON Luminous white lamp.	When on, it indicates that the main switch is on.
	PLUG FOR OPTION	



SYMBOL	FUNCTION	DESCRIPTION
	FLASHING LIGHT + SIREN COLUMN	The red flashing light indicates an alarm has occurred or a safety device has activated. The yellow flashing light indicates the closing machine is in stand-by waiting for product. The green flashing light indicates the closing machine is rotating. The siren sends a sound signal every time an alarm or a safety device is activated, when the closing machine re-starts in automatic mode or when the closing machine operates in manual mode (Jog, Lifting/Lowering).
	OPERATOR PANEL	The operator panel displays the states and alarms and allows the actuation of the various functioning controls of the closing machine.



9.3 OPERATOR PANEL



Each displayed message is associated with a short description of its function and of the operation needed to restore the closing machine standard operating conditions.

9.3.1 OPENING PAGE

When the closing machine is turned on, the operator panel performs its initialization functions.

It is important that during the start up phase, no icon is pressed until the panel displays the opening screen.

The page shows info on the manufacturer and on the installed software release (at the right bottom).

Tap anywhere on the screen to access the “*main*” screen.



Fig. 70



9.3.1.1 TIME AND DATE ADJUSTMENT

In the opening page shows the date and time (at the left bottom).



Fig. 71

It is possible to adjust date and time, proceeding as described below:

- Push on date and time, displayed at the left bottom in the opening page.
- On the screen appears the adjustment page. Press on the values, and by the appeared arrows, adjust date and time.
- Press EXIT to go out from the adjustment page.



Fig. 72



9.3.2 MAIN PAGE

During the closing machine standard operation, the operator panel displays the main page that shows:

- operating mode
- recipe number
- production speed
- closing machine status.
- error messages (active alarms)

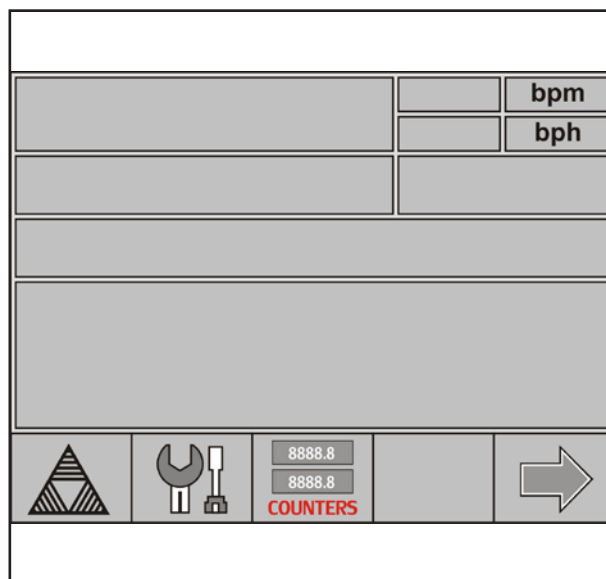


Fig. 73

9.3.2.1 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	INFO Pressing the icon displays the opening screen with information about the <i>AROL S.p.A.</i> . Touching the screen again, you return to the " <i>main</i> " screen.
	LANGUAGE SETTING The language setting page is accessed by pressing this icon.
	COUNTERS Pressing this icon it is possible to enter the page where are displayed the counters on the closing machine.
	KEY FUNCTIONS PAGE ACCESS Pressing the icon, functions page is accessed.



9.3.2.2 OPERATING METHOD

This field indicates the closing machine operating mode.

- **MANUAL:** The closing machine is ready to operate through the manual jog control (*JOG*).
- **AUTOMATIC:** The closing machine is ready for the automatic operation, but not running yet.
- **START:** The closing machine works automatically. If it is stopped, a message explaining the reason is displayed in the machine status field.

9.3.2.3 PRODUCTION SPEED

This field shows an approximate value of the production speed. The calculation is made in *bpm (bottles per minute)* and in *bph (bottles per hour)*.



9.3.2.4 CLOSING MACHINE STATUS

This field displays warning messages on the closing machine operating status.

STATE OF FUNCTIONING	DESCRIPTION
MAXIMUM SPEED	The entrance sensors (B5 and B2) are both dimmed, thus the closing machine is operating at its max. speed.
WAITING FOR BOTTLES	The closing machine is stopped as there is no bottle on the entrance belt. The entrance sensor (B2) must be dimmed before the closing machine is re-started.
EXIT BELT TOO FULL	The exit accumulation sensor (B6) is dimmed, thus the closing machine is stopped waiting for the exit to be cleared.
SLOWDOWN	The exit slowing-down sensor, if installed, is dimmed, thus slowing down the closing machine rotation speed.
WAITING ENABLE	The closing machine is at a standstill waiting for a signal of consent to the rotation, coming from an external source.
EMPTYING	From a bottle lack, it indicates the closing machine is still rotating for a set number of steps necessary to close the containers still under the heads.
WAITING FOR CAPS	The machine is stopped as there is no cap in the caps chute. The cap lack sensor (B1) must be dimmed before the closing machine is re-started.
SYNCHRONIZED SPEED	The closing machine operates in synchronous mode with the set outer reference.
MACHINE IN BY-PASS	The by-pass mode has been selected to exclude the closing machine by the work cycle.
WAITING FOR STEAM	The closing machine is stopped waiting for the actioning of the steam device. Wait for device activation before restarting the closing machine.



9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS)

The active alarm field displays warning messages on possible problems during the closing machine operation. If there are several alarms, the most important one is displayed first, and once the problem explained in the displayed message has been solved, the other present alarms are displayed, till all found problems are solved.

The "Message table" lists all possible alarms and their solutions.



The panel displays only the messages of the closing machine described hereby.

MESSAGE		DESCRIPTION	RESET MODE	
3	ERROR 10 SAFETY CIRCUIT (KM0)	Probable remote control switch failure (KM0)	Switch the closing machine off and check the re. component.	
4	ERROR 11 SAFETY CIRCUIT (KM1)	Probable failure of the closing machine safety remote control switch (KM1)		
6	CHANNEL SAFETY CIRCUIT ALARM 1 SAFETY CIRCUIT	Probable failure of one of the two safety circuit canals.	It is necessary to check the correct operating of the safety sensors and all the others components of the safety chain.	
7	CHANNEL SAFETY CIRCUIT ALARM 2 SAFETY CIRCUIT			
8	ERROR 15 SAFETY CIRCUIT	Interruption of the safety device sequence.	Failure of the monitoring signal of safeties devices.	
9	ERROR 20 SAFETY CIRCUIT	Sometimes it can be displayed because the doors are not correctly closed.	It is necessary to check the correct operating of the mobile guards safeties.	
10	WAITING FOR PROFIBUS ACTIVATION	Wait for the complete inizialization of the profibus circuit.		
15	PUSH THE RESET BUTTON (SB6/SB6A)	All safety devices have been reset.	Press the safety reset button to continue with the closing machine operation.	
16	CAPS CHUTE FULL (B12)	The sensor (B12) noticed a caps accumulation in the caps chute.		



MESSAGE		DESCRIPTION	RESET MODE	
17	LOW AIR PRESSURE (SP1)	A pressure lack in the compressed air circuit does not enable the closing machine correct operation.	Check the presence of air in the pneumatic circuit and possibly the correct operation of the solenoid valves (YV0) and (YV1) or of the pressure switch (SP1). Remove the message pressing the button <i>RESET</i> or <i>START</i> .	
18	CAPS SORTER LATERAL DOOR OPEN (SQ41A)	The sorter side door is open. This can be due to an excessive pressure by the caps for a probable caps chute clogging.	Clear the caps chute and the sorter outlet, close the door. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
19	CAPS SORTER UPPER DOOR OPEN (SQ41B)	The sorter closing door is open.	Close the door. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
20	HEIGHT MAXIMUM LIMIT REACHED (SQ5)	The messages are displayed while adjusting the height and indicate that the limit has been reached and thus it is impossible to continue.		
21	HEIGHT MINIMUM LIMIT REACHED (SQ6)			
22	ADJUSTEMENT HEIGHT REACHED	The stop placed on the sample bottle (if installed) has been reached while adjusting the closing machine.		
23	CAPS CHECK INTERVENTION (B10)	It indicates the intervention of the cap presence check device. If the condition does not stop the closing machine (see parameter 300) the displayed message flashes, while if the alarm stops the closing machine it is fixed.	Remove the message pressing the button <i>RESET</i> or <i>START</i> .	



MESSAGE		DESCRIPTION	RESET MODE	
24	FRONT PANEL EMERGENCY PRESSED (SB10)	The emergency push button on the indicated panel is pressed.	Release the emergency button.	
25	REAR PANEL EMERGENCY PRESSED (SB10B)		Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
26	FOIL CHECK INTERVENTION	It indicates the intervention of the FOIL presence check device.		
27	HEIGHT MAXIMUM REGULATION LIMIT REACHED	The stroke end value has been reached during the height automatic adjustment.		
28	HEIGHT MINIMUM REGULATION LIMIT REACHED			
29	RELEASE THE HANDLES FASTENING ANTI-TWIST COLUMNS (SQ64A/SQ64B)	The handles fastening the anti torsion columns are locked	Unlock the handles.	
30	MACHINE MOTOR OVERLOAD (QM1)	The magneto-thermal switch protecting the specified motor has activated.	Check the reason of the excessive stress on the motor, reset the switch inside the electric board. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
31	HEADS MOTOR OVERLOAD (QM2)			
32	CAPS SORTER MOTOR OVERLOAD (QM4)			
33	HEIGHT REGULATION MOTOR OVERLOAD (QM6)			
34	CAPS STORAGE TANK OVERLOAD			
35	BELT MOTOR OVERLOAD (QM13)			
38	AIR PUMP MOTOR OVERLOAD			
39	HEADS MOTOR FAN OVERLOAD (QM2F)			
40	PANEL ACCESSORIES OVERLOAD (QF17)			
41	SUCTION MOTOR OVERLOAD (QM15)			



MESSAGE		DESCRIPTION	RESET MODE	
42	MACHINE MOTOR DRIVE ALARM (U1)	The frequency converter controlling the specified motor has generated an error message.	Refer to the frequency converter instruction manual to check the cause.	
43	HEADS MOTOR DRIVE ALARM (U2)			
44	CAPS SORTER MOTOR DRIVE ALARM (U4)			
45	BELT MOTOR DRIVE ALARM (U13)			
46	HEIGHT REGULATION MOTOR DRIVE ALARM (U6)			
47	CAPS ELEVATOR MOTOR OVERLOAD	The magneto-thermal switch protecting the specified motor has activated.	Check the reason of the excessive stress on the motor, reset the switch inside the electric board. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
48	CAPS BELT MOTOR OVERLOAD			
49	CAPS BELT MOTOR INVERTER ALARM	The frequency converter controlling the specified motor has generated an error message.	Refer to the frequency converter instruction manual to check the cause.	
50	AIR PUMP MOTOR DRIVE ALARM			
51	ENTRANCE FEED SCREW SAFETY SENSOR (SQ1)	A sensor on the specified driving component for the bottles has activated. This is due to a clogging. If the indication flashes, the component has restored its operating condition. If the indication does not flash, remove the clogging manually.	Check no container hinders the passage of other containers. Remove the clogging, press the <i>RESET</i> button and the <i>START</i> button to re-start the closing machine.	
52	INFEED STAR-WHEEL SAFETY SENSOR (SQ2)			
53	OUTFEED STAR-WHEEL SAFETY SENSOR (SQ3)			
54	TRANSFER STAR-WHEEL SAFETY SENSOR (SQ4)			
56	BELT SAFETY SENSOR (SQ70)	The sensor signalling a faulty operation of the belt tensioning device (SQ70) has activated.		



MESSAGE		DESCRIPTION	RESET MODE	
57	BOTTLE TOO HIGH (B16)	The sensor (B16) indicates there is a bottle exceeding the height limit because the cap is wrongly positioned.		
58	CAPS SORTER SAFETY SENSOR (SQ91)	and the marking of the caps position along the belt from the entrance photocell/camera	Restore the safety conditions. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	 
60	RIGHT-FRONT DOOR OPEN (SQ12)	There door or lid has been opened, so that the closing machine cannot operate standard.	Close the door in question. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
61	LEFT-FRONT DOOR OPEN (SQ13)			
62	RIGHT-REAR DOOR OPEN (SQ14)			
63	LEFT-REAR DOOR OPEN (SQ15)			
64	FRONT DOOR OPEN (SQ12/SQ13)			
65	REAR DOOR OPEN (SQ14/SQ15)			
66	ENTRANCE TUNNEL OPEN (SQ16/SQ16B)			
67	EXIT TUNNEL OPEN (SQ17/SQ17B)			
68	RIGHT SIDE DOOR OPEN (SQ22)			
69	LEFT SIDE DOOR OPEN (SQ23)			
71	VOLTAGE DISCHARGE INTERVENTION	Indicate the intervention of the voltage discharge inside the electrical board.	Verify the cause of the intervention. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
73	DUST EXTRACTOR FULL	It indicates that the dust recovery tank is full.	Empty the tank. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	



MESSAGE		DESCRIPTION	RESET MODE	
74	OVERTURNED CAPS CHECK INTERVENTION	It indicates that there is an overturned cap in the pick & place.	Remove the caps as indicated in the relevant chapter. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
77	HEAD SAFETY SWITCH INTERVENTION	The sensor (SQ90) indicates that a bottle has remained locked under the head.	Remove the bottle left locked. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
82	MINIMUM CAPS LEVEL IN CAPS STORAGE TANK	The sensor in the storage tank reveals a lack of caps The closing machine operating is not stopped.	Fill the tank till dimming the sensor.	
83	FALLEN BOTTLE ALARM (B23)	It indicates that at the closing machine entry there is a bottle lying.	Adjust the bottle position. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
86	LINE EMERGENCY PRESSED	The remote emergency button on the line has been pressed.	Release the emergency button. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
87	BOTTLE PRESENCE SENSOR CHECK INTERVENTION	It indicates the intervention of the check system of the correct operating of the bottles presence sensor.	Check the correct operating of the sensor.	 
88	CAP CHECK DEVICE ALARM	It indicates that the correct capping device is in alarm mode.	Check the device alarm condition on the panel.	
89	CAP CHECK DEVICE STOP	It indicates the stop of the cap inspection device.	Verify the intervention of the device on the operator panel.	
90	MACHINE MOTOR DISCONNECT OFF (QS1)	The disconnect off of the motor has been opened.	Close the disconnect off contact. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	



MESSAGE		DESCRIPTION	RESET MODE	
91	MACHINE MOTOR FAN DISCONNECT OFF	The disconnect off of the motor has been opened.	Close the disconnect off contact. Press the <i>RESET</i> button and the <i>START</i> to re-start the closing machine.	
92	HEADS MOTOR DISCONNECT OFF			
93	HEADS MOTOR FAN DISCONNECT OFF (QS2)			
94	HEADS MOTOR BRAKE DISCONNECT OFF			
95	CAPS SORTER MOTOR DISCONNECT OFF (QS4)			
96	CAPS SORTER MOTOR FAN DISCONNECT OFF			
97	BOTTLES CONVEYOR BELT DISCONNECT OFF (QS13)			
130	MACHINE MOTOR FAN OVERLOAD (QM1F)	The magneto-thermal switch protecting the specified motor has activated.	Check the cause of the excess of effort on the motor; after the resolution reset the switch inside the electric panel	
131	COOLING UNIT OVERLOAD			
132	STERILIZATION OVERLOAD			
140	PLC BATTERY FAULT	It indicates that the PLC battery is in abnormal condition.	It is necessary to check the battery connection and eventually, replace it.	
141	I/O MODULE FAULT	It indicates that one of the entrance/exit modules of the PLC is in alarm.	Check the correct operating of the module.	



9.3.2.6 ALARM MESSAGES HISTORIAN

9.3.2.6.1 VISUALIZATION



It is possible to visualize and store the intervention and resets of the alarm messages historian on an external support.

To see the list, proceed as follows:

- Press on alarms visualization field.

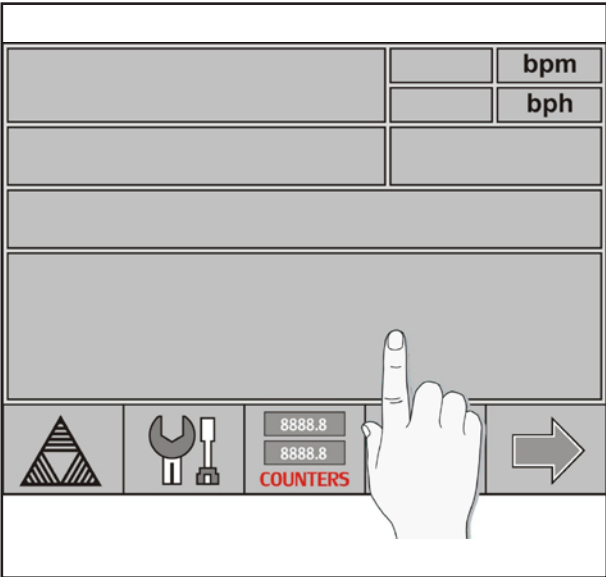


Fig. 74

- The page "*alarms message historian*" appears, where it is possible to see when alarms intervened and when they has been reset.
- Press  to exit the "#ii#alarms message historian#ii#" page.

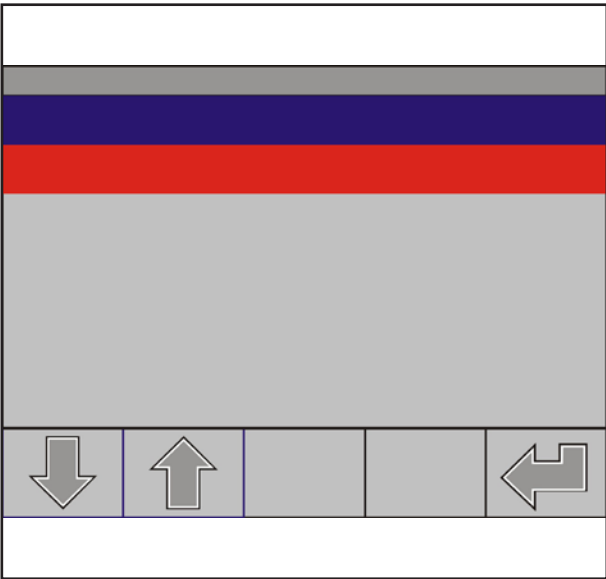


Fig. 75



9.3.2.6.2 DATA STORAGE ON OUTER SUPPORT



It is possible to visualize and store the intervention and resets of the alarm messages historian on an external support.

To save the list, proceed as follows:

- Set the memory card to receive the data from the panel.



Contact the *AROL S.p.A.* Assistance Service, to get the information about the setting of the memory card.

- Turn the door lock switch on *OFF* position.
- Open the electric board.
- Insert the memory card into the USB port, placed in the rear part of the panel.
- Close the electric board.
- Turn the door lock switch on *ON* position.
- The panel automatically stores data on the memory card.
- Wait for the panel to complete the storage operation.
- When the operation is over on the display will appear the relevant message.



Fig. 76

- Press  to exit the "#ii#alarms message historian#ii#" page.
- Turn the door lock switch on *OFF* position.
- Open the electric board.
- Remove the memory card from the USB port, placed in the rear part of the panel.
- Close the electric board.
- Turn the door lock switch on *ON* position.

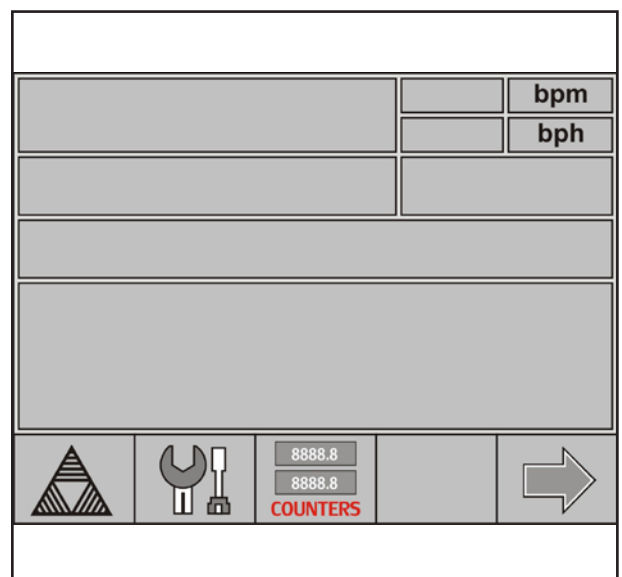


Fig. 77



9.3.2.6.3 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	PREVIOUS LINE Pressing the icon, you can scroll to the previous line.
	NEXT LINE Pressing the icon, you can scroll to the next line.
EXIT	EXIT Pressing the icon, you can exit from the visualization page of the " <i>alarms message historian</i> ".



9.3.3 COUNTERS



It is possible to select any time in the page display counters.

To access the display counters, you must:

- in the "main" page, press .

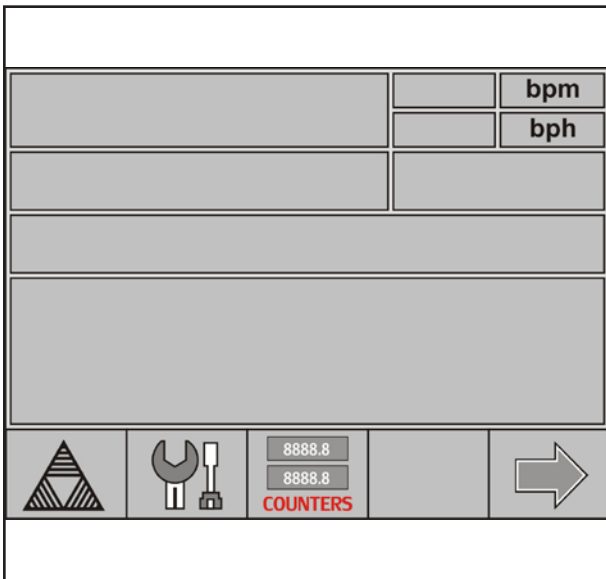


Fig. 78

- In the "machine rotation" page are displayed in the following fields:
 - TOTAL: indicates the total number of hours of work the closing machine. The field is not reset.
 - BATCH: indicates the partial number of hours of work the closing machine. You can reset the count, holding at least 3 seconds for

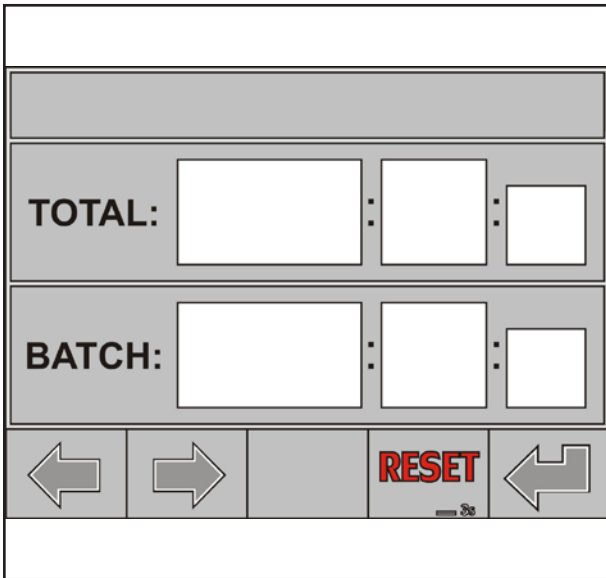


Fig. 79

- In the "bottle" page are displayed in the following fields:
 - TOTAL: indicates the total number of capped bottles during the automatic gearwork cycles of the closing machine. The field is not reset.
 - BATCH: indicates the capped bottles partial number. You can reset the count, holding at least 3 seconds for



- To return to the "main" page, press the  key.

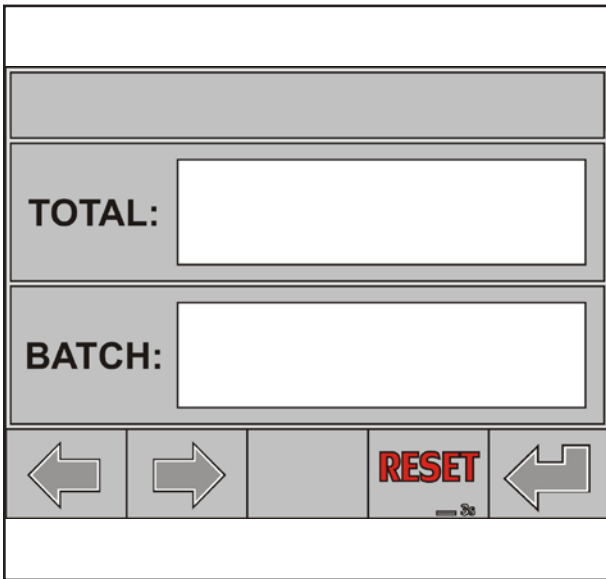
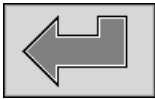


Fig. 80



9.3.3.1 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	COUNTERS Pressing this icon it is possible to enter the page where are displayed the counters on the closing machine.
	RESET Press the icon for at least 3 seconds, you can reset the counter BATCH.
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.4 LANGUAGE SELECTION



The displaying language for all writings in the operator panel can be selected at any time.

To access the language configuration page, is necessary:

- from the "main" page, press .

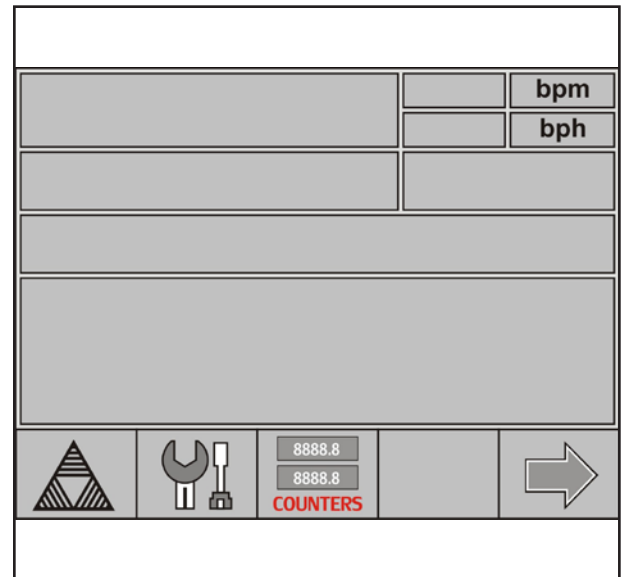


Fig. 81

- Press the key corresponding to the desired language.
- To return to the "main" page, press the  key.

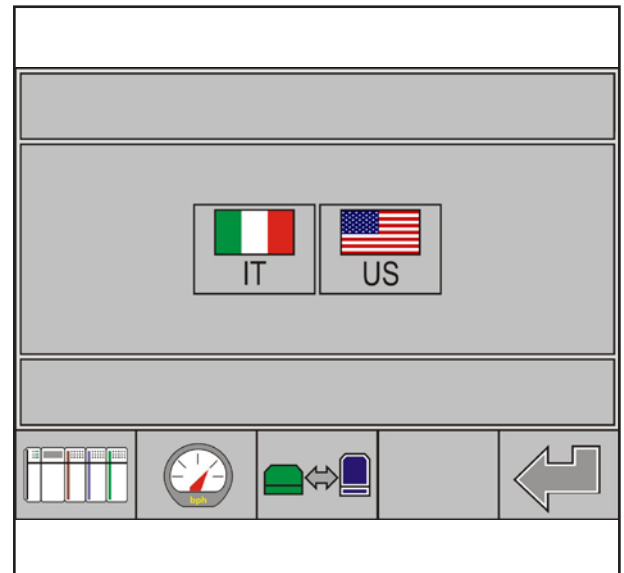
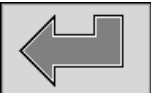


Fig. 82

9.3.4.1 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	LANGUAGE SETTING The language setting page is accessed by pressing this icon.
	LANGUAGE SELECTION The selected language is set on the display pressing the icon for the wished language.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.5 PLC STATUS VISUALISATION



it is possible to see the state of entrances and exit of the PLC, proceeding as described below:

- from the "main" page, press .

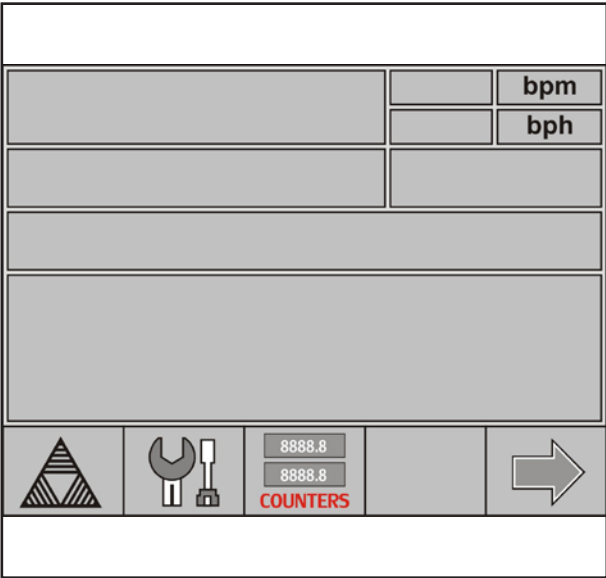


Fig. 83

- Press  to enter into the "input/output status".
- In this page are displayed the the status of the entrances and exit of the PLC.



This is a displaying page only. It is NOT possible to modify any status.

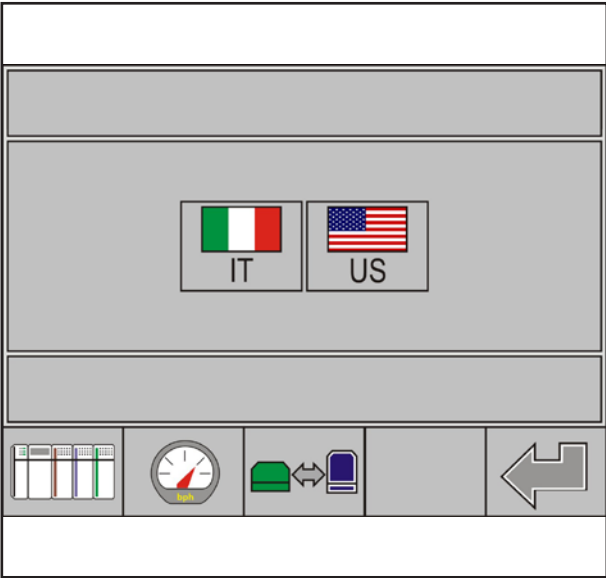


Fig. 84

- Each square indicates the state of the entrances and exit of the PLC.
- For each square there is the relevant PLC address of the represented entrance and exit.
- Each cell represents the address of the relevant bit, **as described in the electric diagram**.
- Press  or  to enter the other pages of state visualisation.
- To return to the "main" page, press the  key.

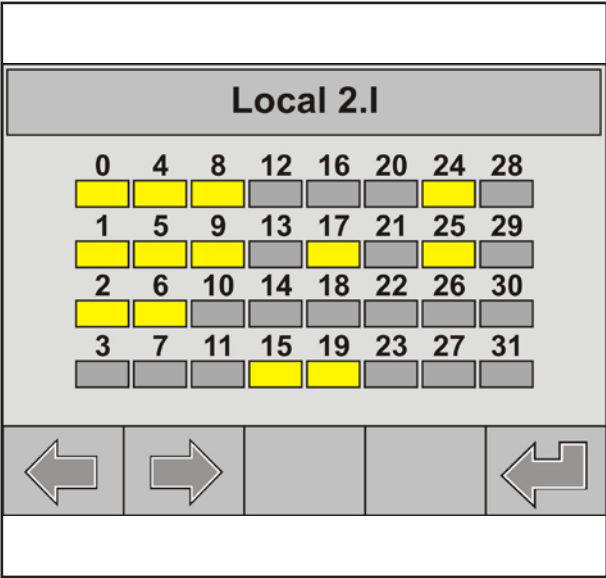
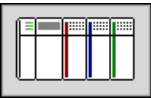
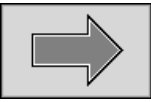
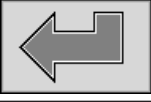


Fig. 85



9.3.5.1 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	PLC STATUS Pressing the icon the status PLC page is accessed.
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.6 SPEED REFERENCE



It is possible to set the speed references of the closing machine, proceeding as follows:

- from the "main" page, press .

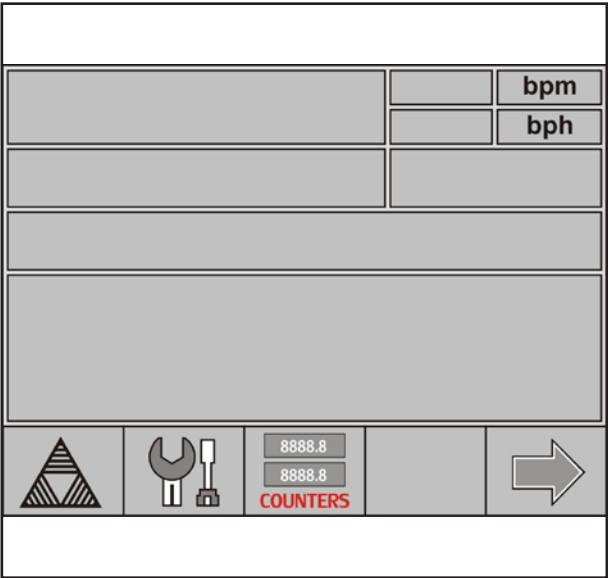


Fig. 86

- Press  to enter into the page of closing machine speed adjustment.

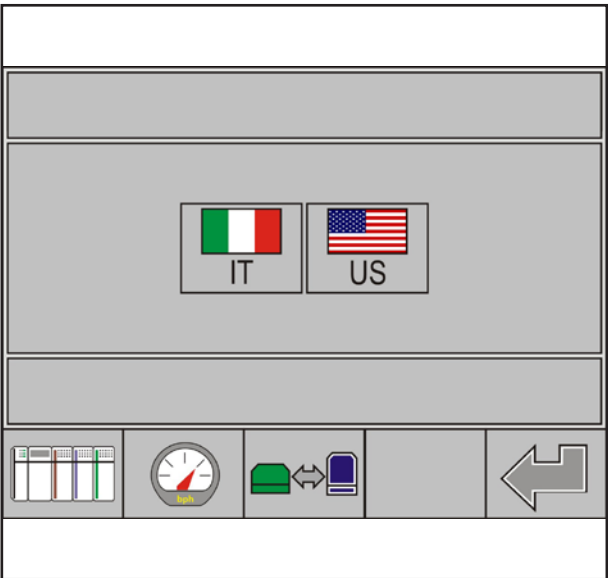


Fig. 87

- Press the appropriate field and enter the desired value.

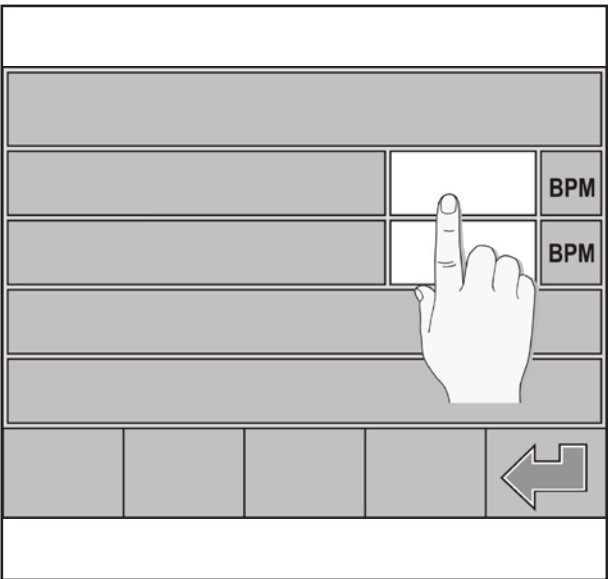


Fig. 88



- Type on the keyboard appeared, the new value to be set.
- Confirm by pressing the *ENTER* icon, on the keyboard.

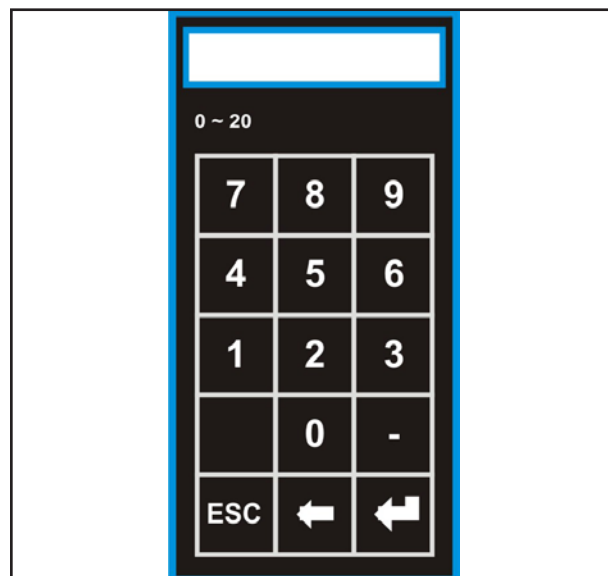


Fig. 89



It is possible to set a value between the minimum speed and the maximum production speed.

- To return to the "main" page, press the  key.



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.

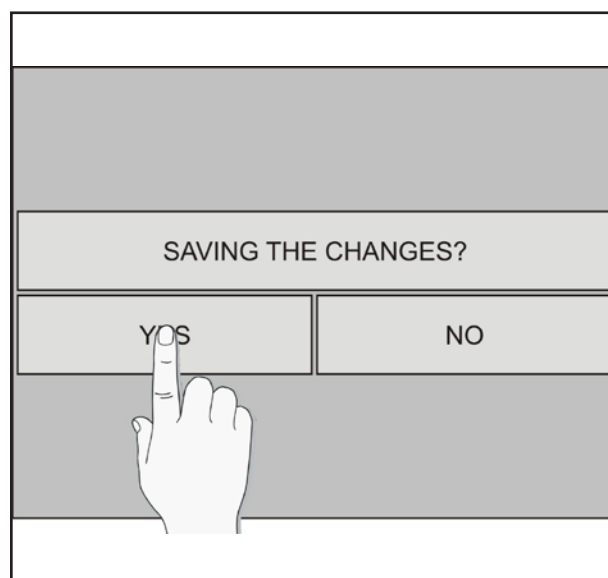


Fig. 90



9.3.6.1 DESCRIPTION OF THE SPEED REFERENCE

The table below contains descriptions of the parameters of speed.

PARAMETER	DESCRIPTION
PRODUCTION SPEED	The parameter is referring to the closing machine rotation speed, when the entrance sensors B2 and B5 are both engaged.
MINIMUM SPEED	The parameter is referring to the closing machine rotation speed, when the automatic start up sensor B2 only is engaged. The minimum value that could be set is equal to $\frac{1}{3}$ of the maximum speed (4500bph).

9.3.6.2 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.7 PRODUCT RECIPES



9.3.7.1 ENTER THE PRODUCT RECIPE SELECTION PAGE

It is possible to select a recipe previously stored, proceeding as follows:

- Make sure that the *AUT/MAN* switch is set to *MAN*;
- from the "*main*" page, press .

- Press  to enter the page for displaying recipes.



In case of access to the page "*product recipe selection*" without registration, the display shows an error message.

Close the error message pressing the icon key "X" and carry out the log-in.

- Press the "NO USER LOGGED" field to access the login page.

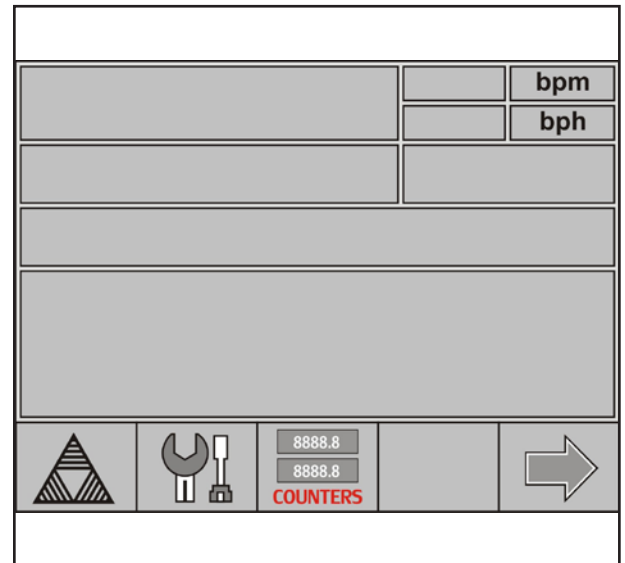


Fig. 91

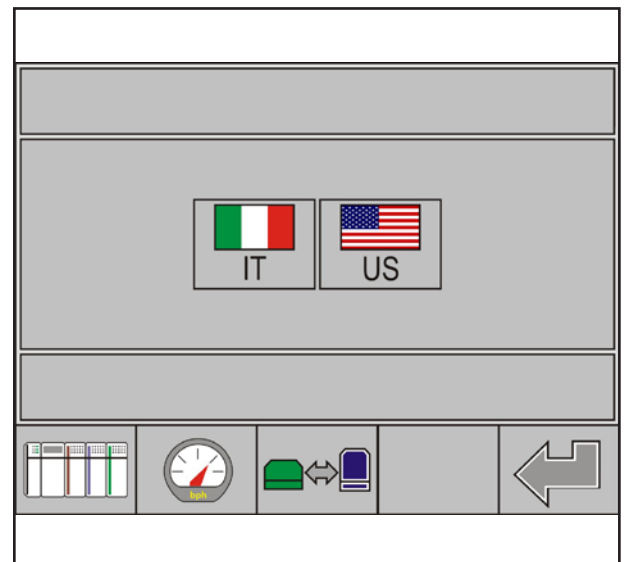


Fig. 92

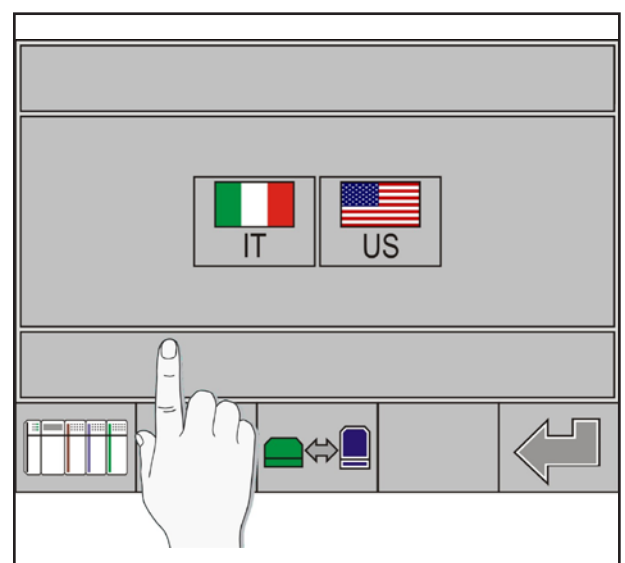


Fig. 93



- On the display will appear a keyboard on which you must enter the user name and password. (See sub-chapter 9.3.7.1.1 **U S E R S ACCESS** page 92).

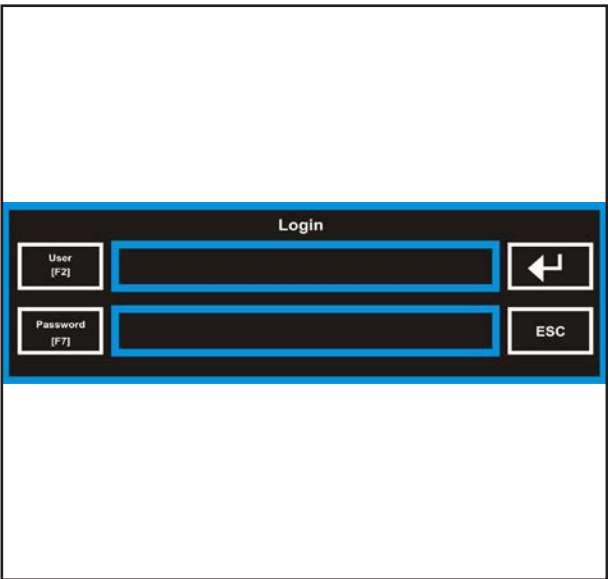


Fig. 94

- Press the *USER* icon on the display and type, on the keyboard that will appear, the user name.

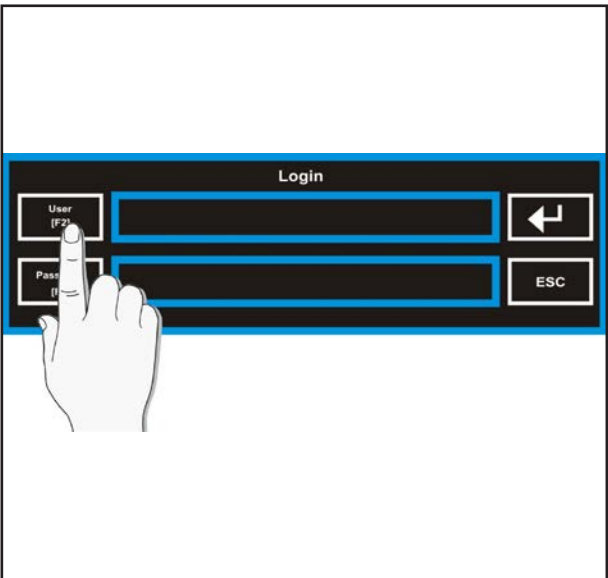


Fig. 95

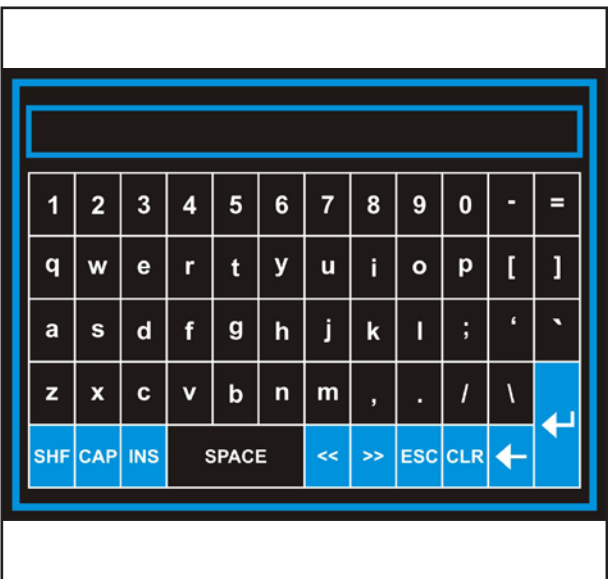


Fig. 96



- Press the *PASSWORD* icon on the display and type, on the keyboard that will appear, the password.

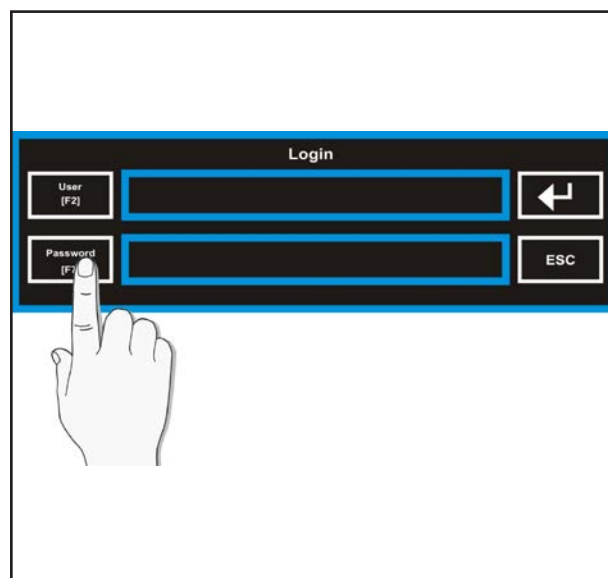


Fig. 97

- Once the user name and password are typed, press the *ENTER* icon for login.
- The display shows the name of the user who logged on.
- Press  to enter to the "product recipe selection" page.
- To return to the "main" page, press the  key.

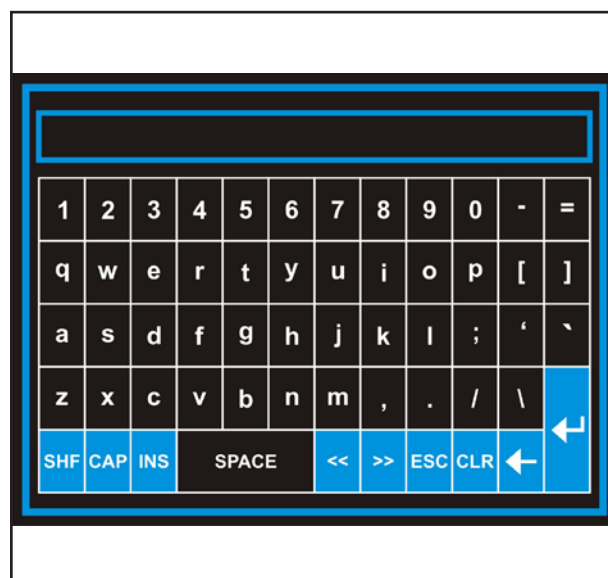


Fig. 98



Once the  icon is pressed, on the display will appear the message "Logout confirmation?". Select the option you want to confirm the exit from the page.

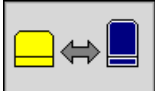
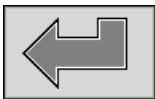


9.3.7.1.1 USERS ACCESS

According to the registered user it is possible to carry out certain operations.

USER	ACCESS
OPERATOR	Accessing as " <i>operator</i> " user it is possible to recall recipes previously stored.
MAINTENANCE MAN	Accessing as " <i>maintenance man</i> " user it is possible to modify parameters of the recipes and create new ones.

9.3.7.1.2 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	PRODUCT RECIPE SELECTION Pressing the icon, the " <i>product recipe selection</i> " page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.7.2 SET RECIPE RECALL

It is possible to select a recipe previously stored, proceeding as follows:

- Enter the "*product recipe selection*" page, proceeding as described in sub-chapter 9.3.7.1
ENTER THE PRODUCT RECIPE SELECTION
PAGE page 89.

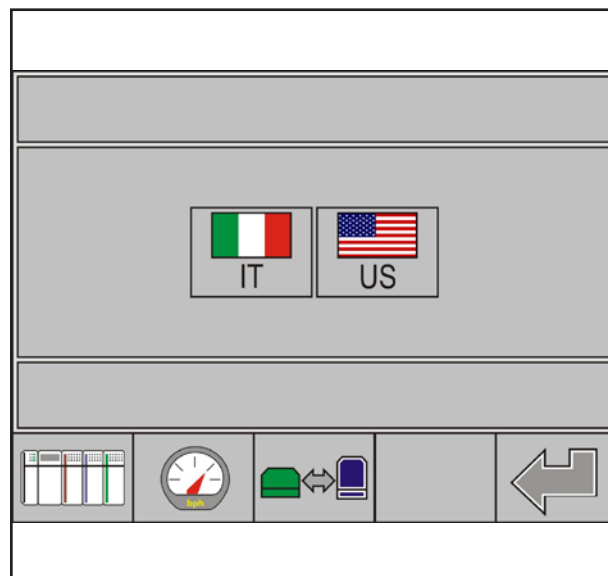


Fig. 99

- Press on the appropriate field on the product recipes.

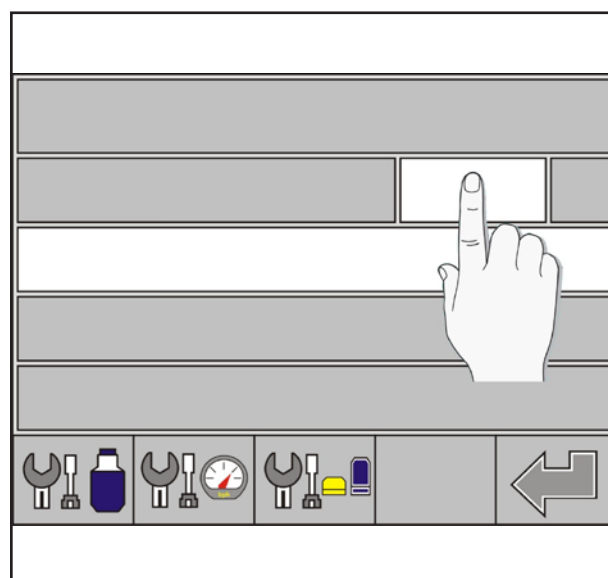


Fig. 100

- Type, on the keyboard appeared, the value relevant to the recipe to recall and confirm the typing pressing the *ENTER* key.
- Once the *ENTER* icon is pressed, the closing machine will automatically set on the values previously stored.

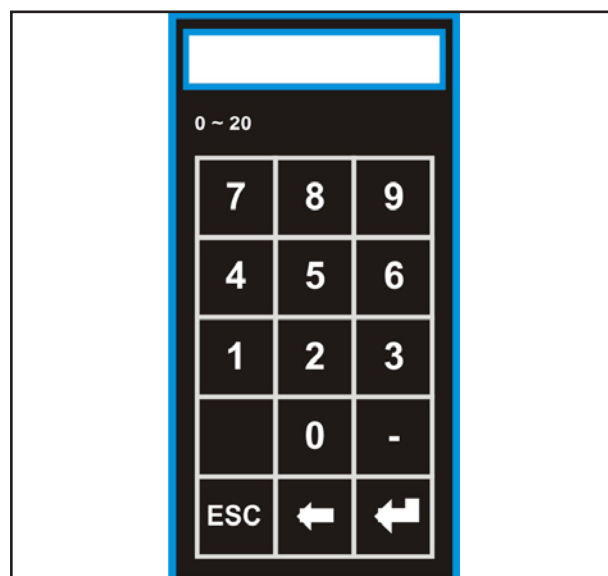


Fig. 101



- To return to the previous page, press .

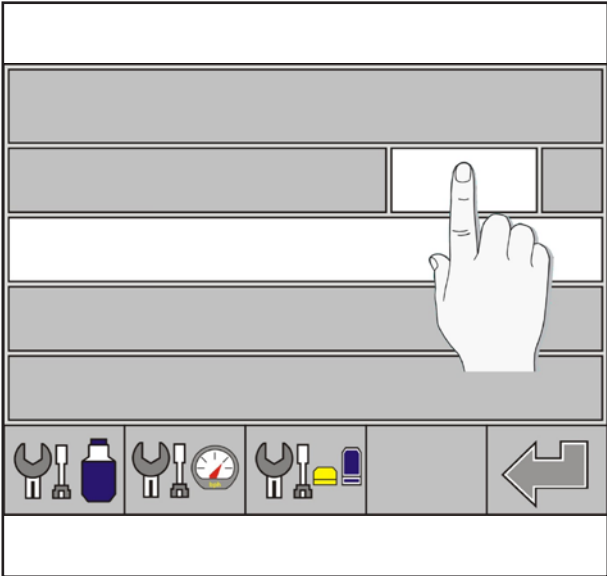


Fig. 102



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.

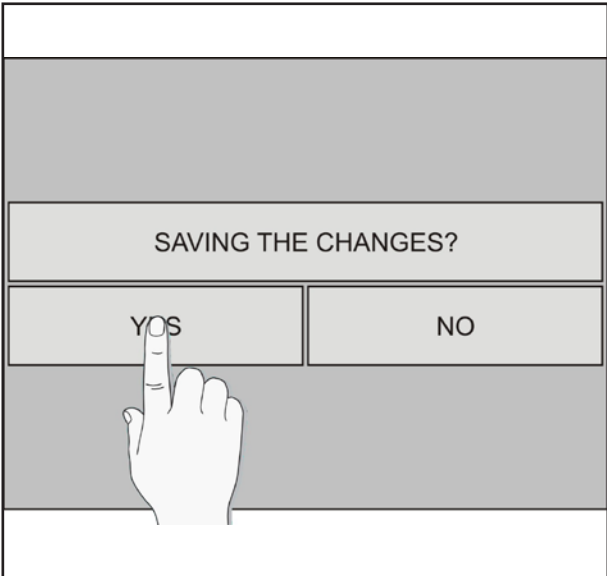


Fig. 103



9.3.7.3 MODIFICATION OF THE CONFIGURATION PARAMETERS

It is possible to modify the configuration parameters of one recipe, proceeding as follows:

- recall the recipe to modify, following the instructions in sub-chapter 9.3.7.2 S E T RECIPE RECALL page 93.



To modify the set up of a recipe it is necessary to access as "*maintenance man*".

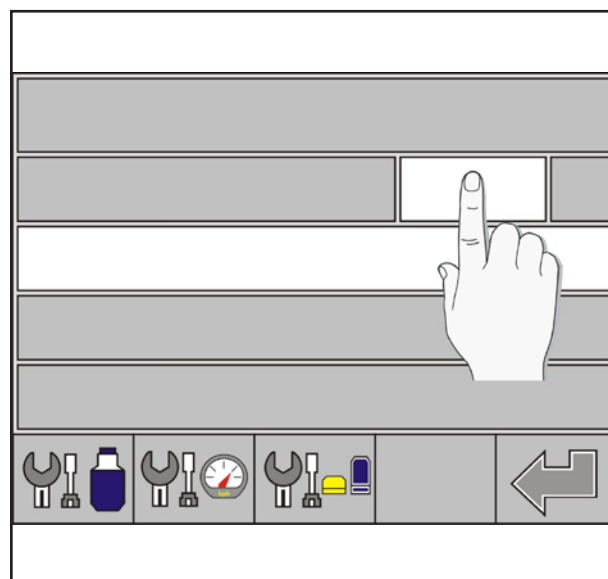


Fig. 104

- Press  to access pages referring to the parameters of the closing machine.
- Following the instructions listed below, access parameters set up pages and carry out the necessary modifications.



9.3.7.3.1 NUMERICAL PARAMETERS

To change the set up numerical parameters of the closing machine, it is necessary to:

- Press on the field of value to modify.
- Type on the keyboard appeared, the new value to be set.
- Confirm by pressing the *ENTER* icon, on the keyboard.

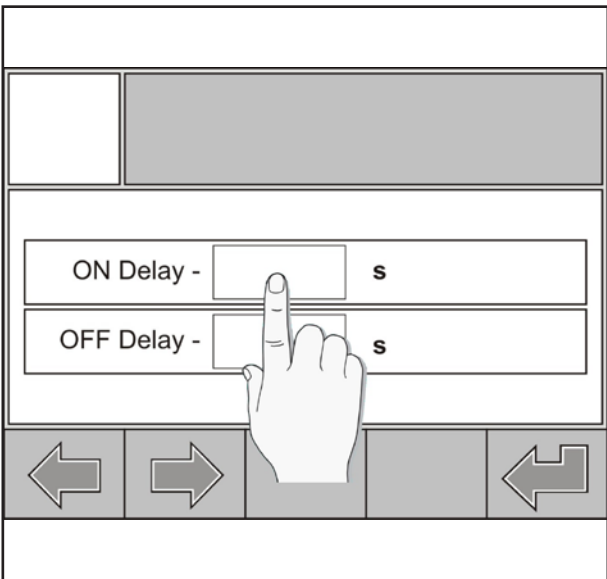


Fig. 105

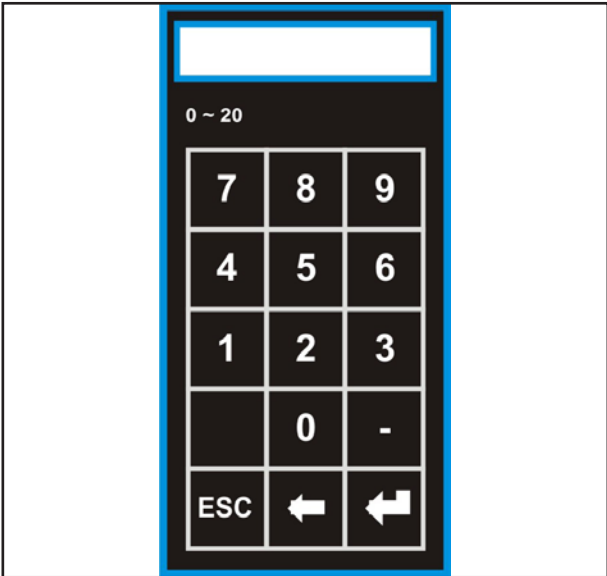


Fig. 106

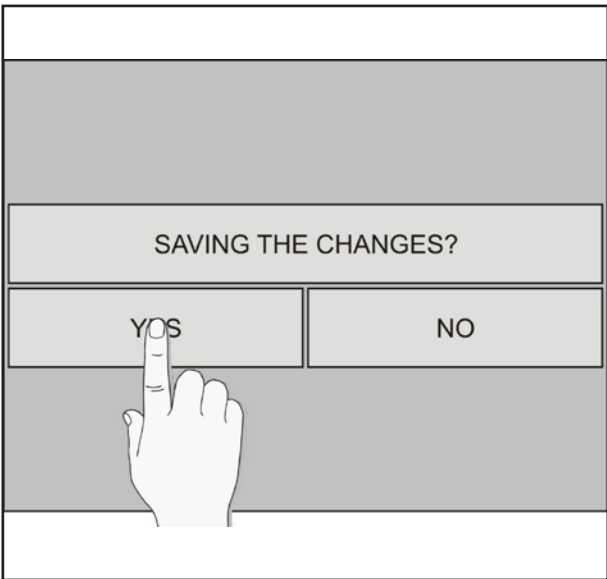


Fig. 107



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.



9.3.7.3.2 ENGAGEMENT PARAMETERS

Pressing on the appropriate icon it is possible to engage (ON) or disengage (OFF) the function referring to the parameter.

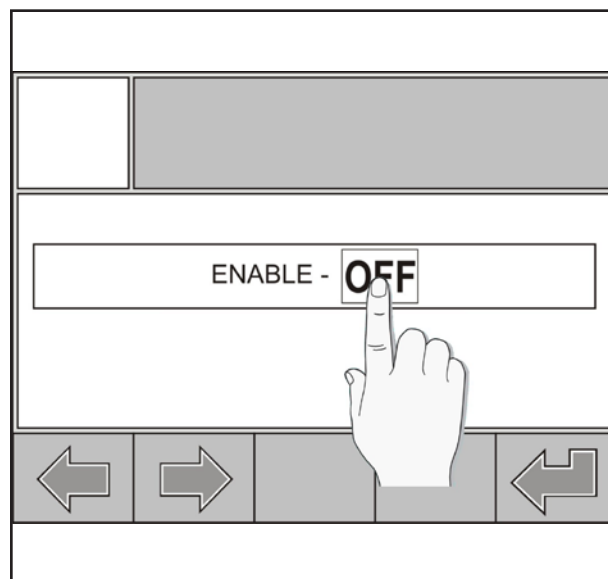


Fig. 108



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.

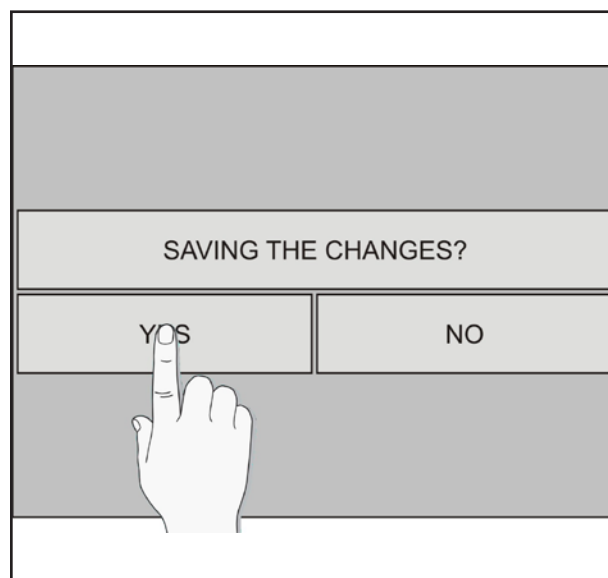


Fig. 109



9.3.7.3.3 PARAMETERS WITH PRE SET CHOICE

To set the operating mode, press one time on the desired icon.

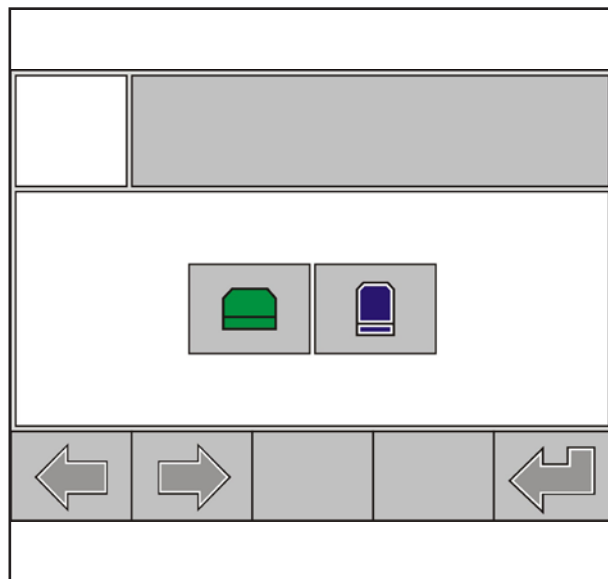


Fig. 110



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.

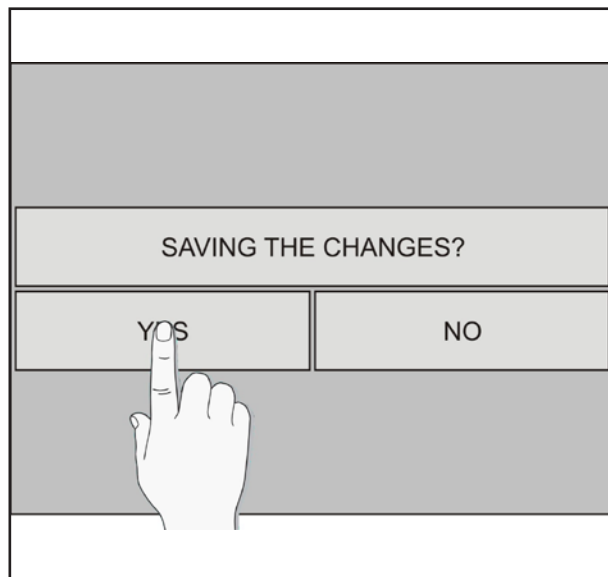


Fig. 111



9.3.7.3.4 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	CLOSING MACHINE PARAMETERS SETTING The parameter setting and displaying page of the closing machine is accessed by pressing this icon.
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.7.3.5 SETUP PARAMETERS LIST



The panel displays only the parameters of the closing machine described hereby.

CLOSING MACHINE PARAMETERS SETTING				
PARAMETER		DESCRIPTION		
100	MISSING CAPS SENSOR (B1)	The cap lack sensor (B1) keep its signal active as long as there are caps in the caps chute.	O N delay	It enables to ignore the gap between caps during the closing machine operation. Typical time 3.0 s
		The lack of signal involves stopping the closing machine. The message is displayed: Waiting for caps.	O F F delay	It enables to ignore the gap between caps during the closing machine operation. Typical value 1.0 s Entering a too high value may result in non-capped bottles.
101	AUTOMATIC START SENSOR (B2)	The automatic start sensor (B2) keep its signal active as long as there are containers at the closing machine entrance.	O N delay	It delays the closing machine start so as to restart with a wider container storage. Typical time 3.0 s
		The lack of signal involves stopping the closing machine. The message is displayed: Waiting for bottles.	O F F delay	It must be set at a value higher than the time between containers passing in the production line. Typical value 1.0 s



CLOSING MACHINE PARAMETERS SETTING				
PARAMETER		DESCRIPTION		
102	MAXIMUM SPEED SENSOR (B5)	The max. speed sensor (B5) reads the presence of containers accumulating at the closing machine entrance to give the consent to the closing machine max. speed rotation. The message is displayed: Maximum speed.	O N delay	It must be set at a value enabling to ignore the simple passage of containers. Typical value 1.0 s
			O F F delay	It must be set at a low value, so that the closing machine rotation slows down as soon as there is no container accumulation. Typical value 1.0 s
103	EXIT SENSOR (B6)	The exit sensor (B6) reads containers accumulating at the exit of the closing machine, stopping it when the exit belt is full. The message is displayed: Exit belt too full.	O N delay	Set the min. value enabling to ignore the passage of containers. Entering a too high value may result in broken containers at the exit. Typical value 1.0 s
			O F F delay	It sets the delay time for the restart after there is no accumulation at the exit. Typical time 3.0 s
104	CAPS SORTER CHECK SENSOR (B12)	The sorter control sensor (B12) reads caps in the caps chute; if there is no cap, it controls the sorter.	O N delay	Delays the perception of the caps presence and allows the loading of the caps chute. The value depends on the caps chute loading speed. A too low value causes an intermittent operation of the sorter, while a too high value causes a safety activation. Typical time 0.4 s.
			O F F delay	Enter a low value to activate the cap feeding more quickly. Typical time 0.2 s.



CLOSING MACHINE PARAMETERS SETTING

PARAMETER		DESCRIPTION		
105	ELEVATOR CHECK SENSOR (B13/SQ30)	The elevator control sensor (B13/SQ30) reads when there is no cap inside the sorter, thus forcing its control signal for loading. The values of this parameter closely depend on the sorter technology.	O N delay	Typical value 1.0 s
			O F F delay	Typical value 1.0 s
106	CAPS STORAGE TANK LEVEL SENSOR (B14)	The cap level sensor (B14) reads the reaching of the cap min. level inside the tank.	O N delay	Set a value to enable a fast intervention in case of emptying.
			O F F delay	
108	CAPS STORAGE STORAGE BIN SENSOR (B15)	The cap level sensor in the storage tank (B15) reads the reaching of the cap max. level inside the tank. The lack of signal involves stopping the machine. The message is displayed: Caps storage bin full.	O N delay	Set a value to enable a fast intervention in case of accumulation. Typical value 1.0 s
			O F F delay	Set a sufficiently long time to enable the correct movement of the cover on the next tank.
109	CAPS STORAGE TANK LEVEL SENSOR (B14)	The elevator cap level sensor (B14) reads the reaching of the cap min. level inside the elevator drum. The message is displayed: Minimum caps level in elevator tank	O N delay	Set a value to enable a fast intervention in case of emptying. Typical value 1.0 s
			O F F delay	Set a value to enable a fast intervention in case of emptying.



ALARMS AND CONTROLS PARAMETERS				
PARAMETER		DESCRIPTION		
200	MACHINE START	Delay time in the closing machine start between pressing the start push button and the carousel starting its rotation. During this time the closing machine gives a sound signal.	O N delay	Typical value 1.0 s.
201	HEADS STOP	The parameter enables to keep the heads rotating for a certain time after the closing machine has stopped.	O F F delay	The heads usually stops simultaneously with the closing machine. Typical value 0.0 s.
203	BUZZER ALARM DURATION (HA1)	The value of this parameter indicates the duration time of the sound alarm given by the closing machine in failure conditions and at the start.	O N delay	Typical value 3.0 s.
204	MAIN SOLENOID (YV1)	This parameter enables to intervene on the switching-off delay of the solenoid valve controlling air blowing vs. the closing machine stop. Two different values can be set for the switching-off depending on the closing machine operating mode.	man.	It is the value applied with the machine operating manually. Typical value 1.0 s.
			aut.	It is the value applied with the machine operating automatically. Typical value 5.0 s.
205	CAPS SORTER	This parameter enables to indicate the delay time of the cap sorter start vs. the closing machine start.	O N delay	Typical value 1.0 s.
			O F F delay	This parameter enables to indicate the delay time of the cap sorter switching-off vs. a stop of the closing machine waiting. Typical value 30.0 s.



ALARMS AND CONTROLS PARAMETERS

PARAMETER		DESCRIPTION		
206	CAPS ELEVATOR	This parameter enables to adjust the delay times at the start and stop of the cap elevator control signal vs. the cap absence read in the sorter.	O N delay	The delay at the elevator start can be useful for some types of feeders needing to work with few caps. Typical value 0.5 s.
			O F F delay	The delay in the elevator switching-off enables to load more caps in the sorter. Typical value 0.5 s.
212	CAPS VIBRATOR STOP DELAY	The parameter defines the time of delay of vibrator turn off in the tank of the elevator, in comparison to the caps elevation device.	O F F delay	Typical value 5.0 s.
213	CAPS CONVEYOR BELT STOP DELAY	The parameter defines the time of delay of belt turn off, in the elevator tank in comparison to the caps elevation belt stop.	O F F delay	Typical value 5.0 s.
214	OVERTURNED CAPS DISCARDER	The parameter allows to define the times that adjust the overturned caps ejection system operating.	O N delay	It indicates the minimum time necessary to define the device intervention.
			O F F delay	It indicates the duration of the intervention of the overturned caps discarder device.
215	DOORS OPENING ENABLE	The parameter refers to the operation of opening and closing of the doors.	N.	
216	CAPS HEATING CYCLE DURATION.	The parameter allows to define the times that adjust the heating system of the caps.	O N delay	It indicates the necessary time for the caps heating.
			O F F delay	It indicates the time beyond which it is necessary repeat the heating cycle, since the machine stops.
217	CAPS TEMPERATURE TIMEOUT	The parameter indicates the time, starting from machine stop, beyond which it is necessary empty the cap chute.	O F F Delay	Typical value 5.0 m.



ANOMALIES PARAMETERS				
PARAMETER		DESCRIPTION		
300	ERRORS CHECK FOR MACHINE STOP	The parameter defines the number of consecutive errors in the cap control to stop the closing machine automatically. If the value 0 is set, the closing machine does not stop automatically still displaying the cap control. The message is displayed: Caps check intervention.	N.	Typical set value 0.
301	STEPS FOR CAP RELEASE	The parameter defines the distance in machine steps between the bottle presence sensor and the cap release actuator. When the value is 0, the cap is released simultaneously with the bottle presence.	N.	The set value must be adjusted at the closing machine installation.
302	STEPS FOR 2ND CAP RELEASE	This parameter defines the distance in steps of the machine between the bottle presence sensor and the release activation for the second cap supply chute. When the value is 0, the cap is released simultaneously with the bottle presence.	N.	The set value must be adjusted at the closing machine installation.
303	STEPS FOR 3RD CAP RELEASE	This parameter defines the distance in steps of the machine between the bottle presence sensor and the release activation for the third cap supply chute. When the value is 0, the cap is released simultaneously with the bottle presence.	N.	The set value must be adjusted at the closing machine installation.



ANOMALIES PARAMETERS				
PARAMETER		DESCRIPTION		
304	STEPS FOR CAP CHECK	This parameter defines the distance in steps of the machine between the bottle presence sensor and the release activation for the cap supply chute.	N.	The set value must be adjusted at the closing machine installation.
305	STEPS FOR EMPTYING CYCLE	This parameter defines the emptying, namely the number of machine steps necessary to keep the turret rotating from the lack of bottles on the conveyor belt, so as to close the containers still under the heads. The message is displayed: Emptying.	N.	The set value must be adjusted at the closing machine installation.
311	CAPS TEMPERATURE	The parameter defines the temperature of detecting sensor.	actual	The value indicates the not set temperature of the sensor.
			set	The value indicates the limit temperature below which appears on the operator panel the error message.



AUXILIARY PARAMETERS				
PARAMETER		DESCRIPTION		
405	SYNCHRONIZED SPEED	The parameter allows the rotation of the closure carousel (at maximum speed) synchronized with the reference given by an analog signal coming from the external line.	ON	
			OFF	
411	BOTTLE-STOP	This parameter activates the operation of the device to lock the containers at the closing machine entry. If enabled, its operation depends on the presence of containers at the closing machine entry; if not enabled, the "bottle-stop" device is never working.	OPEN	OPEN Pressing the relevant button, the device allows the flow of the containers.
			AUTO	AUTO Pressing the relevant button, the device allows the passage of containers according to their flow.
			CLOSE	CLOSE Pressing the relevant button, the device does not allow the passage of containers.
412	BOTTLES DISCARDER UNIT	The parameter activates the operation of the device to reject incorrectly capped containers.	ON	If enabled, its operation depends on the capping control devices installed on the closing machine.
			OFF	
413	CLEANING CYCLE DURATION	The parameter indicates the effective duration of the washing cycle.	ON	
415	HEADS ROTATION	The parameter engages the heads rotation motor.	ON	
416	CAPS SORTER	The parameter engages the caps sorter rotation in automatic mode.	OFF	
			ON	If this parameter is enabled, when the closing machine starts up, automatically will start up also the caps sorter.
417	BOTTLES CONVEYOR BELT	The parameter engages the bottles transfer belt motor.	ON	
			OFF	
418	BOTTLES CONVEYOR BELT BY-PASS	The parameter engages the BY-PASS mode operating of the bottles transfer belt.	ON	
			OFF	



AUXILIARY PARAMETERS				
PARAMETER		DESCRIPTION		
420	MACHINE BY-PASS	It engages or disengages the by-pass mode to exclude the closing machine from the work cycle.	ON	
			OFF	
422	ENCODER ZEROING	Pressing the button it is possible to get the zeroing of the encoder position.	ON	For the zeroing it is necessary to press the button for 3 seconds. The zeroing operation must be carried out after a new adjustment between the star-wheel of the Master machine and the worm of the Slave closing machine.
			OFF	
432	BELT SAFETY SENSOR	Engage or disengage the control for the belt breaking.	ON	
			OFF	
433	CAPS HEATER	Engages or not the caps heating cycle.	OFF	

RECIPE PARAMETERS				
PARAMETER		DESCRIPTION		
500	CAPS SIZE SELECTION	It allows to select the format to proceed.		
501	BOTTLE FORMAT SELECTION	It allows to select the format to proceed.		
503	LINER PRESENCE SENSOR SETTING	It allows to set the sensor that controls the cap presence.		
504	SIZE SELECTION	It allows to select the closing machine version desired.		

MEMORISATION PARAMETERS				
PARAMETER		DESCRIPTION		
900	SAVING THE CHANGES? YES/NO	It allows to store the modification carried out to the recipe set into the recipes archive.	YES	The modification will be stored into the recipes archive and used again to the next recall of the proceeded recipe.
			NO	The modification will not be stored into the recipes archive: to the next recall, the recipe proceeded at the moment, the modified parameters will be replaced by the values previously stored.



9.3.7.4 COMPONENTS SPEED REFERENCES SET UP

It is possible to modify the configuration parameters of one recipe, proceeding as follows:

- recall the recipe to modify, following the instructions in sub-chapter 9.3.7.2 S E T RECIPE RECALL page 93.



To modify the set up of a recipe it is necessary to access as "maintenance man".

- Press  to access pages referring to the components speed references.

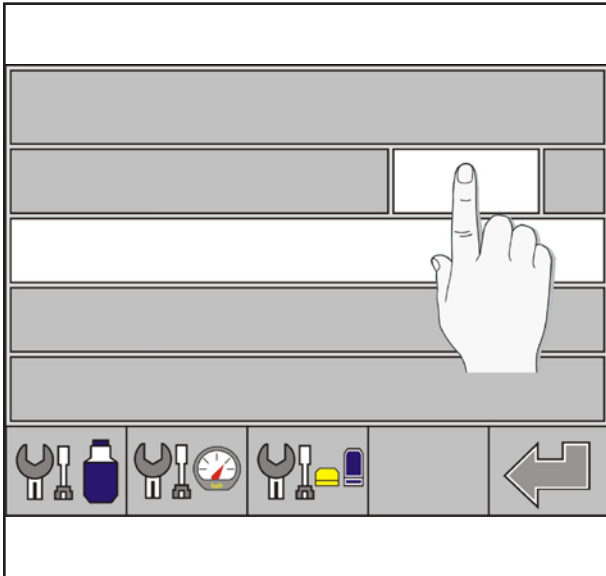


Fig. 112

- Press on the field of value to modify.
- Type on the keyboard appeared, the new value to be set.
- Confirm by pressing the *ENTER* icon, on the keyboard.
- To return to the previous page, press .



Once the  icon is pressed, on the display will appear the message "Save the changes?". Select the desired option to return to the previous page.

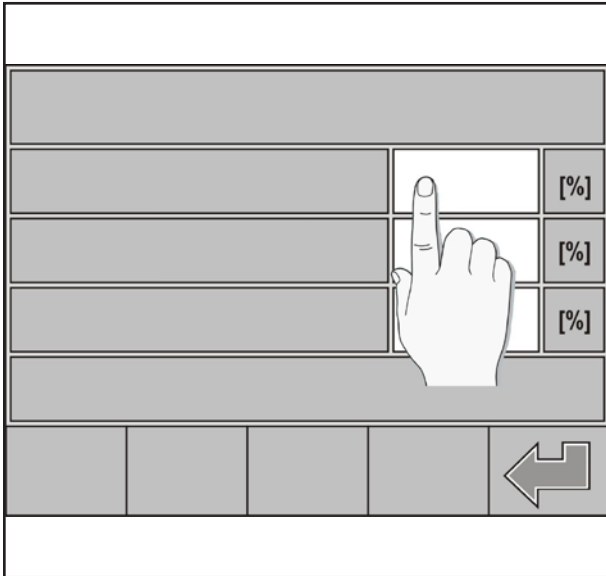


Fig. 113

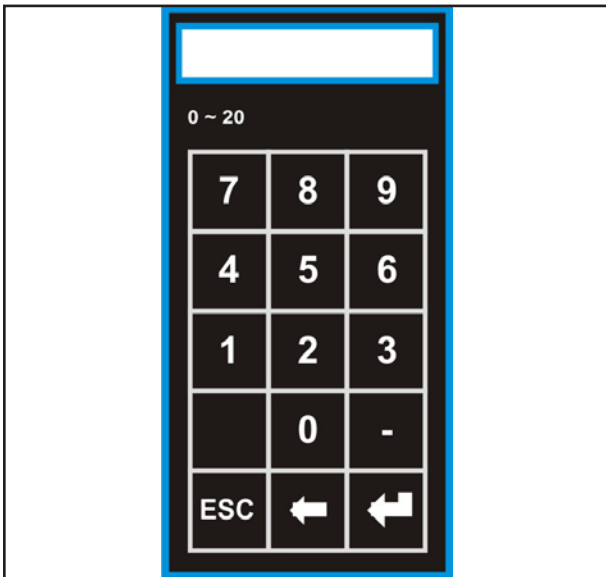


Fig. 114



9.3.7.4.1 LIST OF COMPONENTS SPEED REFERENCES

REFERENCE	DESCRIPTION
HEADS MOTOR REFERENCE	The parameter allows to adjust the rotation speed of the heads rotation motor, according to the closing machine main motor rotation speed. Inserting a negative value in the heads rotation speed reference field, it is possible to get a reverse heads rotation.
CAPS SORTER MOTOR REFERENCE	The parameter allows to adjust the rotation speed of the caps sorter motor rotation. In case of vibrating caps sorter, this parameter allows to increase or decrease the vibrations frequency.
CAPS SORTER MOTOR REFERENCE REDUCTION	The parameter identifies the percentage of the rotation speed reduction in comparison to the set value in "caps sorter motor reference", of the caps sorter. This condition is verified when the caps sorter maximum speed sensor B29 (when provided) is engaged.
CAPS STORAGE TANK REFERENCE	The parameter allows to increase or decrease the vibration frequency of the supplementary tank.
BELT MOTOR REFERENCE	The parameter allows to adjust the rotation speed of the belt rotation motor, according to the closing machine main motor rotation speed.
BELT MOTOR BY-PASS REFERENCE	The parameter allows to adjust the rotation speed of the belt rotation motor, according to the closing machine main motor rotation speed, when the closing machine is in by-pass mode (exclusion from the work cycle).
EXTERNAL REFERENCE	The parameter allows to adjust the analogical exit value (used for the outer synchronism) according to the closing machine main motor rotation speed.



Increasing or decreasing the reference percentage (%), it is possible to adjust the motor speed relevant to the changed value, according to the closing machine main motor speed.



9.3.7.4.2 ENTRANCE SYNCHRONISM SPEED REFERENCE

It is possible to set speed references of the closing machine synchronism, proceeding as follows:

- recall the recipe to modify, following the instructions in sub-chapter 9.3.7.2 S E T RECIPE RECALL page 93.



To modify the set up of a recipe it is necessary to access as "maintenance man".

- Press  to access pages referring to the components speed references.
- Pressing on  or , scroll through the pages until the page "Entrance synchronism speed reference".

- Press on the field of value to modify.
 - ANALOG REFERENCE Insert in fields minimum and maximum analog values of the analog value of the closing machine entrance reference. Field (mA - V)
 - PRODUCTION SPEED Insert in fields the production speed minimum and maximum value, which the closing machine will turn, according to the analog reference received at the entrance. Field (BPH)



Pressing on the unit of measurement, it is possible to vary the value V to mA and vice versa

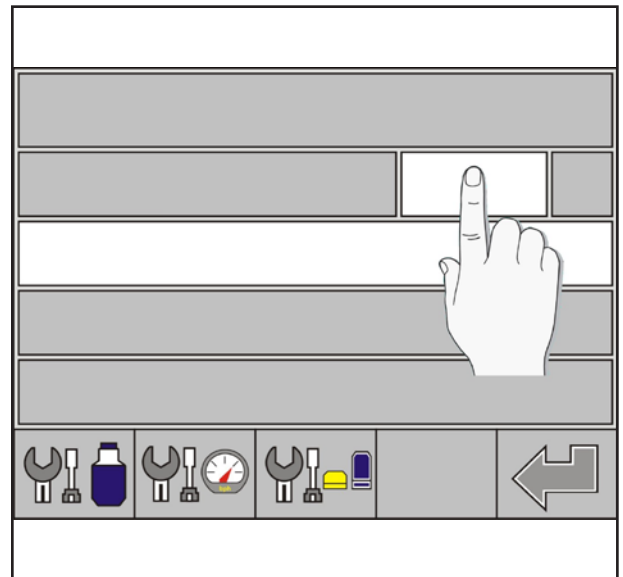


Fig. 115

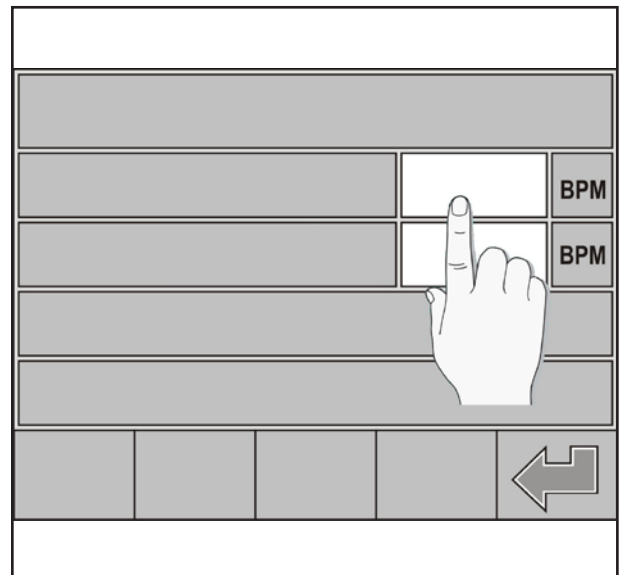


Fig. 116

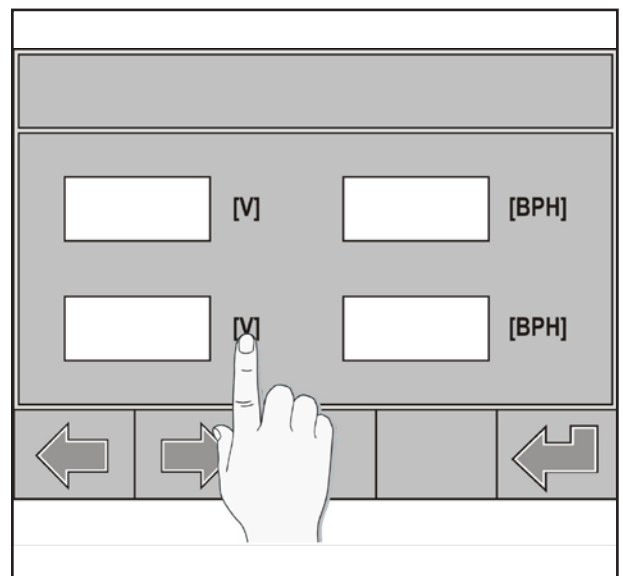


Fig. 117



9.3.7.4.3 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	COMPONENTS SPEED REFERENCES SET UP Press the icon to access the setup page of the speed references of the components of the closing machine.
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	PREVIOUS LEVEL PAGE RETURN Pressing the icon, you can return to the previous level page.



9.3.8 CLOSING MACHINE CONTROLS



On the operating panel there are the icons to control some closing machine functions.

To enter the "*controls pages*" it is necessary to:

- from "*main*" page press .

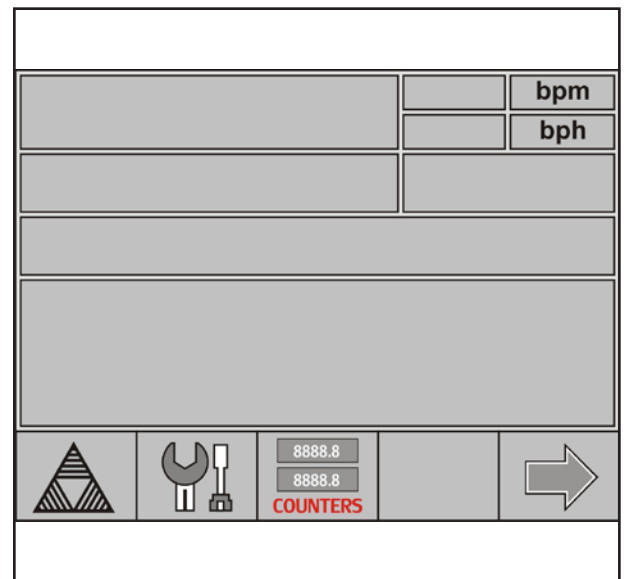


Fig. 118

- According to the operation to do, press on the relevant icon.
- To return to the "*main*" page, press .

In the next subchapter you can find a brief description of the functions associated with the icons.

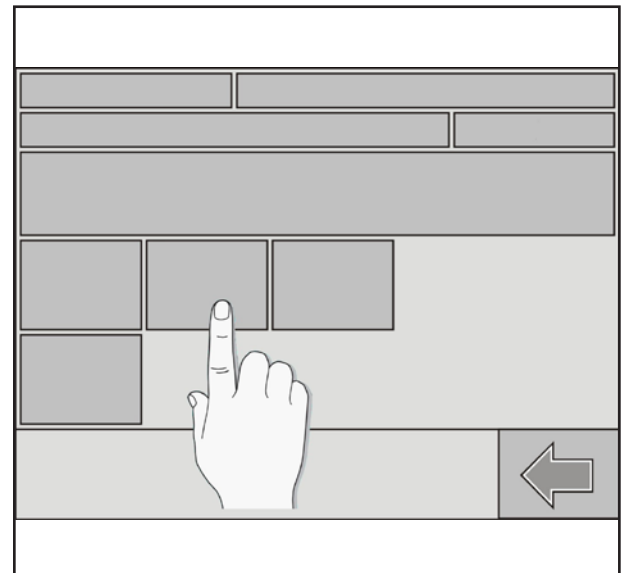
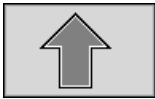
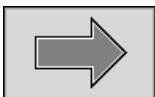
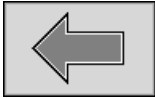


Fig. 119



9.3.8.1 DESCRIPTION OF THE ICONS FUNCTIONS

ICON	DESCRIPTION
	CLOSURE CAROUSEL LIFTING The closure carousel lifting device is operated by holding this icon pressed. Releasing it will stop rising. It operates only if the switch is on <i>MAN</i> .
	CLOSURE CAROUSEL LOWERING The closure carousel lowering device is operated by holding this icon pressed. Releasing it, it stops. It operates only if the switch is on <i>MAN</i> .
	NEXT PAGE Pressing the icon the next page is accessed.
	PREVIOUS PAGE Pressing the icon the previous page is accessed.
	CAPS ORIENTATOR in MANUAL position The cap chute loading is activated in manual mode by pressing this icon. Releasing it, the motor turns off. The icon enables the function on or off when the closing machine stops.



9.4 PRELIMINARY CHECKS BEFORE STARTING



Before starting the closing machine up, check the following:

- Check that all other safety devices installed on the closing machine are operating properly. (See also sub-chapter 5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE page 32).
- Check that the mobile interlocked guards are closed.
- Check all closing machine components are lubricated (See Chapter 11.6 LUBRICATION page 213).
- Make sure the specific installed equipment are suitable for the size to be processed.
- Check the correspondence between container and cork to be treated.



Each set of equipment is marked as indicated in the relevant tables enclosed in this manual. Each mark corresponds to a size to be processed.

- Check the height of the closure carousel according to the container height.
- Check the caps sorter is sufficiently full.
- Check that the working pressure is correct (the pressure is indicated on the pressure gauges).



9.5 PREPARATION FOR STARTING

9.5.1 CLOSING MACHINE HEIGHT ADJUSTMENT



For the adjustment and check, operate as described here below:

- Set the switch *AUT/MAN* in the position *MAN*.
- Release the handles fastening the anti-torsion columns.



Fig. 120



Fig. 121



- Release the screw locking the central column.
- Close interlocked mobile guards and restore the operation of the safety circuit.



Fig. 122

- From "main" page of the operator panel press



Pressing  or , the closure carousel moves immediately.

- Press  or  to lift or lower the closure carousel as long as the indicator corresponds to the value indicated on the millimeter rod.



The height reference values are specified in the EQUIPMENT COLOR TABLE enclosed in this manual.

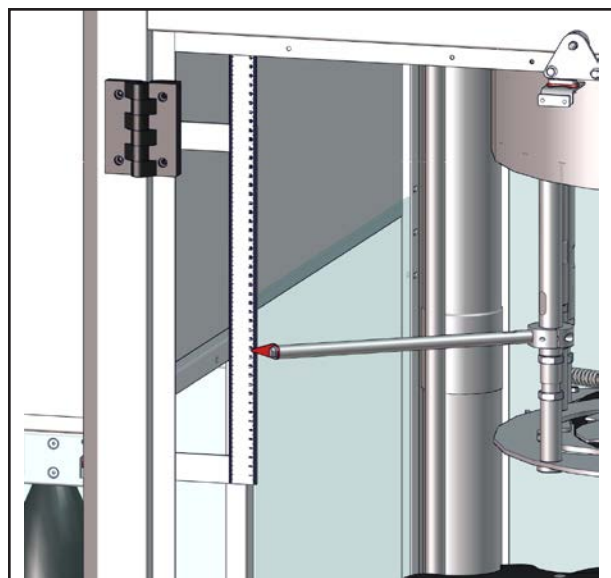


Fig. 123



- Lock the fixing screw of the central column.



Fig. 124

- Lock the handles of the anti-torsion columns.



Fig. 125

- Close interlocked mobile guards and restore the operation of the safety circuit.
- At the adjustment end, turn the switch *AUT/MAN* to *AUT* and remove the key.



Fig. 126



9.5.2 SPEED ADJUSTMENT OF THE CLOSING MACHINE



Adjust the production speed by the operator panel as indicated in subchapter 9.3.6
REFERENCE page 86.

S P E E D



The set value is achieved by the closing machine only when the minimum and maximum speed photocells are engaged. If this does not happen, the speed of the closing machine would be equal to the minimum speed previously set.



9.5.3 CAPS SORTER LOADING



On the caps sorter is installed a caps level sensor having the task to signal to the automatic elevator when it is necessary to put a new quantity of caps.

For correct operation, the device levelling rod must be adjusted so that it is about 5 mm above the height of an only layer of caps.

For this adjustment, loosen the grub screw on the levelling rod bush.

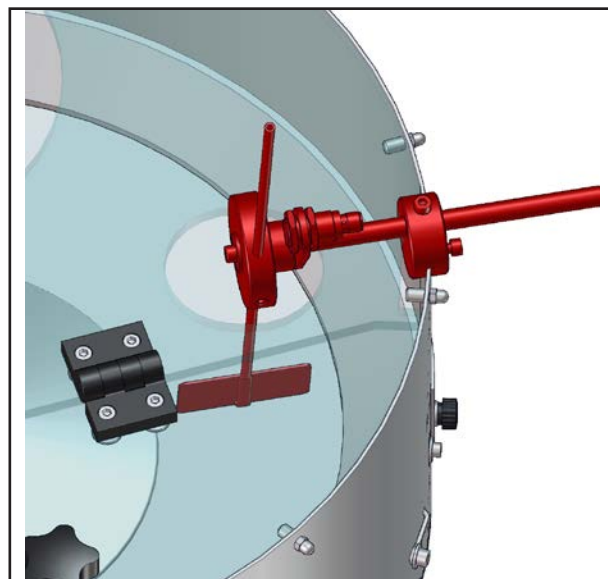


Fig. 127

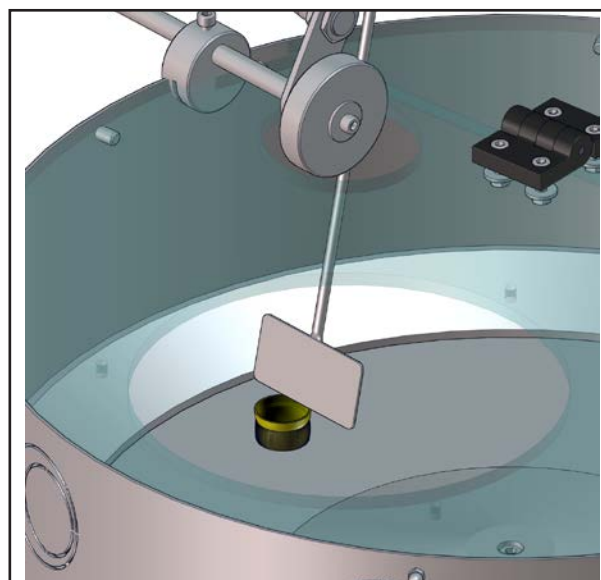


Fig. 128

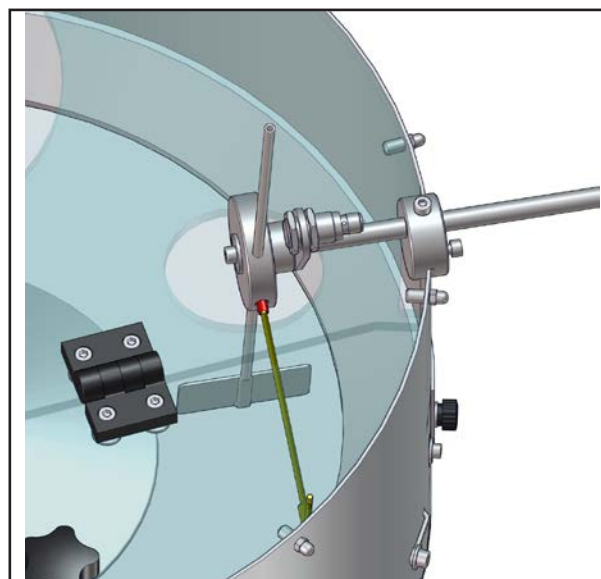


Fig. 129



The levelling rod must be positioned so that there is a distance of one cap between it and the hopper ring.

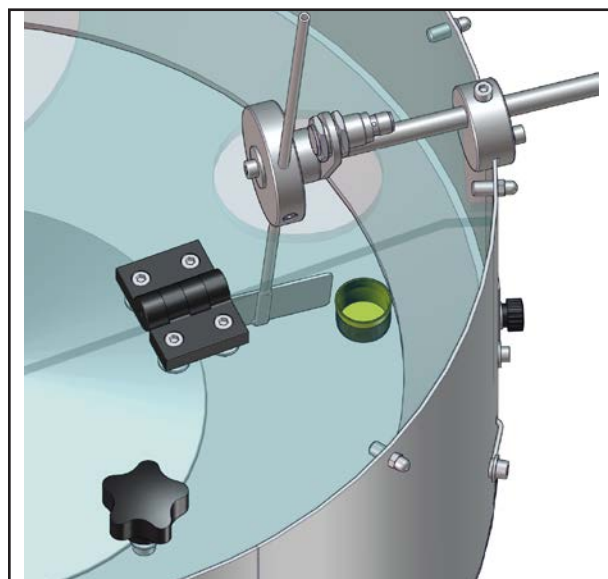


Fig. 130

For this adjustment, loosen the fixing screw on the rod fixing bush.

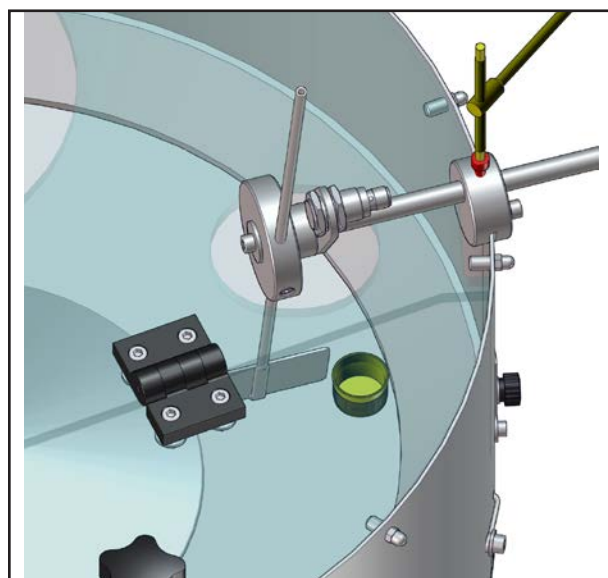


Fig. 131



For correct operation of the device, with empty orientator the sensor must be oriented so as to be in the center of the levelling rod.

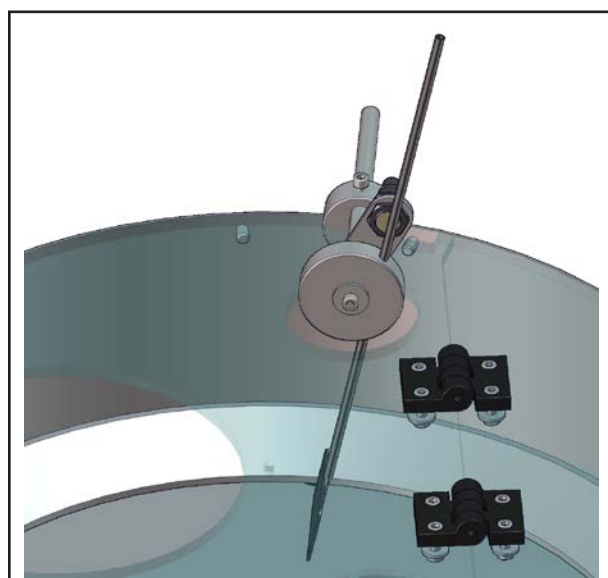


Fig. 132



9.6 EMPTYING OF THE CAPS DISTRIBUTION SYSTEM

9.6.1 CAPS SORTER EMPTYING



To discharge the caps sorter it is necessary to carry out the following operations:

- Set the switch *AUT/MAN* in the position *MAN*.
- From "*main*" page of the operator panel press .
- Press  to set the operation of the caps sorter in manual mode and to hold it till totally empty.

9.6.2 CAPS CHUTE DISCHARGE



To discharge the caps chute, operate as follows:

- Insert a collection container under the caps chute, in proximity of the unloading door.
- Loosen the knob securing the unloading door.

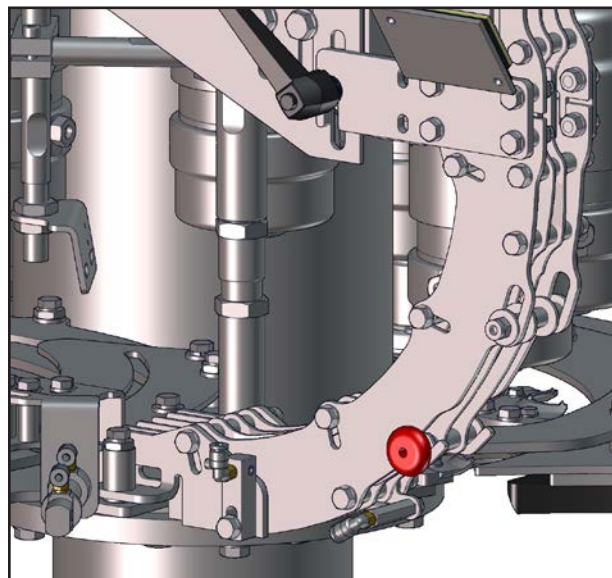


Fig. 133

- Push up the unloading door so as to release it from the fixing seat.
- Open the unloading door.
- If necessary, empty the cap sorter.
- Empty the cap chute.
- Restore the position of the unloading door and fasten by screwing the fixing knob.

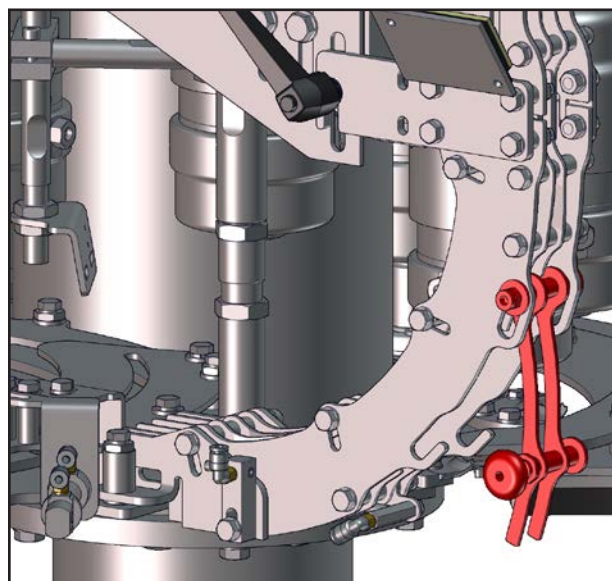


Fig. 134



9.6.3 OVERTURNED CAPS EJECTION



The caps chute is provided with a manual device **MRS** (Manual Rejection System) for ejecting the overturned caps.

If caps are orientated in an incorrect way into the caps chute, a selection star-wheel stops the descent.

To discharge the overturned cap from the caps chute, it is necessary to push on the knob of the discharge device in such way to let the cap go out.

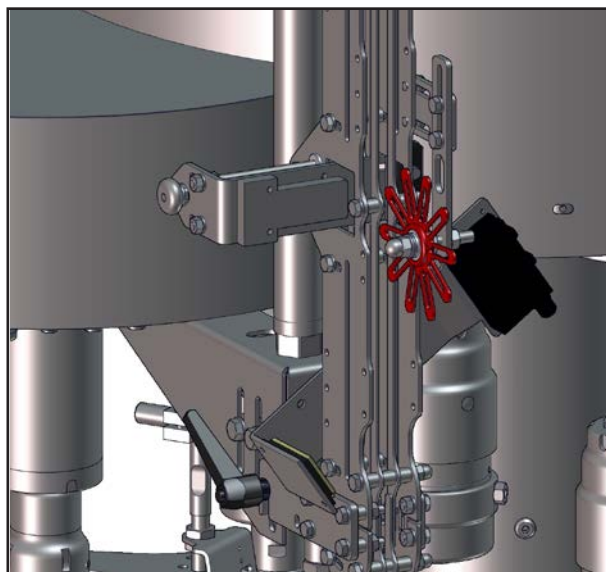


Fig. 135

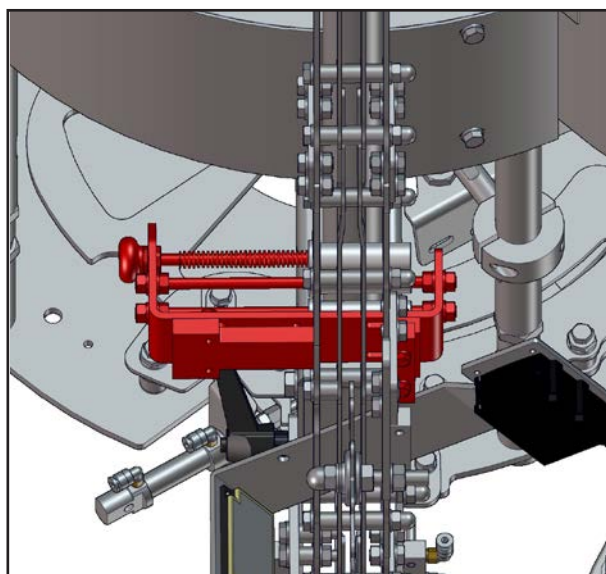


Fig. 136

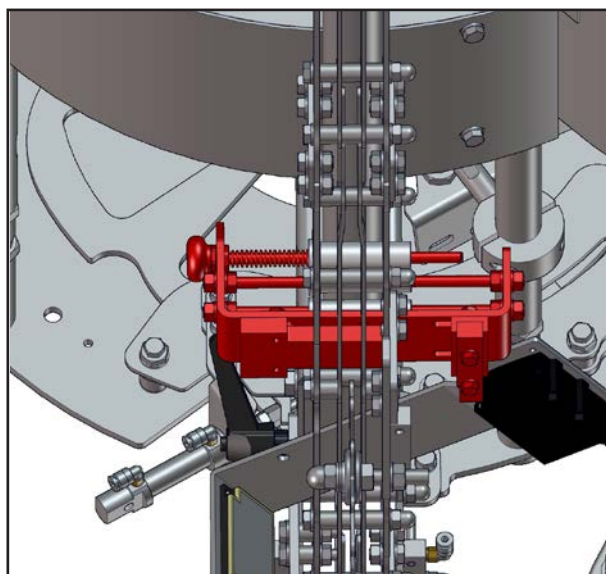


Fig. 137



9.7 START-UP



For start-up, do the following:

- rotate the door lock isolating switch to *ON*, the white *POWER ON* lamp turns on.
- Check that the remaining indicator lights on the panel are off and that there are no error messages on the operator panel.
- To restore any errors, proceed as indicated in the subchapter 9.10 STOP DUE TO THE TRIP OF THE SAFETY AND PROTECTION DEVICES OF THE CLOSING MACHINE page 126.
- Make sure that the *AUT/MAN* switch is set to *MAN*;
- Select on the operator panel the recipe referring to the format to proceed as indicated in subchapter 9.3.7 PRODUCT RECIPES page 89.
- From "*main*" page of the operator panel press .
- Press  to manually operate the caps sorter; keep pressed until the caps feeding chute is full.
- Rotate the *AUT/MAN* switch to *AUT*.
- Prepare the closing machine for start by pressing the green *START* button.
- When a certain number of containers accumulates at the closing machine entry, a photocell enables to start the closing machine at a set speed.



A photocell is installed on the exit belt and it stops the closing machine rotation if the containers accumulate.

This stop type is signalled by a flashing indicator and must be considered temporary with automatic re-start.

As soon as the accumulation is cleared the closing machine gives a sound signal and then re-starts automatically.



9.8 WORKING CYCLE STOP



Press the red *STOP* button to stop the closing machine.

9.9 EMERGENCY STOP



The emergency stop is carried out as follows:

- press one of the red mushroom *EMERGENCY* buttons on the control panel.
 - All the closing machine functions are stopped.
 - The intermittent red light placed on the light column located on top of the control panel turns on and the siren issues an audible signal for 3 seconds.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 **ERROR MESSAGES (ACTIVE ALARMS)** page 71.
 - Press the luminous blue *RESET* button to restore the operation of the safety circuit.

9.10 STOP DUE TO THE TRIP OF THE SAFETY AND PROTECTION DEVICES OF THE CLOSING MACHINE

9.10.1 STOP DUE TO OPENING OF THE INTERLOCKED MOVABLE GUARDS



The closing machine stops when:

- The interlocked movable guards remain open during operation.
 - The luminous white lamp *POWER ON* remains lit.
 - The intermittent red light placed on the light column located on top of the control panel turns on and the siren issues an audible signal for 3 seconds.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 **ERROR MESSAGES (ACTIVE ALARMS)** page 71).
 - Close the Interlocked moveable guards On the control panel, the luminous blue *RESET* button lights up.
 - Press the luminous blue *RESET* button to restore the operation of the safety circuit.



9.10.2 STOP BY PROTECTION ACTIVATION OF THE CLOSING MACHINE



When the functional safeties trip, the closing machine stops waiting for their reset.

- The intermittent red light placed on the light column located on top of the control panel turns on and the siren issues an audible signal for 3 seconds.
- The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.
- On the control panel, the luminous blue *RESET* button lights up.
- Reset the tripped functional safety proceeding as shown in subchapter 11.7.4 TRANSPORT EQUIPMENT TIMING page 236.
- Press the *RESET* blue button to resume the closing machine operation.

9.10.3 STOP BY ELECTRICAL PROTECTION ACTIVATION



When the electrical protection intervene the closing machine stops, waiting the reset of the protection activated.

- The intermittent red light placed on the light column located on top of the control panel turns on and the siren issues an audible signal for 3 seconds.
- The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.



Before restoring the closing machine operation, it is necessary the intervention of a electrical maintenance operator.



9.10.4 STOP WITH AUTOMATIC RESTART



The closing machine stops when:

- the check for no cap in the caps chute has activated:
 - The intermittent yellow light placed on the light column located on top of the control panel turns on.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.



The stop takes place only when the lack of caps in the caps chute check device is present.

- The check at the conveyor belt entry has intervened for no container at the entry.
 - The intermittent yellow light placed on the light column located on top of the control panel turns on.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.
 - After restoring the container flow in the conveyor belt, the closing machine gives a sound signal of the duration of 1 second and then re-starts automatically.
- The check at the conveyor belt exit has intervened for an excessive presence of containers.
 - The intermittent yellow light placed on the light column located on top of the control panel turns on.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.
 - After the container accumulation on the conveyor belt has cleared, the closing machine gives a sound signal of the duration of 1 second and then re-starts automatically.
- When lack the signal external rotation.
 - The intermittent yellow light placed on the light column located on top of the control panel turns on.
 - The relevant message is displayed. See sub-chapter 9.3.2.5 ERROR MESSAGES (ACTIVE ALARMS) page 71.
 - Once the cause is removed, the closing machine beeps for 1 second and then automatically restarts.



9.11 CLOSING MACHINE CLEANING



The determination of the frequency of cleaning operations during the closing machine continuous use is to be checked by the user, depending on the environment is being operated.

The following subchapter describes the cleaning intervals of the closing machine and its components, when considering their use in a clean industrial production area.

9.11.1 CLOSING MACHINE CLEANING INTERVALS

The following table shows the cleaning intervals of the closing machine, expressed in hours.

Cleaning	Work shift (hours)		
	1 shift	2 shifts	3 shifts
Closing machine external washing. See sub-chapter 9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 130.	8	16	40
External washing of the closure heads. See sub-chapter 9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 130.	8	16	40
External washing of the closure grippers. See sub-chapter 9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 130.	8	16	40
Bottle conveying group cleaning. See sub-chapter 9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 130.	8	16	40
Cork/cap sorter cleaning. See sub-chapter 9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 130.	120	120	120



9.11.2 EXTERNAL WASHING OF THE CLOSING MACHINE



The description below concerns: cleaning areas, cleaning systems and specifications, products allowed for cleaning the closing machine and its components.

Legend:

- 1 Closure system direction area
- 2 Upper area of the closure carousel



The direction of the jet coming from cleaning system must be directed downwards. Do not direct air jet towards area "1".

- 3 Closure area
- 4 Bottle-guide and conveying area
- 5 Basement area

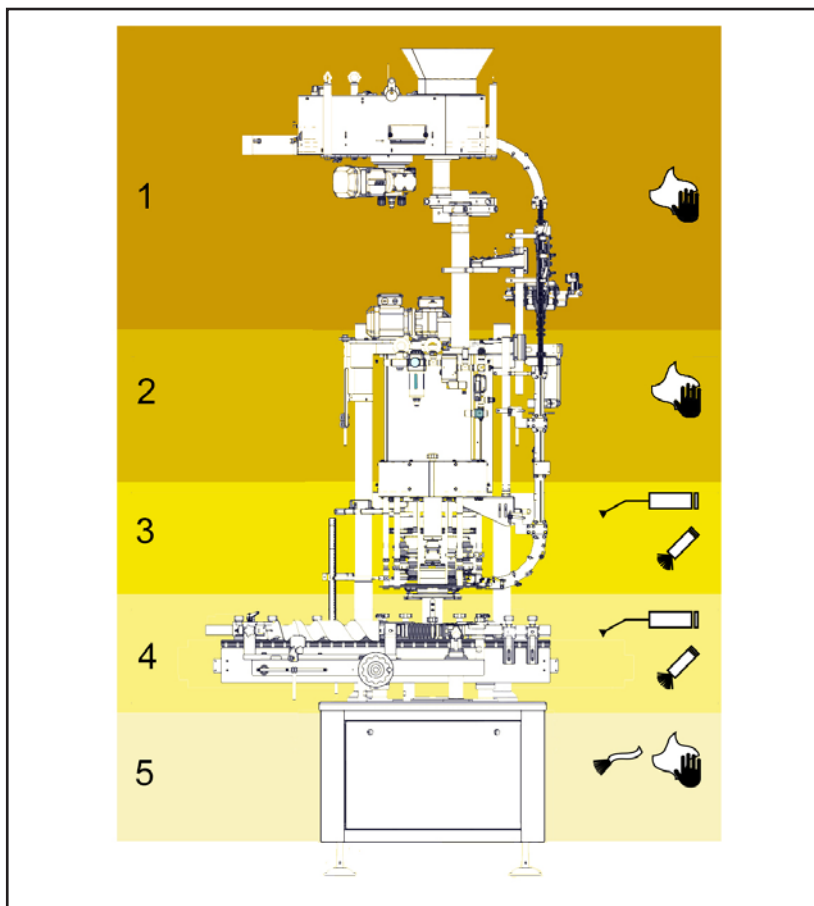
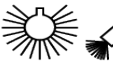
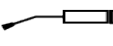


Fig. 138

CLEANING SYMBOL		MAX PRESSURE OF THE CLEANING [bar]	DISTANCE FROM THE SURFACE [mm]	WASHING SYSTEM TILTING [°]
	fixed squirting nozzles or spray balls	≤ 5	≥ 250	
	washing hose	≤ 2	≥ 250	
	cloth	-	-	-
	washing lance	≤ 20	≥ 500	



PRODUCT ZONE 3 - 4	CONCENTRATION	MAX. TEMP. (°C)	pH
Alkaline solutions of hydroxide sodium (NaOH) and/or potassium (KOH)	1 ÷ 2 [%]	35	12 ÷ 13
Acid detergents containing nitric acid (HNO ₃)	1 ÷ 2 [%]	35	1 ÷ 2
Acid detergents containing phosphoric acid (H ₃ PO ₄)	1 ÷ 2 [%]	35	1 ÷ 2
Disinfectant non chlorinated (Aldehyde, quaternary ammonium)	1 ÷ 2 [%]	35	2 ÷ 6

PRODUCT ZONE 1 - 2 - 5	MAX. TEMP. (°C)	pH
Water	35	
Neutral detergents	35	5 ÷ 9



Before starting production, the closing machine should be suitably rinsed, to remove all the detergents used, then dried up.

It is absolutely forbidden to clean the closing machine with:

- steam.



Before carrying out cleaning, you should refer to the product data sheets relating to detergents.



IMPORTANT : AROL S.p.A. supplies instructions for washing in order to clean the parts are remove coarse grease stains, powder and scale. On the other hand, it does not guarantee the hygiene of the surfaces, which must be checked by the user depending on the type of treated products, pollutants and disinfection, sanitisation or sterilisation procedures.



Detergents residuals or residuals of liquids used for the cleaning of the closing machine must be disposed according the laws in force of the country in which it is used.



9.11.2.1 AUTOCLAVE CLOSURE HEADS STERILIZATION

The closure heads thanks to their specific materials, following the instructions listed below, can be subjected to a sterilization process in autoclave.

The frequency of the sterilization in autoclave depends on the specifications and needs of the bottling process and must be determined by the user according to his target and experiences. *AROL S.p.A.* suggests, anyway, that closure heads in bottling lines of sensitive products must be subjected to a sterilization process at least every time one proceeds to their maintenance, manipulation or after a long inactivity period.

AROL S.p.A. suggest that these procedures must be stricter by the increasing the frequency of the sterilization cycles, considering some conditions linked to the context and specific needs or targets that only the costumer can define.

9.11.2.2 CLOSURE HEADS STERILIZATION PROCEDURE

To carry out a sterilization cycle, it is necessary:

- Carry out the ordinary maintenance and closure heads components lubrication.
- Insert the closure heads previously cleaned, into the fitting sacks.
- Close and hermetically seal the sack containing the cleaned closure head.
- Position the closure head into the autoclave.
- Close the autoclave door.
- Carry out a 20 minutes duration sterilization cycle having a pressure of 1 bar and a temperature of 121 °C.
- After the sterilization cycle, remove the closure head from the autoclave.



The closure head must be removed from the sack **ONLY** in a sterile environment.

Remove the envelope bag. Install the closure head on the closing machine.



10 SIZE CHANGE



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The closing machine has been designed to handle containers and caps / corks as indicated in the relevant tables enclosed in this manual.



Every container and/or cap must be processed exclusively with the equipment there indicated.

Each series of equipment is marked according to the size to be processed:

- the container guide equipment are marked with coloured inserts.
- The cap guide equipment, in case of multiple sizes, are marked with an identification initials.

In order to run properly and without obstructions the format change operations, it is necessary to raise the closing turntable at its maximum height.



10.11.1 WORM



10.11.1.1 REMOVAL

To remove proceed in the following way:

- Loosen the fastening handle.
- Slide the worm support pin out.
- Slide the worm out.

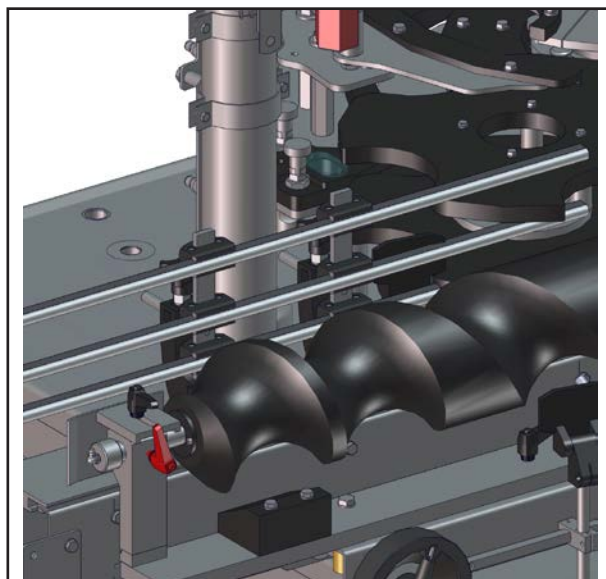


Fig. 139

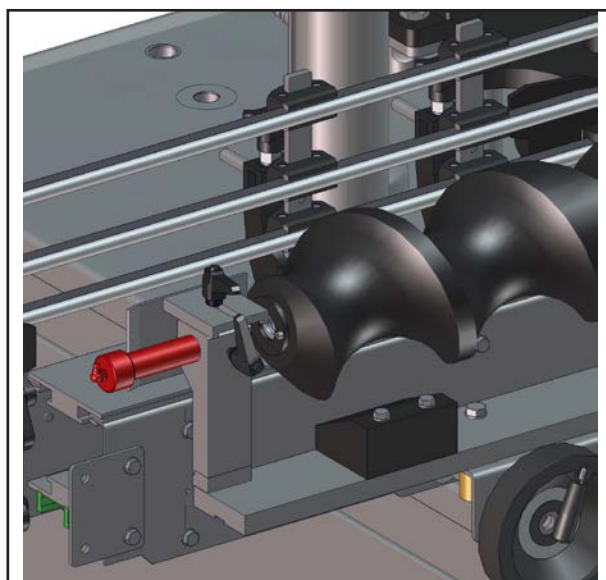


Fig. 140

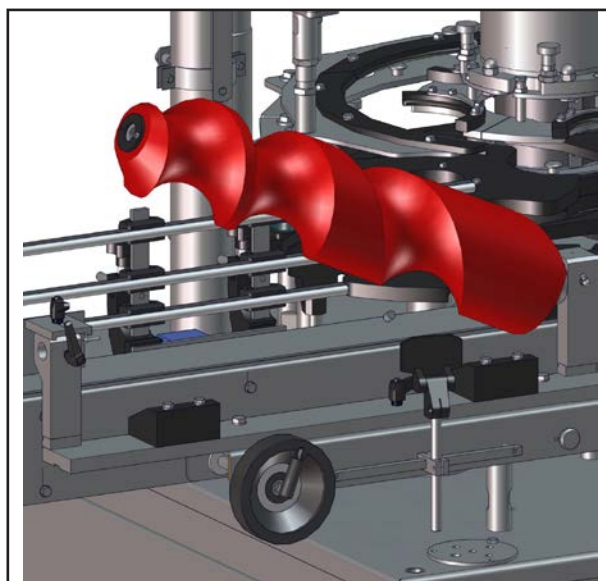


Fig. 141

10.11.1.2 INSTALLATION

For installation proceed as follows:

- Place the worm on the transmission shaft so that the centering pin is correctly positioned in the seat of the worm central shaft.

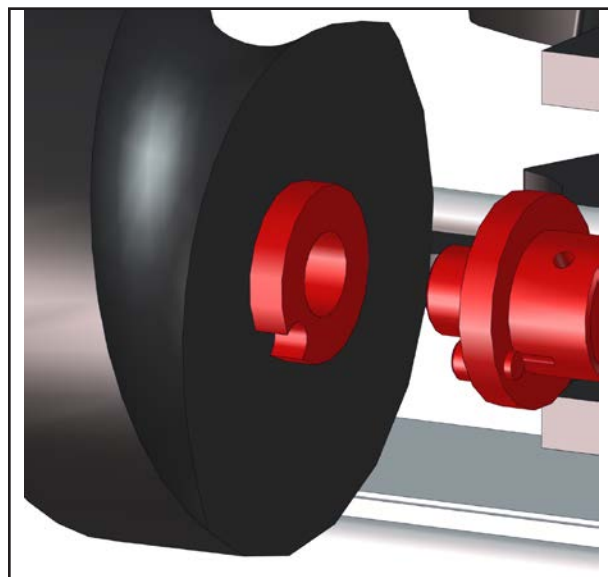


Fig. 142

- Positioning the worm support pin

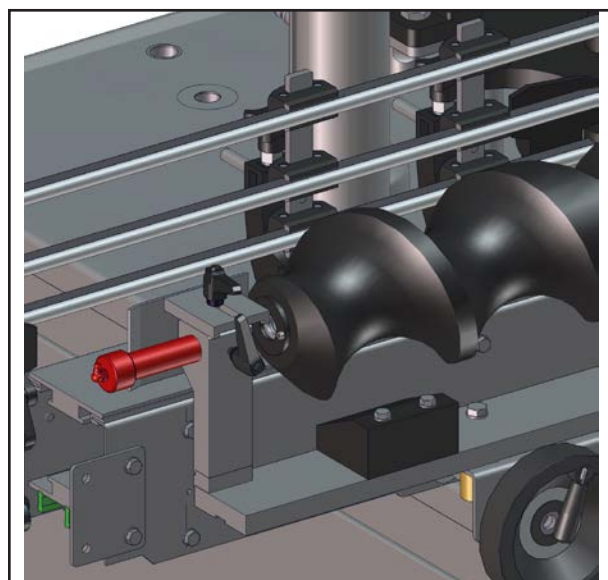


Fig. 143

- Screw the fixing handle making sure that the same is inserted into the housing of the worm support pin.

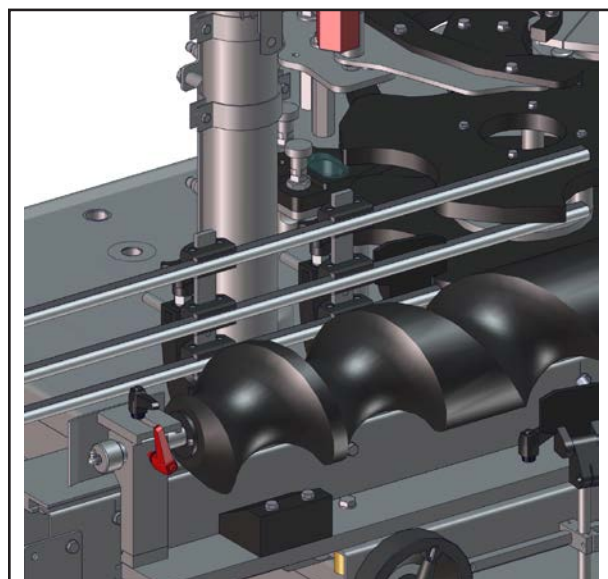


Fig. 144



Depending on the bottle to be treated need to adjust the position of the worm acting in the following way:

- Loosen the fastening handle.

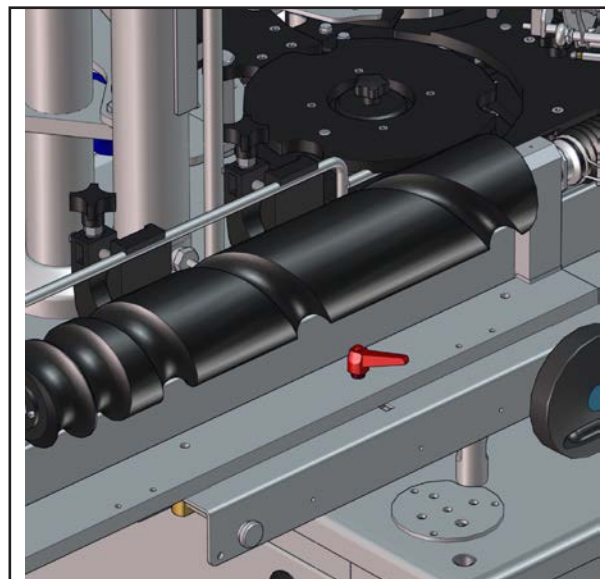


Fig. 145

- turn the handwheel to move the worm.

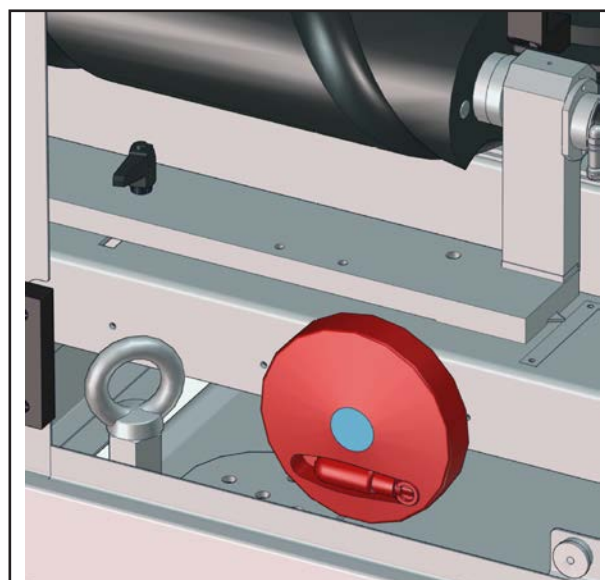


Fig. 146

- the reference index must be aligned to the reference measure on the millimeter rod



The reference measurements of worm position are shown in the EQUIPMENT COLOUR TABLE enclosed in this manual.

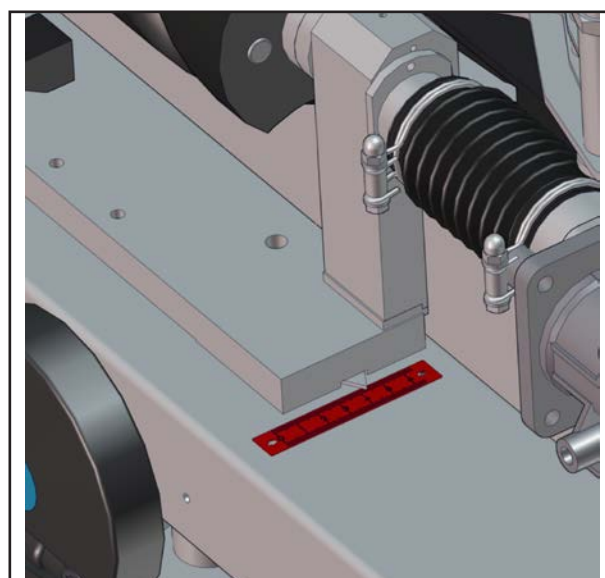


Fig. 147

Further, adjust the containers guide position operating as follows:

- loosen the knobs of the direct passage external guide support.

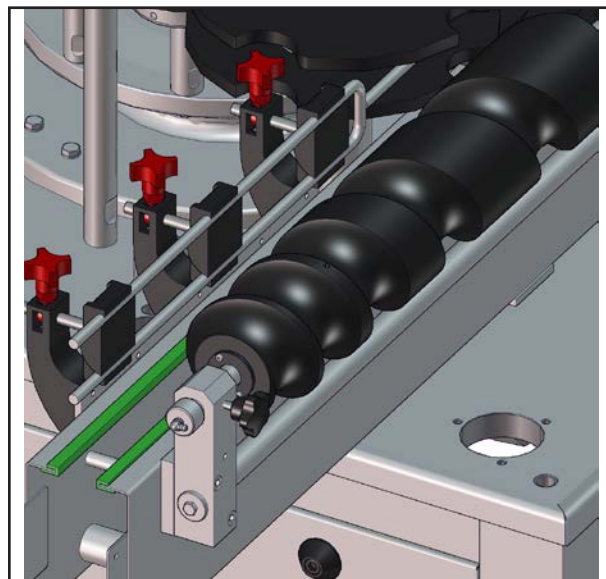


Fig. 148

- After having fitted a container in a worm cavity, position the container guide so that it is at about 2 mm from the container.
- After adjustment, tighten the knobs of the guide support.

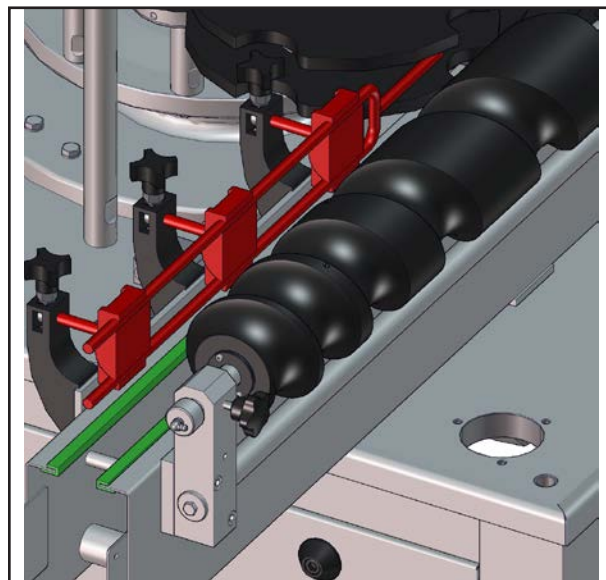


Fig. 149



10.11.2 LATERAL GUIDES

10.11.2.1 REMOVAL



To remove proceed in the following way:

- Unscrew the screws fastening the guides.

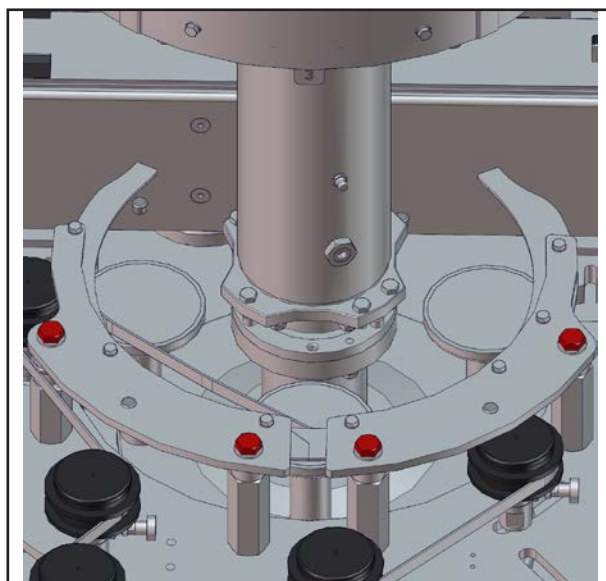


Fig. 150

- Lift and extract the external guides.

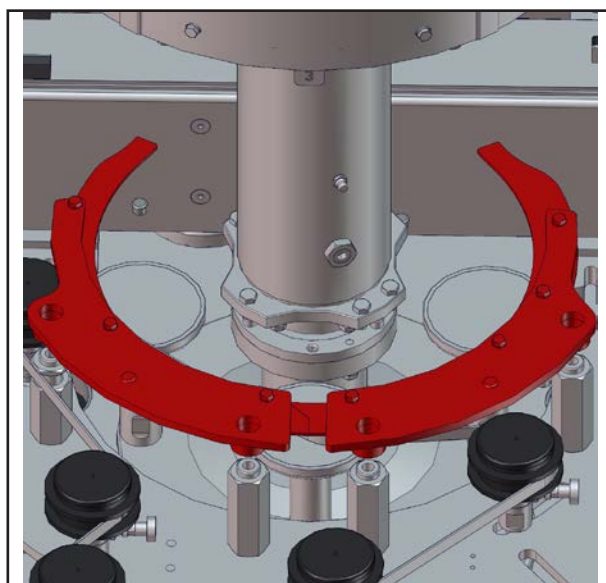


Fig. 151



10.11.2.2 INSTALLATION



For installation proceed as follows:

- Position the upper lateral guides on the support columns.

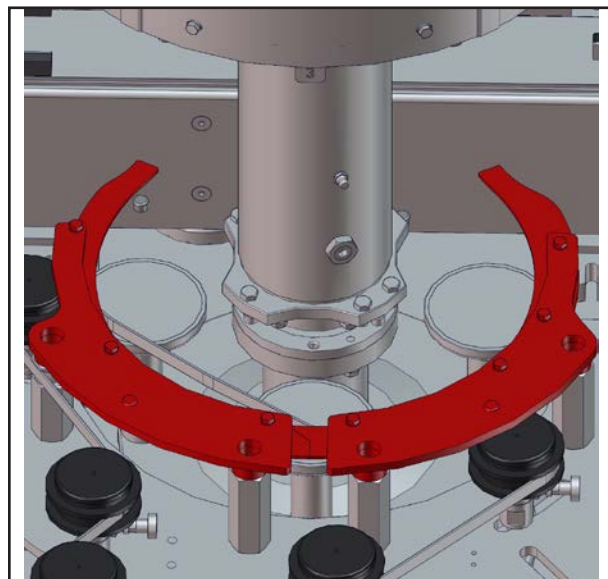


Fig. 152

- Fasten the upper lateral guides by the fastening screws.

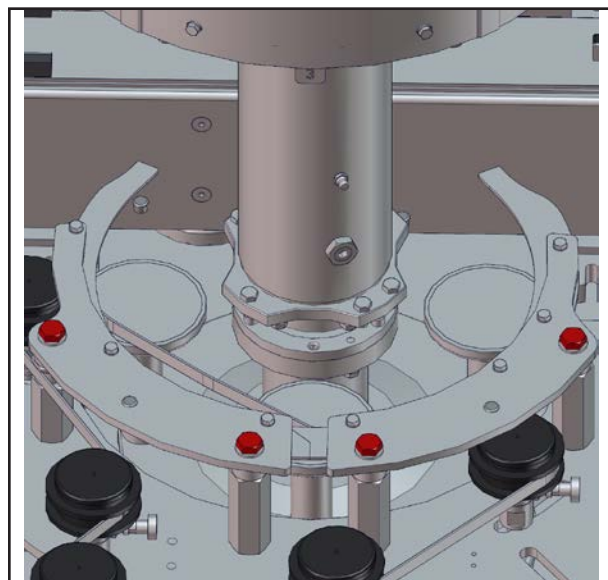


Fig. 153



10.11.3 CENTRAL STAR-WHEEL



10.11.3.1 REMOVAL

To remove proceed in the following way:

- Make the closing machine rotate in manual jog until a semi star-wheel is totally in the the closure carousel rear part.

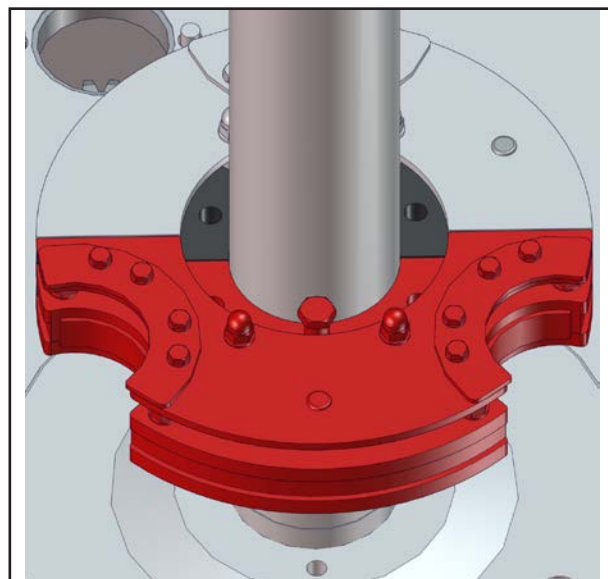


Fig. 154

- Unscrew and remove the fasten knobs.

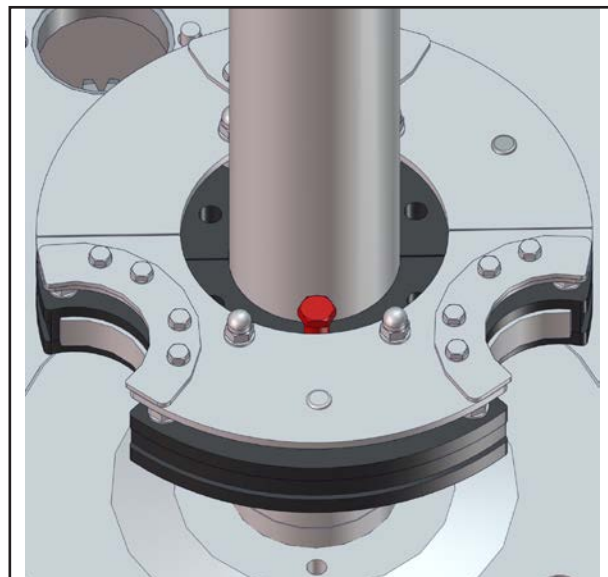


Fig. 155

- Slide the star-wheel out removing it from the centering pins
- Repeat the described above operations also for the second half star-wheel.

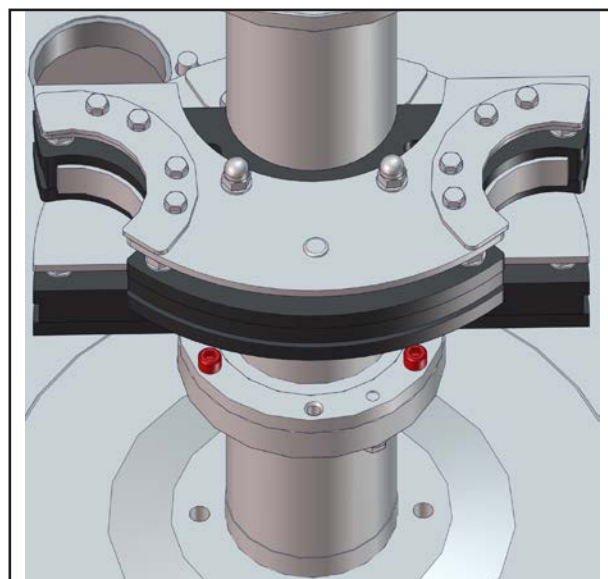


Fig. 156

10.11.3.2 INSTALLATION

For installation proceed as follows:

- Position the half star-wheel on the centering pins.

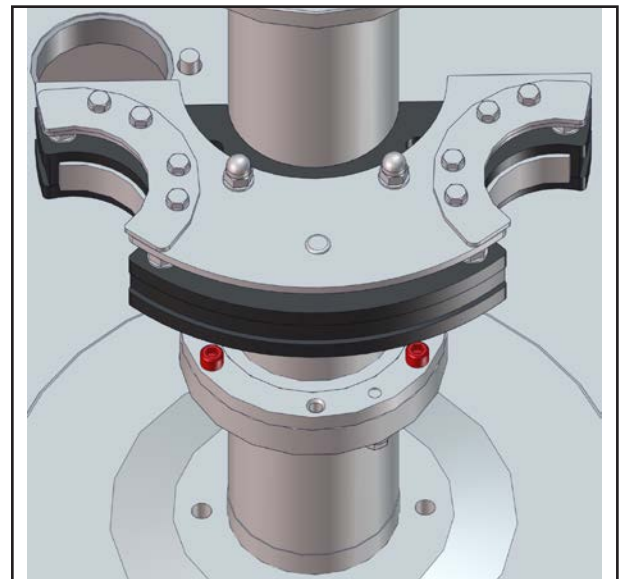


Fig. 157

- Lock the half star-wheel by fixing knob.



Fig. 158

- Make the closing machine rotate in manual jog until the seat of the the second semi star-wheel is totally in the the closure carousel rear part
- Repeat the above steps for the second half star-wheel.



If it is necessary to adjust the height of the star-wheel, loosen the locking grub screw and adjust the height of the star-wheel holder ring. When the adjustment is completed, lock the locking grub screw

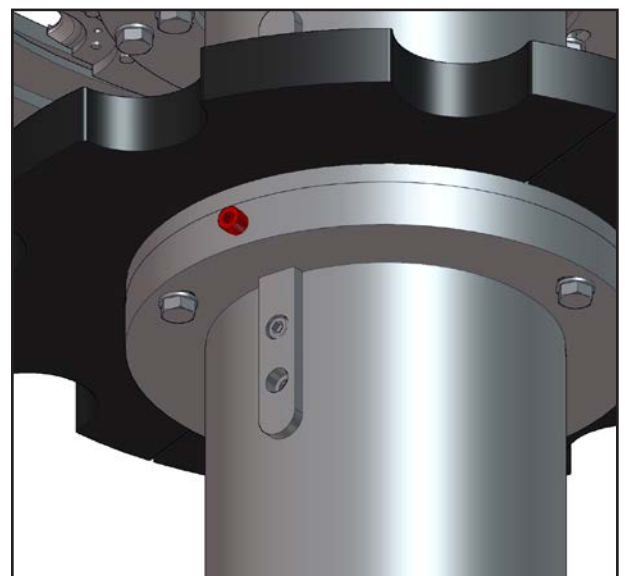


Fig. 159



10.11.4 CONVEYOR



10.11.4.1 REMOVAL

To remove proceed in the following way:

- Untighten conveyor fastening knobs.

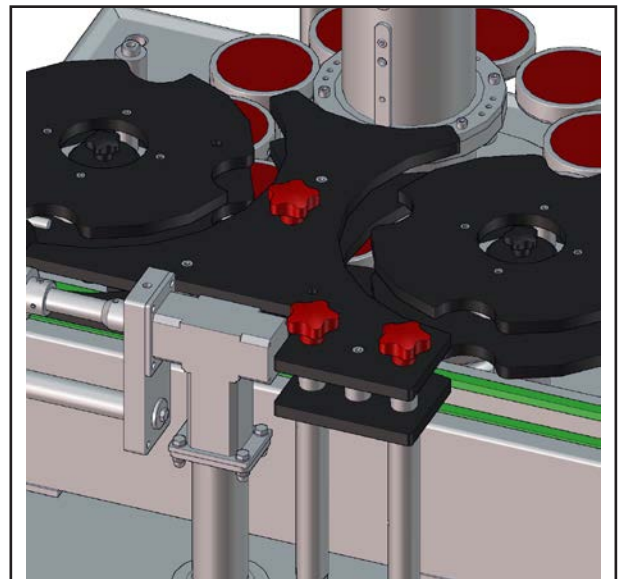


Fig. 160

- Lift and extract the conveyor

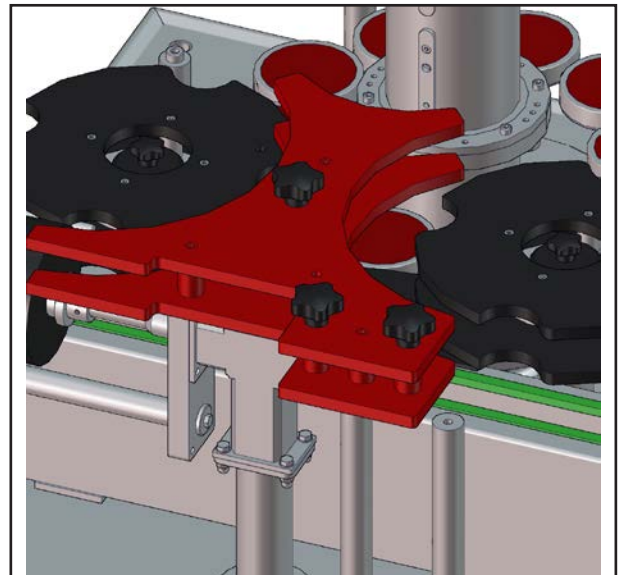


Fig. 161

10.11.4.2 INSTALLATION

For installation proceed as follows:

- Position the conveyor in its suitable seats.
- Fasten the conveyor by knobs.



10.11.5 ENTRY OR EXIT STAR-WHEEL



10.11.5.1 REMOVAL

To remove proceed in the following way:

- Unscrew the screw fastening
- Remove the star-wheel removing it from the centering pin.

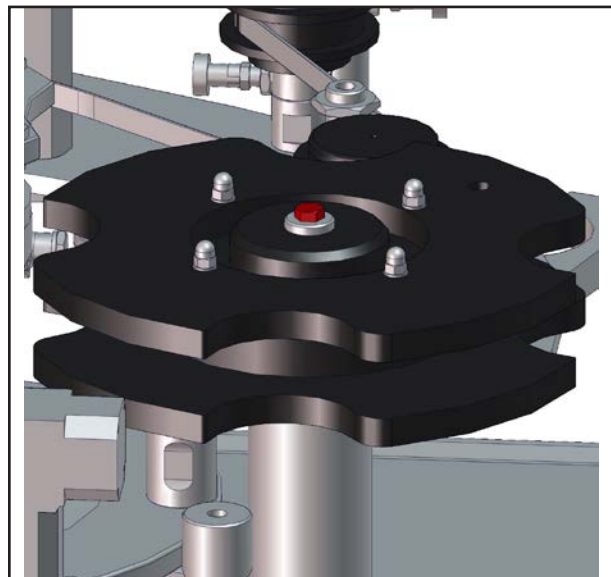


Fig. 162

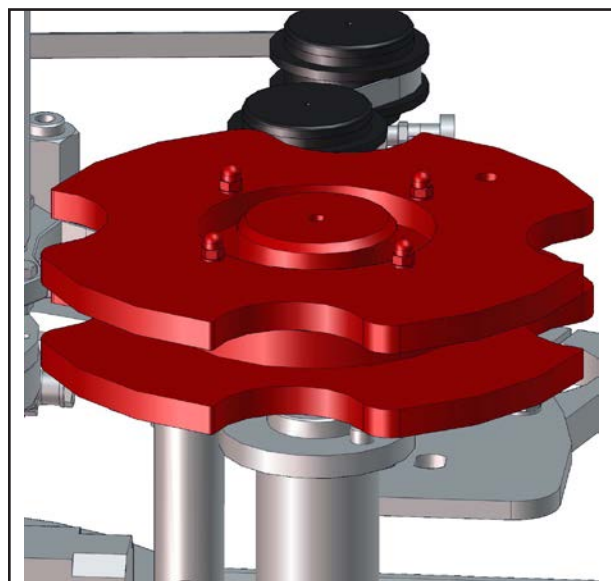


Fig. 163



10.11.5.2 INSTALLATION

For installation proceed as follows:

- Fit the star-wheel in its centering pins.

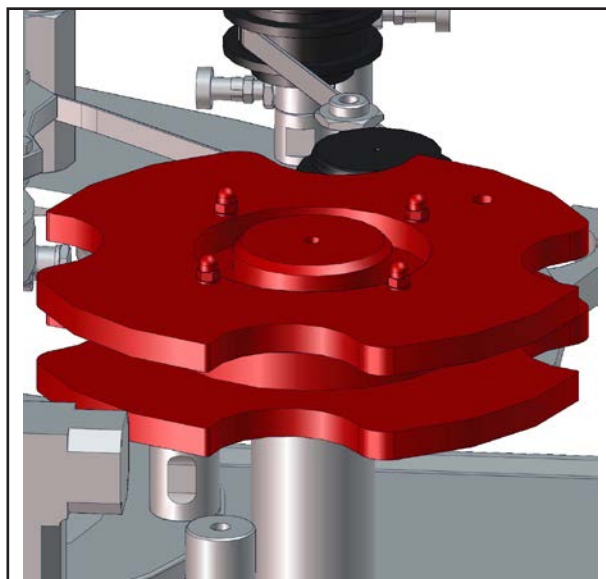


Fig. 164

- Lock the star-wheel by the fastening screw.

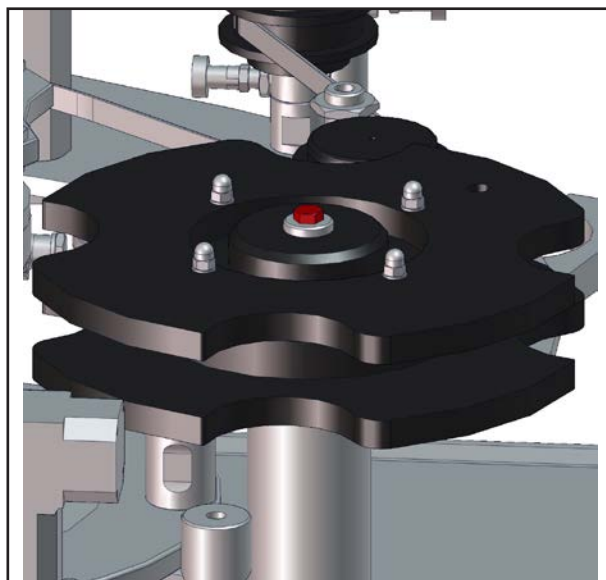


Fig. 165

10.12 CAP SIZE CHANGE



Having to change the format of the caps to be treated, you must:

- replace the closure grippers. (See sub-chapter 10.12.1 REMOVAL AND INSTALLATION OF THE CLOSURE GRIPPER page 151).

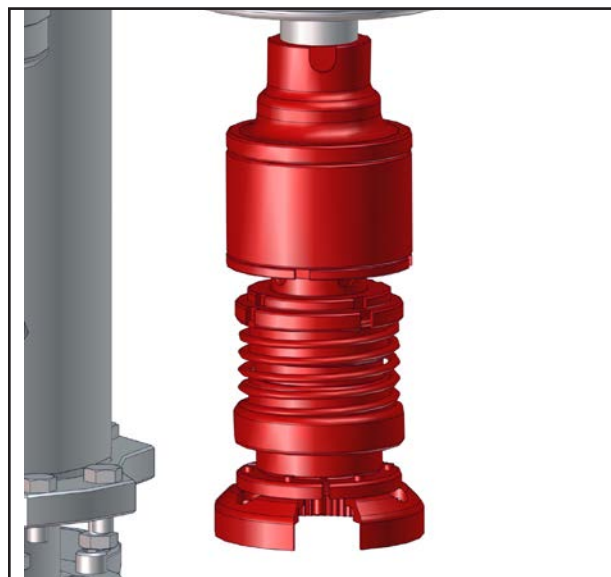


Fig. 166

- Replace the closure heads, if provided. (See sub-chapter 10.12.2 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS page 156).

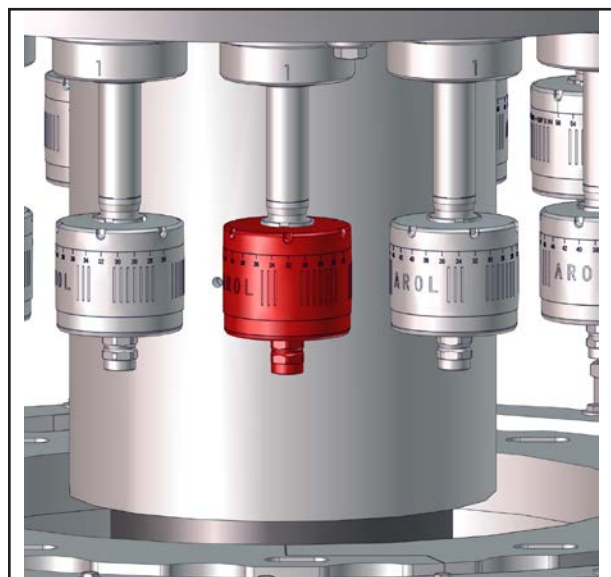


Fig. 167

- Replace the caps chute, if provided. (See sub-chapter 10.12.3 PICK AND PLACE CAPS CHUTE page 161).

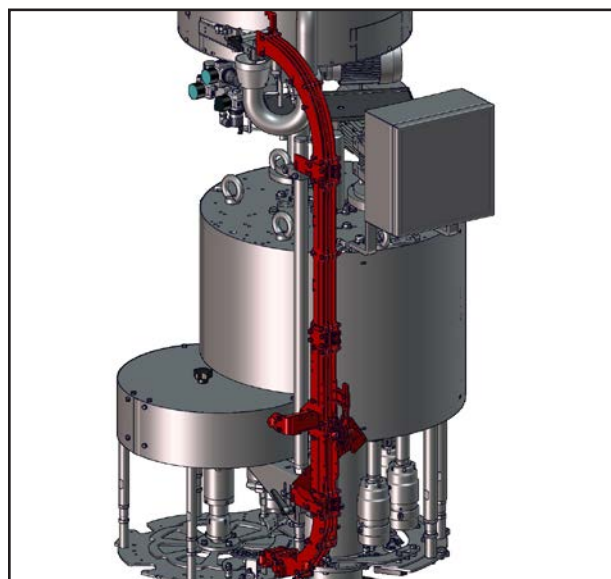


Fig. 168



- Replace the caps transfer group on the transfer disk, if provided. (See sub-chapter 10.12.4 CAPS TRANSFER GROUP page 167).

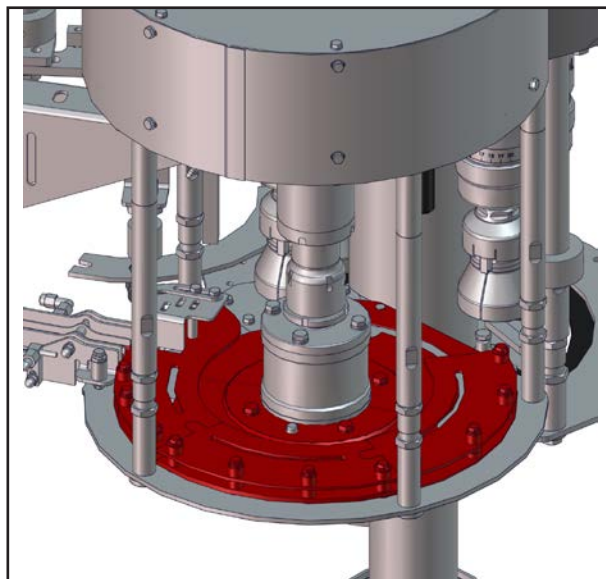


Fig. 169

- Replace the caps selection units, if provided. (See sub-chapter 10.12.5 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER page 171).

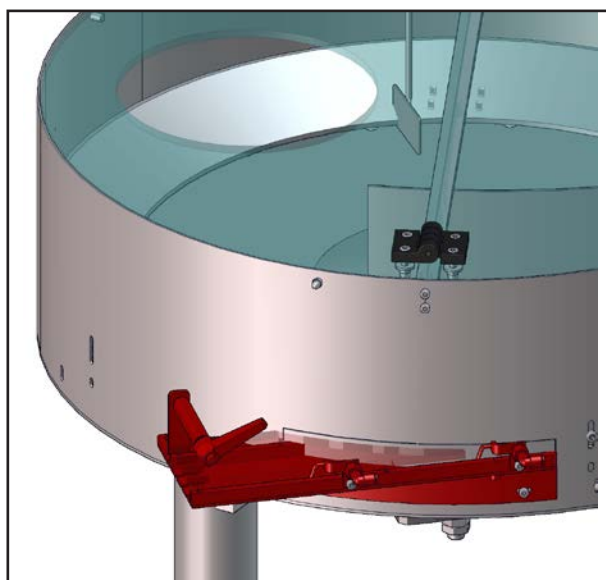


Fig. 170



- According to the cap to process, it is necessary to adjust the position of the internal bulkhead.
- Lift the fixing knob, turn it of 90° in such way to avoid the return in its seat.

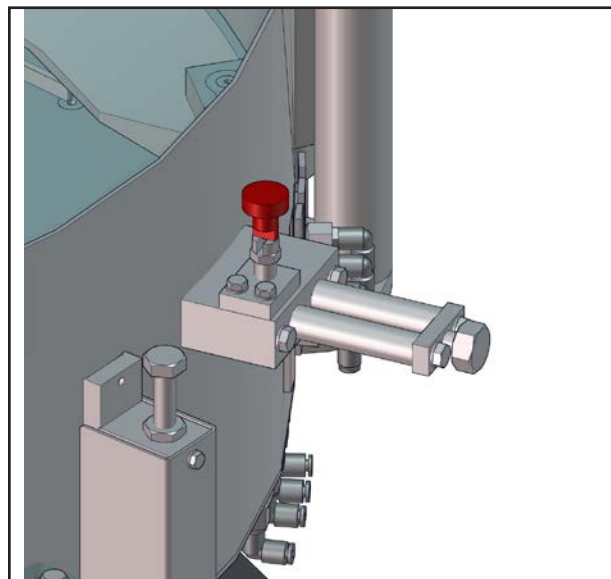


Fig. 171

- carry out on the knob in such way to move the adjustable bulkhead.
- The adjustable bulkhead must be correctly lead the cap into the exit mouth according to the diameter of the cap itself.

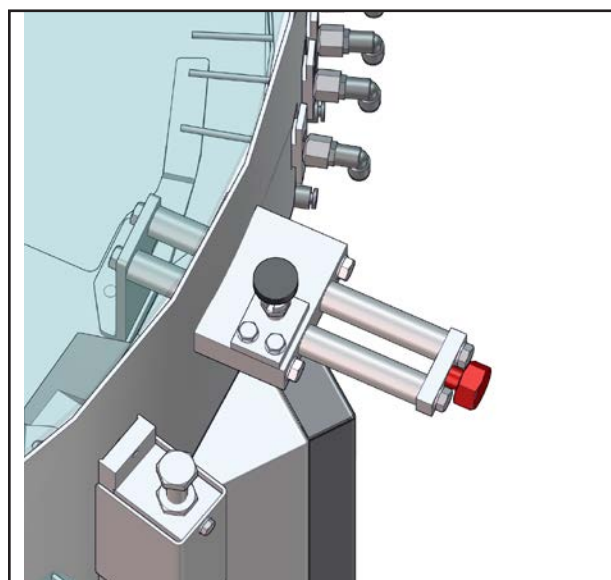


Fig. 172

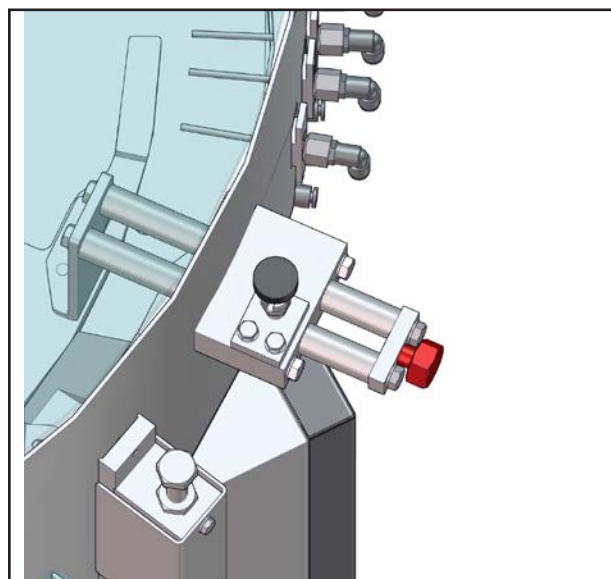


Fig. 173



- Turn the fastening knob of 90° and restore it in locking position in such way to fasten the adjustable balkhead

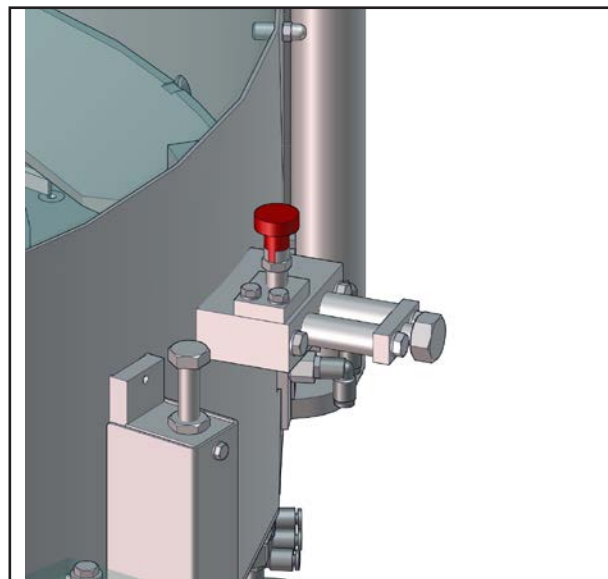


Fig. 174

- According to the cap to process, it is necessary to adjust the position of the caps stop device.

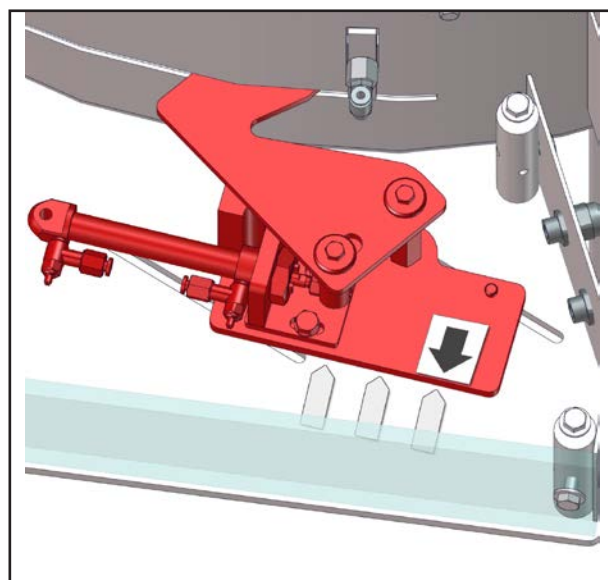


Fig. 175

- Pull the fixing knob and rotate it of 90° in such way to avoid the return in its seat.

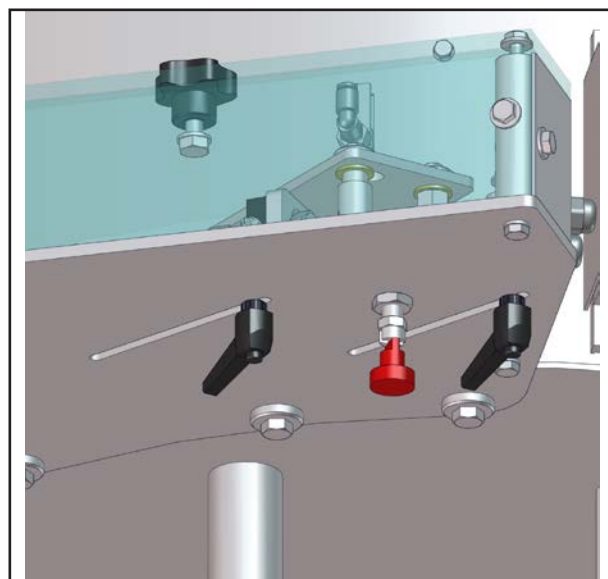


Fig. 176



- Loosen the return handles of the device.

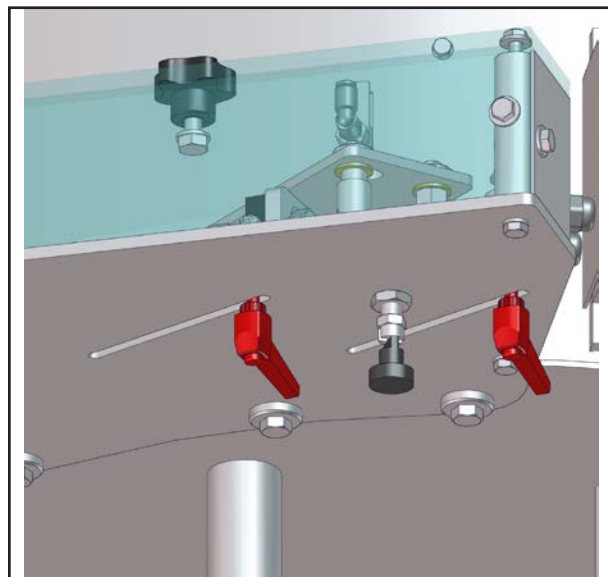


Fig. 177

- Adjust the position of the caps stop device.

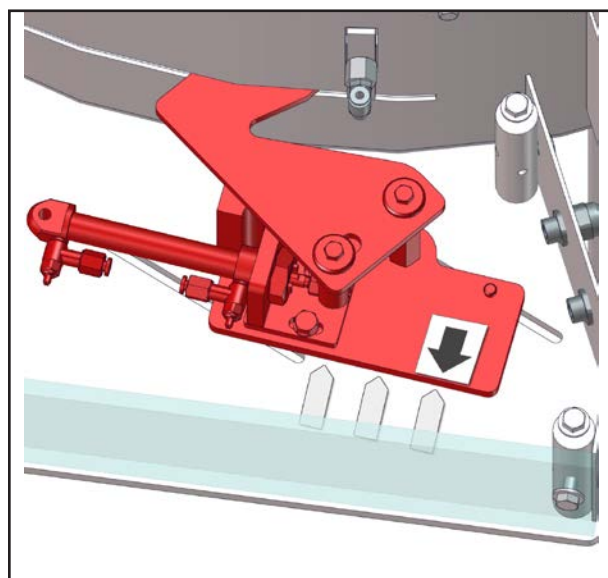


Fig. 178

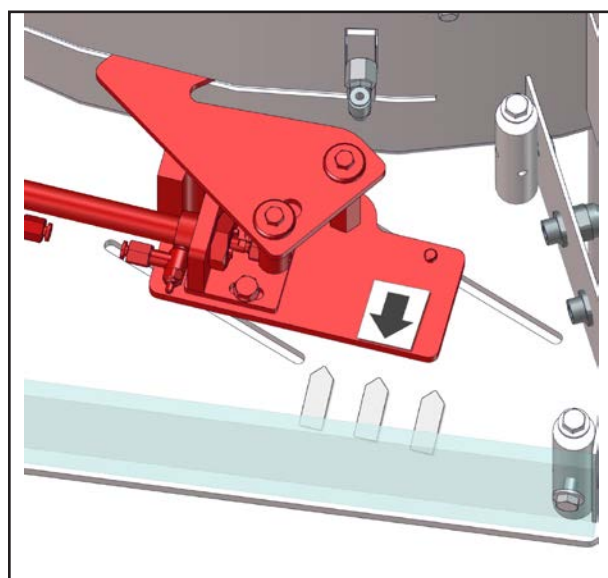


Fig. 179



- After the adjustment, lock the handles

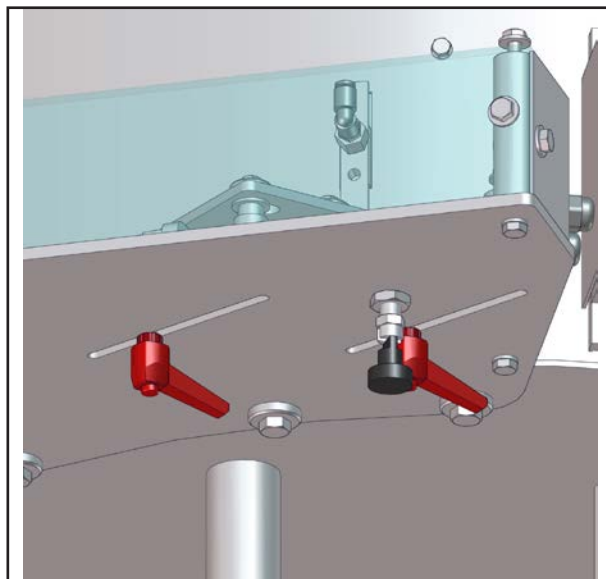


Fig. 180

- Turn the fixing knob of 90° and restor it in locking position.

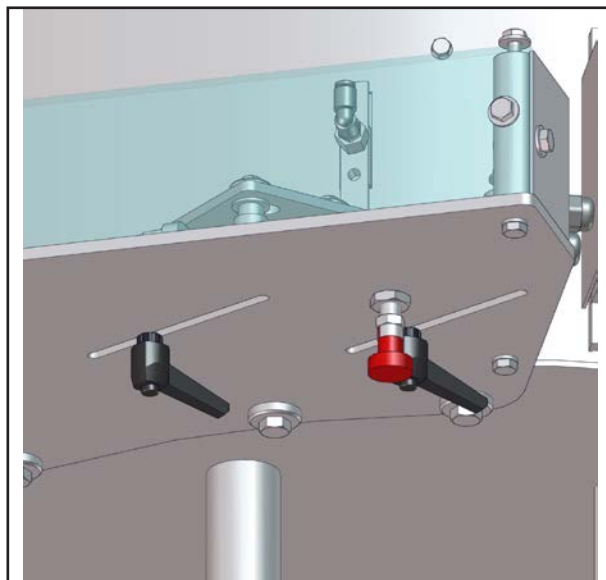


Fig. 181



10.12.1 REMOVAL AND INSTALLATION OF THE CLOSURE GRIPPER

The table below shows the dedicated tools for disassembly and assembly of the closure gripper.

	TOOL
	Hook wrench 40-42
	Hook wrench slim

10.12.1.1 REMOVAL OF THE CLOSURE GRIPPER

To remove the closure gripper, proceed as follows:

- fit the slim hook wrench in the hole of the ejector guide bush.



Fig. 182

- Insert the 40-42 hook wrench into the seats of the gripper holder bush.

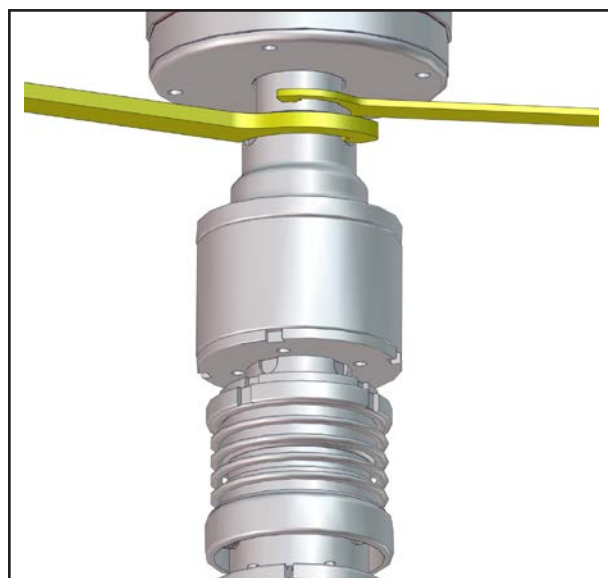


Fig. 183



- By means of the two wrench, loosen the closure gripper; If the gripper is locked it is enough give a light stroke on the handle of the 40-42 hook wrench.
- Remove the closure gripper unscrewing by hand.



Fig. 184



10.12.1.2 INSTALLATION OF THE CLOSURE GRIPPER

To install the closure gripper, proceed as follows:

- Tighten by hand the closure gripper on the ejector guide bush.



Fig. 185

- fit the slim hook wrench in the hole of the ejector guide bush.



Fig. 186

- Insert the 40-42 hook wrench into the seats of the gripper holder bush.
- Using the two wrenches, lock the closure gripper.

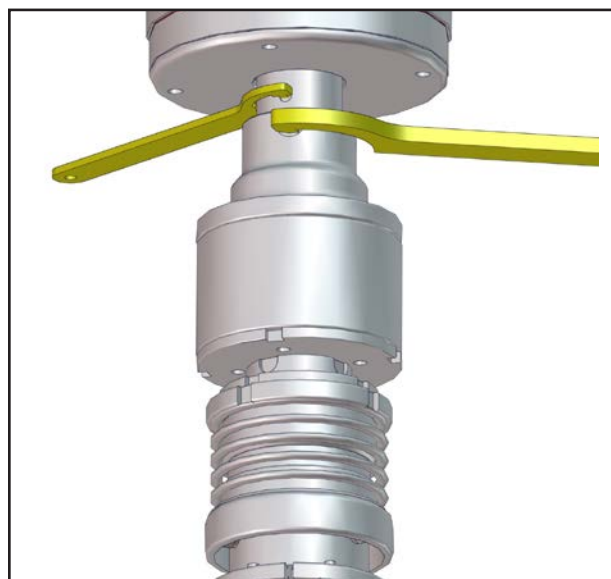


Fig. 187



10.12.1.3 GRIPPER REMOVAL FROM THE COMPENSATION HEAD

To remove the gripper from the compensating head it is necessary to proceed as follows:

- remove the closure gripper complete with the capping head as described in paragraph 10.12.1.1 REMOVAL OF THE CLOSURE GRIPPER page 151.
- Insert the 58-62 hook wrench into the seat of the compensator threaded ring.

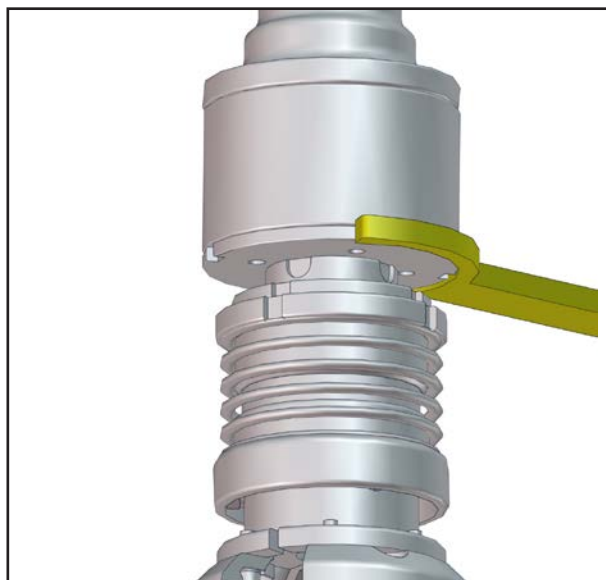


Fig. 188

- Insert the 40-42 hook wrench into the seat of the terminal holder bush.
- By means of the two wrench, loosen the closure gripper; If the gripper is locked it is enough give a light stroke on the handle of the 40-42 hook wrench.

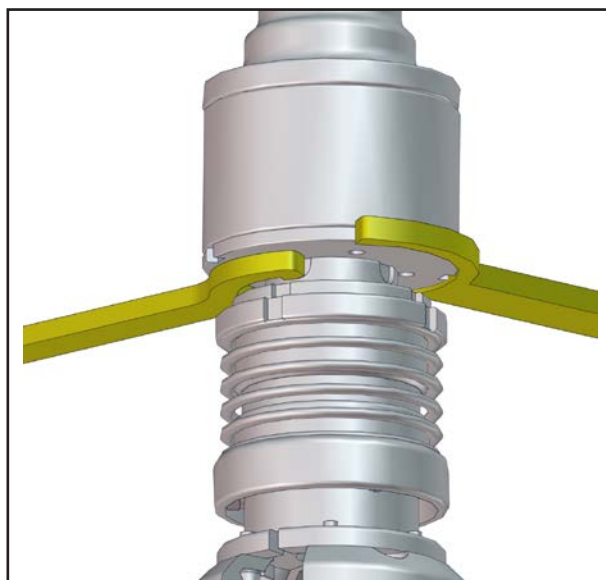


Fig. 189



- Remove the closure gripper from the compensation body



Remount proceeding in the reverse way to what described for the dismounting.



Fig. 190



10.12.2 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS

The table below shows the dedicated tools for disassembly and assembly of the closure head.

TOOLS	
	Hook wrench 110-115
	Hook wrench 45-50

10.12.2.1 "VP750" CLOSURE HEADS REMOVAL



Protect the centering device upper surface with rubber or similar material.

To remove the closure head from the spindle, proceed as follows:

- Remove the closure gripper from the closure head.
- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.

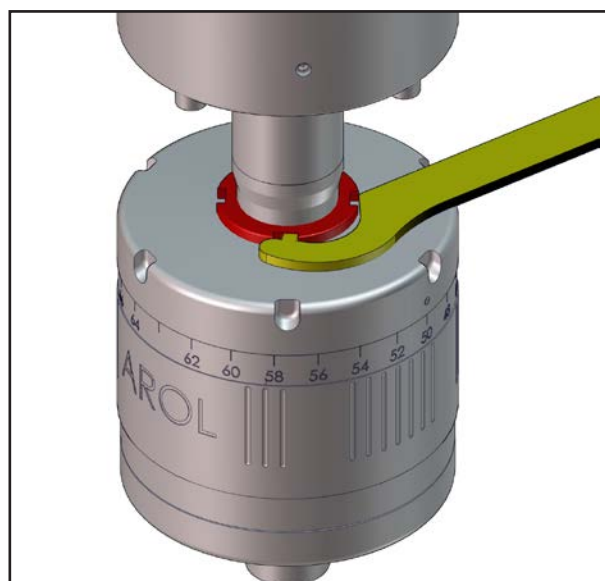


Fig. 191



- Unscrew the head fastening ring nut with the 45-50 hook wrench till releasing the closure head.

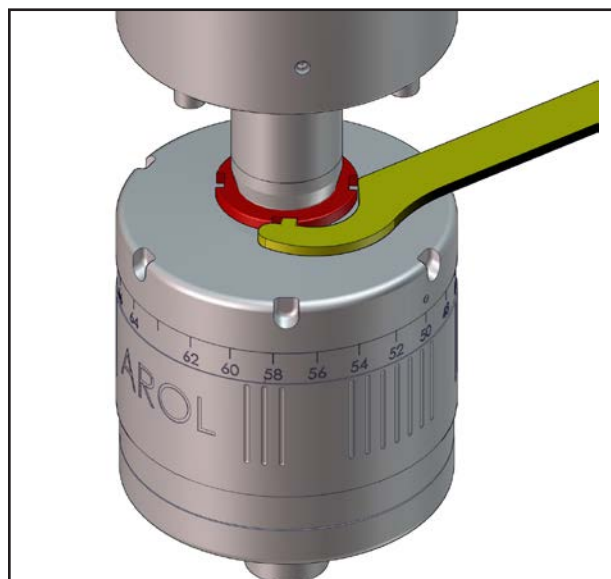


Fig. 192

- Insert the 110-115 hook wrench in the seats of the closure head.

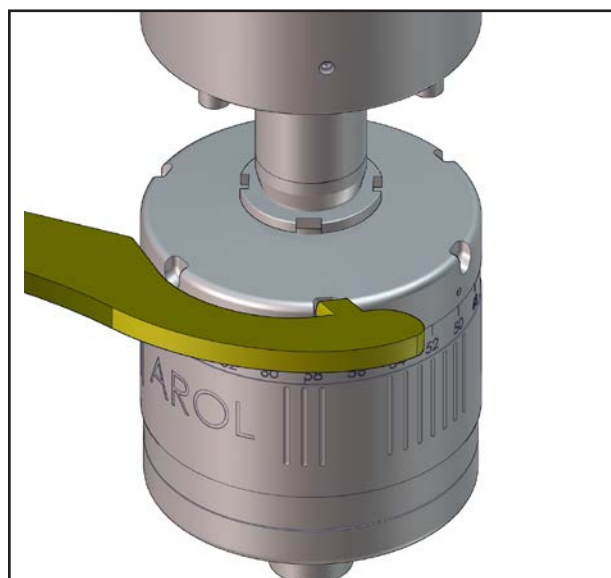


Fig. 193

- Unscrew the closure head with the 110-115 hook wrench and remove it.

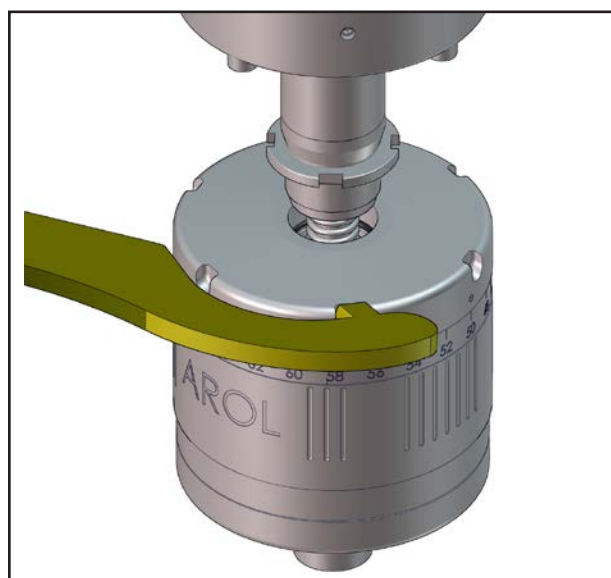


Fig. 194



- If the upper ejector unit has not come out from the spindle together with the head, remove it manually.

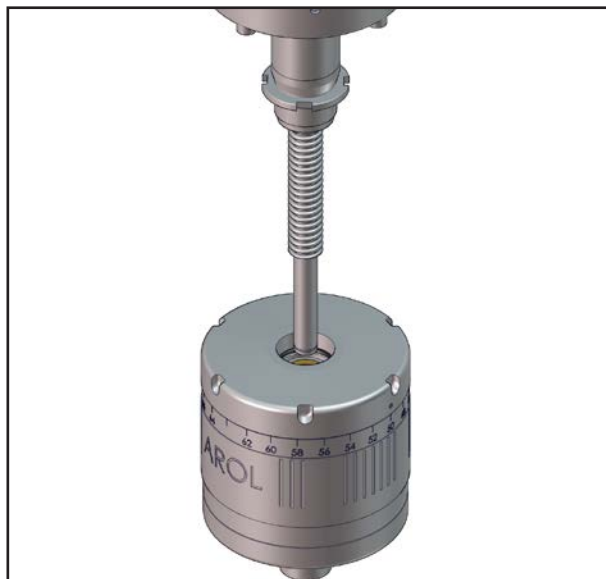


Fig. 195



10.12.2.2 CLOSURE HEADS INSTALLATION "VP750"

To install the closure head on the spindle, proceed as follows:

- Fit the closure upper together with the ejector unit inside the spindle.



Fig. 196

- Screw the head on the spindle, manually as far as possible.



Fig. 197

- Insert the 110-115 hook wrench in the seats of the closure head.
- Complete the screw of the closure head to the bottom of the thread with the 110-115 hook wrench.

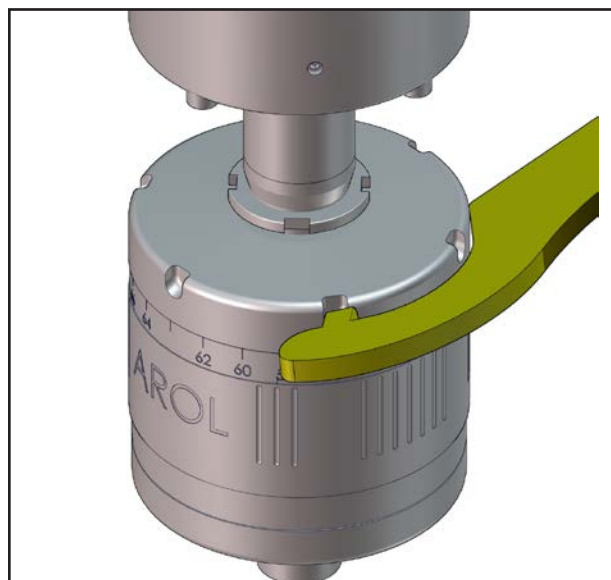


Fig. 198



- Tighten the head fixing ring-nut on the spindle by hand until possible to.



Fig. 199

- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.
- Complete the screw of the head fixing ring-nut with the 45-50 hook wrench.

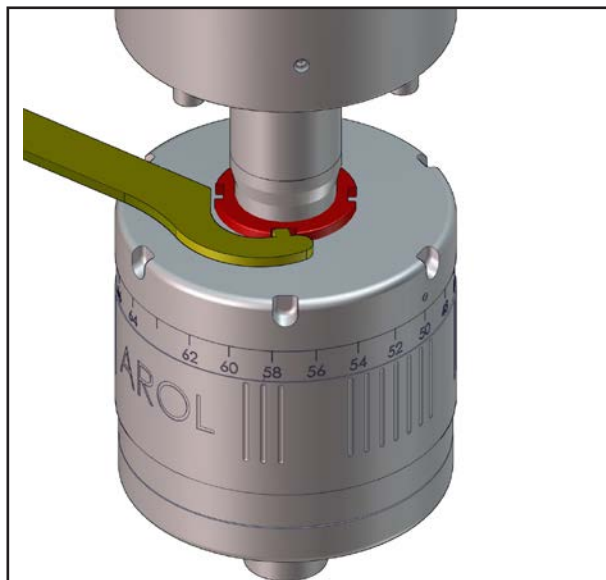


Fig. 200

10.12.3 PICK AND PLACE CAPS CHUTE



10.12.3.1 CAPS CHUTE REMOVAL

Remove the caps chute in the following way:

- Disconnect the compressed air fitting.
- Disconnect the electrical connection from the caps presence photocell (one or more).



Fig. 201

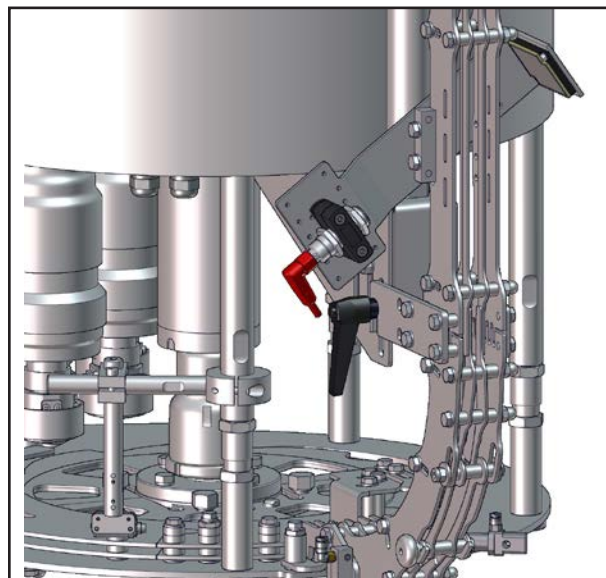


Fig. 202



- Unscrew the bolts that fasten the caps chute to the base disk.

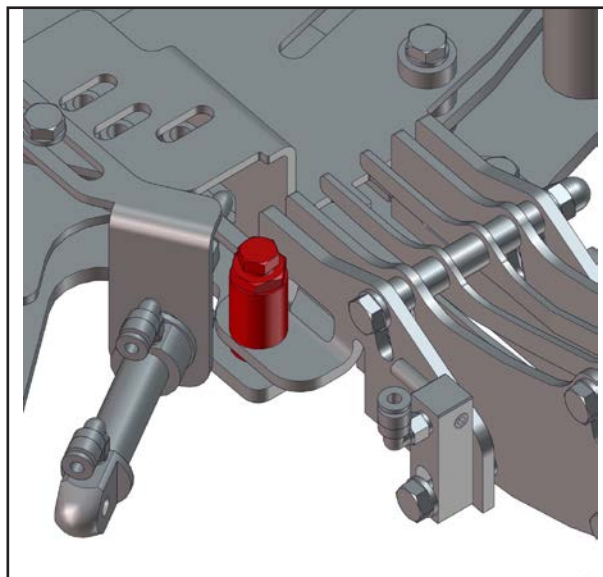


Fig. 203

- Loosen the threaded fasteners located on the support chute.

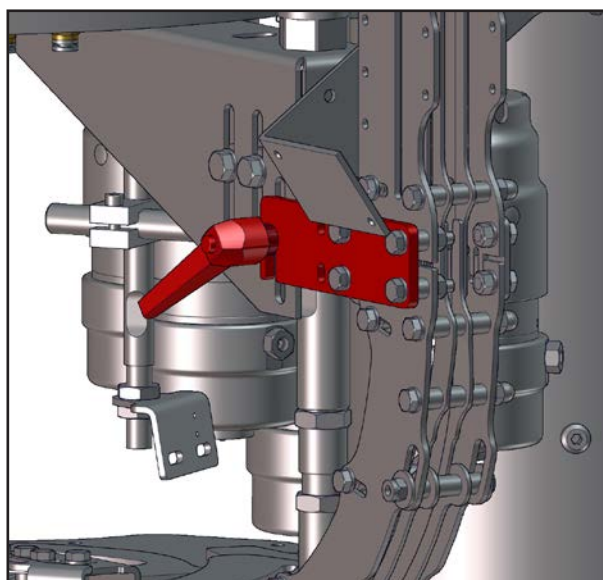


Fig. 204

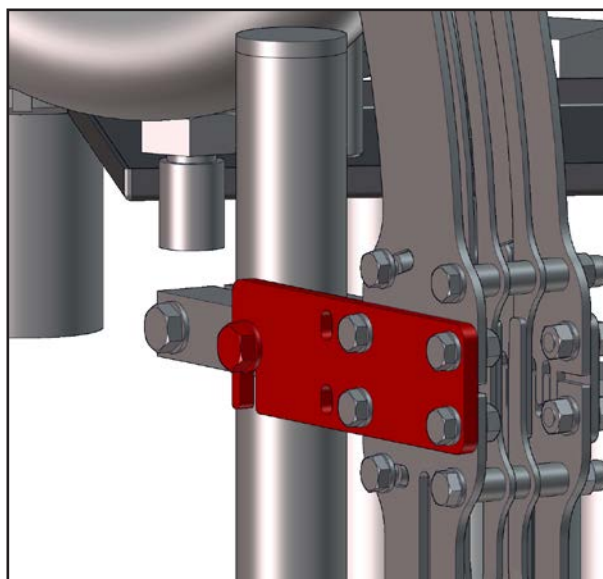


Fig. 205



- Loosen the threaded fastening placed on the caps sorter

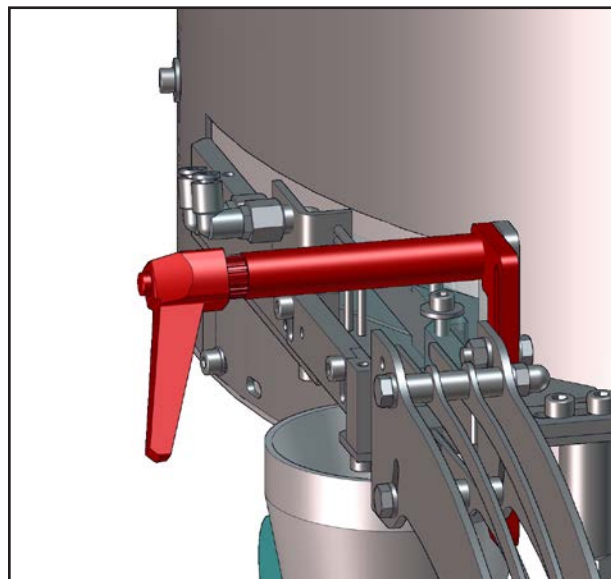


Fig. 206

- Lift the caps chute complete, so as to extract it from fasteners and remove it.

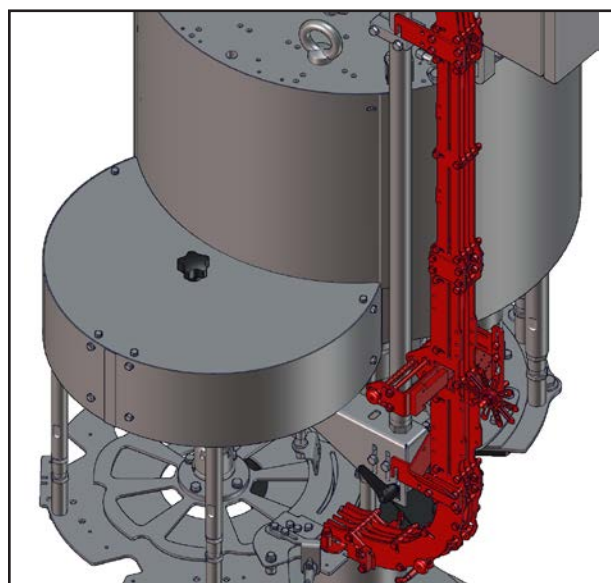


Fig. 207



10.12.3.2 CAPS CHUTE INSTALLATION

To install the caps chute on the machine it is necessary:

- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.

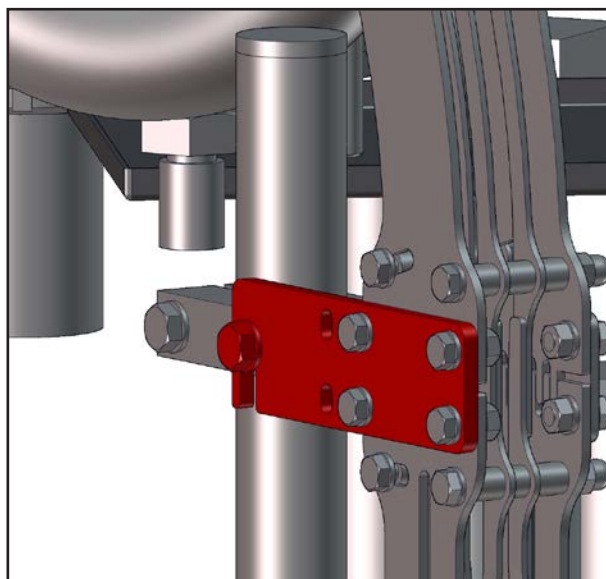


Fig. 208

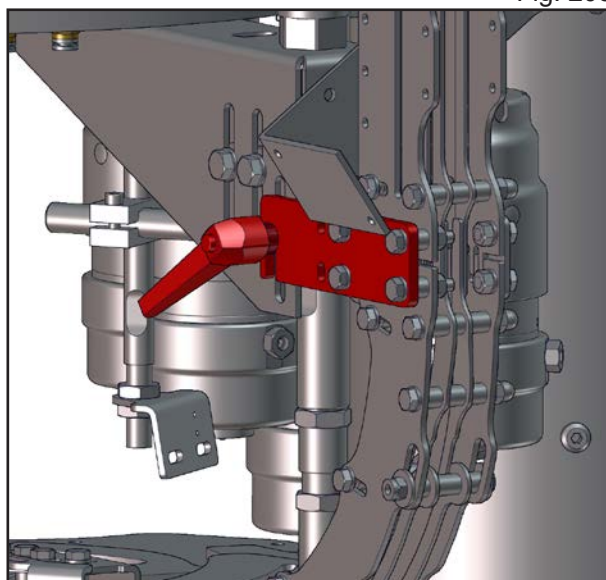


Fig. 209

- Loosen the threaded fastening placed on the caps sorter

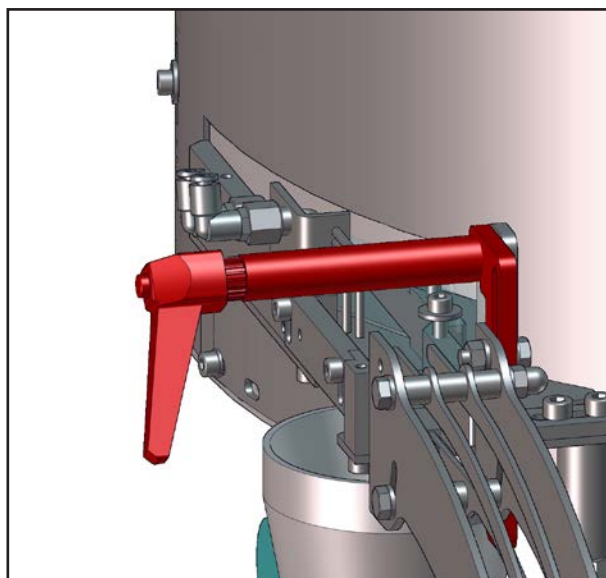


Fig. 210



- Check that the centering pin of the caps chute fits into the seat of a basic disk.

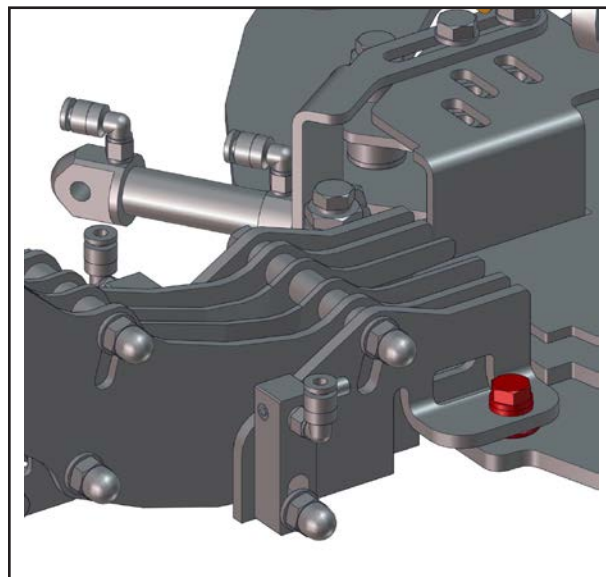


Fig. 211

- Fasten the caps chute to the base disk with the fastening bolt.

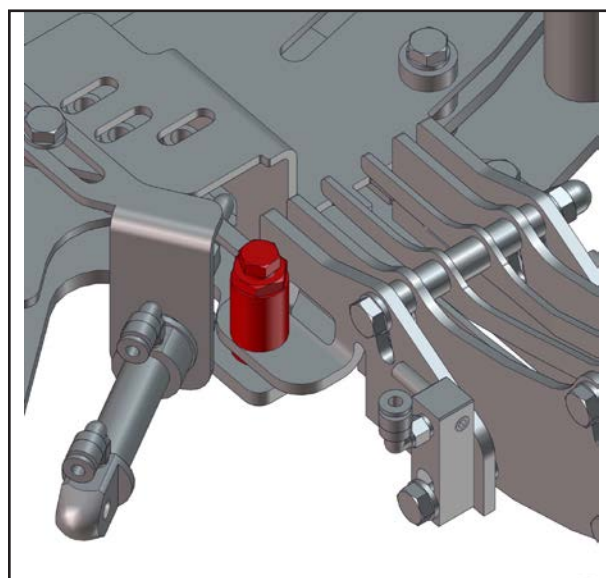


Fig. 212

- Lock threaded fasteners located on the support chute.
- Lock the threaded fastener located on the caps sorter
- Connect the electrical connection of the caps presence photocell (one or more).

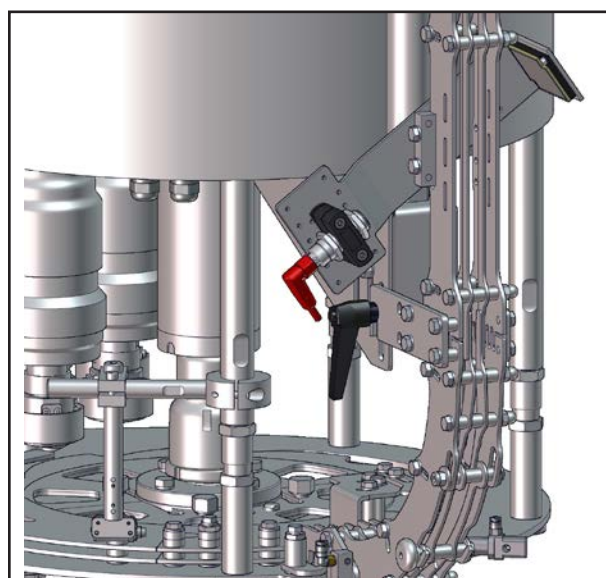


Fig. 213



- Connect the air compressing fitting to them connections.



Fig. 214

10.12.4 CAPS TRANSFER GROUP



10.12.4.1 CAPS GUIDE SECTORS

10.12.4.1.1 REMOVAL

- Disconnect the caps stopping cylinder actioning air fittings.
- Unscrew bolts that fastening the caps guide sectors of the base disc
- Lift and remove the caps guide sectors.



Fig. 215

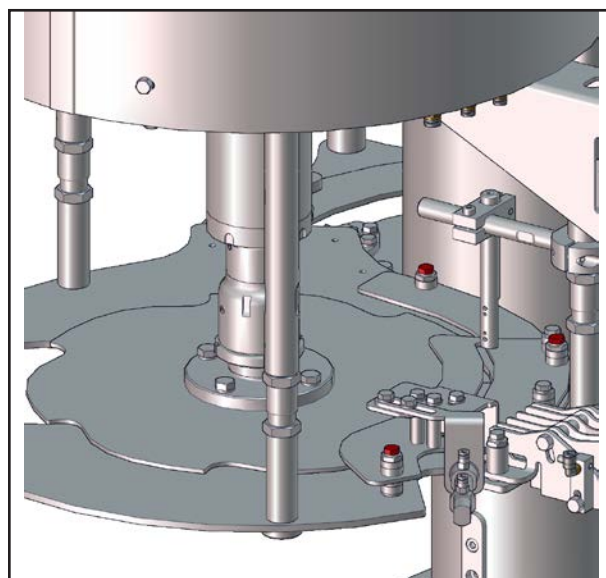


Fig. 216

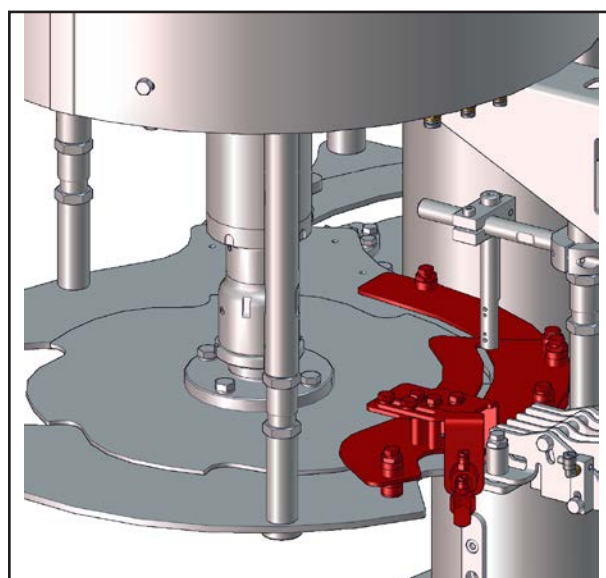


Fig. 217



10.12.4.1.2 INSTALLATION

Proceed as follows:

- Place the caps guide sectors on the base disk.

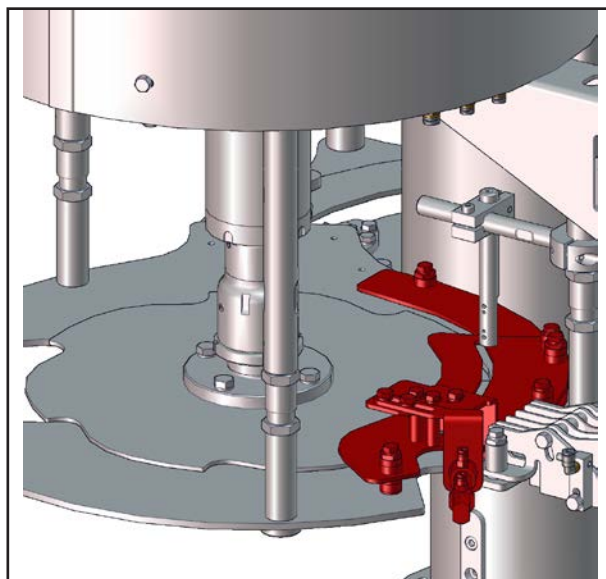


Fig. 218

- Screw the bolts to the base disc.

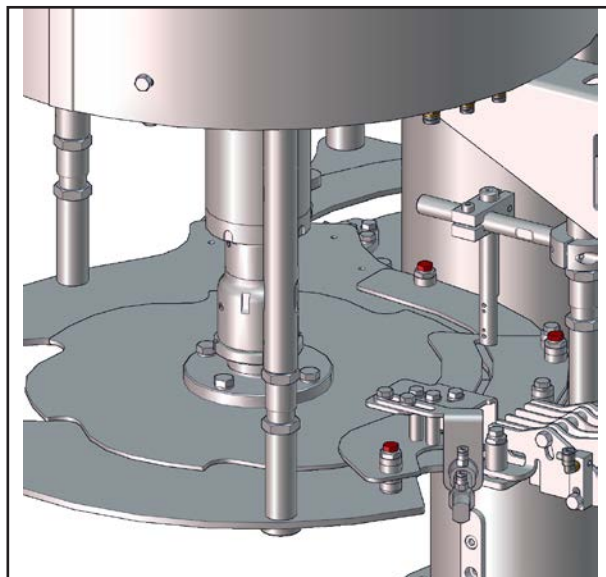


Fig. 219

- Connect the air compressing fitting to them connections.



Fig. 220

10.12.4.2 CAPS TRANSFER STAR-WHEEL

10.12.4.2.1 REMOVAL

- Remove the caps guide sectors as described in subsection 10.12.4.1 CAPS GUIDE SECTORS page 167.

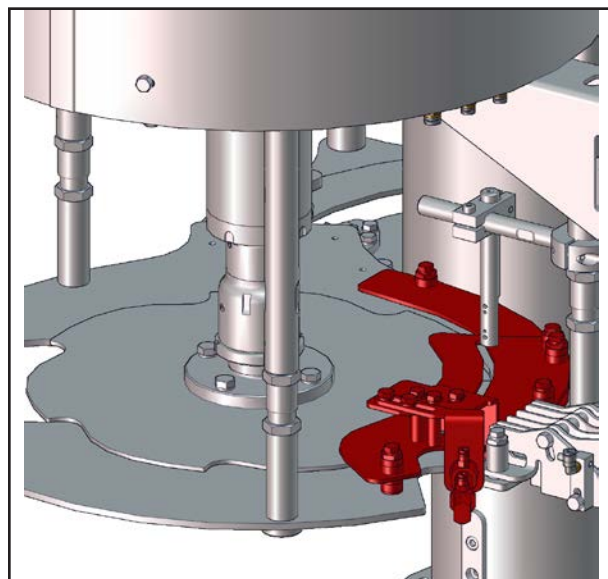


Fig. 221

- Unscrew the star-wheel fastening bolts

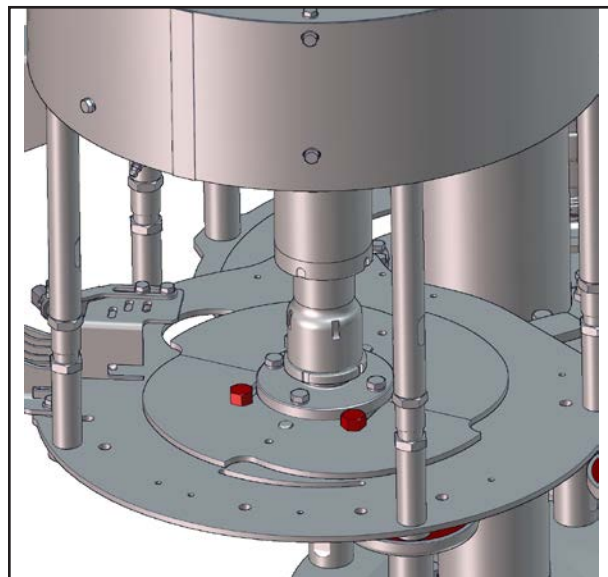


Fig. 222

- Lift the half star-wheel so that they come out from the centering pins
- Repeat the described above operations also for the second half star-wheel.

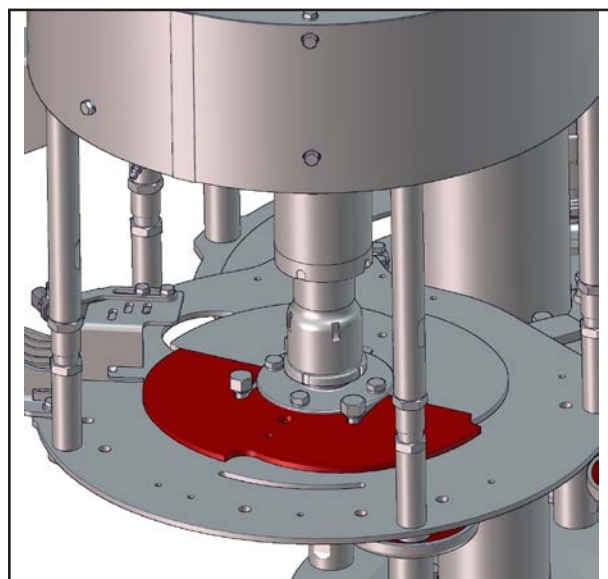


Fig. 223



10.12.4.2.2 INSTALLATION

Proceed as follows:

- insert the half star-wheel into the centering pin

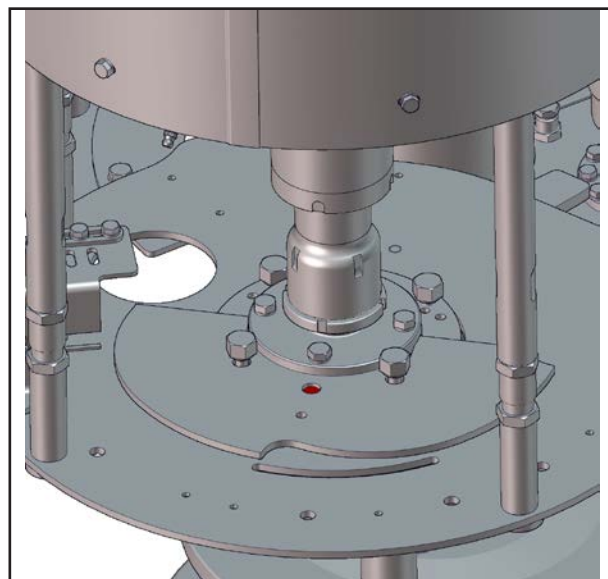


Fig. 224

- Firmly screw the fastening bolts.
- Repeat the above described operation also for the installation of the second star-wheel

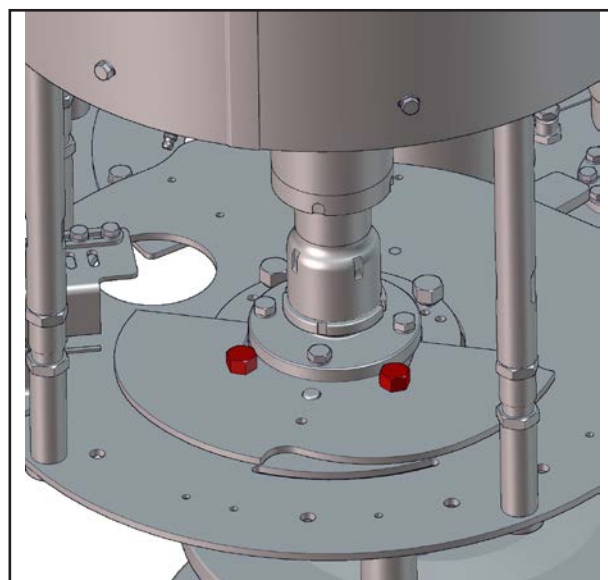


Fig. 225

- Install the caps guide sectors as indicated in subchapter 10.12.4.1 CAPS GUIDE SECTORS page 167.

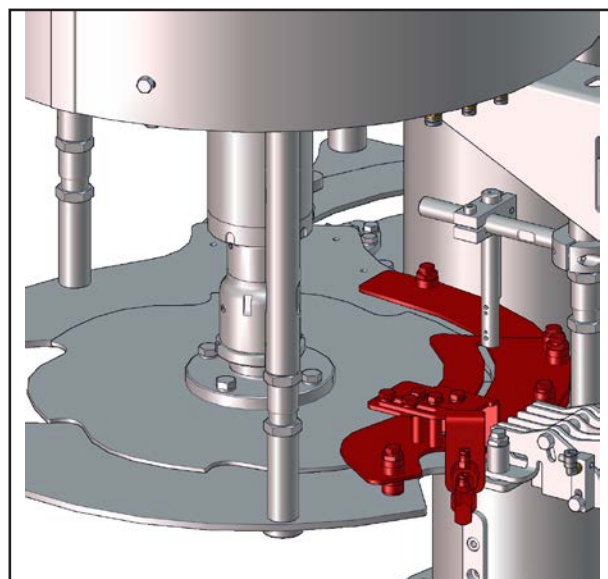


Fig. 226



10.12.5 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

10.12.5.1 CAPS EXIT MOUTH



10.12.5.1.1 REMOVAL

To remove proceed as follows:

- disconnect the pneumatic connections from the caps exit-mouth.
- Unscrew and remove the fasten knobs of the caps exit-mouth.

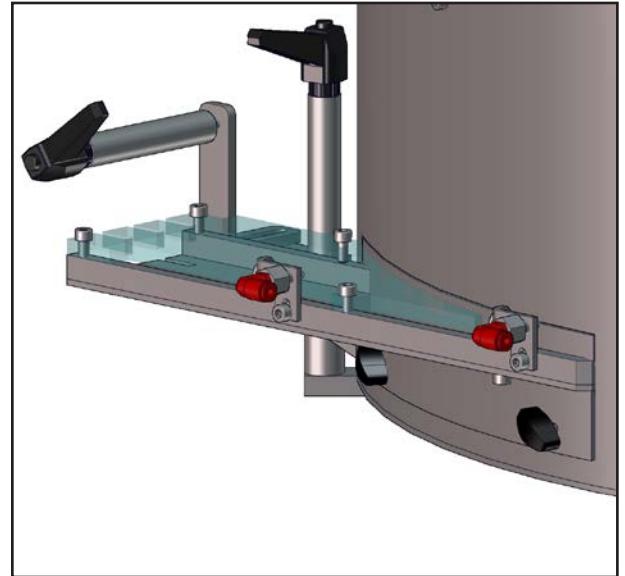


Fig. 227

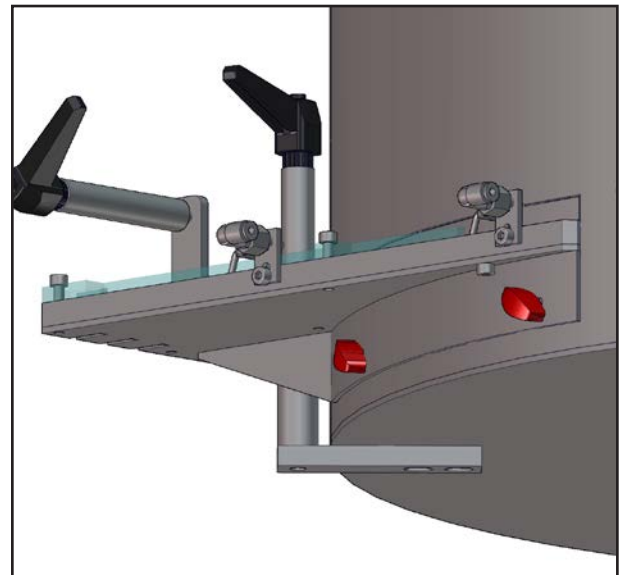


Fig. 228



- Unscrew and remove the return handle that fixes the caps exit-mouth to the caps sorter.

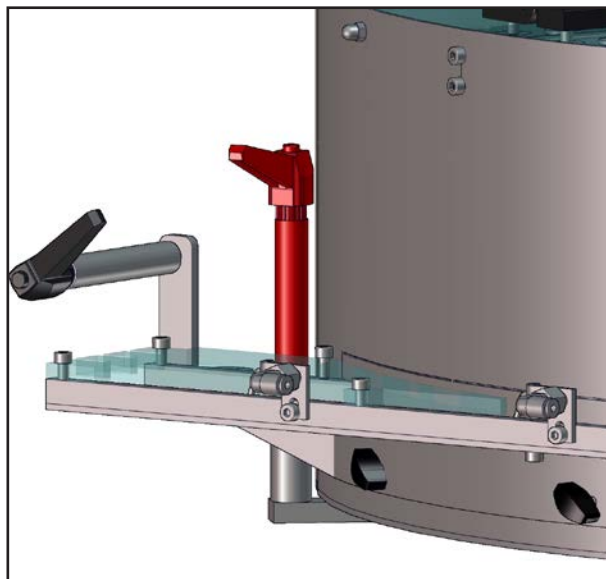


Fig. 229

- Remove the whole caps exit-mouth.

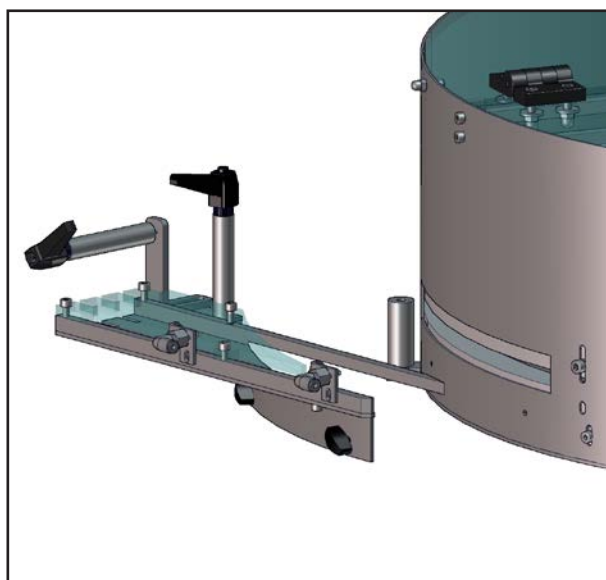


Fig. 230



10.12.5.1.2 INSTALLATION

To install proceed as follows:

- Position the exit mouth in its seat and fix with the fasten knobs.

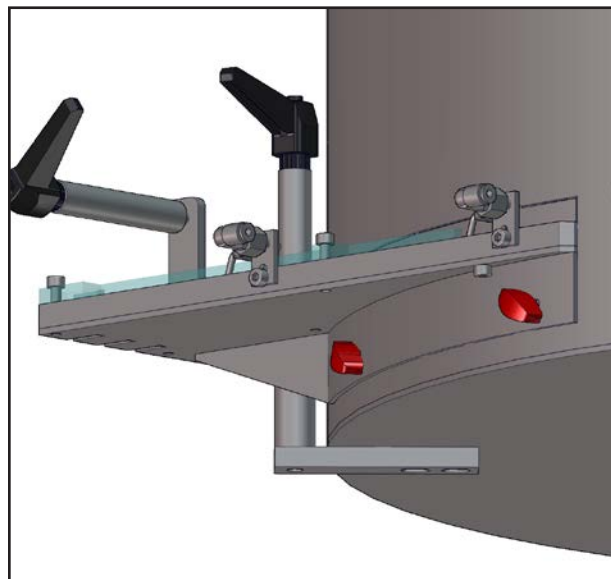


Fig. 231

- Tighten the return handle that fixes the caps exit-mouth to the caps sorter.

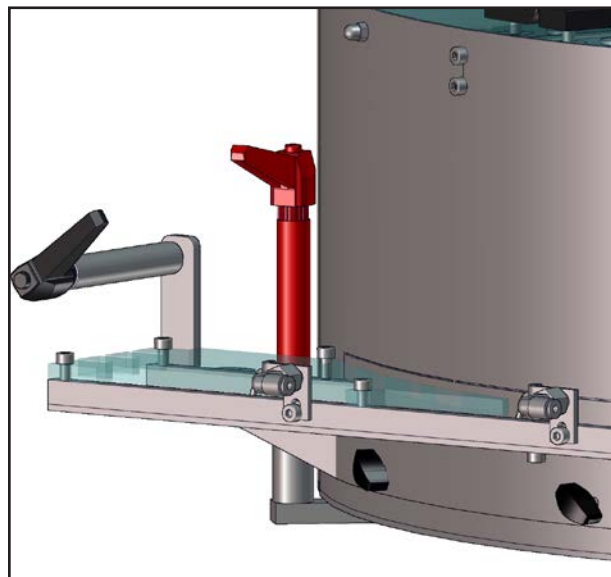


Fig. 232

- Connect the pneumatic connections to the caps exit-mouth.

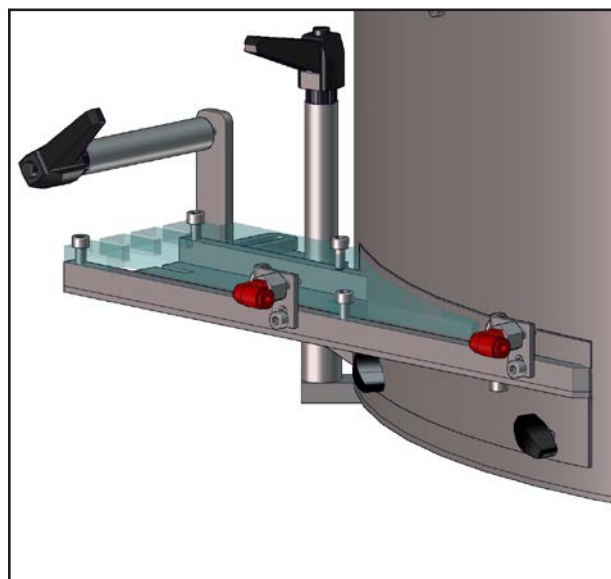


Fig. 233



10.12.5.2 ADJUSTMENT OF THE CAPS SORTER FOR SIZE CHANGE



- If varies the height of the cap to be treated must be adjust the height of the inner hopper ring in the following way:
 - loosen the screws located on the outer perimeter of the hopper.
- Adjust the height of the hopper ring in such a way between the new cap to be treated and the ring there is a clearance of about 3 mm.
- Tighten the screws previously loosened.

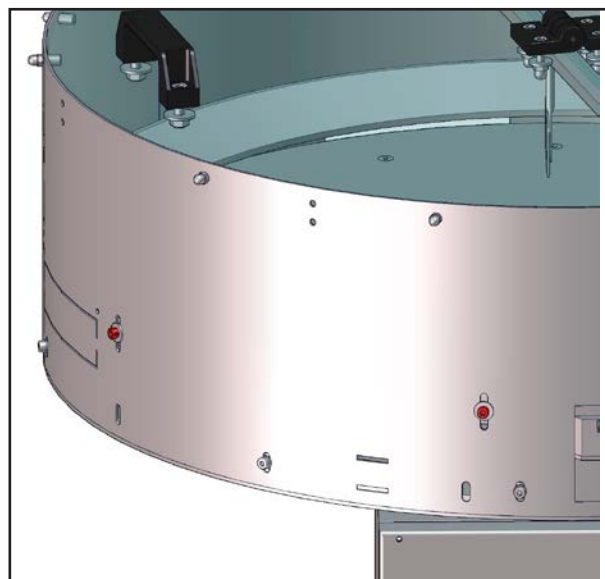


Fig. 234

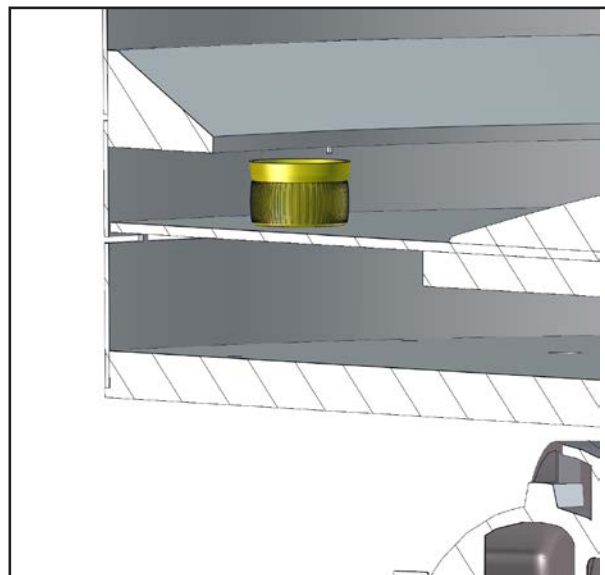


Fig. 235

- On the caps sorter there is a door that allows, thanks to the centrifugal force, the ejection of caps without seaming and seaming on them.
- If varies the height of the cap to be treated must adjust the height of the adjustable door in the following way:
 - loosen the fixing knobs.

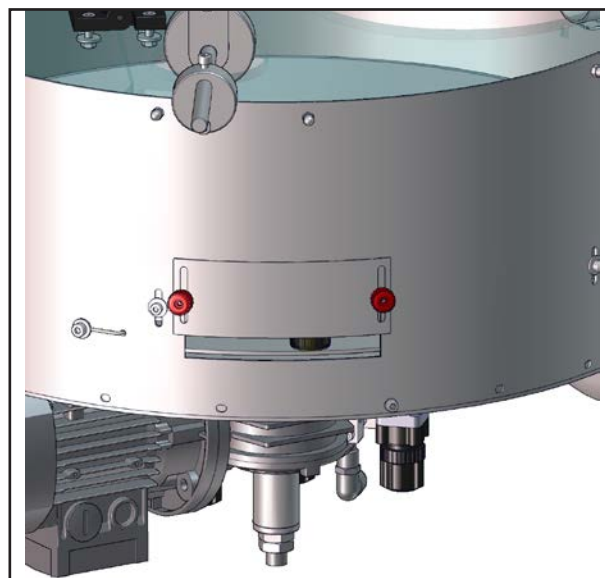


Fig. 236



- Adjust the height of the adjustable door so that they can be expelled only the caps without seal.
- After adjustment, tighten the fixing knobs.

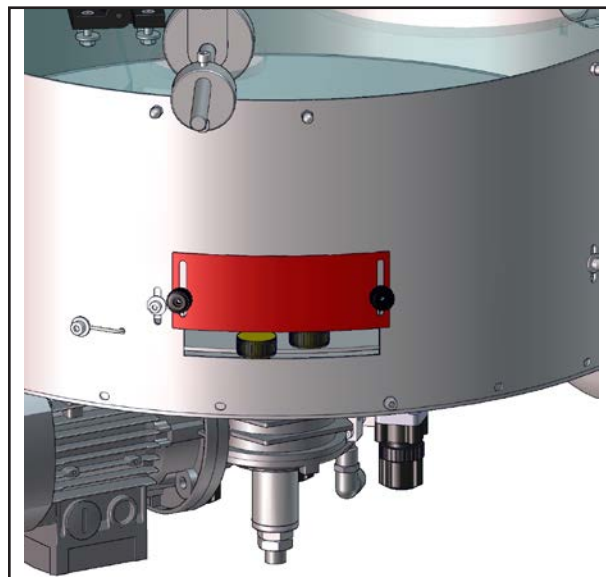


Fig. 237

If the cap is not correctly shifted by the nozzles it is sufficient to increase or decrease the air pressure by the fitting pressure reducers.

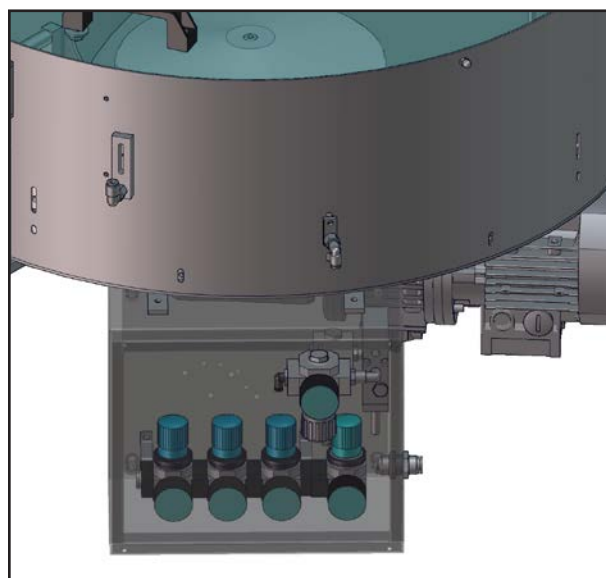


Fig. 238



10.13 ADJUSTMENTS OF THE CLOSURE HEAD

10.13.1 CLOSURE HEAD "VP750"

The table below shows the necessary tools for adjustment of the closure head.

TOOLS	
	Torque adjustment gripper
	Torque adjustment wrench

10.13.1.1 TIGHTENING TORQUE ADJUSTMENT

This step can be carried out with the head installed on the closing machine.

The adjustment is made by a graduated scale etched on the magnets adjustment threaded ring (in Pound x Inch) and a reference etched on the head central body.

To carry out the adjustment, proceed as follows:



Fig. 239

10.13.1.1.1 TORQUE ADJUSTMENT WITH HEAD ON THE SPINDLE

- Insert the suitable torque adjustment gripper on the magnets adjustment threaded ring

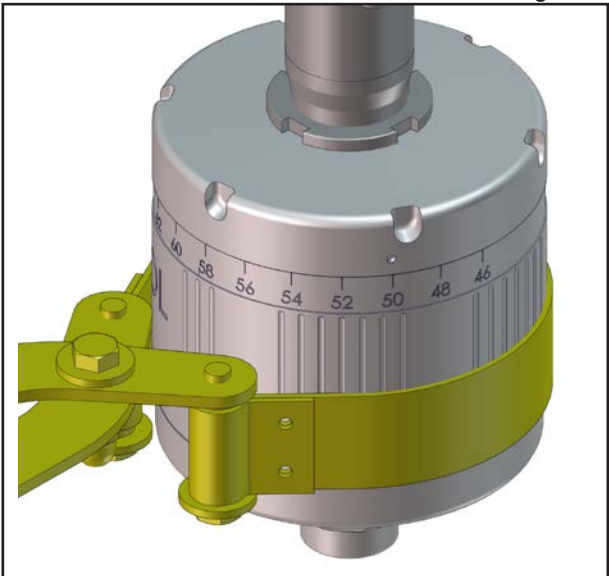


Fig. 240



- Grip the torque adjustment gripper in such way that the belt adhere to the threaded ring and turn magnets adjustment threaded ring until to obtain the alignment between the desired torque value and the mark impressed on the central body



It is also possible to carry out the tightening torque adjustment manually without the using of the couple adjustment gripper.



The adjustment must be made on all closure heads with the same tightening torque.

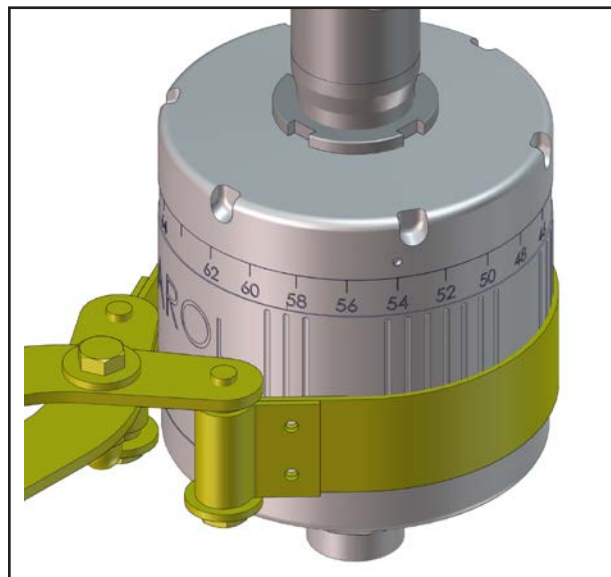


Fig. 241

10.13.1.1.2 BENCH TORQUE ADJUSTMENT

- Position the torque adjustment wrench into a jaw
- Insert the head body on the torque adjustment wrench

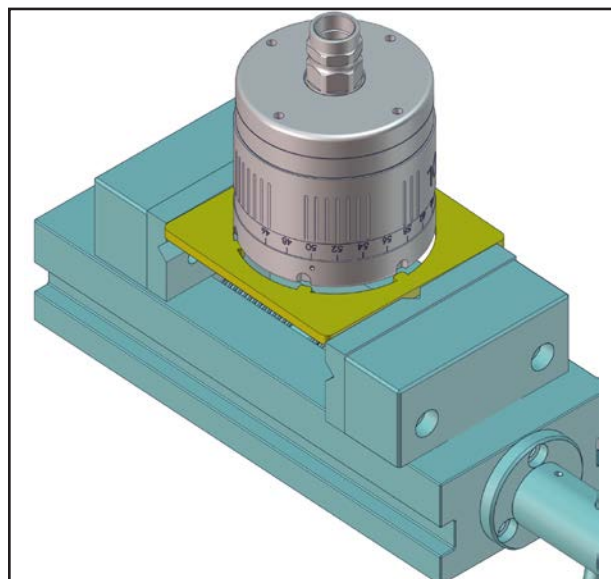


Fig. 242



- Insert the suitable torque adjustment gripper on the magnets adjustment threaded ring

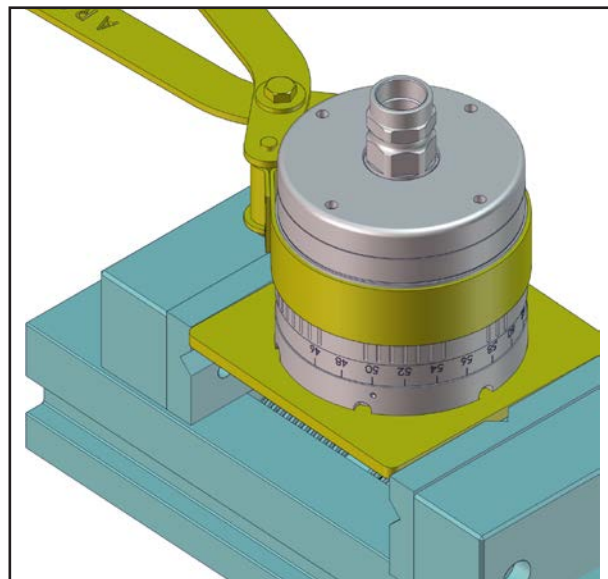


Fig. 243

- Grip the torque adjustment gripper in such way that the belt adhere to the threaded ring and turn magnets adjustment threaded ring until to obtain the alignment between the desired torque value and the mark impressed on the central body



It is also possible to carry out the tightening torque adjustment manually without the using of the couple adjustment gripper.



The adjustment must be made on all closure heads with the same tightening torque.

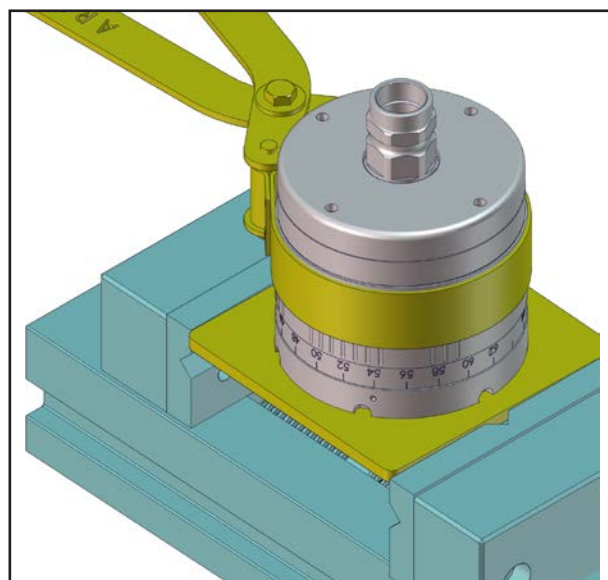


Fig. 244



10.13.2 CLOSURE APPLICATION DATA TABLE

To facilitate the correct collection of the information on the closure, at the end of this manual and for example purposes only, is attached a table useful for the purpose.



The user shall anyway refer to his own quality department and to the specifications given by the cap manufacturers, following them.

For the compilation of attached table, proceed as follows:

- indicate in the table heading:
 - the model of the closing machine
 - the serial number
 - the used container type
 - the used cap type
 - the set static torque
 - the number of bottles per minute
 - the type of compensating spring
 - the ejector spring type
 - the type of closure terminal
- indicate in the “data collection” table:
 - the number of bottles relevant to the closure head taken at the exit of the closing machine
 - the value indicated by the torquemeter checking each single bottle and, in the last column, the average value.



Photocopy the enclosure table to get the number of modules required for collection of data over time.



10.14 CLOSURE GRIPPER ADJUSTMENTS

The table below shows the necessary tools for adjustment of the closure gripper.

TOOLS	
	Hook wrench 16-20

10.14.1 "VP750" CLOSURE GRIPPER

When it is necessary to adjust the gripper pressure on the cap, proceed as described below:

- remove the closure gripper from the closure head.
- Loosen the grub screws of the spring adjustment threaded ring with a 3 mm Allen wrench.

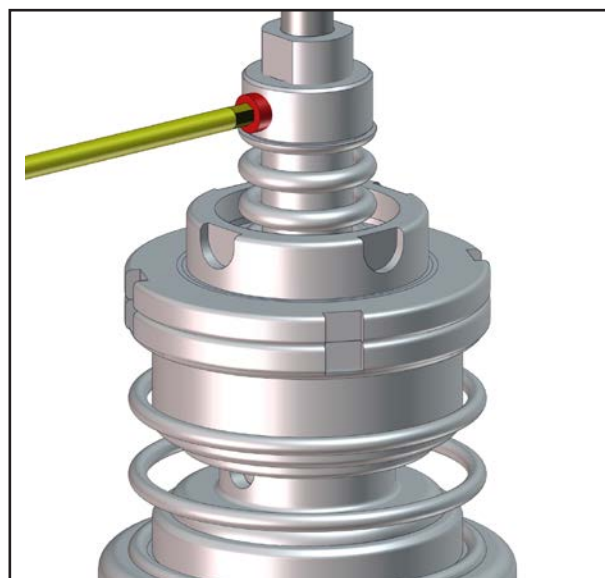


Fig. 245

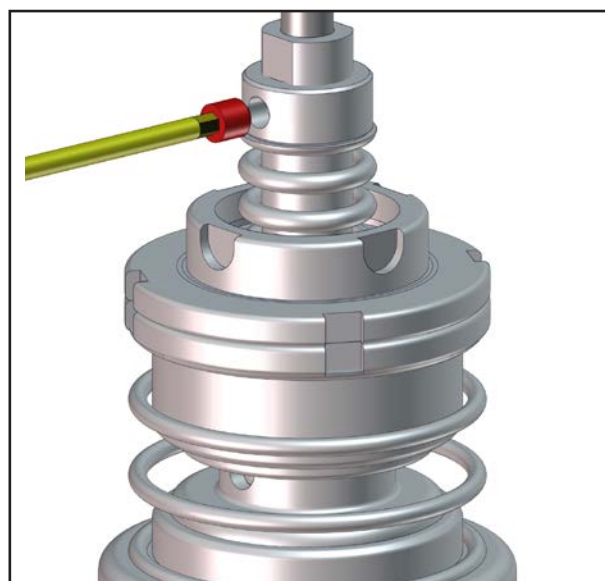


Fig. 246



- Insert the 16-20 hook wrench in the seats of the spring adjustment threaded ring and:
 - loosen the threaded ring to decrease the pressure of the gripper.
 - screw the threaded ring to increase the pressure of the gripper.

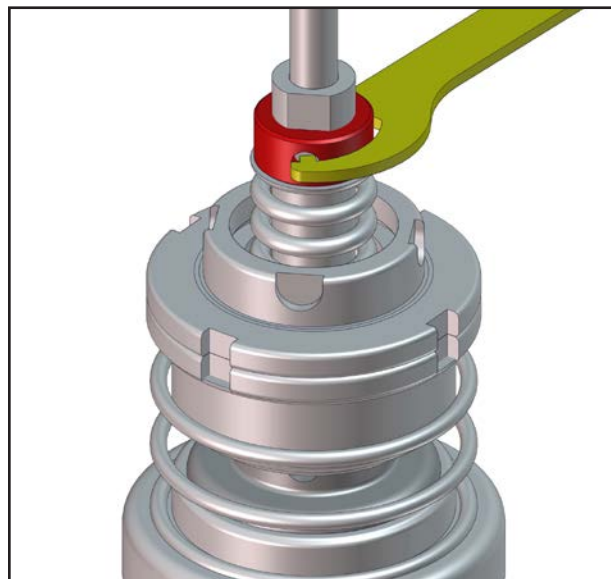


Fig. 247



11 MAINTENANCE OF THE CLOSING MACHINE

11.1 MAINTENANCE GENERAL INFORMATION



11.1.1 MAINTENANCE SCHEDULE

From the constructive point of view, interventions concern mechanical, electrical and pneumatic parts.

The envisaged operations are grouped according to work time intervals. Each intervention or group of interventions can concern about mechanical, electrical and pneumatic aspects.

The ordinary maintenance is divided in two categories:

- scheduled ordinary maintenance (or preventive).
- Maintenance according to the conditions (caused by wear and tear).

The scheduled ordinary maintenance (so called periodical or preventive) includes inspections, checks and interventions that, to prevent stops and faults, keep under control:

- the lubrication state of the closing machine.
- The state of parts that can be affected by wear and tear.

The maintenance according to special conditions recommends a regular inspection of components, to check their conditions and perform the maintenance only if it is required.

11.1.2 MAINTENANCE INTERVALS



The following tables show the maintenance deadlines and the operations needed for a correct operation and long life of the closing machine.

The interventions described below have been arranged at intervals of time expressed in hours.

Interval (hours)	Working in 1 shift	Working in 2 shifts	Working in 3 shifts
8	-	-	-
40	1 week	-	-
120	3 weeks	-	1 week
250	6 weeks	3 weeks	2 weeks
500	3 months	6 weeks	1 month
1000	6 months	3 months	2 months
3000	12 months	9 months	6 months
6000*	*	*	12 months

(*) every 6000 hours or anyway once a year.



11.1.3 GENERAL INDICATIONS ON THE SCHEDULED MAINTENANCE PLAN

A logbook of the scheduled maintenance and lubrication interventions must be arranged according to the maintenance indications in the sub-chapters below and to the closing machine working conditions.

Example:

MAINTENANCE SCHEDULED AT 1000 HOURS					
INTERVENTION	DATE	MAINTENANCE MAN	SIGNATURE	INTERVENTION RESULT	NOTES
Check of the height sliding of the closure carousel.	XX/XX/XX	Mario Rossi		positive	
Control gear lubrication	XX/XX/XX	Mario Rossi		lubricated	

11.2 MAINTENANCE AND LUBRICATION PLANNING



Interval	Intervention type
40 hours	Ordinary maintenance
120 hours	lubrification
	Ordinary maintenance
250 hours	lubrification
500 hours	lubrification
	Ordinary maintenance
1000 hours	lubrification
	Ordinary maintenance
3000 hours	Ordinary maintenance
5000 hours	Ordinary maintenance
6000 hours	lubrification
	Ordinary maintenance
12000 hours	Ordinary maintenance
18000 hours	Ordinary maintenance



11.3 CLOSURE HEAD WEARING PARTS VP750



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours			
	3000	6000	12000	18000
Ejector rod supporting spring	x			
Central radial bearing		x		
Thrust bearing on the ejector rod		x		
Magnetic rings			x	
O-ring		x		
V-ring		x		



Service the closing head following the instructions in sub-chapter 11.5.3
OF THE CLOSURE HEAD page 194.

MAINTENANCE

11.4 CLOSURE GRIPPER WEARING PARTS "VP750"



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours		
	3000	6000	12000
Open gripper spring	x		
Caps supporting plate spring.	x		
Cap gripper ring	x		
Thrust bearing		x	
Snap ring		x	
Gripper guide Allen screws		x	
IGUS bushes cap stop star-wheel			x
Cap gripper			x
Gripper opening pin			x
Gripper opening pin rod			x
Pin guide bush			x



Carry out the gripper overhaul, following the instructions in the sub-chapter 11.5.4
CLOSURE GRIPPER MAINTENANCE page 198.



11.5 SCHEDULED MAINTENANCE



11.5.1 ROUTINE MAINTENANCE

11.5.1.1 EVERY 40 WORKING HOURS

OPERATION	AIM	NOTES
Check the operating of the the safety devices.  	With closing machine at a standstill, operate the safety devices manually (interlocked mobile guards and emergency buttons) and make sure that the red warning light <i>SAFETY DEVICE TRIPPED</i> on the control panel switches on; the display should show the message describing the safety device that was triggered. After the safety devices have activated and with the closing machine in automatic operation mode, check that, pressing the green button <i>START</i> the machine does not start and the red flashing light flashes. The siren sends a sound signal of 3 seconds duration.	
Pneumatic system check.  	Check the air supply pressure to the various utilization points. Check by the gauges, that the pressure reducers are adjusted at the work pressure (the pressure is indicated on the pressure gauges) Drain the possible condensate collected under the discharge filter. Checking the filter clogging. If necessary, replace it.	



11.5.1.2 EVERY 500 WORKING HOURS

OPERATION	AIM	NOTES
Lubrication of the closure gripper. 	Lubricate the components of the closure gripper	See sub-chapter 11.6.7 CLOSURE GRIPPER LUBRICATION page 222.
Pneumatic system check. 	Check the correct operation and possible leaks (from fittings, lines, etc.): ▪ verify the air pressure at the closing machine entry. ▪ Supplying the pneumatic system with compressed air, check there is no leak, broken or disconnected pipe. ▪ Check that the closing machine pressure gauges indicate the applied pressure on the suitable data plate.	
Check of the bottle anti-rotation device belt. 	Check the wear of the bottle anti-rotation belt and, if necessary, replace it.	For the replacement, See sub-chapter 11.5.6 B O T T L E S ANTI-ROTATION DEVICE page 207.



11.5.1.3 EVERY 1000 WORKING HOURS

OPERATION	AIM	NOTES
Check of the height sliding of the closure carousel.  	Check the correct vertical sliding, making the closure carousel slide from the min. to the max. value, so as to ensure a correct component sliding in time.	
Equipment timing check.  	Check the timing of the whole equipment and, if necessary, re-time it.	See sub-chapter 11.7.4 TRANSPORT EQUIPMENT TIMING page 236.
Electric connection check.   	Carefully check all the connectors and connection cables and if necessary replace all the damaged parts.	
Sight and sound check of possible noises, clearances and unbalancing.    	Check, starting and stopping the closing machine several times without bottles, for possible noises, faults, clearances and vibrations in the motor-drive components (motors, reducers, gears, bearings). Replace possible worn components. Lubricate the new installed components. Clean and lubricate the disassembled components. Report any anomalies you need to contact <i>Technical Support</i> .	
Check of the head control spindles.  	Check the clearance of the spindle pads.	See sub-chapter 11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT page 192.



11.5.1.4 EVERY 5000 WORKING HOURS

OPERATION	AIM	NOTES
Check the wear of the rubber sectors of the central star-wheel.  	Check the wear of the rubber sectors of the central star-wheel and change it, if necessary.	See sub-chapter 11.5.5 S T A R - W H E E L ANTIROTATION SECTORS page 205.

11.5.1.5 EVERY 6000 WORKING HOURS

OPERATION	AIM	NOTES
Check of the head control spindles.  	Check the clearance of the spindle pads.	See sub-chapter 11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT page 192.
Overhaul of the closure grippers. 	Overhaul the gripper completely.	See sub-chapter 11.5.4 CLOSURE GRIPPER MAINTENANCE page 198.



11.5.1.6 EVERY 12000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the cork inserting pistons. 	Complete overhaul of pistons, including replacement of: ▪ Bearings. ▪ Rotating and static sealing. ▪ Scraper. ▪ Snap rings. ▪ Upper pressure roller. ▪ Lower pressure roller. ▪ Bearing fixing nut. ▪ Piston-guide pads. ▪ Pad fixing plates. ▪ Hardened bush for lower roller. ▪ Main sealing. ▪ Spindle key. ▪ Head fixing threaded ring. ▪ Roller rotation pin. ▪ Spindle closing threaded ring. ▪ Spindle gear. ▪ Gear connection sleeve.	For the overhaul of the piston, contact the <i>Technical Service</i>
Check the wear of the cap/cork driving punch unit cam race. 	Visually check the wear of the roller sliding race.	See sub-chapter 11.7.1 SPINDLES CONTROL CAM WEAR CHECK page 231.
Closing machine general check. 	Check the correct operation of the closing machine; if faults are found, contact the <i>Technical Service</i>.	



11.5.1.7 EVERY 18000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the transfer disk shaft unit. 	Overhaul the transfer disk by replacing the components below: ▪ Bearings. ▪ Snap rings. ▪ Seals. ▪ Cork stopping cylinder.	For the cork transfer disk overhaul, you need to contact <i>Technical Support</i> .
Overhaul of the star-wheel shaft unit. 	Overhaul the star-wheel unit by replacing the components below: ▪ Bearings. ▪ Snap rings. ▪ MIM seal rings.	For the overhaul of the star-wheel shaft, contact the <i>Technical Service</i> .
Overhaul of the worm unit. 	Overhaul the worm unit by replacing the components below: ▪ Angular idler unit shaft bearing. ▪ Angular idler shaft seal ring. ▪ Snap rings. ▪ Joint protection bellows and unit O-Ring. ▪ Joint. ▪ Worm supporting bearings. ▪ MIM seal rings and worm supporting O-Ring ▪ Worm bearings	For the overhaul of the worm unit, contact the <i>Technical Service</i> .
Overhaul of the worm idler unit. 	Overhaul the worm idler unit by replacing the components below: ▪ Idler gears. ▪ Bearings.	For the overhaul of the worm idler unit, contact the <i>Technical Service</i> .
Overhaul of the heads rotation motor group. 	Overhaul by replacing the following components: ▪ Bearings. ▪ Seals. ▪ Movement drive gears.	Contact the <i>Technical Service</i> .



11.5.1.8 EVERY 24000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the cork inserting pistons. 	Replacement of the inner sleeve group.	For the overhaul of the piston, contact the <i>Technical Service</i>



11.5.2 HEADS CONTROL SPINDLES MAINTENANCE



11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT

To adjust the pad clearance, operate as follows:

- make the closing machine rotate in jog until the spindle to be adjusted is in the flat (rear) section of the cam.
- Lift the ends of the pads fixing safety plates so as to release the bolts.

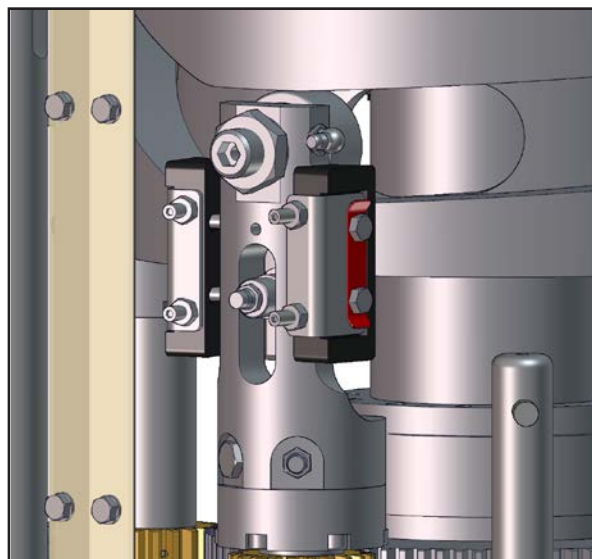


Fig. 248

- Loosen the fixing bolts of both pads.

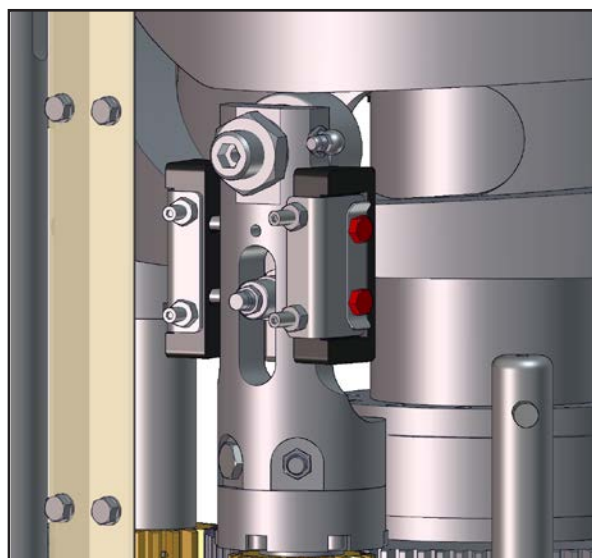


Fig. 249

- Loosen nuts and adjustment grub screws for both pads.

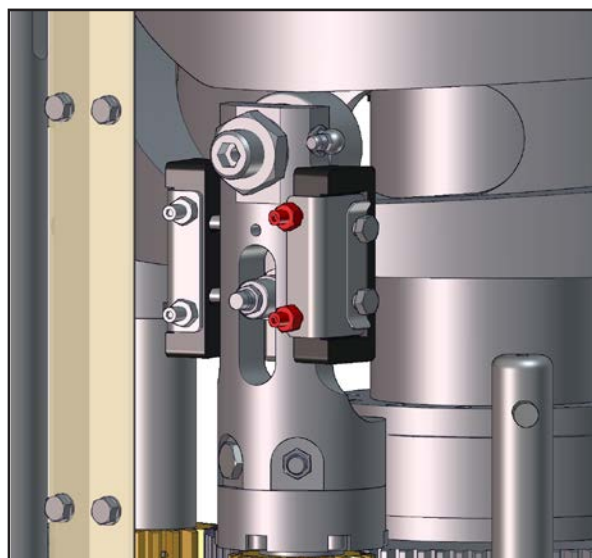


Fig. 250



- Fit a 0.1 mm shim between the two pads and the spindles driving cam.

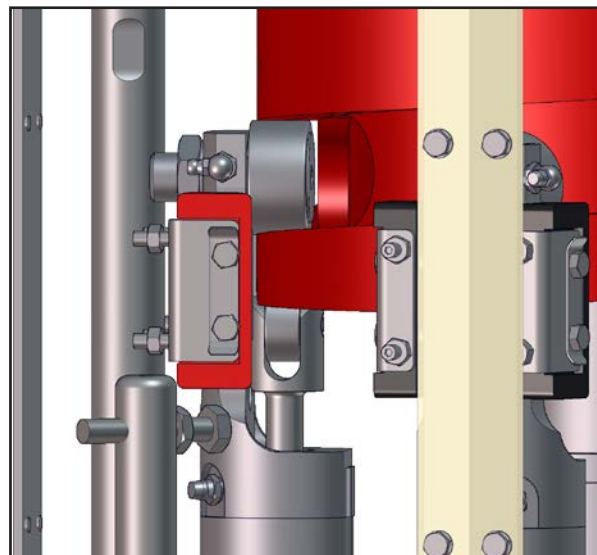


Fig. 251

- Push the pads against the shim.
- Register the position of the head driving piston so that the distance between the pad and the cam is the same for both pads.
- Verify the distances and lock the previously loosened bolts.
- Screw the adjusting screws until they are in contact with the pads.
- Screw all the grub screws locking nuts.

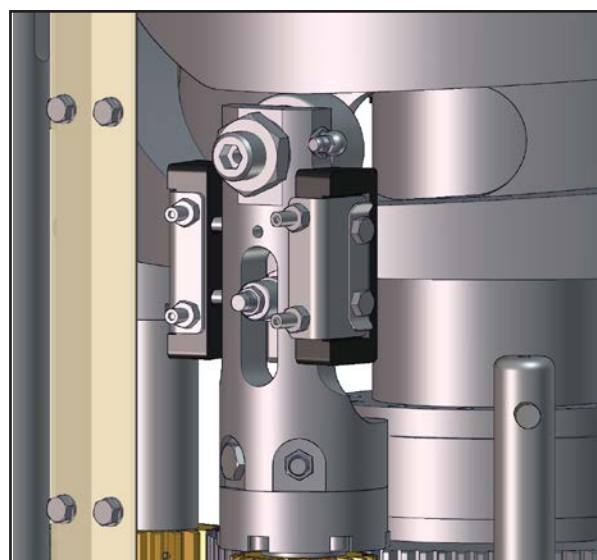


Fig. 252



A clearance between 0.1 and 0.2 mm must be kept all along the race cam.

This condition must be checked on the cam circumference in at least four diametrically opposed points (e.g. rear and front parts, left and right side).

- After adjustment, again bend the ends of the pads fixing safety plates so as to prevent that bolts may loosen.

Repeat the operation for the other possible spindles to be adjusted.



11.5.3 MAINTENANCE OF THE CLOSURE HEAD

11.5.3.1 CLOSURE HEAD "VP750"

11.5.3.1.1 GRIPPER OPENING UNIT SPRING OVERHAUL

When it is necessary to remove or replace the gripper opening unit spring, it is necessary to remove the head from the closing machine and proceed as described below:

- Slide the gripper opening unit out complete with spring from the head upper part



Fig. 253

- Slide the spring out from the gripper opening unit



Fig. 254



11.5.3.1.2 MAGNETIC RING CHECKING

Check using a torquemeter that the torque corresponds (with a tolerance $\pm 5\%$) to the static torque value set on the slotted threaded ring.

If the tolerance is higher than the one indicated above, replace the magnetic rings.



Contact the *TECHNICAL SERVICE* for info.



11.5.3.1.3 THRUST BEARING ON THE EJECTOR ROD OVERHAUL

When you need to overhaul the thrust bearing, proceed as follows:

- Remove the snap ring with a snap ring pliers.



Fig. 255

- Remove the bearing fixing washer.



Fig. 256

- Remove the 3 parts of the thrust bearing.
- Remount, proceeding inversely to that described for disassembly..
-



Fig. 257



11.5.3.1.4 CLOSURE HEAD INSTALLATION

Remount, proceeding inversely to that described for disassembly.



Before reassemble the closure head, clean and grease all components.



11.5.4 CLOSURE GRIPPER MAINTENANCE

11.5.4.1 SPRINGS REMOVAL

The table below shows the necessary tools for maintenance of the closure gripper.

TOOLS	
	Hook wrench 16-20
	Hook wrench 45-50

11.5.4.1.1 GRIPPER PRESSURE ADJUSTMENT SPRING REMOVAL

If it is necessary remove the gripper controlling spring, proceed as described below:

- remove the closure gripper from the closure head.
- When the gripper is removed and closed, push the central rod to the end (the inner cam is in contact with the 3 sectors of the gripper).



Fig. 258

- Measured with a caliper, and record the distance between the nut and the pin guide bush.

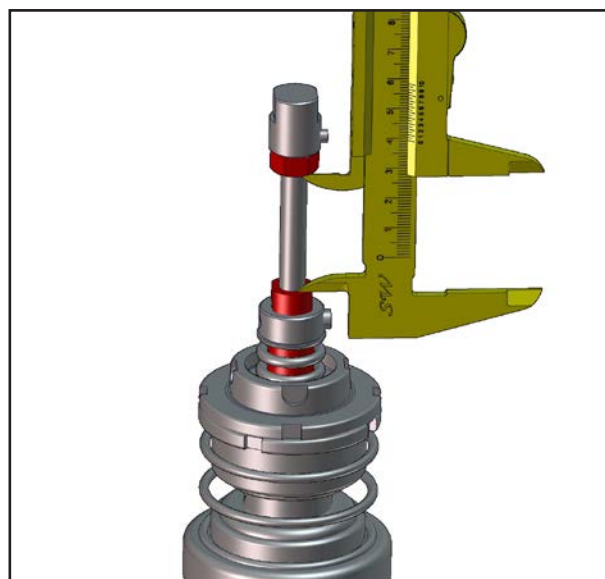


Fig. 259



- Unscrew the locking grub screw with a 2 mm Allen wrench.

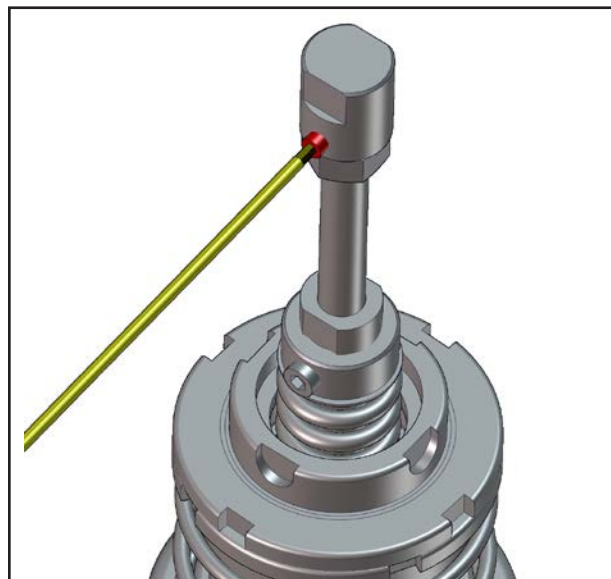


Fig. 260

- Insert the 13 mm hexagonal wrench in the seat of the gripper control pin.

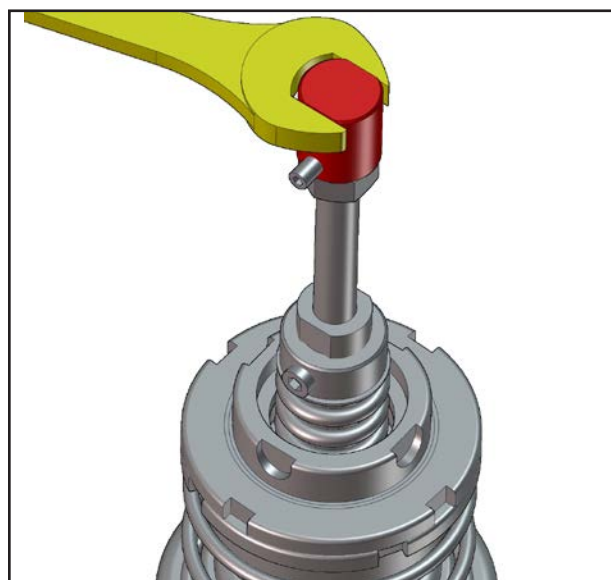


Fig. 261

- Insert the 13 mm hexagonal wrench on the fixing nut.



Fig. 262



- Unscrew the gripper control pin and the fixing nut.



Fig. 263

- Measured with a caliper, and record the distance between the pin guide bush and the spring adjustment threaded ring.

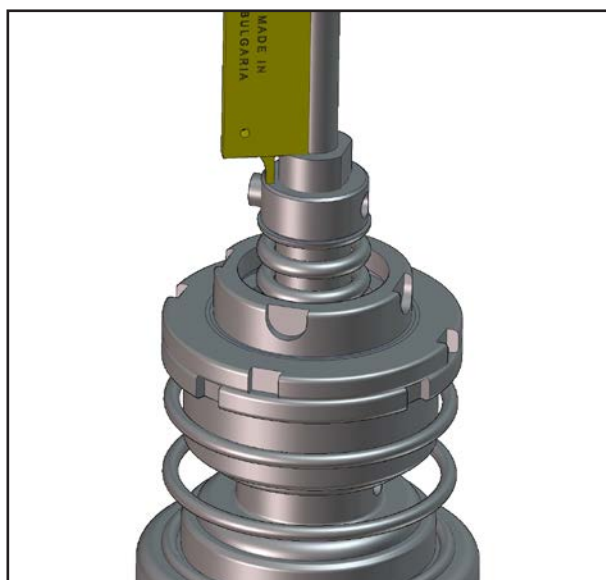


Fig. 264

- Loosen the grub screws of the spring adjustment threaded ring with a 3 mm Allen wrench.

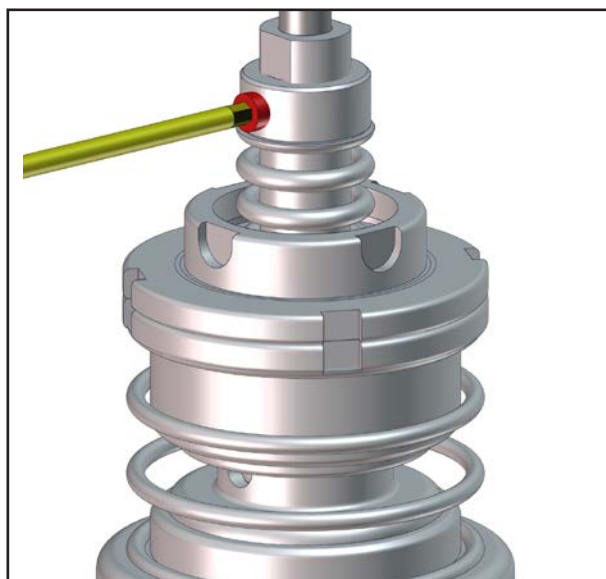


Fig. 265



- Insert the 16-20 hook wrench in the seats of the spring adjustment threaded ring and unscrew the threaded ring to remove it.

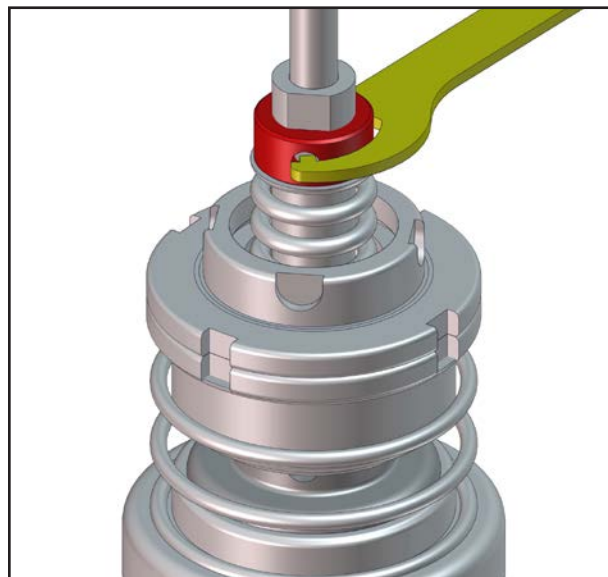


Fig. 266

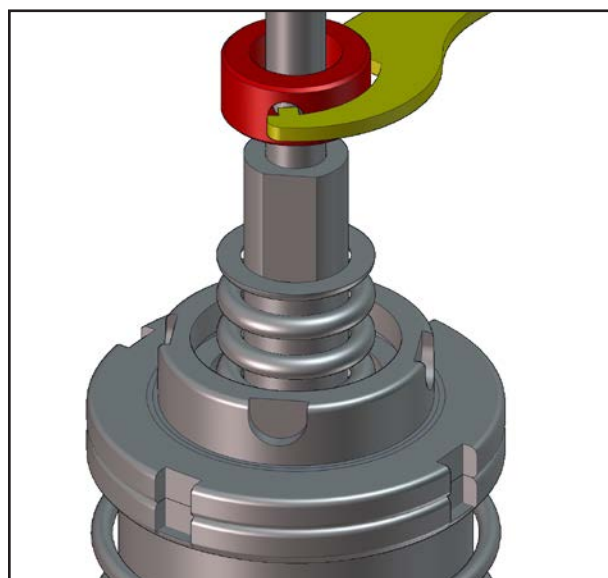


Fig. 267

- Slide out the gripper control spring and, if necessary, replace it.

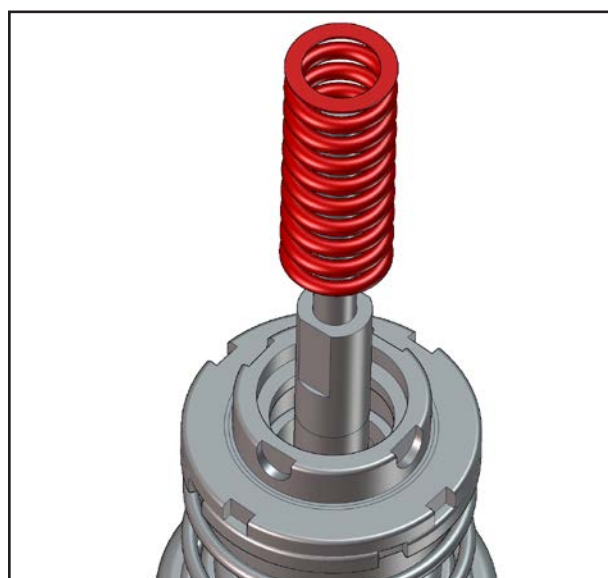


Fig. 268



11.5.4.1.2 COMPENSATING SPRING REMOVAL

If it is necessary to remove the gripper compensating spring, proceed as described below:

- remove the closure gripper from the closure head.
- Insert the 45-50 hook wrench into seats of the screws cover threaded ring.

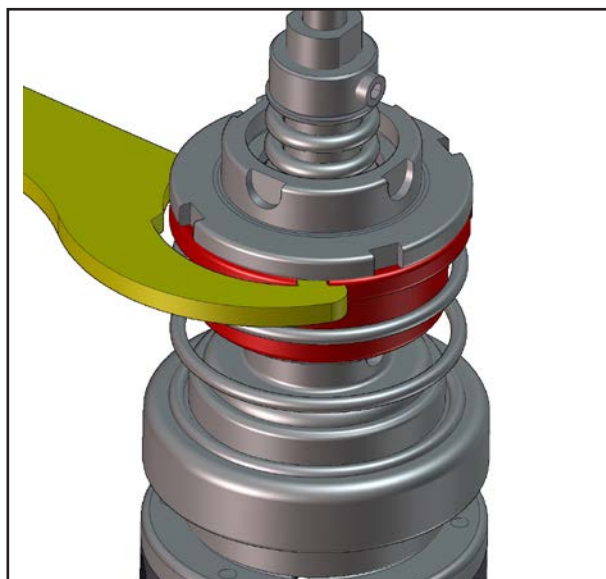


Fig. 269

- Insert the 45-50 hook wrench into seats of the closure threaded ring.

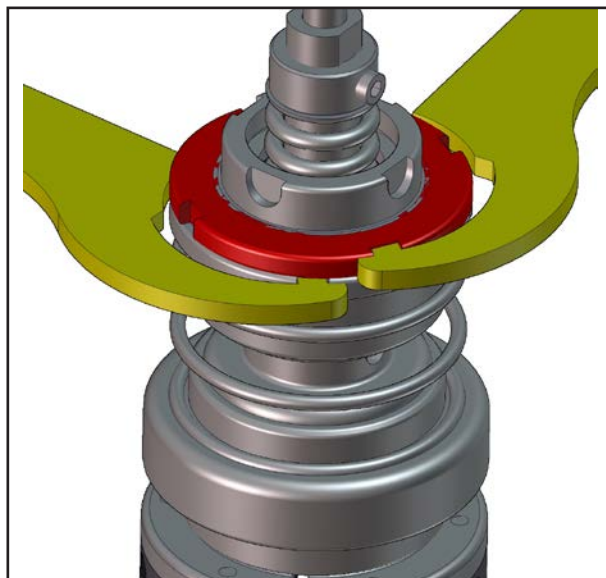


Fig. 270

- Unscrew the closure threaded ring; if the threaded ring is blocked, lightly strike the wrench handle.

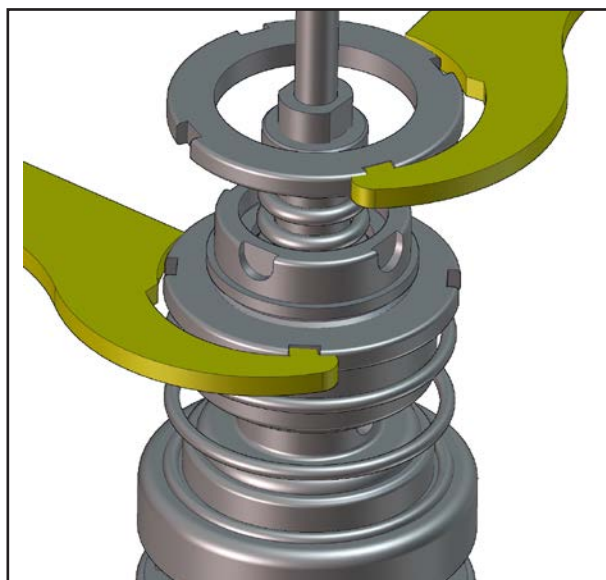


Fig. 271



- Holding the compensating spring pressed and using the 45-50 hook wrench, unscrew the screws cover threaded ring.

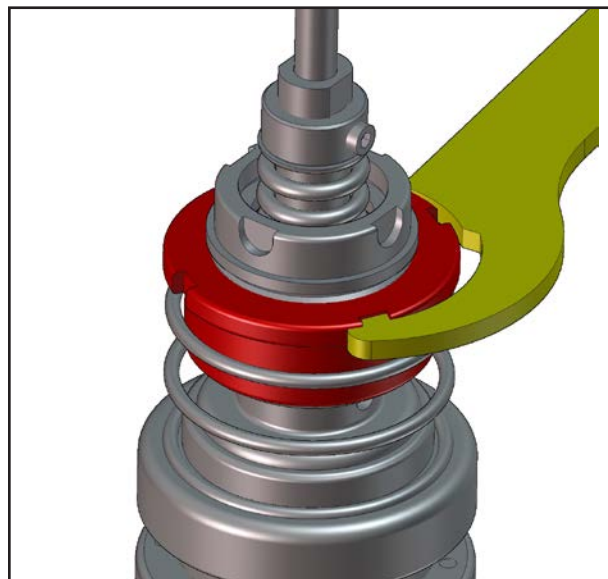


Fig. 272

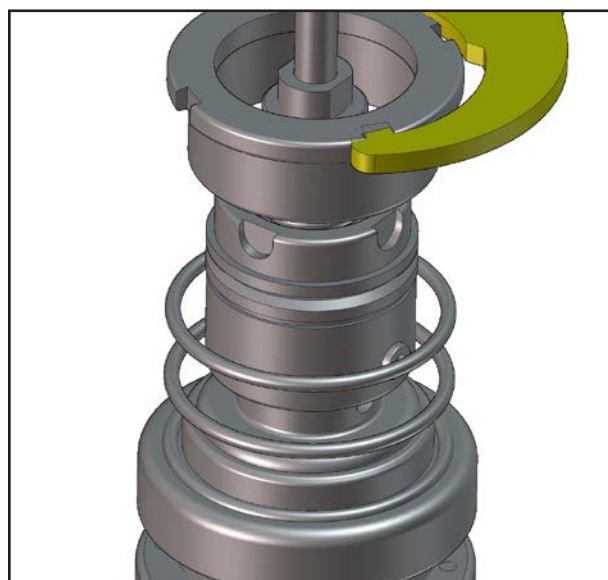


Fig. 273



- Remove the compensating spring.



Fig. 274



The springs, indicated below, are available as standard (1 kg_f = 9.81 N).

Spring	kg _f	Spring	kg _f
	4		8
	(1 cut on the spiral)		(1 cut on the spiral)

11.5.5 STAR-WHEEL ANTIROTATION SECTORS



To remove proceed in the following way:

- Remove the central star-wheel (see the sub-chapter 10.11.3.1 REMOVAL page 140).
- Unscrew and remove the fastening screws to free the star-wheel.



Fig. 275

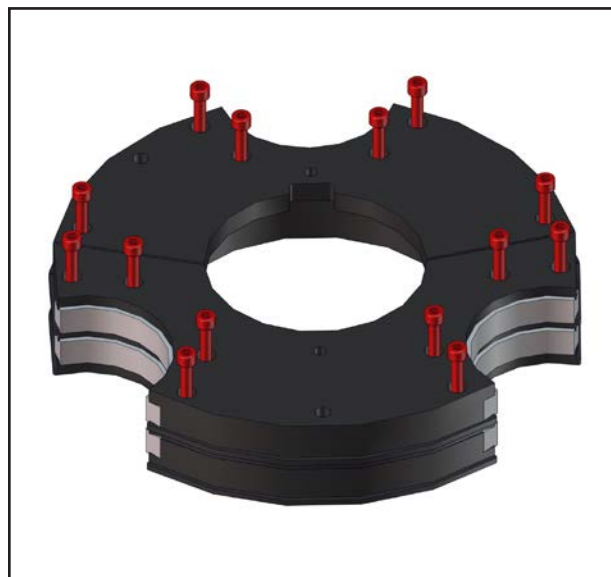


Fig. 276

- Remove the upper and lower part of the star-wheel.

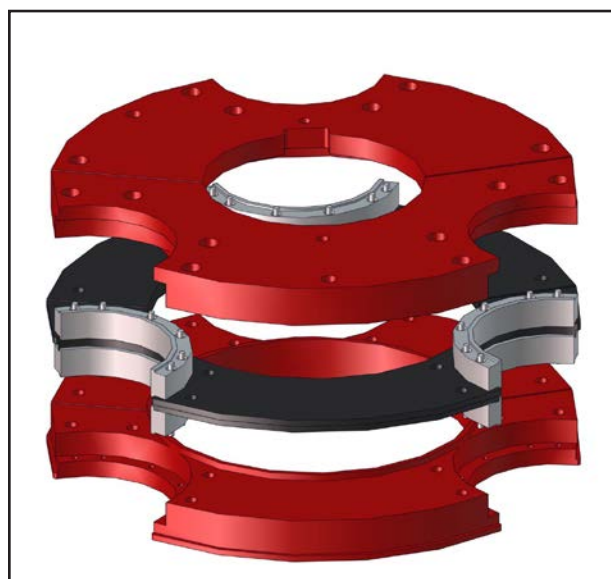


Fig. 277



- Remove the rubber sectors from the centering pins and replace them.



Fig. 278

- Refit, in reverse order to that described above.



Fig. 279



11.5.6 BOTTLES ANTI-ROTATION DEVICE



11.5.6.1 REMOVAL

To remove the containers antirotation belt, proceed as follows:

- Remove the later guides as described in paragraph 10.11.2 LATERAL GUIDES page 138.



The red piping that links the storage to the anti-rotation cylinder can contain residual pressure.

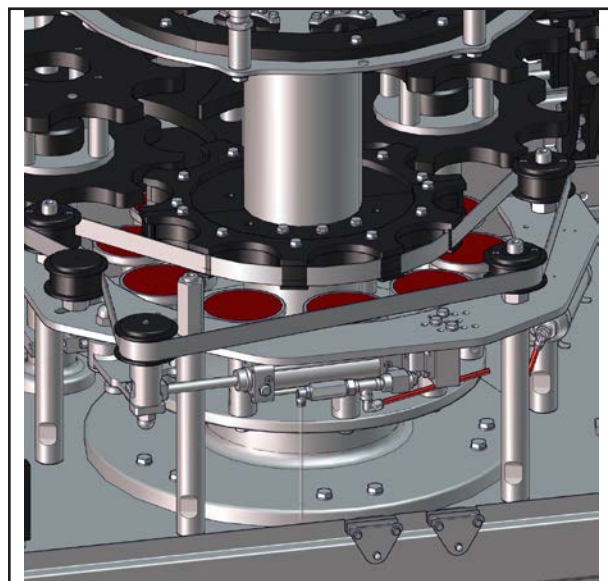


Fig. 280

- Manually close the cylinder stem.

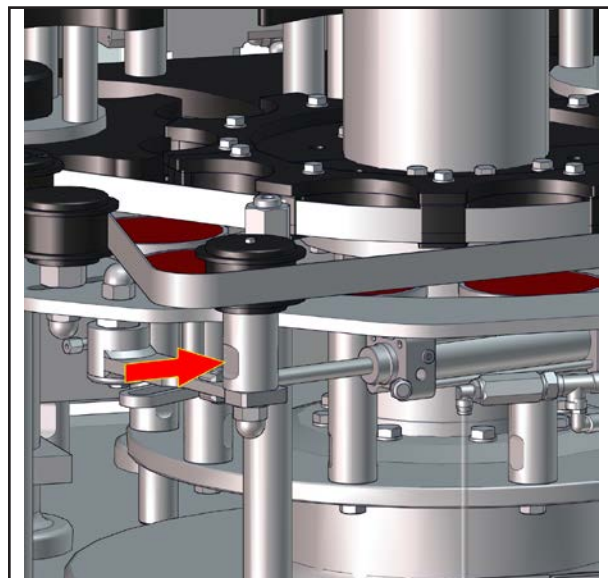


Fig. 281



- Remove the container antirotation belt from the pulleys.

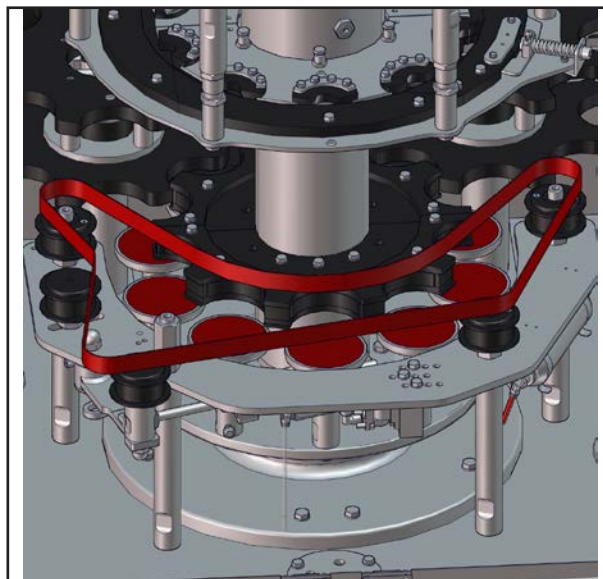


Fig. 282

11.5.6.2 INSTALLATION

For installation of the container antirotation belt, proceed as follows:

- position the belt on the pulleys.
- Manually open the anti-rotation cylinder stem in such way to have the pulley close to the anti-rotation belt.

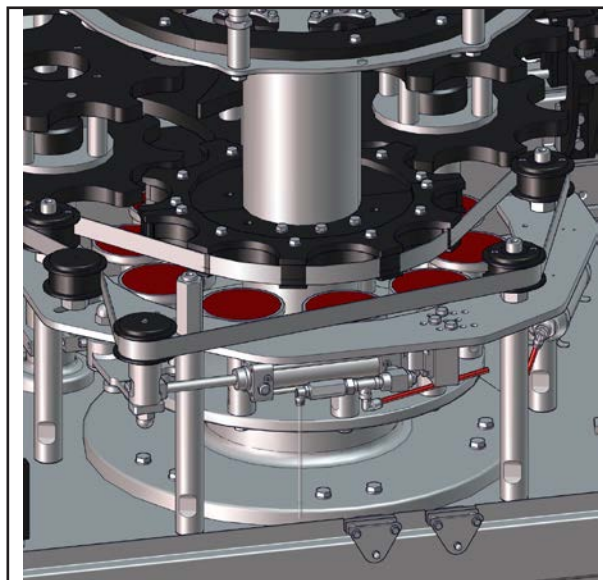


Fig. 283

- Install the lateral guides as described in paragraph. 10.11.2 LATERAL GUIDES page 138.
- Close interlocked mobile guards and restore the operation of the safety circuit.



11.5.7 BOTTLE ANTIROTATION DEVICE ADJUSTMENT

For the adjustment operate as described here below:

- Pull the fasten spindles and turn them of 90° in such way to avoid the return in their seats.

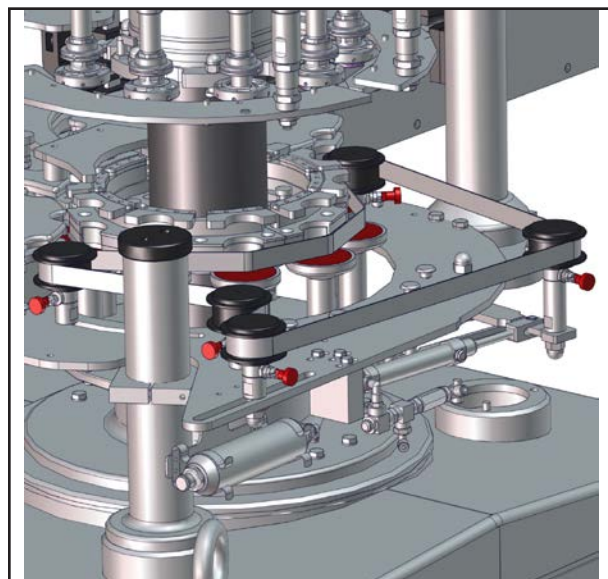


Fig. 284

- Adjust the locking group position according to the container to be proceed.
- Lift or lower the containers locking group.



The height reference values are specified in the EQUIPMENT COLOR TABLE enclosed in this manual.

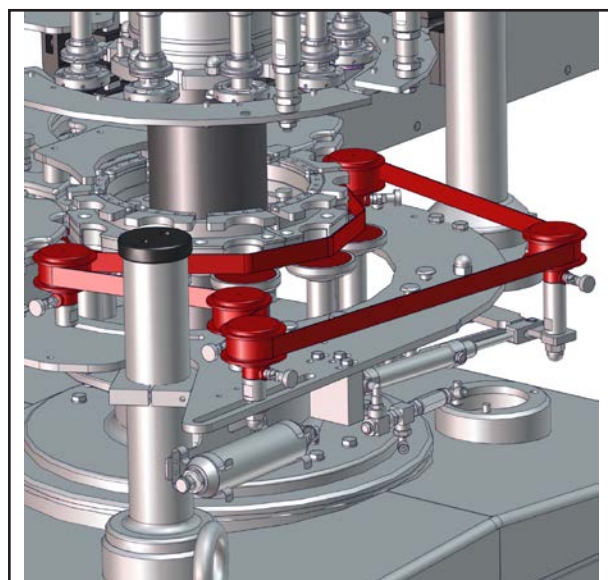


Fig. 285

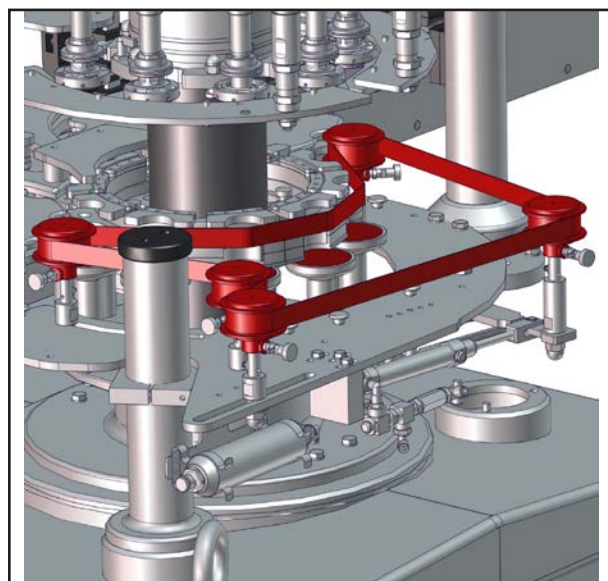


Fig. 286



- Turn the fasten spindles of 90° and put them in locking position.

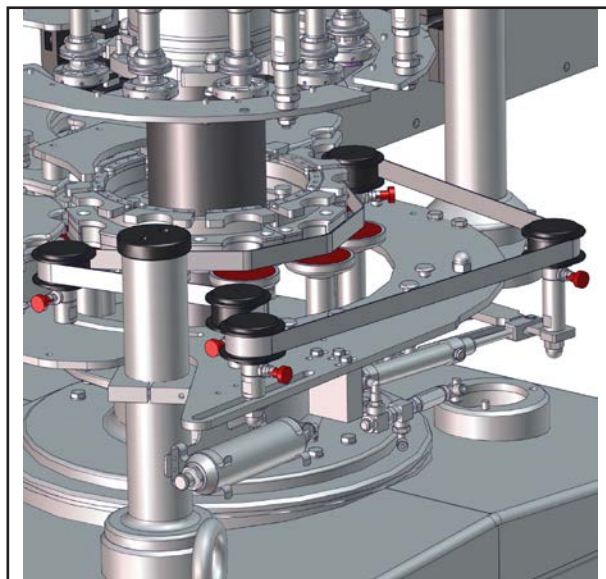


Fig. 287



11.5.8 MAINTENANCE OF THE "FLAT" CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

If there is an anomalous distribution, it is necessary to carry out the following operations:

- check that the nozzles located on the caps exit-mouth are not obstructed and are in the correct position to push the caps into the exit-mouth.

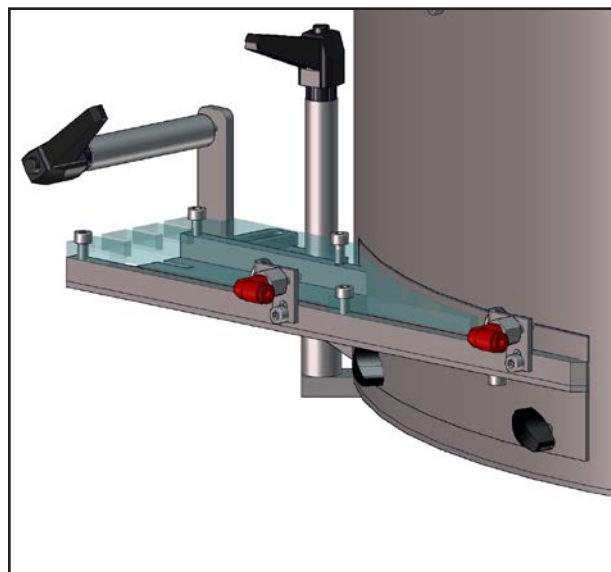


Fig. 288

- Check that the nozzles located on the hopper circumference are not obstructed.

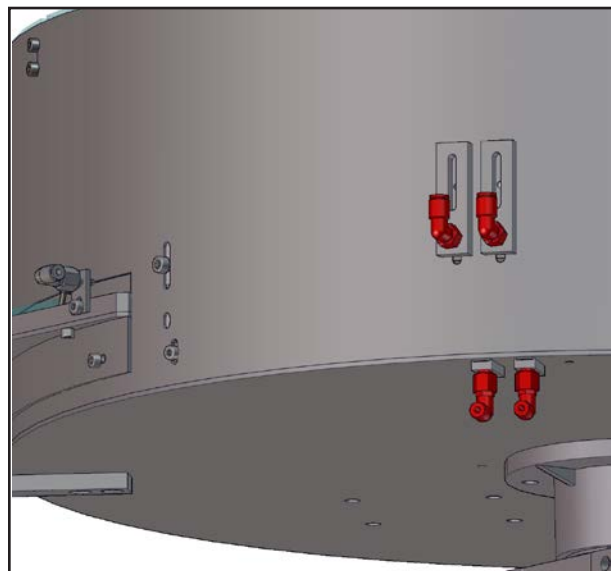


Fig. 289



11.5.9 SEW REDUCERS MAINTENANCE

11.5.9.1 INSPECTION INTERVALS AND MAINTENANCE

TIME INTERVAL	OPERATION
Every 3.000 machine hours, at least every 6 months.	Check oil and oil level.
	Check the seals visually for leakage.
	For gear units with a torque arm: checks the rubber pad and change it, if necessary.
Depending on the operating conditions, every 3 years at the latest. (*).	Change mineral oil.
	Change the bearings grease (recommended)
	Replace the sealing ring (do not install it the same track).
Depending on the operating conditions, every 3 years at the latest. (*).	Change synthetic oil.
	Change the bearings grease (recommended)
	Replace the sealing ring (do not install it the same track).
Variable (depending on external factors).	Touch up or renew the surface/anticorrosion coating.
(*) According to oil temperature.	

11.5.9.2 SEALING RING REPLACEMENT



De-energize the gearmotor and secure it to prevent it from being switched on inadvertently!



Reduction unit surface must allow to lean hands (danger of burn due to oil at high temperature)!

When replace the sealing ring, ensure that there is a sufficient grease reservoir between the dust lip and protective lip.

If you use double sealing rings, the space has to be filled 1/3 with grease.



11.6 LUBRICATION



11.6.1 GENERAL INFORMATION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



Before starting the closing machine, check the lubrication of all components described in the following subchapters.

If the closing machine is not used for a long time, lubricate it again.

Mineral oils, when not used for over six months, lose their properties and must be changed.

11.6.2 LUBRICANT TYPE-APPROVAL ACCORDING TO NSF

The type-approval according to NSF includes two categories: NSF - H1 and NSF - H2.

- **H1** indicates *Food grade lubricants*, that can thus be used in all friction points of machines and systems for the food and pharmaceutical industries, where occasional, and technically unavoidable, contacts can occur between lubricant and food.
- **H2** indicates lubricants that are recommended for the general use in the food and pharmaceutical industries, supposing that a contact with the food is totally to be excluded.



11.6.3 TYPES OF LUBRICANTS

AROL S.p.A., after careful studies and functional tests, recommends to use only the lubricants indicated in the table below to lubricate its machines:



The lubricants indicated in the table are classified **NSF-H1**.

SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
A		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
B		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth-PAO	AIK	220	265-295	-25	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
C		Paraliq P150		Synth+MO		150		-15	100
D		Klubersummit FG46		Synth-PAO	[130]	46		-45	135
		L0346FG		MO	[100]	46		-21	
		Purity AW 46		HT/MO	[108]	46		-15	
E		Purity FG Synth	2	Synth-PAO	Complex sulfonate calcium	46	294	-45	200
F		Klüberpaste UH1 84-201	1	Synth-PAO	PTFE	200			
		P1900	1	MO	AIK	85	290-340	-30	300
G		Barrierta L55 / 1	1	Synth-PFPE	PTFE	400		-40	260
		Krytox GPL 226	2	Synth-PFPE	PTFE [155]	240		-30	288
H		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295		
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
I		Purity MF Spray		HT/MO	[150]	151		-9	150



SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
I		Purity MF Spray		HT/MO	[150]	151		-9	150
J		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth-PAO	AlK	100	310-340	-40	150
K		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
L		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
M		Kluberylub NH1-11-222	2	MO	AlK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth-PAO	AlK	220	265-295	-25	120
		Kluberoil NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
N		Kluberoil 4 UH1-220N		Synth-PAO	[150]	220		-25	120
		L1122FG		Synth-PAO / MO	[142]	197		-33	
		Purity FG Gear Oil 220		HT/MO	[92]	206		-18	
O		G4501	1	Synth-PAO	AlK	100	310-340	-40	150
P		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
Q		Purity FG EP 150		MO	[95]	144		-18	247
R		Kluberoil NH1 4-220N Spray		Synth-PAO	[150]	220		-30	120
S		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
		Kluberoil NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
T		Cassida Grease GTS 2	2	Synth-PAO	Complex sulfonate calcium	46	265-295	-45	170
U		Kluberoil NH1 1-17		MO					



11.6.3.1 LUBRICANT RE-ORDERING

The lubricants used for the machine can be directly ordered to *AROL S.p.A.* or to the local distributors. See the table here below for the international addresses.

	http://www.klueber.com
	http://www.molykote.com
	http://www.petro-canada.ca
	http://www.krytox.com
	http://www.fuchs-oil.com

11.6.4 LEGEND FOR LUBRICATION SCHEMES

SYMBOL	DESCRIPTION
	Lubricate with manual pump for grease
	Lubricate with brush for grease
	Lubrication glass with disposable cartridge
	Lubricate with spray
	Lubricate with manual pump for oil
	Lubrication glass



11.6.5 POINTS TO BE LUBRICATED



To protect the closing machine from wear, seizure or other severe damage to the mechanisms, it is necessary to lubricate and grease periodically all indicated points.



AROL S.p.A. guarantees the maintenance intervals indicated hereby and the correct operation of the mechanisms exclusively with the recommended lubricants.

11.6.5.1 INSTRUCTIONS FOR READING THE "LUBRICATION POINT LIST" TABLE

The following pages show:

- the description table (*Lubrication point table*) relevant to time intervals, methods and used lubricants for the maintenance interventions on the closing machine herein.
- the position of the lubricating points (*Lubrication point scheme*) described herein.

The table "*Lubrication point list*" is designed to find immediately the operations to be carried out for the closing machine lubrication.

The "*Lubrication point scheme*" indicates the position of the points described in the table on the closing machine.

The points marked by a double frame indicate that the lubrication is carried out in a centralized way. Thus, the quantity of lubricant indicated in the table must be filled; then, the system will distribute the correct quantities to the different components.

POINT CODE	The column indicates the reference of the point to be lubricated, indicated on the closing machine next to the greaser and in the " <i>Lubrication point scheme</i> "
POINT DESCRIPTION	This column briefly describes the greasing point.
LUBRICANT TYPE	The reference letter for the lubricant to be used is indicated in this column. See the table for the lubricant type to be used.
METHOD	This column shows the tool by which the relevant greasing point must be greased.
INLET NUMBER [NO.]	It indicates the number of greasing points on the closing machine.
QUANTITY/POINT [gr.]	It indicates the lubricant quantity to be fed in each greasing point.
TOTAL QUANTITY [gr.]	It indicates the total lubricant quantity to be used.
INTERVAL [h]	It indicates every how many hours the different lubrication points must be greased.



After the lubrication of each points by grease, check that there are not leaks from seal gaskets or from the ditribution pipes.



11.6.5.2 LUBRICATION POINT LIST AND SCHEMES

LUBRICATION POINTS LIST

Order Ack.	S/N	Version	Reference	Typ	Model	P.D.	H.N.
	17478	0		M	EURO VP "I.E.S."	288	3
Point	Description	Lubr. Type	Lubr. Method	Number of points	Qty per point [g]	Total qty [g]	Lubr. Int. [h]
51A	Worm pin	B		1	5	5	250
2	Transfer disc bush	M		1	20	20	500
3	Heads rotation motor reducer	M		1	10	10	500
6	Central bearings or central slewing bearing	M		1	50	50	500
8A	Piston upper roller	A		3	6	18	500
8B	Piston lower roller	A		3	3	9	500
8C	Piston fork bearing	A		3	6	18	500
8D	Piston liner	A		3	15	45	500
16A	Lower central bearing or column slewing bearing.	B		1	30	30	500
16B	Upper central bearing or column slewing bearing.	B		1	30	30	500
52	Worm angular transmission	B		1	5	5	500
53	Reducer support	B		1	10	10	500
55	Star-wheel shaft (2 units)	B		2	25	50	500
4	Pistons controlling cam	M					1000
7	Sliding bush	M		1	5	5	1000
11	Heads rotation gearing	M					1000
19	Turret rotation gearing	B					1000
54	Bottles antirotation device	B		1	5	5	1000
56	Transmission gears	B					1000
1	Lifting screw	N		1	120	120	3000 *

Rotating point

* Value that depends on environmental conditions: check into use and maintenance manual

Fig. 290 Lubrication point table

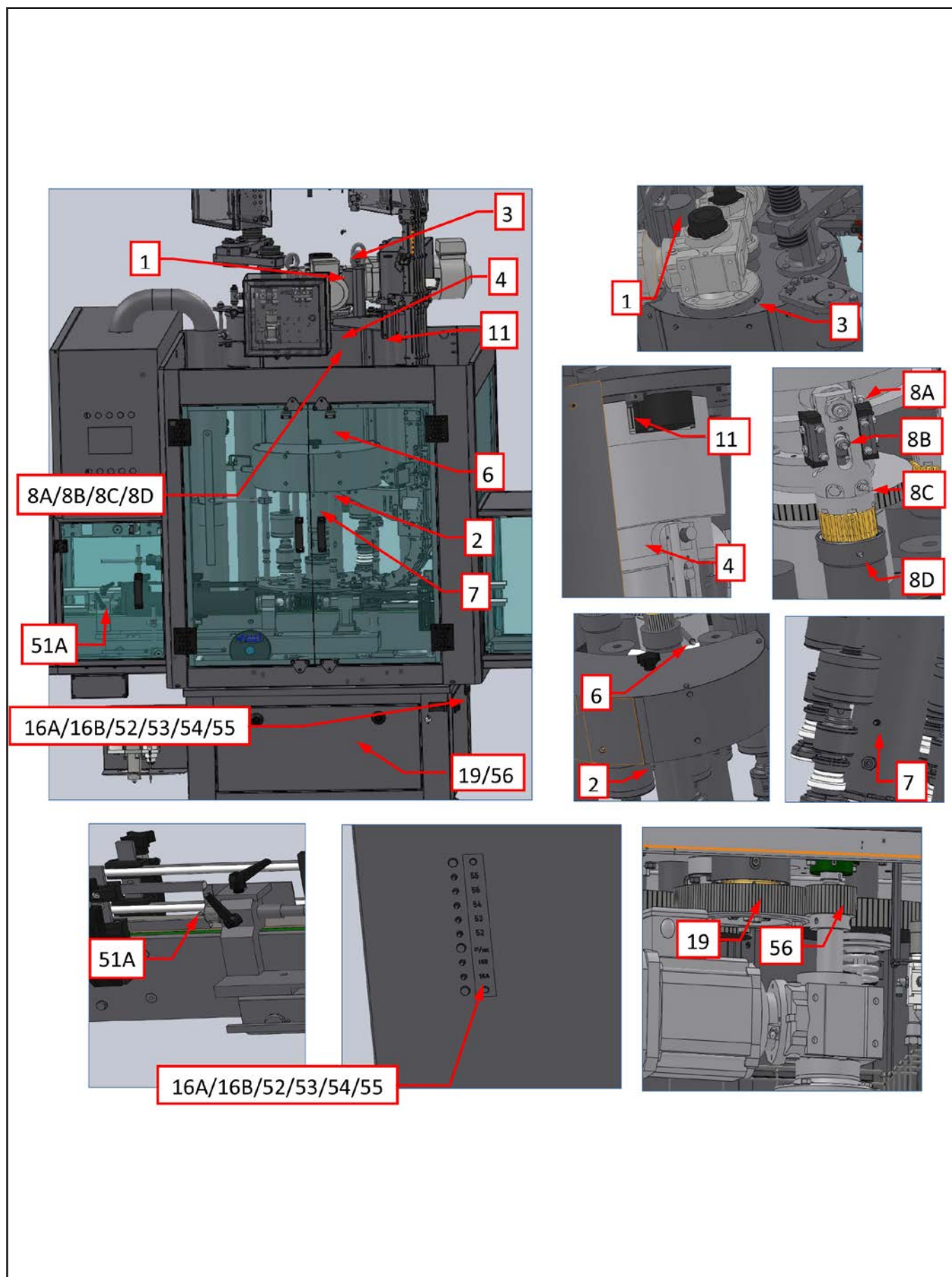


Fig. 291 Lubrication point scheme



11.6.6 LUBRICANT CARTRIDGE



At the closing machine delivery, the lubricant cartridge is not fit, thus it is not active. Before starting the closing machine, install and activate the cartridge.

11.6.6.1 LUBRICATOR DESCRIPTION

The cartridge of the lubricant is composed of:

- an adjustment screw (1).
- A body containing the lubricant (2).



The lubricant cartridge may provide lubrication intervals from 1 to 12 months. The interval is defined according to the alignment of the screw with the number on the ring nut of the cartridge.

AROL S.p.A. suggests for the lubrication of its machine to rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 292

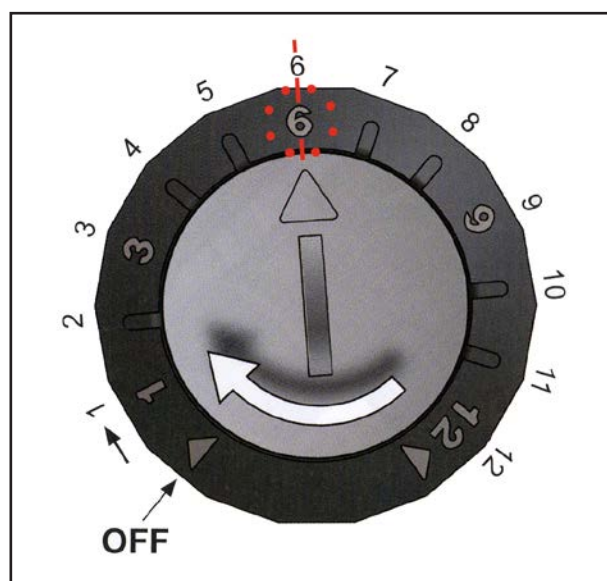


Fig. 293



During the supply period, the lubricator must not be removed to maintain the pressure inside it.



11.6.6.2 CARTRIDGE REPLACEMENT

When the cartridge must be replaced, operate as follows:

- Remove the plug from the discharge opening.
- Screw by hand the lubricating cartridge on the suitable coupling.

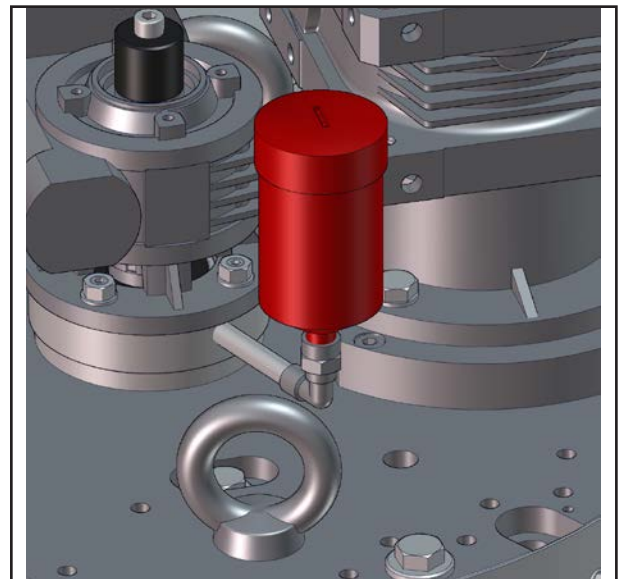


Fig. 294

- Rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 295

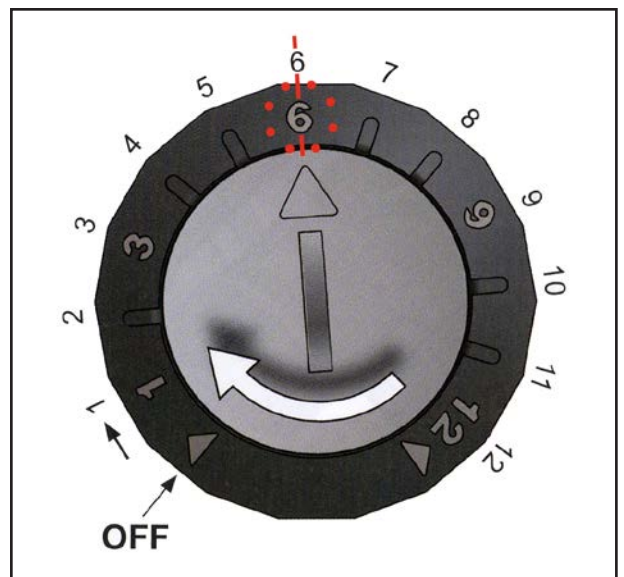


Fig. 296



11.6.7 CLOSURE GRIPPER LUBRICATION



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

Use lubricant "K" type.

11.6.7.1 "VP750" CLOSURE GRIPPER

- Clean the gripper holding bush and apply a layer of lubricant on the inner surface. Use a brush.

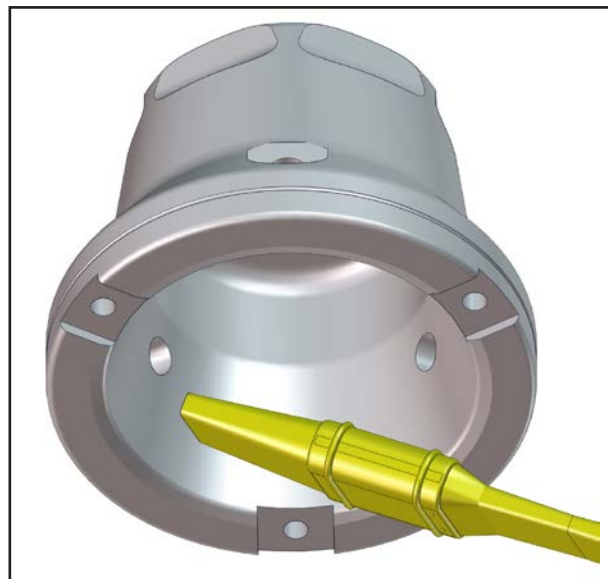


Fig. 297

- Apply a layer of lubricant on the gripper opening pin. Use a brush.

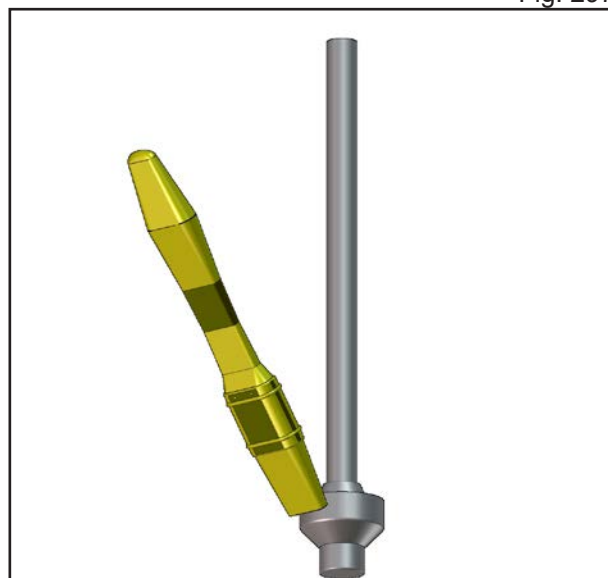


Fig. 298



- Apply a layer of lubricant on the surface of the cap gripper sectors. Use a brush.

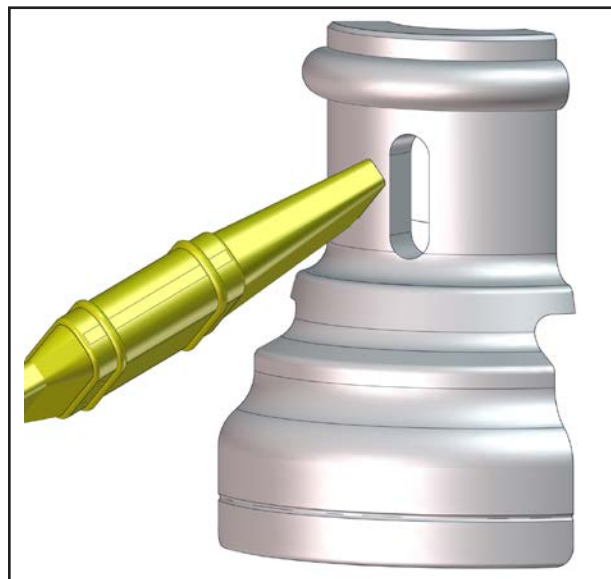


Fig. 299

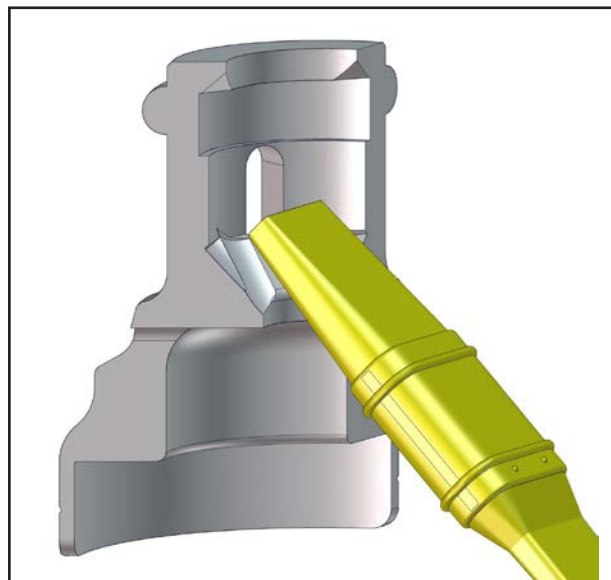


Fig. 300

- Apply a layer of lubricant on the gripper opening pin guide bush. Use a brush.

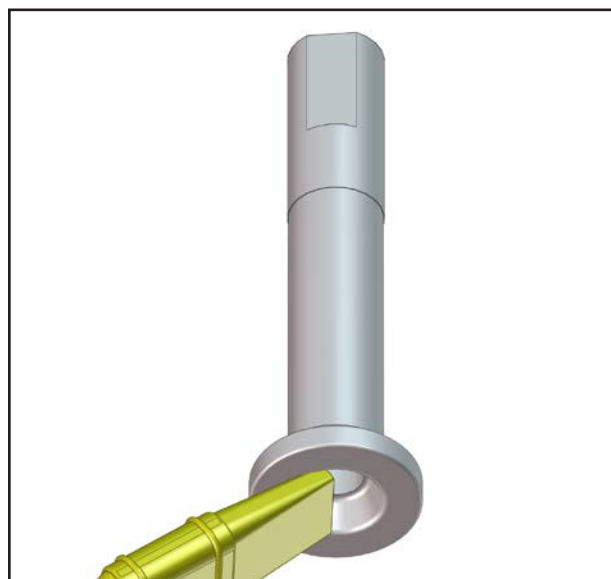


Fig. 301



- Apply a layer of lubricant on the pin guide bush with gripper reassembled. Use a brush.



Fig. 302

- Apply a layer of lubricant on the gripper control spring clamp and the remaining parts with gripper reassembled. Use a brush.

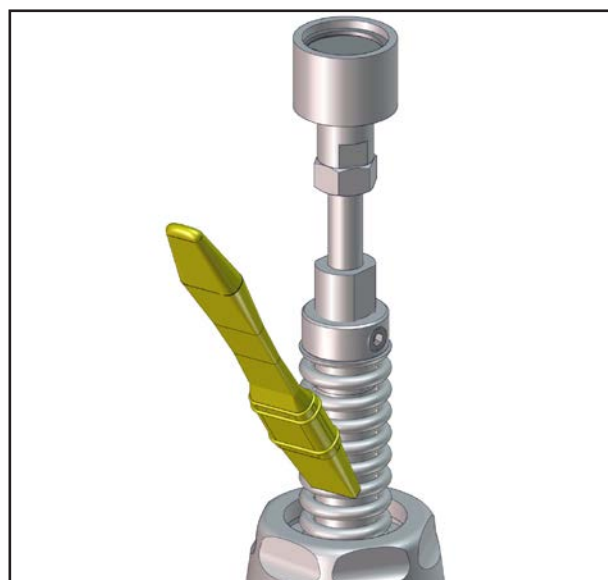


Fig. 303



11.6.8 SEW REDUCERS LUBRICATION



De-energize the gearmotor and secure it to prevent it from being switched on inadvertently!



Reduction unit surface must allow to lean hands (danger of burn due to oil at high temperature)!

11.6.8.1 GENERAL INFORMATION

The identification plate shown here next is applied on the ratiomotor. On this plate there are all the references and all the indispensable indications for its correct identification.



Do not intermix synthetic lubricants and do not mix synthetic and mineral lubricants together! The standard lubricant is mineral oil.

The reducers are supplied with a lubricant fill adapted for the specific reducer and mounting position. The decisive factor is the mounting position.



Fig. 304

11.6.8.2 CHECKING THE OIL LEVEL

For reducers with an oil level plug:

- remove the plug.
- Check the fill level.
- If necessary, top and screw the plug.

11.6.8.3 CHECKING THE OIL

Checking the oil, proceed as described below:

- Remove a little oil from the oil drain plug.
- Check the oil consistency (viscosity and color).
- If the oil is heavily contaminated, you may want to change it, even if you have not yet reached the end of the expected range (See sub-chapter 11.6.8.4 OIL REPLACEMENT page 226).



11.6.8.4 OIL REPLACEMENT



Carry out the oil replacement only when the reducer reaches the operating temperature, as the cold oil does not allow a correct emptying.

Carry out the oil replacement, proceeding in the following way:

- put a vessel under the discharge plug.
- Remove the level plug, drain plug with or without spring and discharge plug.
- Let the oil completely discharge.
- Screw the discharge plug.
- Pour in new oil of the same type through the vent hole.



Do not intermix synthetic lubricants and do not mix synthetic and mineral lubricants together! If necessary please contact the Technical Service.

- Pour in the volume of oil in accordance with the mounting position (See sub-chapter 11.6.8.7 LUBRICANT FILL QUANTITIES page 230) or as specified on the nameplate.
- Check oil and oil level. (See sub-chapter 11.6.8.2 CHECKING THE OIL LEVEL page 225)
- Screw the pipe plug with or without spring.



The exhausted oil must be disposed according to the environment protection norms.



11.6.8.5 LUBRICANT TABLE

11.6.8.5.1 LEGEND FOR LUBRICANT TABLE

The lubricant table on the following page shows the permitted lubricants for SEW-EURODRIVE reducers. Please note the following legend to the lubricant table.

Abbreviations used, meaning of shading and notes:

Abbrev.	Description
CLP	mineral oil
CLP PG	poliglicole (W reducer, conforms to USDA-H1)
CLP HC	synthetic hydrocarbons
E	Ester oil (water pollution danger category WGK 1)
HCE	synthetic hydrocarbons + ester oil (USDA-H1 certification)
HLP	hydraulic oil
SYNTHETIC	synthetic lubricant (= synthetic anti-friction bearing grease)
MINERAL	mineral lubricant (= mineral-based anti-friction bearing grease)
(1)	helical-worm gear units with polyglycols oil. Please contact SEW EURODRIVE
(2)	special lubricant for Spiroplan® gear units only
(3)	we recommend SEW EURODRIVE $f_b \geq 1..2$
(4)	pay attention to critical starting behavior at low temperature!
(5)	low-viscosity grease
(6)	ambient temperature
	lubricant for the food industry (compatible with food products)
	biodegradable oil (lubricant for use in agriculture, forestry and water resources)



11.6.8.5.2 BEARINGS GREASES

The bearings of the bearings and motors are given a factory-fill with the greases listed below.

SEW-EURODRIVE recommends regreasing bearings with a grease fill at the same time as changing the oil.

	ambient temperature	Manufacturer	Type	
Bearing for reducers	-20 °C ... +60 °C		Mobilux EP 2	MINERAL
	-40 °C ... +80 °C		Mobiltemp SHC 100	SYNTHETIC
Bearing for motors	-20 °C ... +80 °C		Unirex EQ3	MINERAL
	-20 °C ... +60 °C		Alvania RL3	MINERAL
	-45 °C ... -25 °C		Aero Shell Grease 16	SYNTHETIC
	+80 °C ... +100 °C		Barrierta L55/2	SYNTHETIC
Special greases for bearings for reducers:				
	-30 °C ... +40 °C		Eural Grease EP 2	SYNTHETIC
	-20 °C ... +40 °C		Aralube BAB EP2	SYNTHETIC



The following grease quantities are required:

- for fast-running bearings (motor and reducer input side): fill the cavities between the rolling elements one third full with grease.
- For slow-running bearings (in reducer and at reducer output side): fill the cavities between the rolling elements two thirds full with grease.



11.6.8.6 LUBRICANTS CHARACTERISTICS TABLE

 S... (HS...)							R..., K... (HK..., F..., S... (HS))		
Temperature °C									
	SYNTHETIC	MINERAL			SYNTHETIC	MINERAL			
	standard 0...+40	-20...+60	(4) -30...+80	(4) -40...+10	-20...+10	-25...+20	(6) -40...0	(6) -30...+40	-20...+40
 DIN (ISO)	CLP (CC)	CLP PG	CLP HC		CLP (CC)	CLP PG	CLP HC	HCE 	E 
ISO; NLGI	VG 680	(1) VG680	VG 460	VG 150	VG 150 VG 100	(1) VG 220	VG 32	VG 460	VG 460
	Mobilgear 636		Mobil SHC 634	Mobil SHC 629	Mobilgear 627	Mobil Glygoyle 30	Mobil SHC 624		
	Shell Omala 680	Shell Tivela S 680	Shell Omala HD 460	Shell Omala HD 150	Shell Omala 100	Shell Tivela S 220		Shell Cassida Fluid GL460	
	Klüberoil GEM 1-680	Klübersynth GH 6-680	Klübersynth EG 4-460	Klübersynth EG 4-150	Klüberoil GEM 1-150	Klübersynth GH 6-220	KlüberSummit HySyn FG-32	Klüberoil 4UH1 -460N	Klüberbio CA2 460
	Aral Degol BG 680				Aral Degol BG 100			Aral Eural Gear 460	Aral Degol BAB 460
	BP Energol GR-XP 680	BP Energyn SG-XP 680			BP Energol GR-XP 100				
	Tribol 1100/680	Tribol 800/680			Tribol 1100/100	Tribol 800/220			
	Meropa 680	Synlube CLP 680	Pinnacle EP460	Pinnacle EP150	Meropa 100	Synlube CLP 220	Cetus PAO 46		
	Optigear BM680				Optigear BM100	Optiflex A 220		Optileb GT 460	Optisynt BS 460
	Renolin CLP 680				Renolin CLP 150				



11.6.8.7 LUBRICANT FILL QUANTITIES

The specified fill quantities are **guideline values**. The precise values vary depending on the number of stages and gear ratio. When filling, it is essential to check the **oil level plug since it indicates the precise oil capacity**.

The following tables show guide values for lubricant fill quantities in relation to the mounting position M1... M6.

S...						
Gear-box type	Lubricant quantity in liters					
	M1	M2	M3(1)	M4	M5	M6
4,49	0,25	0,40	0,50	0,55	0,40	0,40
S47	0,35	0,80	0,70 / 0,90	1,00	0,80	0,80
S57	0,50	1,20	1,00 / 1,20	1,45	1,30	1,30
S67	1,00	2,00	2,20 / 3,10	3,10	2,60	2,60
S77	1,90	4,20	3,70 / 5,40	5,90	4,40	4,40
S87	3,30	8,10	6,90 / 10,4	11,30	8,40	8,40
S97	6,80	15,00	13,40 / 18,00	21,80	17,00	17,00

SF...						
Gear-box type	Lubricant quantity in liters					
	M1	M2	M3(1)	M4	M5	M6
SF37	0,25	0,40	0,50	0,55	0,40	0,40
SF47	0,40	0,90	0,90 / 1,05	1,05	1,00	1,00
SF57	0,50	1,20	1,00 / 1,50	1,55	1,40	1,40
SF67	1,00	2,20	2,30 / 3,00	3,20	2,70	2,70
SF77	1,90	4,10	3,90 / 5,80	6,50	4,90	4,90
SF87	3,80	8,00	7,10 / 10,10	12,00	9,10	9,10
SF97	7,40	15,00	13,80 / 18,80	22,60	18,00	18,00

SA..., SH..., SAF..., SAZ..., SHA...						
Gear-box type	Lubricant quantity in liters					
	M1	M2	M3(1)	M4	M5	M6
S...37	0,25	0,40	0,50	0,50	0,40	0,40
S...47	0,40	0,80	0,70 / 0,90	1,00	0,80	0,80
S...57	0,50	1,10	1,00 / 1,50	1,50	1,20	1,20
S...67	1,00	2,00	1,80 / 2,60	2,90	2,50	2,50
S...77	1,80	3,90	3,60 / 5,00	5,80	4,50	4,50
S...87	3,80	7,40	6,00 / 8,70	10,80	8,00	8,00
S...97	7,00	14,00	11,40 / 16,00	20,50	15,70	15,70

(1) In the combined reducers, the largest reducer should be filled with amount of lubricant greater.



11.7 EXTRAORDINARY MAINTENANCE

11.7.1 SPINDLES CONTROL CAM WEAR CHECK



To check the piston control cam race wear proceed as follows:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Remove the safety guard.
- Check the conditions of the spindles control cam in the cap application point. The worn part is usually the roller sliding race; if the wear is such to form a difference over 1 mm the cam must be replaced.
- Check the wear along the whole sliding race and above all in the area where the piston reaches its bottom dead center.

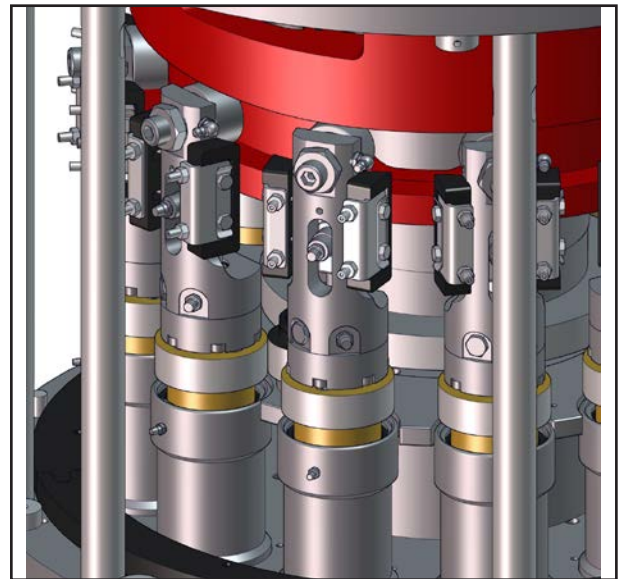


Fig. 305

To replace the cam, contact the *Technical Service*.



11.7.2 REPLACEMENT OF THE BOTTLE SUPPORTING PLATES

To replace the bottle supporting plates, proceed as follows:

- insert a flat head screwdriver or a similar tool between the plate and its support.

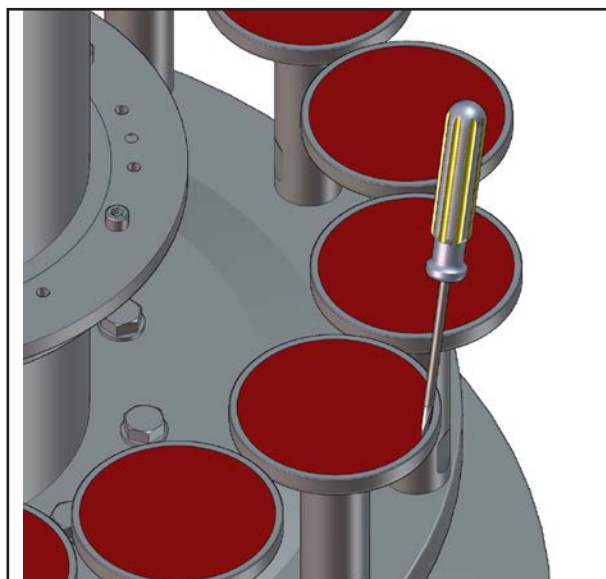


Fig. 306

- Prise the plate out of the support using flat head screwdriver or similar tool.
- Complete the plate removal from the support.

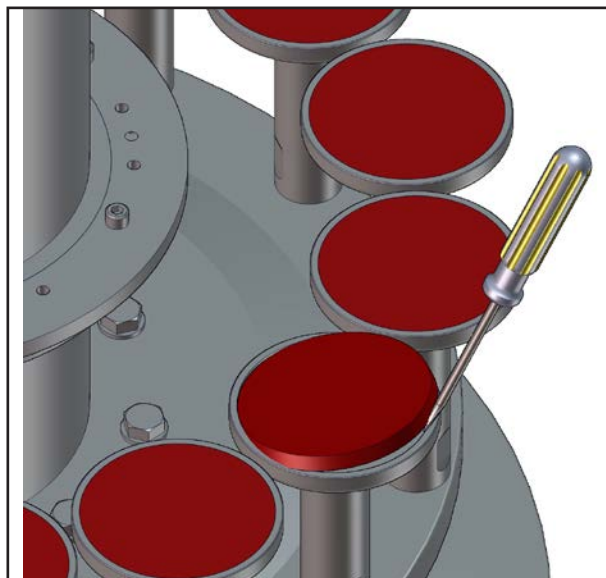


Fig. 307

- Position the new plate in the support seat.

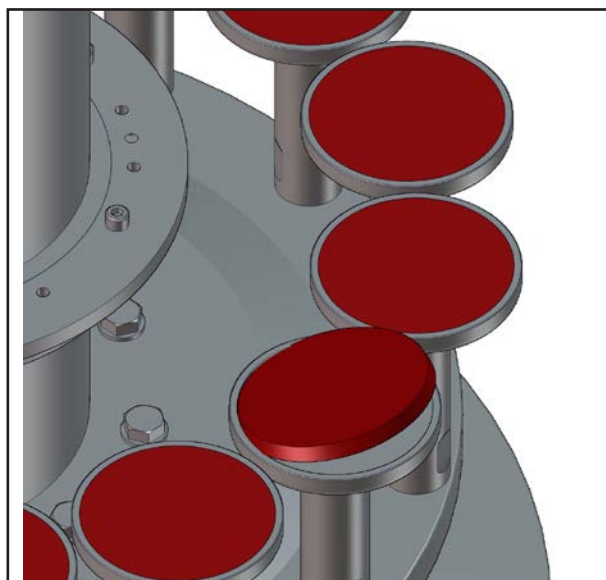


Fig. 308



- Complete the plate insertion inside the support, using a rubber hammer and beating on the plate side

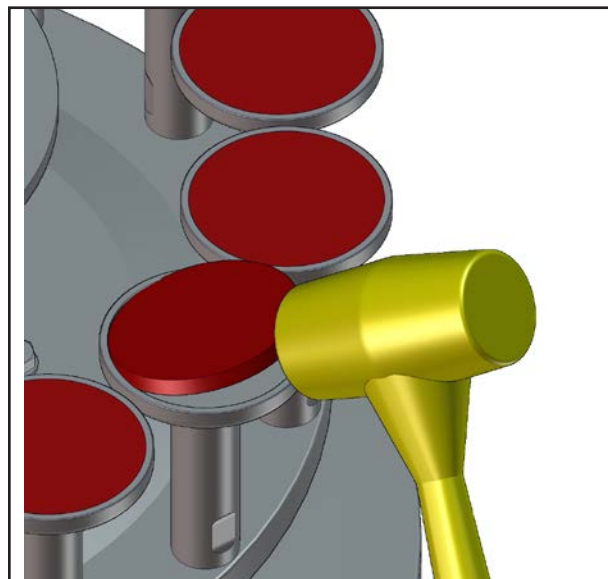


Fig. 309



11.7.3 TIMING OF THE PHASE SENSOR OF THE TRANSFER DISK



To obtain a correct distribution of the caps must be positioned in an appropriate way the phase sensor.

Is sufficient, acting with the jog, bring the closing machine in a situation where the tip of the transfer star-wheel is located 1 mm beyond the central axis of the caps chute.

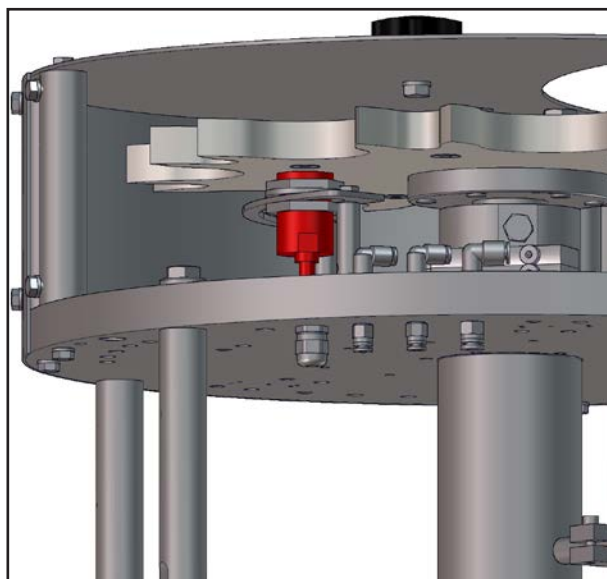


Fig. 310

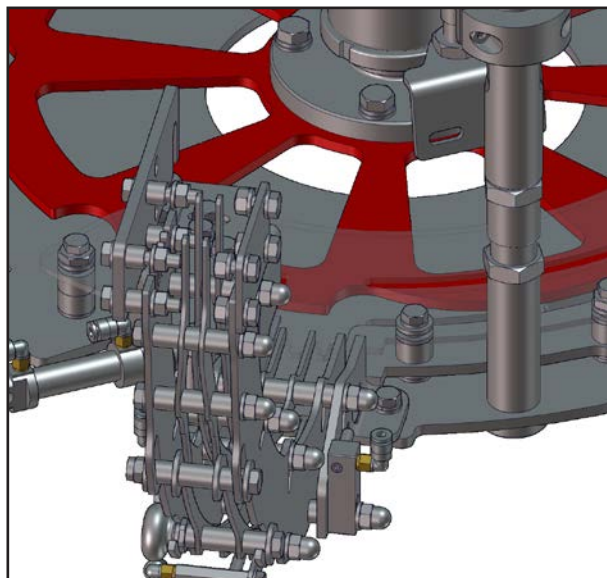


Fig. 311

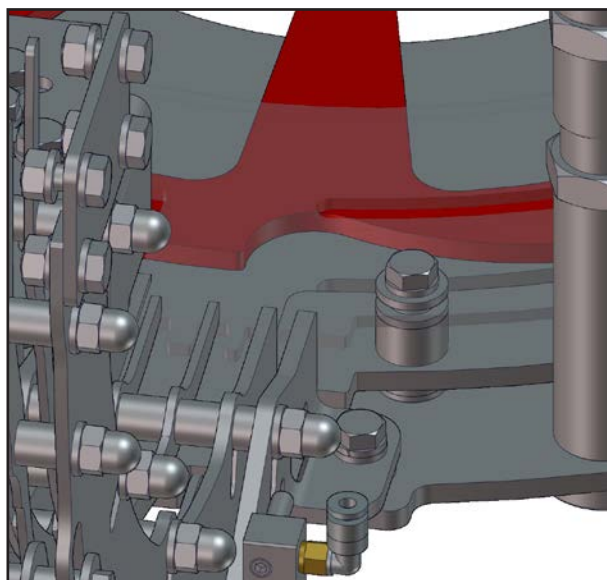


Fig. 312



Now, adjust the timing sensor so that it is at the start of the reading of the screw embedded in the gear of the transfer disk (At the reading start, the sensor LED lights up.).



Remembers that in case of continuous flow of bottles the solenoid valve must remain energized.

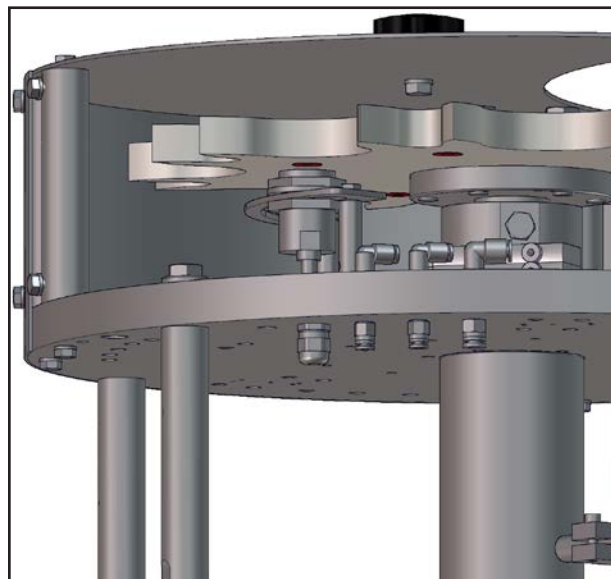


Fig. 313



11.7.4 TRANSPORT EQUIPMENT TIMING



11.7.4.1 ALIGNMENT OF THE TRANSFER STAR-WHEEL WITH THE CLOSURE HEADS

If the transfer star-wheel and the closure heads are not perfectly aligned, the complete transfer disk must be timed, operating as follows:

- Set the switch *AUT/MAN* in the position *MAN*.
- Make the closing machine jog pressing the outer *JOG* button, till a cavity of the central star-wheel is aligned with the closing machine center and the closure head center.
- If the centers are not aligned:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Remove the top cover of the transfer disk.



Fig. 314

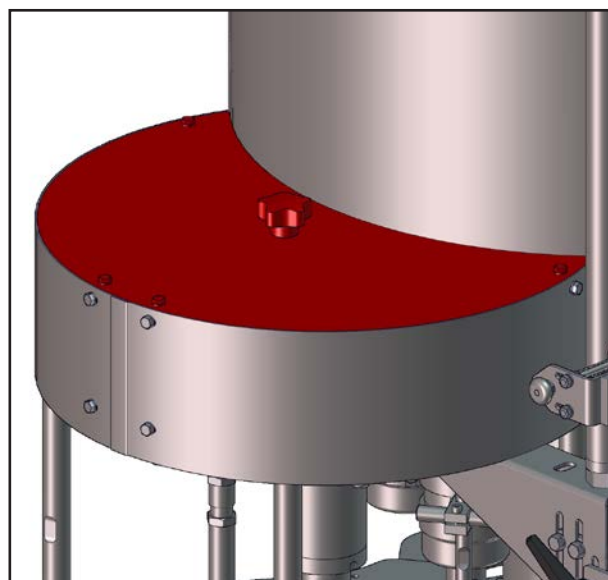


Fig. 315

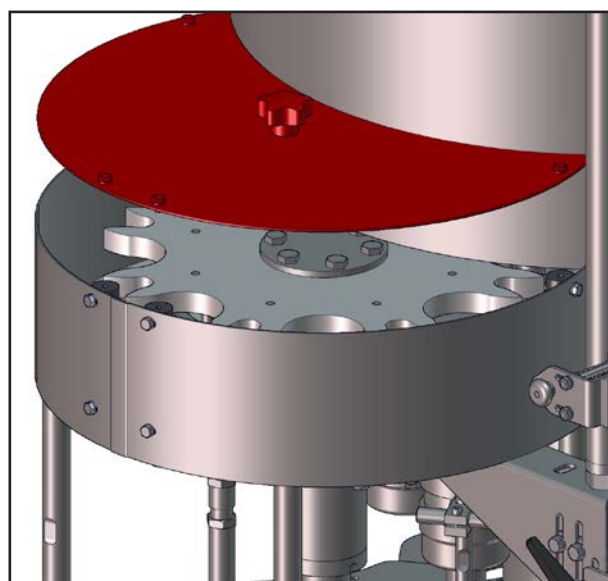


Fig. 316



- Loosen the bolts that secure the disk rotating gear.

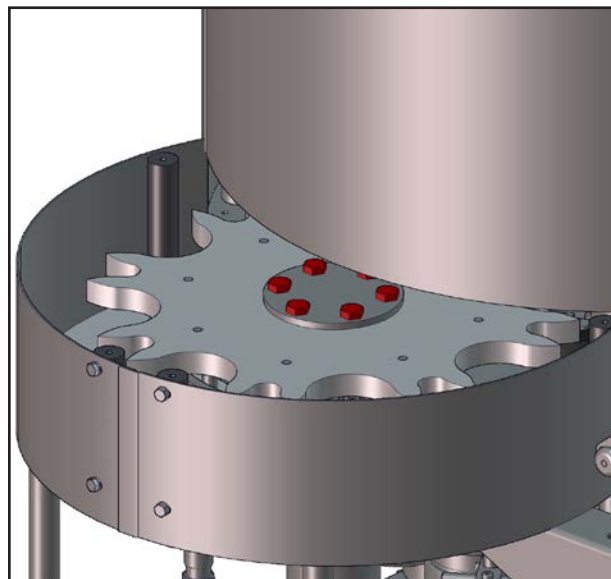


Fig. 317

- Rotate the caps transfer star-wheel till getting the required alignment.
- Strongly lock the bolts that secure the disk rotating gear.

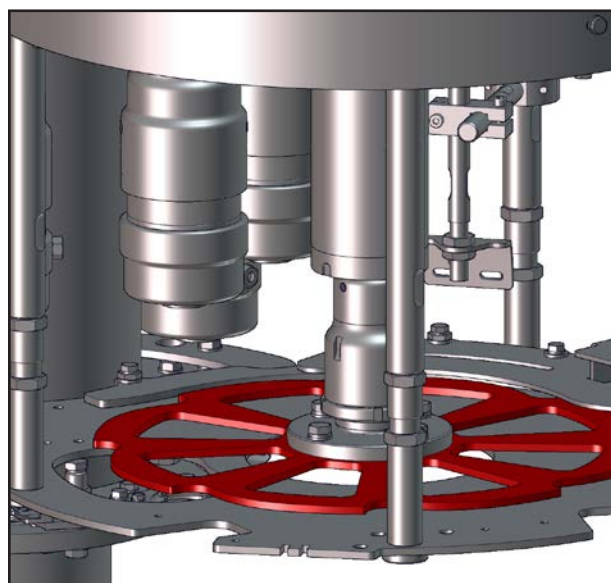


Fig. 318

- Replace the top cover of the transfer disk.

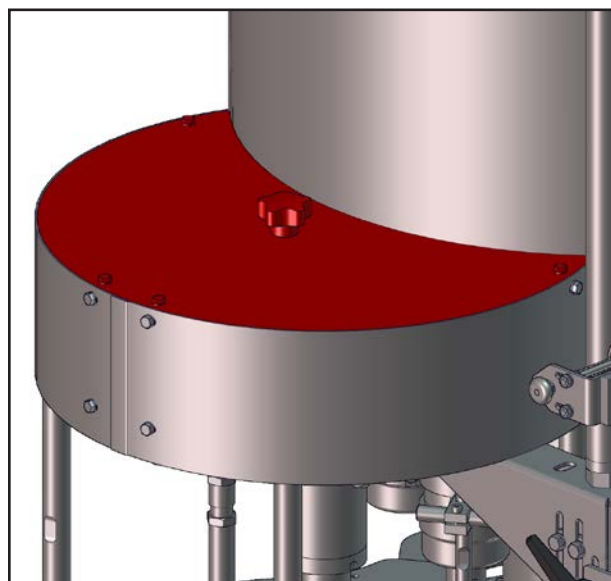


Fig. 319



11.7.4.2 ALIGNMENT BETWEEN BOTTLE CENTERING DEVICE AND CENTRAL STAR-WHEEL

The centering device must always be aligned with a cavity of the central star-wheel.

If the bottle centering device must be axially aligned with the star-wheel cavity, operate as follows:

- Set the switch *AUT/MAN* in the position *MAN*.
- Fit a bottle in the centering device cavity and then in the central star-wheel cavity.
- Check the vertical alignment between centering device axis and star-wheel axis.

- If the alignment is not correct:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- loosen the fixing screws.
- Manually rotate the star-wheel support complete with star-wheel, till getting a correct alignment between a centering device cavity and a star-wheel cavity.
- At the end, tighten the fixing screws strongly again.
- Remove the bottle.
- Close interlocked mobile guards and restore the operation of the safety circuit.
- At the adjustment end, turn the switch *AUT/MAN* to *AUT* and remove the key.



Fig. 320

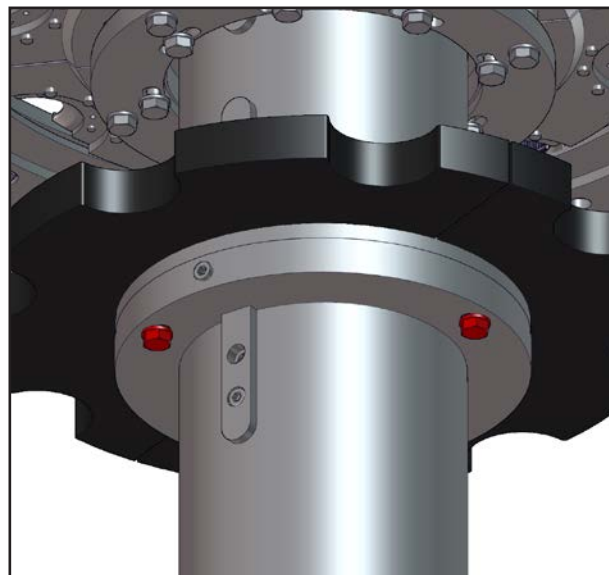


Fig. 321

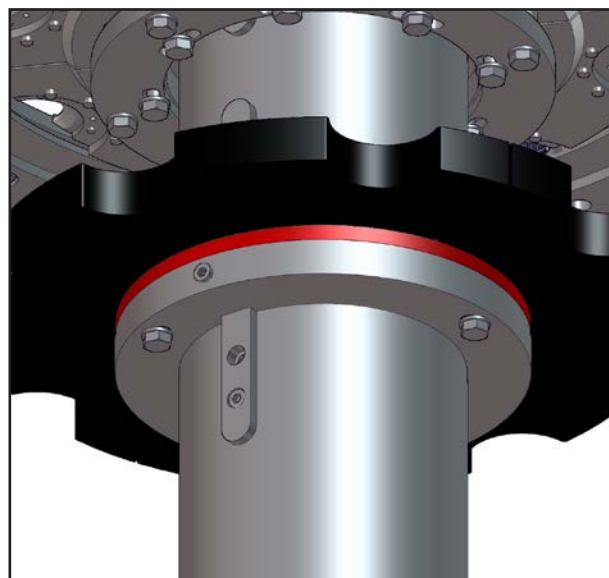


Fig. 322



11.7.4.3 TIMING OF THE ENTRY OR EXIT STAR-WHEEL WITH THE CENTRAL STAR-WHEEL

If the entry or the exit star-wheel is not timed with the central star-wheel, it is necessary:

- Set the switch *AUT/MAN* in the position *MAN*.
- Check that the exit and entrance star-wheel friction device is not inserted; if necessary, proceed as follows:
 - Position again the entry or exit star-wheel, rotating it manually around itself till the clutch device returns to its original seat (a click is heard).
- Close interlocked mobile guards and restore the operation of the safety circuit.
- Make the closing machine jog pressing the outer *JOG* button, till a cavity of the central star-wheel is aligned with the imaginary junction line between central star-wheel and star-wheel to be timed.
- Check that the cavity of the star-wheel to be timed is on the same axis of the cavity of the central star-wheel.



Fig. 323

If the star-wheel is not in the correct position after these steps, it is necessary:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Holding the star-wheel still to prevent it from rotating, loosen the locking bolt.
- Rotate the star-wheel till the cavity is on the same axis of the cavity of the central star-wheel.

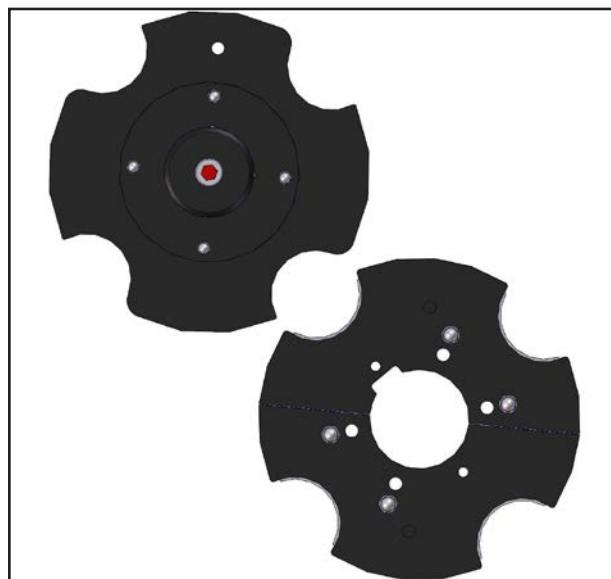


Fig. 324



- Fitting a bottle manually, check that it rotates freely in the seats formed by the cavities of the star-wheels.



Fig. 325

- Remount and screw the fixing central bolt.



Fig. 326

- Close interlocked mobile guards and restore the operation of the safety circuit.
- At the adjustment end, turn the switch *AUT/MAN* to *AUT* and remove the key.



- Strongly lock the fixing bolt that was previously loosened remount the star-wheel fixing bush and the fixing knob.

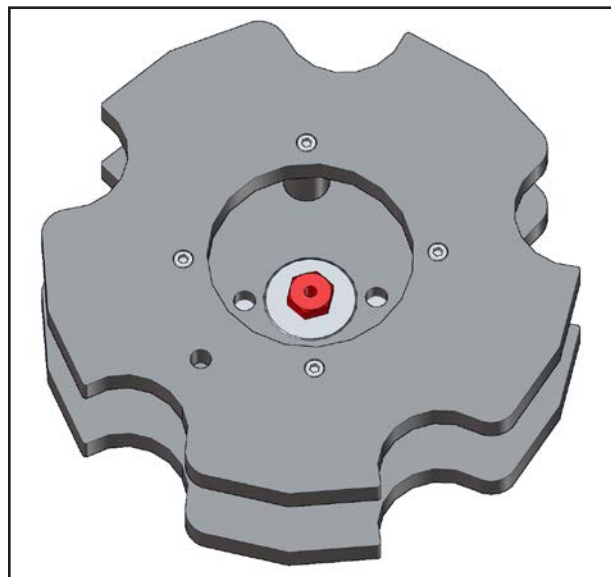


Fig. 327

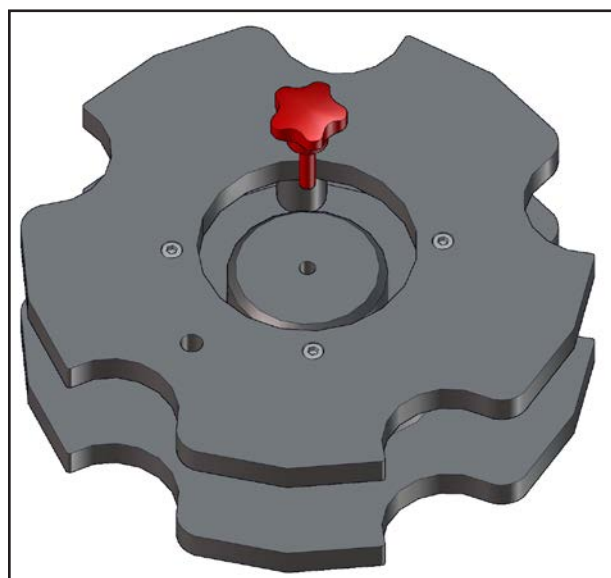


Fig. 328

- Close interlocked mobile guards and restore the operation of the safety circuit.
- At the adjustment end, turn the switch *AUT/MAN* to *AUT* and remove the key.



11.7.4.4 WORM TIMING



The worm is supplied already timed with the remaining bottles body guide equipment and therefore requires no further timing.

In case you need a timing of new worm, successively supplied as an accessory, it is necessary:

- Set the switch *AUT/MAN* in the position *MAN*.
- Press the outer *JOG* button, make the closing machine jog till a cavity of the star-wheel is aligned with the edge of the conveyor.

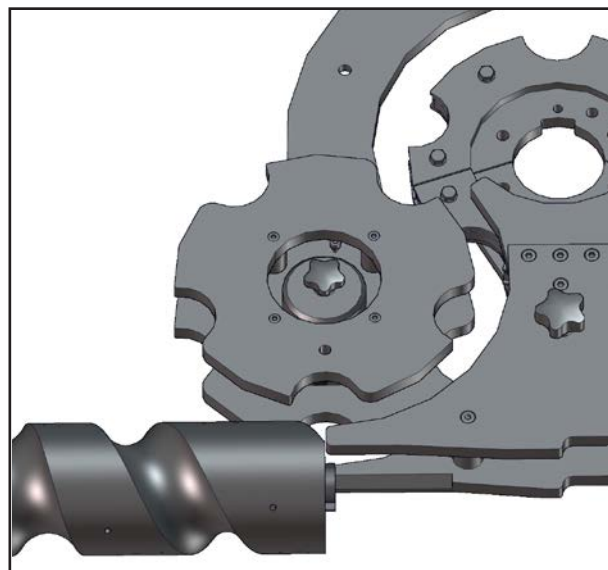


Fig. 329



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Loosen the fixing grub screws.



On the worm there are fastening grub screws, one of which with tapered point, to lock the worm aligned, while the others feature a flat point. To make it easy to recognize the tapered point grub screw, it is closed with a black plastic cap.



Fig. 330

- Manually rotate the worm till an exact bottle-holding cavity is obtained.
- Check the whole fitting a bottle in the cavity of the star-wheel. The container should not force either the worm or the star-wheel.

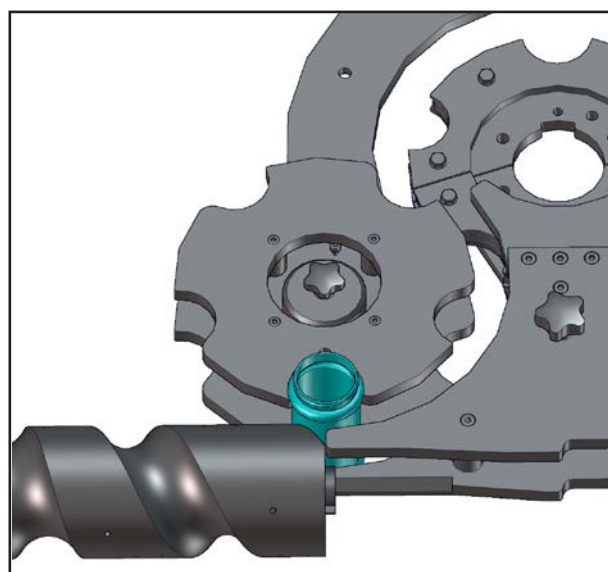


Fig. 331



- Lock all flat point grub screws.
- Check again that the container does not efforts.
- Remove the worm from the closing machine, taking care not to lose the timing.
- Remove the black plastic cap and unscrew the tapered point grub screw.
- Make an imprint for the tapered point grub screw, on the worm shaft by about 3 mm, drilling using a bit (column-type where feasible) and $\varnothing$ 6.5 bit.

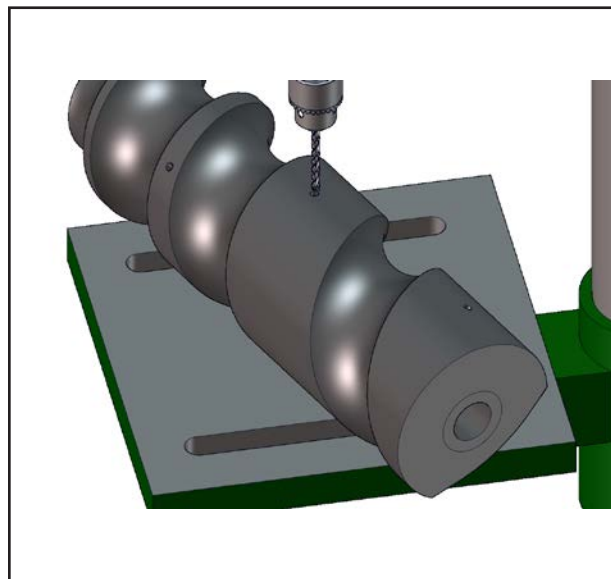


Fig. 332

- Tighten the tapered point grub screw in the suitable seat till the bottom.
- Replace the black plastic cap of the tapered point grub screw.
- Install again the worm on the closing machine and check again the correct timing.



If the timing is not correct, carry it out again making a new mark for the tapered point grub screw next to one of the holes housing a flat point grub screw.

- Close interlocked mobile guards and restore the operation of the safety circuit.
- At the adjustment end, turn the switch *AUT/MAN* to *AUT* and remove the key.



12 FAULTS AND REMEDIES

The closing machine has a red light for *SAFETY DEVICE ACTIVATION* and an operator panel which displays the error messages and the operation of safety devices.



For all faults not signalled by the red light indicator or by the operator panel start identifying the unit where such faults have occurred.

The list of the faults in the following pages is useful to detect as fastest as possible their reasons and possible solutions.

The research and the interventions must be carried out methodically (from the most probable to the less probable reason) and in logical sequence.

Proceed by steps: a first intervention, a test on the obtained result, then passage to the next intervention.

The list is referred to the closing machine. If the machine is synchronized with a line, refer to the manuals of the other machines composing the system.

The main closing machine units are:

- Motor-drive
- Containers transfer
- Closure unit
- Caps/corks distribution.
- Caps/corks feeding.



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



FAULT	CAUSE	REMEDY
The closing machine does not start or stops while operating.	The red <i>SAFETY TRIP</i> lamp is lit on the electric panel. In addition, the message related to the tripped safety appears on the display.	Check if the <i>EMERGENCY</i> push button is pressed. If so, reset it: Re-start the closing machine. Make sure that the interlocked movable guards are securely closed. Re-start the closing machine. Activation of the protections of the closing machine. Remove possible broken containers and restore the equipment timing. Close the Interlocked moveable guards Re-start the closing machine.
	Activation of the magneto-thermal protections of one or more motors. The message related to the tripped safety appears on the display.	Find the overload reason (seizure of the bearings or reducers, foreign matters in the teeth of the movement drive gears). Turn the main switch off, open the control panel and press the possible activated magneto-thermal switch. Close the panel and press again the main switch. Re-start the closing machine.
	The start push button has not been pressed.	Press the green <i>START</i> button.
The closing machine does not start or stops while operating.	No container at the entry.	Check the reasons hindering the container flow.  As soon as the containers arrive near the worm the closing machine starts. If the containers arrive normally to the entry, check the efficiency of the closing machine start photocells.
	Container accumulation at the exit.	Check the reasons hindering the container flow.  As soon as the clogging is cleared, the closing machine starts. If there is no clogging at the exit, check the closing machine stop efficiency.



FAULT	CAUSE	REMEDY
By pressing the  or  , the closure carousel does not move vertically.	The magneto-thermal relay of the closure carousel lifting motor has activated. The display shows the message on the activated safety device.	Identify the reason for the overload (interference with the container conveying equipment, foreign matters in the lifting mechanism, other). Turn the main switch off, open the control panel and press the possible activated magneto-thermal switch. Close the panel and press again the main switch. Re-start the closing machine.
	Interference with the container conveying equipment, foreign matters in the lifting mechanism, other.	Identify the reason for the overload (interference with the container conveying equipment, foreign matters in the lifting mechanism, other).
	The fastening screw of the sliding bush has not been loosened.	Loosen the fastening screw.
	Min. or max. limit of closing carousel vertical range reached.	Make sure the closing machine has been preset for the container size to be processed. If so contact <i>Technical Assistance Service</i> .
Pressing  , the closure carousel lowers instead of lifting.	Wrong sequence of the phases.	Reverse the sequence of the steps of the supply line as shown in subchapter 9.1.2 VERIFICATION OF CORRECT PHASE SEQUENCE on page 61.
Pressing  , the closure carousel lifts instead of lowering.		
The containers break or are badly guided in the worm.	Incorrect positioning of the container guide.	Adjust the position of the container guide.
The containers break or are badly guided in the loading step.	The loading star-wheel is not timed with the worm.	Position again the star-wheel, rotating it manually around itself till the clutch device returns to its original seat (a click is heard).
	The loading star-wheel is not timed with the central star-wheel.	Time the loading star-wheel with the central star-wheel and the worm with the loading star-wheel.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.



FAULT	CAUSE	REMEDY
The containers break in the closure step.	The closure carousel is too low.	Position the closure carousel at the correct height for the container to be processed.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.
The containers break or are badly guided in the unloading step.	The unloading star-wheel is not timed with the central star-wheel.	Time the unloading star-wheel with the central star-wheel.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.
Caps are not descending from the sorter.	The sorter exit-mouth is clogged.	Remove possible caps, cap parts or foreign matters. Check on the pressure gauges if the pressure of the air jets is correct.
	Caps are not correctly sized or are strained.	Use caps with the suitable size and even features.
The cap screws badly on the bottle.	Closure gripper not exact.	Use the gripper suitable for the cap to be processed.
	Cap fragments in the closure gripper.	Remove the fragments from the closure gripper inside.
	Bottle-neck guides not suitable for the bottle size to be processed.	Replace the guides in the next with those suitable for the bottle size to be processed.
	Broken compensating spring.	Replace the spring with a spare one. For the replacement see the relative sub-chapter of the closure head.
	Container threads/cap not compatible.	Check that the bottle to be processed and the used cap correspond exactly.



FAULT	CAUSE	REMEDY
The cap is not totally screwed.	Insufficient magnetic torque.	Increase the head tightening torque. For the adjustment see the relative sub-chapter of the capping head.
	The cap slips in the gripper.	Verify the wear of the gripper and eventually replace it with a spare one. Verify that the gripper is suitable for the cap to be processed, eventually replace it with a suitable one.
	The bottles rotate around themselves.	Check the wear of the centering device teeth.
	The bottles rotate around themselves.	Tension the belt. Check the pressure values on the pressure gauge and possibly check the air distribution unit. If the belt is worn, replace it
	Closure head springs not working.	Push the head body upwards. The elastic reaction of the compensating spring must be felt. For the replacement see the relative sub-chapter of the capping head.
Problems in the cap distribution.	The caps fall from the closure head.	Insufficient gripper pressure. Adjust the gripper pressure as described in relevant sub-chapter about the closure gripper. Worn cap gripper ring; replace it with a new one.
	The caps do not enter the closure heads.	Make sure the mode switch <i>AUT/MAN</i> is on <i>MAN</i> . Press the <i>START</i> button till the led of the sensor on the transfer disk switches on. Check that the container presence detection sensor at the entry of the conveyor belt detects their presence. The cap consent cylinder must retract; otherwise, check the control air pressure and the operation of the consent solenoid valve. Check the perfect alignment between transfer star-wheel and closure heads.



13 SCRAPPING AND DISPOSING



As there are in force different norms of observation, it is necessary to follow the prescriptions fixed by laws and authorities of the states where the closing machine will be installed.

Proper precautions must be taken for out-of-order and scrapping of the closing machine in order to avoid possible dangerous situations.

In particular, follow these general instructions:

- disconnect the closing machine from the electric network and check that no residual voltage exists.
- Empty all the lubrication points of oil paying attention in not dispersing and dispose it according the laws in forces in the country where the closing machine has been used.
- Lock all the moving parts of the closing machine, making sure that there are no possibilities of movement, also due to accidental collision during the transfer operation.
- With the help of qualified and authorized personnel proceed to the dismounting of components and the differentiated collection of materials (steel, plastic, electrical components, wires, etc.) in order to dispose everything according to the indications forseen by the laws in force.



Always use security clothes during the operations that require direct intervention on the closing machine.



CUSTOMER					DATE		
Closing Machine Model					Serial number		
Container dimensions					Container Type		
Cap dimensions					Caps Type		
Head static torque					Bottles/min		
Compensating spring kg.					Gripper/chuck type		
N° Head	DETECTED TORQUE					AVERAGE Lbs x inch	NOTES
	Lbs x inch	Lbs x inch	Lbs x inch	Lbs x inch	Lbs x inch		
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